Spatial scale space from hemigroup axioms: affine covariance and shape adaptation

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Abstract

In dimension $d \geq 2$ an aperture has a shape, which shape-adapted smoothing fits to the image. For the affine Gaussian scale space neither kernel nor cascade is a choice. Under hemigroup axioms both are: a family is given by a group G of linear maps it respects and an order W on scale matrices along which coarser records further smooth finer ones. The main result describes the families of three groups (general linear, similarities, positive diagonal) along the ray, stable and Loewner orders, completely in some cases, by bounds or examples in others. For the general linear group, an order with one step from a positive definite matrix that widens the aperture in some directions and leaves it unchanged in others admits only Gaussian families. Shape-adapted apertures at fixed least smoothing are such steps from the first stage. Among linear images of rotation-invariant laws with a positive transform only the Gaussian reaches them by further smoothing. Every other such kernel measures the image afresh there, as the iteration of shape adaptation does anyway. Along a chain of orders from the rays ($\alpha = 0$) to the Loewner order ($\alpha = \infty$), which allow faster changes of shape as α grows, a non-Gaussian kernel pays in speed of shape change. For $0 < \alpha, \gamma < 2$ linear images of the isotropic γ-stable law cascade along the order of index α exactly when $\gamma \geq \alpha$; from $\alpha = 2$ on only the Gaussian does. Linear images of a Matérn law, or of a law with a zero-free decaying rational transform in $|\omega|^2$, cascade at most along rays. With the similarities, non-Gaussian families cascade along the Loewner order. The results are machine-checked in Lean 4, assuming five published theorems.

1 Introduction

A scale-space representation measures an image through an aperture whose size is the scale. A measurement is convolution with a probability kernel, and the record at a coarser scale is computed from the record at a finer one by a further convolution, the *cascade*. In dimension $d \geq 2$ the aperture also has a shape. The affine Gaussian scale space of Lindeberg and Gårding makes the shape a parameter. Its kernel at the matrix Σ is the Gaussian with covariance Σ, the image is smoothed once for each Σ, and an invertible linear change B of the image coordinates moves the parameter by $\Sigma \mapsto B\Sigma B^{\mathsf{T}}$ ([39], eqs. (12)–(16), pp. 393–394; [40], §4). Its use is shape adaptation. The iteration of Lindeberg and Gårding fits the shape of the aperture at a point to the second-moment matrix of the image there, so that what is measured is the same for every affine image of a pattern ([39], pp. 395–396; [40], §§5.2 and 6.2). The nonlinear filter of Nitzberg and Shiota likewise shapes a Gaussian kernel at each point by the gradients of the image around it, to enhance edges and corners ([47], eqs. (3), (3') and (4), p. 827). The Gaussian is also what the classical axioms of scale space select ("Relation to scale-space theory" below). This paper asks which affine scale spaces exist under other axioms, incomparable with the classical ones (Remark 3.15), and what each of them can do for shape adaptation.

What is an affine scale space? In the affine Gaussian scale space the cascade needs no choice. The kernel is the Gaussian, and the Gaussian of covariance Σ' is the Gaussian of covariance Σ convolved with the Gaussian of covariance $\Sigma' - \Sigma$ whenever that difference is positive semidefinite. So every widening of the aperture is a further smoothing, and there is nothing to choose. Under other axioms the kernel need not be the Gaussian, and then the cascade becomes a choice. A rotation-invariant kernel of the kind the axioms below allow, a self-decomposable one in the sense defined below, can be carried to every elliptical aperture by the linear maps of the image. For $d \geq 2$ and a kernel other than the Gaussian, the family it gives cascades along every zoom, a change of size that keeps the shape, and never along every change of shape (Propositions 4.5 and 4.4(3)). So the name "affine scale space" leaves two things open: which changes of aperture must be further smoothings, and which linear maps of the image the family must respect. Several answers are reasonable. The changes of aperture can be the zooms alone or every widening, and the maps all invertible linear maps, the rotations with the dilations, or the independent rescalings of the axes of a sampling grid. The name already has more than one sense in the literature. Lindeberg's affine scale space is the family of Gaussians of every covariance matrix, closed under the invertible linear maps of the plane ([36], (15.17), p. 388). Weickert's affine Gaussian scale space is the Gaussians of covariance tD for one fixed matrix D ([63], §1.2.6, p. 13). The affine invariant scale spaces of Alvarez, Guichard, Lions and Morel and of Sapiro and Tannenbaum are nonlinear evolutions with a scalar time ([1], Thm. 5, p. 229; [56], §6, p. 32). This paper takes the name in Lindeberg's sense with the kernel left free, and its question is a classification over the two choices.

The two choices. In dimension d an aperture is indexed by a positive semidefinite matrix Σ, proportional to its covariance in the Gaussian case, which fixes its size and its shape together. A scalar scale is only one part of this parameter. A dilation $x \mapsto \lambda x$ of the image, $\lambda > 0$, moves Σ to $\lambda^2\Sigma$, along its *ray*, and changes the size of the aperture alone. The dilations form a subgroup of the invertible linear maps, and a general map A moves Σ to $A\Sigma A^{\mathsf{T}}$, changing the shape as well. A family has a kernel $\mu_{\Sigma,\Sigma'}$ for each *stage*, from an aperture Σ to a wider one Σ'. The two choices fix which stages there are and how they transform.

- The *covariance group* G is a closed group of invertible linear maps of $\mathbb{R}^d$ that contains the dilations. The family is covariant under it: for $A \in G$ the kernel of the stage from $A\Sigma A^{\mathsf{T}}$ to $A\Sigma' A^{\mathsf{T}}$ is the image under A of the kernel of the stage from Σ to Σ'. Every family commutes with the translations, so the affine maps of the image that it respects are those whose linear part lies in G. Covariance lets a measurement made at one shape be carried to another by the image's own change of coordinates.

- The *cascade order* W is a partial order $\leq_W$ on the positive semidefinite matrices that is invariant under G and relates each Σ to its multiples $c\,\Sigma$, $c \geq 1$, and 0 to every matrix. Its pairs $\Sigma \leq_W \Sigma'$ are the stages of the family, and along them the stages compose, $\mu_{\Sigma,\Sigma''} = \mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}$ for $\Sigma \leq_W \Sigma' \leq_W \Sigma''$. So W is the set of the changes of aperture along which the record at the coarser aperture is a further smoothing of the record at the finer one. Off W the family has no stage, so a change of aperture need not be a further smoothing. A change with no increment that is a probability measure is a *re-measurement* of the original image.

Two orders bound the ones studied here. The *ray order* relates Σ only to its multiples $c\,\Sigma$, $c \geq 1$, and 0 to every matrix, so a stage grows the aperture and keeps its shape. The *Loewner order* relates Σ to Σ' whenever $\Sigma' - \Sigma$ is positive semidefinite, so a stage shrinks the aperture in no direction. One definition covers every pair, the (G, W) cascade scheme of Definition 3.1: kernels $\mu_{\Sigma,\Sigma'}$, indexed by the pairs $\Sigma \leq_W \Sigma'$, that cascade along W, depend continuously on the pair, are not trivial from 0 to I, and are covariant under G. The aperture at Σ is the kernel $\mu_\Sigma := \mu_{0,\Sigma}$ of the stage from 0, the kernel *from scale zero*. The affine Gaussian scale space is

a family of the largest group, the general linear group $\mathrm{GL}(d)$, along the largest of these orders, the Loewner order.

The axioms. Scale-space axioms usually require the cascade to be a semigroup in one scale parameter, as those of Pauwels, Van Gool, Fiddelaers and Moons do [49], so that one parameter indexes the stages and the apertures alike. The axioms used here require only that the stages compose. With one scale parameter, as on the line, and $\mu_{s,t}$ the kernel of the stage from scale s to scale t, this is $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ for $r \leq s \leq t$, and how a stage depends on its two scales is left free. Such a family is a *convolution hemigroup* ([22], Def. 2.14.16, p. 380). On the line the axioms are those of the author's paper on the line [15], the *line paper* below (Definition 2.1). They require positive, mass-preserving stages that cascade, commute with translations and the reflection and depend continuously on the scales, and a dilation of the signal may only relabel the scales. Its sequel [17], the *cone paper*, describes the line's kernels further, and the *isotropic paper* [16] carries the theory to $\mathbb{R}^d$ for apertures that have a size and no shape. What is used from the three is restated in §2.

A kernel is a probability measure, and the classification on the line, like the study of the apertures that follows, rests on the theory of such measures in probability. So from here on the paper uses the vocabulary of that theory: a kernel is a *law*, and its Fourier transform, the transfer function of the smoothing, is the characteristic function of the law, its *transform* for short. Scale, stage and cascade keep their meaning, and "kernel" stays in use for a law in its role as an aperture or as a stage. On the line the families are known (Theorem 2.3). In a suitable parametrization of the scales, the kernels from scale zero are the dilations of one symmetric *self-decomposable* law. For every $0 < c < 1$ such a law is the convolution of its own image under $x \mapsto cx$ with another law ([58], Def. 15.1, p. 90). In a cascade this is the relation between two scales: the kernel at the finer scale is a contracted copy of the kernel at the coarser one, and the other law is the stage between them. Every such law other than the point mass occurs. Among them are the symmetric stable laws, with transforms $e^{-a|\omega|^p}$, $a > 0$, $0 < p \leq 2$, the Gaussian at $p = 2$ ([58], Thm. 14.14, p. 86), and the Matérn laws, with transforms $(1+\omega^2)^{-\gamma}$, $\gamma > 0$, which carry the name of the Matérn covariances of spatial statistics ([44], (2.4.5)–(2.4.7), pp. 17–18; [66], §9, pp. 447–448; [41], §2.1, eqs. (1)–(2), p. 426). Carried to every elliptical aperture by the linear maps of the image, a rotation-invariant self-decomposable law gives the family called its *affine orbit*. The orbits of the isotropic stable laws, with transforms $e^{-|\omega|^\alpha}$ of index $0 < \alpha < 2$, cascade along the changes of shape that are slow enough for the growth in size, and those of the Matérn laws along none (Propositions 4.7 and 4.17).

The classification. The main result of the paper is the classification of the families in dimension $d \geq 2$ over the two choices, Table 2 in §4.3. It takes three groups: $\mathrm{GL}(d)$; the similarities $\mathrm{S}(d)$, the rotations and reflections composed with the dilations; and the positive diagonal group $\mathrm{D}_+(d)$, which rescales the axes of a sampling grid independently. It sets them against three kinds of order: the ray order, the Loewner order and, between them, the *stable orders* $\leq_\alpha$, $0 < \alpha < 2$. Between positive definite matrices the stages of $\leq_\alpha$ are exactly the steps along which the isotropic stable law of index α, carried to every shape by linear maps, cascades (Proposition 4.7). For each group and each order the table says what is known of the families, and cites the statements that prove it. For the general linear group the description is complete along the rays, along the stable orders and along the Loewner order. In the other cells it is a bound, examples or a partial description (§5). The families are monotone in both choices. A family of a larger group, or along a larger order, restricts to a family of every smaller pair, with the same apertures (Lemma 3.3(1)). So the families grow as the group shrinks and as the order shrinks towards the rays, and several of these inclusions are strict. The affine Gaussian scale space is the answer for the largest group with the largest order. The rest of the table says what each relaxation buys, a smaller group or a smaller order. For shape adaptation its main

reading is the price of a kernel other than the Gaussian, kernel by kernel: along which changes of shape the record stays a further smoothing, and how fast the shape may change along them. The paper reads the table in three parts.

The general linear group with the Loewner order. For $d \geq 2$ this pair admits the affine Gaussian scale spaces alone, on kernels and on operators (Theorems 3.6 and 3.13). On the line the same axioms admit every symmetric self-decomposable law. One step decides it. The axioms make every aperture a linear image of the aperture at the identity, which is rotation invariant (Lemma 3.9). Call a step $\Sigma \to \Sigma'$ from a positive definite Σ *rank-deficient* when it widens the aperture in some directions and leaves it unchanged in the others, that is, when $\Sigma' - \Sigma$ is positive semidefinite, singular and not zero. The increment of one such step is a probability measure only for the Gaussian (Theorem 3.7). A linear map carries such a step to one from the identity, so for the general linear group every order of the scheme that contains one rank-deficient step admits only Gaussian families (Proposition 4.23(3)). The theorem takes a positive definite base, where a rank-deficient step is one with $\Sigma' - \Sigma$ singular. The base must not be 0: from $\Sigma = 0$ every step $0 \to P$ has the increment μ_P.

Shape adaptation takes such a step. In the iteration of Lindeberg and Gårding the first stage is isotropic, at wI, and the adapted aperture is $\Sigma_w = wM$, with M the shape matrix that the second-moment matrix of the image gives at the current pass, divided by its least eigenvalue ([39], p. 396, step 2; [40], §6.2). So Σ_w keeps the least eigenvalue w, and for $M \neq I$ the step $\Sigma_w - wI = w(M - I)$ is rank-deficient (Proposition 4.23(1)). Among the affine orbits of the rotation-invariant laws with a positive transform only the Gaussian reaches Σ_w by a further smoothing of the first stage (Proposition 4.23(2)). For every other kernel the record at Σ_w is a new convolution of the original image, which each pass of the iteration computes in any case (§4.1). A stable orbit reaches the adapted shape by a further smoothing only with a wider minor axis, and a Matérn orbit not at all (§4.8). So, as far as the apertures of the practice are concerned, a non-Gaussian kernel loses only the cascade between the stages of the iteration, which the iteration does not use. How the selection of the shape behaves under such a kernel is not treated (§5).

The general linear group from the rays to the Loewner order. This part carries the reading for shape adaptation: the price of a kernel other than the Gaussian, kernel by kernel. Between the rays and the Loewner order the kernels that keep a cascade decrease. Along the rays alone the families of the general linear group are the affine orbits of the rotation-invariant self-decomposable laws (Propositions 3.4(3) and 4.5). Read on the multiples of the identity they are, up to a change of the scale coordinate, the isotropic families of the isotropic paper (Proposition A.3). A chain of orders indexed by $0 \leq \alpha \leq \infty$ runs from the rays to the Loewner order, through the stable orders $\leq_\alpha$, $0 < \alpha < 2$ (Proposition 4.12). Along it the families of the general linear group shrink from all the affine orbits of the rays to the Gaussian (Theorem 4.21). For $0 < \alpha, \gamma < 2$ the orbit of the isotropic stable law of index γ cascades along the order of index α exactly when $\gamma \geq \alpha$, and from $\alpha = 2$ on only the Gaussian does. Each order limits how fast the shape may change as the scale grows, and Table 3 gives the limit for each family it lists. The orbits whose transform decays like a power of the frequency keep at most the rays, so every change of shape is a re-measurement for them (Proposition 4.15). The Matérn orbits are among them (Proposition 4.17). So are the orbits of a law whose transform is a decaying rational function of $\langle \omega, \Sigma\omega \rangle$ without zeros on $[0, \infty)$ (Proposition 4.18). These are the kernels computed by finitely many elliptic solves in the shape's operator $-\nabla \cdot \Sigma \nabla$.

The orders between the rays and the Loewner order are a question of matrix analysis and of the theory of Lie semigroups, and from §4.4 on the paper uses their words. An order invariant under $\mathrm{GL}(d)$ is read off the set of the relative steps $\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ that it allows (Proposition 4.2). A *cone field* assigns to each Σ the cone $\Sigma^{1/2}C\Sigma^{1/2}$ of allowed directions, for a closed convex

cone C of symmetric matrices invariant under rotations and containing no line. It generates the order in which two matrices are related when a curve that moves inside the cones joins them, as in the theory of Lawson and of Hilgert, Hofmann and Lawson ([34], Def. 1.6, p. 277, and Thm. 4.9, p. 290; [23], Def. VI.1.7, p. 493, and Thm. VI.1.16, p. 500). These are exactly the closed invariant partial orders whose steps are closed under fractional powers (Theorem 4.3), and the order of index α in the chain above is that of a cone C_α (Proposition 4.12).

Smaller groups. A smaller group admits more families, and several of the inclusions are strict (§4.9). Along the rays the isotropic members of every row are the self-decomposable laws its group allows. Along the general linear row they shrink to the Gaussian, while the similarity row keeps every rotation-invariant one in every column, the Loewner column included (Proposition F.3(4)), by a family that sees only the volume of the aperture. For the similarities with the Loewner order non-Gaussian families survive (Proposition H.1(3)). The exponent (§2.2) of their non-Gaussian part changes under a rotation by at most a factor of four times the eccentricity of the aperture, the ratio of its extreme eigenvalues (Proposition 4.25(2)). So the rigidity of the general linear group with the Loewner order needs both: the Loewner order alone does not force the Gaussian, and the general linear group alone does not either (Proposition H.1(3) and Theorem 4.21(3)). For the positive diagonal group a family that cascades along each axis is, at the diagonal apertures, a product of self-decomposable laws on the line, one for each axis (Proposition F.2). Such families gain anisotropy along the axes as fast as the Gaussian does and cannot turn. Along the rays a family of any group is a continuous covariant field of self-decomposable laws over the shapes (Proposition 3.4). The price of a smaller group is shape selection that transforms with the image, which needs a group that carries every shape to every other (§4.1).

Scope. The axioms are stated on operators and on their kernels, and for every pair (G, W) the two forms describe the same families (Lemmas A.1 and A.2). So each statement on the kernel families of a pair is one on its operator families. The isotropic theory in $\mathbb{R}^d$ is that of the isotropic paper, used through the six statements restated in §2.3, and five more cited where they are used. The paper is theory and makes no empirical claim about images. How a shape is selected, how the iteration behaves and singularity theory over the larger group are not treated, and §5 names the limits of the results. The computation of these families on grids is left to a further paper, in preparation.

Relation to scale-space theory. The affine Gaussian scale space is classical, and so is its derivation from axioms. Iijima derived it in 1962 and 1963, as Weickert, Ishikawa and Imiya report ([64], p. 241). His scale is a positive definite 2×2 matrix, his transformations are linear, translation invariant, positive and closed under every invertible linear map, and any two of them compose to a third. Lindeberg obtains the affine Gaussian scale space from a semigroup in a scalar scale with the non-enhancement of local extrema ([37], Thm. 5, p. 42). Theorem 3.6 rests on other axioms. Its cascade is asked for only from a matrix to a larger one in the Loewner order, and its increments need not belong to the family. It holds for every $d \geq 2$, and by Theorem 3.7 one rank-deficient step is enough. Neither earlier result implies it, and it implies neither of them (Remark 3.15). The morphological axiomatics of Alvarez, Guichard, Lions and Morel keeps the scale a scalar, and a linear map of the image is absorbed there by a change of that scalar ([1], p. 206). The α-scale spaces of Pauwels, Van Gool, Fiddelaers and Moons and of Duits, Florack, de Graaf and ter Haar Romeny admit the stable laws ([49], p. 694; [14], p. 277). The general linear group with the Loewner order removes them. For $\gamma < 2$ the orders of the chain up to index γ keep the stable law of index γ (Theorem 4.21). §3.5 sets the axiomatics side by side.

Anisotropy is often counted among the minor modifications of the Gaussian scale space, those that can be "easily accounted for" (Duits, Felsberg, Florack and Platel, [13], p. 494). For a kernel other than the Gaussian, §4 gives the cost of a change of shape. In the scale-space and vision work read for this paper, neither the order of a linear family nor its group is posed as a choice. Lindeberg states the order as the meaning of a larger matrix scale (§3.1), and in feature detection the group follows the application. Tuytelaars and Mikolajczyk choose between scale-invariant and affine-invariant detectors by the deformations expected, on experimental evidence ([61], §7.2, p. 258). For nonlinear evolutions the group is chosen. Olver, Sapiro and Tannenbaum classify the evolutions invariant under a prescribed group ([48], §4, p. 183), and Sapiro and Tannenbaum note that other groups give other scale spaces ([56], p. 36).

Relation to probability and to matrix analysis. In probability the core of Theorem 3.6 is Kelker's lemma, that a rotation-invariant law with independent coordinates is normal ([31], Lemma 5, p. 423). Through the singular matrix $I - e_1 e_1^{\mathsf{T}}$ the cascade makes the first coordinate of μ_I independent of the others. Rotation invariance makes every coordinate so, and Kelker's lemma then makes μ_I normal (Remark 3.14). Theorem 3.7 reads one rank-deficient step from a positive definite matrix alone. Its increment is any probability measure, and the transforms at its two ends need not be those of laws. The affine Gaussian scale space is also a convolution semigroup over the cone $\mathrm{PSD}(d)$, $\mu_\Sigma * \mu_{\Sigma'} = \mu_{\Sigma+\Sigma'}$. With the covariance Σ at Σ it is the canonical $\mathrm{PSD}(d)$-parameter convolution semigroup of Pedersen and Sato ([50], Def. 2.1 and the paragraph after it, p. 505; [51], Def. 3.5 and Ex. 3.6, p. 546). Their setting assumes additivity and has no covariance group. Within it, for $d \geq 2$, finite second moments and a covariance at most Σ at each Σ force Gaussian laws ([51], Prop. 5.4, p. 556), and a covariance equal to Σ leaves only the canonical semigroup, up to drift (Rem. 5.5, p. 557). That is a Gaussian rigidity under other hypotheses than those of Theorem 3.6: additivity over the whole cone, moments and a covariance bound, in place of covariance under $\mathrm{GL}(d)$ and the cascade. The scheme assumes neither additivity nor a semigroup, and for the general linear group with the Loewner order Theorem 3.6 derives the additivity, and the Gaussian with it, from covariance and the cascade. §4.6 also uses the operator-self-decomposable laws of Urbanik, in which the contraction $x \mapsto cx$ of self-decomposability is replaced by the flow e^{-tB} of a matrix B [62]. Its cones of such matrices belong to the setting of Jurek and Mason ([30], p. 85 and Prop. 3.3.6, p. 86). The paragraphs headed "Prior art" in §4.4 and §4.6 say what is taken from Lawson, from Hilgert, Hofmann and Lawson and from Jurek and Mason. The Matérn orbit is the anisotropic Matérn covariance of spatial statistics, in which the shape matrix is a parameter of the covariance ([41], §3.4, p. 437). No change of that matrix other than a rescaling is a further smoothing (Proposition 4.17).

Plan of the paper. §2 restates what is used from the line paper, the cone paper and the isotropic paper, and the five theorems of the literature that the proofs cite, the *cited facts*. §3 states the (G, W) scheme and treats the general linear group with the Loewner order. On the way it shows that the increments of a continuous cascade along every order with a path condition are *infinitely divisible*, each the n-fold convolution power of a law for every n (Theorem 3.5). §4 gives the classification. It says what an affine scale space is for (§4.1) and states the table of the families of each group along each order (§4.3). Then it treats the invariant orders, the steps an orbit survives, the general linear group along the chain of cones, shape-adapted smoothing and the smaller groups. §5 is the conclusion: it reads the results for shape adaptation and names their limits, and §5.1 collects the open problems. Appendices A–C hold the proofs deferred from §§3 and 4, and the complete proof of Theorem 4.21. Appendices D–H hold further results, among them the orientation-preserving group (Proposition D.1), the stable orbits under a tempering (Proposition E.1), the negative mass of a Matérn step across shapes (Proposition E.4) and the families tied to a frame of axes (Proposition F.1). What is machine-checked, and what it assumes, is said next, and Appendix I holds the detail.

1.1 What is machine-checked

Every definition, lemma, proposition and theorem of §§3–4 and of the appendices has been formalized in the Lean 4 proof assistant, on its mathematical library Mathlib, and its proof is machine-checked; so have the six statements restated from the isotropic paper in §2.3. This subsection says what the verified results rest on, and it can be read without the formal development. A proof marked as a sketch shows the decisive steps, and its complete proof is in the LaTeX sources deposited with this version (Appendix I lists these statements).

Five theorems of the literature are taken as given rather than reproved, four of probability theory and one of matrix analysis. They are the only assumptions of the results beyond the logical foundations of the proof assistant, and §2.4 states each with the page where it was read.

C1 Khintchine's theorem on the limits of null arrays (Theorem 2.10).

C2 The Lévy–Khintchine representation of the infinitely divisible laws on $\mathbb{R}^d$: (i) its existence, (ii) the uniqueness of the triplet and (iii) its converse (Theorem 2.11).

C3 The infinite divisibility of operator-self-decomposable laws with a contracting exponent (Theorem 2.12).

C4 An inequality between the eigenvalues of a product of two positive definite matrices and those of its factors (Theorem 2.13).

C5 The polar criterion, which describes the self-decomposable laws among the infinitely divisible ones by their Lévy measures (Theorem 2.14).

Two of them go beyond the letter of their sources, and §2.4 says how. C1 is stated for laws where its source states it for random variables. C3 joins the last sentence of a theorem of Jurek and Mason to the paragraph of their book that drops its hypothesis on the support of the law, and it reads two further things beyond those two passages. C4 states less than its source, the inequalities without the equality of the totals, which is $\det(MN) = \det M \det N$ and is proved. C2 and C5 are stated as their sources state them. Appendix I lists the statements that rest on each.

The statements of the line paper restated in §2.2, and the two restated inside the proof of Proposition A.3, are proved in its released version, which is cited by version and says what of it is machine-checked. The formal development requires Mathlib and the shared library ScaleSpaceCore, which makes no assumption of its own, and no formal development of the line paper. The formal proofs take two statements of the line paper from ScaleSpaceCore, where they are proved: the gauge of [15, Prop. 6.6], in the proof of Proposition A.3, and the necessity half of Theorem 2.3. ScaleSpaceCore states the latter with the uniqueness clause of C2 as a hypothesis, which this development discharges by C2(ii). The fact of [15, Lem. 6.3] is proved inside the proof of Proposition A.3. The six statements restated from the isotropic paper are proved in its released version, which says what of it is machine-checked, and are machine-checked in the same development, which the two papers share.

Two remarks report computations (Remarks E.5 and H.3), and three figures are drawn by scripts (Figures 1, 2 and 3). Each says that its script is deposited with the paper, and the statements are independent of the computed values.

The formal statements were compared with the printed ones by AI agents, in search of one weaker than printed, true vacuously or true only by a convention of the proof assistant, and the review found none. The statement on the use of AI at the end of the paper says how the work was done, and Appendix I says where the formal development is.

2 What is used from the earlier papers

This section restates, without proof, the statements of the line paper [15] and of the isotropic paper [16] that the later sections use in their statements or in more than one proof, with the notions of the cone paper [17] that they read, and states the five theorems of the literature that the proofs cite. The three papers are cited in their released version 0.1, by the statement numbers of that version. A restated statement carries its released number in its title, and its proof is at that number; a statement used once, inside one proof, is restated in that proof. With the restatements the statements of this paper can be read without the earlier papers, and its proofs rest on theirs. A reader who knows the earlier papers can go to §2.4.

2.1 Notation

As in §1, a probability measure is a *law*, and its transform is its characteristic function, $\widehat{\mu}(\xi) = \int e^{i\langle \xi, x\rangle}\, \mu(dx)$ for a law μ on $\mathbb{R}^d$. For $A \in \mathrm{GL}(d)$, $A_*\mu := \mu \circ A^{-1}$ is the image measure, with $\widehat{A_*\mu}(\xi) = \widehat{\mu}(A^{\mathsf{T}}\xi)$. On functions A acts by $D_A f := |\det A|^{-1} f(A^{-1}\,\cdot)$, which preserves the integral, so that $D_A(\mu * f) = (A_*\mu) * D_A f$. On the line the dilation by $\lambda > 0$ is $D_\lambda f = \lambda^{-1} f(\cdot/\lambda)$, the translation by a is $\tau_a f = f(\cdot - a)$ and the reflection is $Rf = f(-\,\cdot)$. $\mathrm{Sym}(d)$ is the space of real symmetric $d \times d$ matrices, $\mathrm{PSD}(d)$ and $\mathrm{PD}(d)$ the positive semidefinite and the positive definite ones, and $\Sigma \le \Sigma'$ the Loewner order, $\Sigma' - \Sigma \in \mathrm{PSD}(d)$. $\mathrm{O}(d)$ is the orthogonal group, S^{d-1} the unit sphere, σ_{d-1} its surface measure and $|S^{d-1}| = \sigma_{d-1}(S^{d-1})$ its surface area. The frequency has three letters, each following its source. From §3 on, ω is the frequency on $\mathbb{R}^d$, and a proof that needs a second frequency writes it ξ. The statements of the isotropic paper restated in §2.3 write it ξ, as that paper does, and the frequency on the line ω. The cited fact C2 writes it z, as its source does. Table 1, at the end of this section, collects the symbols of §§3–4, each with the statement that defines it, and the list after it names the letters that keep a second meaning.

2.2 The axioms and the classification on the line

The line paper's object is a family of operators on $L^1(\mathbb{R})$ indexed by pairs of scales $s \le t$. The operator $\Phi_{s,t}$ takes the record of a signal at scale s to its record at the coarser scale t. The axioms are these.

Definition 2.1 (symmetric cascade measurement family, [15, Def. 3.1])**.** A family $(\Phi_{s,t})_{0\le s\le t}$ of linear operators on $L^1(\mathbb{R})$ is a *symmetric cascade measurement family* if:

(A1) each $\Phi_{s,t}$ is a bounded operator on $L^1(\mathbb{R})$;

(A2) $\Phi_{s,t}\tau_a = \tau_a \Phi_{s,t}$ for every $a \in \mathbb{R}$;

(A3) $\Phi_{s,t} R = R\, \Phi_{s,t}$;

(A4) $\Phi_{s,t}$ maps nonnegative functions to nonnegative functions;

(A5) $\int \Phi_{s,t} f = \int f$ for every nonnegative $f \in L^1(\mathbb{R})$;

(A6) $\Phi_{t,t} = \mathrm{Id}$, and $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$ whenever $r \le s \le t$;

(A7) for every $f \in L^1(\mathbb{R})$ the map $(s,t) \mapsto \Phi_{s,t} f$ is continuous from $\{0 \le s \le t\}$ to $L^1(\mathbb{R})$;

(A8) there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0,\infty)$ with $D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s,\, S_\lambda t} D_\lambda$ for all $0 \le s \le t$ and all $\lambda > 0$;

(ND) $\Phi_{s,t} \ne \mathrm{Id}$ whenever $s < t$.

Axiom (A6), the *cascade*, requires that the stages compose, and leaves free how the stage from s to t depends on the two scales: the family need not be a semigroup in any scale parameter. Axiom (A8) is the scale invariance of the theory. Dilating the signal changes which scale a stage belongs to, and (A8) requires only that it relabel the scales by an increasing bijection S_λ, without fixing in advance how. Axiom (ND) excludes stages that do nothing. The first consequence of the axioms is that every stage is a convolution.

Lemma 2.2 (from operators to kernels on the line, [15, Lem. 4.1])**.** *Under (A1), (A2), (A4) and (A5), for each pair $0 \le s \le t$ there is a unique probability measure $\nu_{s,t}$ on $\mathbb{R}$ with $\Phi_{s,t} f = \nu_{s,t} * f$ for every $f \in L^1(\mathbb{R})$; under (A3) it is symmetric. Conversely every symmetric probability measure on $\mathbb{R}$ defines an operator satisfying (A1)–(A5).*

The same passage in dimension d, for every group and order of the scheme of §3, is Lemma A.1, and its converse is Lemma A.2; both are proved in Appendix A.

A *gauge* is an increasing bijection χ of $[0,\infty)$, hence continuous, with $\chi(0) = 0$. It labels the scales: the classification below describes a family by a gauge and an exponent, and in a family with gauge χ axiom (A8) holds with $S_\lambda = \chi^{-1}(\lambda\,\chi(\cdot))$, so that a dilation by λ multiplies χ by λ. A function $F : [0,\infty) \to [0,\infty)$ is *admissible* if it has the form [15, (2.3)]

$$F(r) = a_{\mathrm{V}}\, r^2 + \int_0^\infty \big(1 - \cos rx\big)\, k(x)\, \frac{dx}{x}, \tag{2.1}$$

with $a_{\mathrm{V}} \ge 0$ and $k : (0,\infty) \to [0,\infty)$ nonincreasing, $\int_0^1 x\, k(x)\, dx < \infty$ and $\int_1^\infty k(x)\, x^{-1} dx < \infty$. The number a_{V} is the *Gaussian coefficient* and k the *folded profile*. A subscript V marks an object of the line, and an upright subscript I, in §2.3, one of the time-causal theory; the italic I of μ_I is the identity matrix. A law is *infinitely divisible* when, for every n, it is the n-fold convolution of some law μ_n with itself. An admissible F is the exponent of a symmetric infinitely divisible law, and the measure $k(x)\,dx/x$ on $(0,\infty)$ is that law's *Lévy measure* folded onto the half-line. Displacements of size about x arrive at rate $k(x)\,dx/x$, the two directions counted together. The two integrability conditions are those of [15, Lem. 2.15].

Theorem 2.3 (the classification on the line, [15, Thm. 7.3])**.** *A family $(\Phi_{s,t})_{0\le s\le t}$ of operators on $L^1(\mathbb{R})$ satisfies (A1)–(A8) and (ND) if and only if there are a gauge χ and an admissible $F \not\equiv 0$ such that $\Phi_{s,t} f = \nu_{s,t} * f$ with symmetric probability measures $\nu_{s,t}$ on $\mathbb{R}$,*

$$\widehat{\nu}_{s,t}(\omega) = \exp\big[-\big(F(\chi(t)\,\omega) - F(\chi(s)\,\omega)\big)\big].$$

The pair (χ, F) is unique up to the normalization $\chi(1) = 1$.

The uniqueness says that $(c\,\chi, F(c^{-1}\,\cdot))$ gives the same family for every $c > 0$, and nothing else does; the normalization $\chi(1) = 1$ fixes c. In the *canonical gauge*, χ the identity, the kernel from scale zero at scale t has transform $e^{-F(t\omega)}$: it is the dilation by t of one law. A law μ on $\mathbb{R}^d$ is *self-decomposable* if for every $c \in (0,1)$ there is a law ρ_c, the *residual*, with

$$\widehat{\mu}(\xi) = \widehat{\mu}(c\,\xi)\,\widehat{\rho}_c(\xi), \qquad \xi \in \mathbb{R}^d \tag{2.2}$$

([58], Def. 15.1, p. 90; on the line [15, Def. 2.12]). In words, the law is a contracted copy of itself convolved with an independent remainder, for every contraction ratio. This is what makes a cascade along one scale direction possible: the residuals are the stages. The admissible functions are exactly the exponents of the symmetric self-decomposable laws on the line [15, Prop. 2.13].

2.3 What is used from the isotropic paper

The isotropic paper measures an image on $\mathbb{R}^d$ with apertures that have a size and no shape and no preferred direction. In this paper its theory is the isotropic corner of the scheme (Proposition A.3), and the isotropic member μ_I of an affine orbit is described in its terms (§4.6).

The later sections use six of its statements: the axioms, the classification with the two lemmas that reduce it to the line, the description of a law by its polar profile, and one estimate of the exponent. The isotropic paper states its axioms on operators, as the line's with $\mathbb{R}^d$ in place of $\mathbb{R}$ and the orthogonal group in place of the reflection [16, Def. 3.1], and on kernels, the form restated here; in this paper's scheme Lemmas A.1 and A.2 join the two.

Definition 2.4 (isotropic cascade family, [16, Def. 3.2])**.** Fix an integer $d \geq 1$. For $A \in \mathrm{GL}(d)$ write $A_*\mu := \mu \circ A^{-1}$ for the image of a measure μ on $\mathbb{R}^d$, and $\widehat{\mu}(\xi) = \int e^{i\langle \xi, x\rangle}\mu(dx)$, so that $\widehat{A_*\mu}(\xi) = \widehat{\mu}(A^{\mathsf{T}}\xi)$. An *isotropic cascade family in dimension d* is a family $(\mu_{s,t})_{0\leq s\leq t}$ of probability measures on $\mathbb{R}^d$ such that

(I1) *(cascade)* $\mu_{t,t} = \delta_0$, and $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ whenever $r \leq s \leq t$;

(I2) *(continuity)* for every $\xi \in \mathbb{R}^d$ the map $(s,t) \mapsto \widehat{\mu}_{s,t}(\xi)$ is continuous on $\{0 \leq s \leq t\}$;

(I3) *(rotation invariance)* $Q_*\mu_{s,t} = \mu_{s,t}$ for every $Q \in \mathrm{O}(d)$ and every pair;

(I4) *(dilation covariance)* there is a family $(S_\lambda)_{\lambda>0}$ of increasing bijections of $[0,\infty)$ with $(\lambda I)_*\mu_{s,t} = \mu_{S_\lambda s,\, S_\lambda t}$ for every $\lambda > 0$ and every pair;

(ND) *(nondegeneracy)* $\mu_{s,t} \neq \delta_0$ whenever $s < t$.

The cone paper names the class $\mathcal{A}_d$ of the admissible F of (2.1) for which $\xi \mapsto F(|\xi|)$, $\xi \in \mathbb{R}^d$, is the exponent of a self-decomposable law on $\mathbb{R}^d$ [17, Prop. 4.9]. By the next theorem its members other than 0 are the exponents of the isotropic families in $\mathbb{R}^d$. The Gaussian exponent $a_{\mathrm{V}}r^2$ lies in every $\mathcal{A}_d$.

Theorem 2.5 (the isotropic classification in dimension d, [16, Thm. 3.6])**.** *Let $d \geq 1$. A family $(\mu_{s,t})_{0\leq s\leq t}$ of probability measures on $\mathbb{R}^d$ is an isotropic cascade family in dimension d if and only if there are a gauge χ and an $F \in \mathcal{A}_d$ with $F \not\equiv 0$ such that*

$$\widehat{\mu}_{s,t}(\xi) = \exp\big[-\big(F(\chi(t)|\xi|) - F(\chi(s)|\xi|)\big)\big], \qquad 0 \leq s \leq t,\ \xi \in \mathbb{R}^d. \tag{2.3}$$

In particular the kernels from scale zero are $\widehat{\mu}_{0,t}(\xi) = e^{-F(\chi(t)|\xi|)}$. The pair is unique up to $(\chi, F) \mapsto (c\,\chi,\, F(c^{-1}\,\cdot))$ with $c > 0$, in the sense that if a gauge χ' and a real function F' on $[0,\infty)$ with $F'(0) = 0$ also satisfy (2.3), then $\chi' = c\,\chi$ and $F' = F(c^{-1}\,\cdot)$ for some $c > 0$, and $c = 1$ when $\chi(1) = \chi'(1) = 1$.

The classification reduces to the line through the marginals of a family along a line. Two lemmas of the isotropic paper carry the reduction, and the proof of Proposition A.3 uses them too.

Lemma 2.6 (a rotation-invariant law is determined by its marginal, [16, Lem. 3.4])**.** *Let $d \geq 1$, let $e \in \mathbb{R}^d$ be a unit vector and $\pi(x) := \langle e, x\rangle$, and for a probability measure μ on $\mathbb{R}^d$ write $\pi_*\mu$ for its image on $\mathbb{R}$.*

1. *If $Q_*\mu = \mu$ for every $Q \in \mathrm{O}(d)$, then $\kappa := \pi_*\mu$ is a symmetric probability measure on $\mathbb{R}$, and*
$$\widehat{\mu}(\xi) = \widehat{\kappa}(|\xi|), \qquad \xi \in \mathbb{R}^d.$$
In particular $\widehat{\mu}$ is real and radial, and κ does not depend on e.

2. *Under the hypothesis of (1), $\pi_*\mu = \delta_0$ if and only if $\mu = \delta_0$.*

3. *For all probability measures μ, μ' on $\mathbb{R}^d$ and every $\lambda > 0$, $\pi_*(\mu * \mu') = \pi_*\mu * \pi_*\mu'$, $\pi_*\big((\lambda I)_*\mu\big) = (\lambda I)_*(\pi_*\mu)$ on $\mathbb{R}$, and $\pi_*\delta_0 = \delta_0$.*

Lemma 2.7 (the marginals of an isotropic family form a family on the line, [16, Lem. 3.5])**.** *Let $(\mu_{s,t})$ be an isotropic cascade family in dimension d with dilation action (S_λ), let e and π be as in Lemma 2.6, and put*

$$\kappa_{s,t} := \pi_* \mu_{s,t}, \qquad \Phi_{s,t} f := \kappa_{s,t} * f \quad (f \in L^1(\mathbb{R})), \qquad 0 \le s \le t.$$

Then $\widehat{\mu}_{s,t}(\xi) = \widehat{\kappa}_{s,t}(|\xi|)$ for every $\xi \in \mathbb{R}^d$, each $\kappa_{s,t}$ is a symmetric probability measure, and $(\Phi_{s,t})$ is a symmetric cascade measurement family in the sense of Definition 2.1, with the same (S_λ).

The origin data below describe the laws of the class $\mathcal{A}_d$ through their Lévy measures. The Lévy measure of a rotation-invariant self-decomposable law is described by one function of the radius, by the polar criterion of self-decomposability ([58], Thm. 15.10, p. 95), and §4.6 reads its statements through that function.

Some of these laws come from *subordination*, which builds kernels from the Gaussian. Blur with a Gaussian whose variance is drawn at random, and average over the draws. The result is a *Gaussian variance mixture*, the law of a Brownian motion at a random time (Bochner's subordination, [58], §30, p. 197). The Matérn kernel with transform $(1 + t^2|\xi|^2/2)^{-\gamma}$ is one: its variance is drawn from the gamma law with Laplace transform $(1 + t^2\sigma)^{-\gamma}$ [16, Prop. 3.12(2)]. The cone paper draws the variance from the delay laws of the time-causal theory, and describes the law of the variance by a *delay datum* [17, Def. 4.11]: a pair (b_0, k_{I}) with $b_0 \ge 0$ and $k_{\mathrm{I}} : (0,\infty) \to [0,\infty)$ measurable, subject to $\int_0^1 k_{\mathrm{I}}(u)\, du < \infty$ and $\int_1^\infty k_{\mathrm{I}}(u)\, u^{-1} du < \infty$, with exponent

$$F_{\mathrm{I}}(\sigma) = b_0\, \sigma + \int_0^\infty \bigl(1 - e^{-\sigma u}\bigr)\, k_{\mathrm{I}}(u)\, \frac{du}{u}, \qquad \sigma \ge 0.$$

Its polar profile in dimension d is the integral of (2.6) below, which is [17, (4.6)], and in Definition 2.8 the drift b_0 is written a. This paper uses the subordinated data as examples of profiles bounded at the origin (§4.6). The isotropic paper names the function of the radius, and the data that subordination gives, as follows.

Definition 2.8 (isotropic origin data in dimension d, and the polar profile, [16, Def. 3.10])**.** Fix an integer $d \ge 1$ and write σ_{d-1} for the surface measure of the unit sphere $S^{d-1} \subset \mathbb{R}^d$ and $|S^{d-1}| = \sigma_{d-1}(S^{d-1})$ for its surface area. An *isotropic origin datum in dimension d* is a pair (a, k_d) with $a \ge 0$ and $k_d : (0,\infty) \to [0,\infty)$ nonincreasing, subject to

$$\int_0^1 k_d(r)\, r\, dr < \infty \qquad \text{and} \qquad \int_1^\infty k_d(r)\, \frac{dr}{r} < \infty.$$

Its Lévy measure is the rotation-invariant measure ν_d on $\mathbb{R}^d \setminus \{0\}$ with radial density $f_d(r) = k_d(r)\, \bigl(|S^{d-1}|\, r^d\bigr)^{-1}$, so that

$$\int_{\mathbb{R}^d} g\, d\nu_d = \frac{1}{|S^{d-1}|} \int_{S^{d-1}} \sigma_{d-1}(d\eta) \int_0^\infty g(r\eta)\, k_d(r)\, \frac{dr}{r} \tag{2.4}$$

for every nonnegative Borel g vanishing at the origin. Its *exponent* is

$$F_d(\xi) = \tfrac{1}{2} a|\xi|^2 + \int_{\mathbb{R}^d} \bigl(1 - \cos\langle \xi, x\rangle\bigr)\, \nu_d(dx), \qquad \xi \in \mathbb{R}^d, \tag{2.5}$$

and its *law* is the symmetric infinitely divisible law μ on $\mathbb{R}^d$ with $\widehat{\mu} = e^{-F_d}$. We call k_d the datum's *polar profile* and $c := k_d(0+) \in [0,\infty]$ its *profile at the origin*. The datum is *subordinated* when there is a delay datum (a, k_{I}) in the sense of §2.3, with k_{I} nonincreasing, whose polar profile [17, (4.6)] is k_d:

$$k_d(r) = \frac{2}{\Gamma(d/2)} \int_0^\infty v^{d/2-1} e^{-v}\, k_{\mathrm{I}}\Bigl(\frac{r^2}{2v}\Bigr)\, dv, \qquad r > 0. \tag{2.6}$$

An origin datum describes its law, a rotation-invariant symmetric infinitely divisible law, through the coefficient a of its Gaussian part and the polar profile k_d. The data describe exactly the class $\mathcal{A}_d$: a function $F : [0,\infty) \to [0,\infty)$ lies in $\mathcal{A}_d$ if and only if $\xi \mapsto F(|\xi|)$ is the exponent F_d of an isotropic origin datum in dimension d, and the datum is then unique, k_d up to a Lebesgue-null set [16, Prop. 3.11]. In this paper it describes the isotropic member μ_I of an affine orbit by its profile (§4.6). The name refers to the use the isotropic paper makes of it, at the origin of a kernel.

The later sections use one more statement of the isotropic paper.

Lemma 2.9 (a profile bounded at the origin gives a logarithmic exponent, [16, Lem. A.2])**.** *Let $d \geq 1$, let $(0, k_d)$ be an isotropic origin datum in dimension d with Gaussian coefficient 0 and $c = k_d(0+) < \infty$, let F_d be its exponent, and put $C_k := \int_1^\infty k_d(r)\, dr/r$.*

1. (The exponent grows at most like $c \log|\xi|$.) *For every ξ with $|\xi| \geq 1$,*

$$F_d(\xi) \leq c \log|\xi| + \tfrac{13}{4}\, c + 2C_k, \qquad \text{so} \qquad e^{-F_d(\xi)} \geq e^{-13c/4 - 2C_k}\, |\xi|^{-c}.$$

2. (The exponent is logarithmic.) *For every $a > 0$, $F_d(a\xi) - F_d(\xi) \to c \log a$ as $|\xi| \to \infty$. Equivalently, for a unit vector e the function $g(u) := F_d(\sqrt{u}\, e)$, $u \geq 0$, satisfies $g(au) - g(u) \to \frac{c}{2} \log a$ as $u \to \infty$, for every $a > 0$.*

Clause (2) says $e^{-F_d(\xi)} = |\xi|^{-c} L(|\xi|)$ with L *slowly varying*, that is, $L(a\rho)/L(\rho) \to 1$ as $\rho \to \infty$ for every $a > 0$. It is the link from a profile bounded at the origin to an exponent of logarithmic growth, which §4.6 uses.

2.4 The cited facts

The statements of this paper rest on five theorems of the literature that it cites without proving them, four of probability theory and one of matrix analysis. Each is stated below as it is used, with the page of its source, and named C1 to C5, C for cited, to keep them apart from the axioms (A1)–(A8); from here on a cited fact is referred to by its name alone. Inside a cited fact the letters are its source's: A is a matrix, ν a Lévy measure and γ a drift vector, and in C5 B is a Borel set and λ a measure on the unit sphere.

A *null array* is a triangular array of laws in which, for every $\varepsilon > 0$, the largest mass that an entry of row n puts outside the ball of radius ε tends to 0 as n grows. Khintchine's theorem says that the row products of such an array can only converge to an infinitely divisible law. It is what makes the increments of a cascade infinitely divisible in §3.

Theorem 2.10 (C1: Khintchine's theorem, [58], Thm. 9.3, p. 47)**.** *Let $r_n \geq 1$ be integers and $\mu_{n,k}$, $1 \leq k \leq r_n$, $n = 1, 2, \ldots$, probability measures on $\mathbb{R}^d$ with $\lim_{n\to\infty} \max_{1\leq k\leq r_n} \mu_{n,k}(\{|x| > \varepsilon\}) = 0$ for every $\varepsilon > 0$. If $\mu_{n,1} * \cdots * \mu_{n,r_n} * \delta_{-b_n}$ converges weakly to a probability measure μ for some $b_n \in \mathbb{R}^d$, then μ is infinitely divisible.*

Theorem 2.10 is stated in the source for the row sums of independent random variables. The form above restates it through the law of a sum of independent variables, which is the convolution of their laws ([58], p. 8).

The *generating triplet* of an infinitely divisible law is the triple of its Lévy–Khintchine representation: a Gaussian covariance matrix A, a Lévy measure ν and a drift γ. The paper compares the kernels of an affine orbit through their Lévy measures (§4.6), and C2 is what lets a law and its triplet stand for each other.

Theorem 2.11 (C2: the Lévy–Khintchine representation, [58], Thm. 8.1, pp. 37–38)**.** *On $\mathbb{R}^d$:*

(i) An infinitely divisible law μ on $\mathbb{R}^d$ has

$$\widehat{\mu}(z) = \exp\Big[-\tfrac{1}{2}\langle z, Az\rangle + i\langle \gamma, z\rangle + \int_{\mathbb{R}^d} \big(e^{i\langle z,x\rangle} - 1 - i\langle z, x\rangle \mathbf{1}_{\{|x|\le 1\}}\big)\, \nu(dx)\Big]$$

for a symmetric nonnegative definite A, a $\gamma \in \mathbb{R}^d$ and a measure ν with $\nu(\{0\}) = 0$ and $\int (|x|^2 \wedge 1)\, \nu(dx) < \infty$.

(ii) The triplet (A, ν, γ) is unique.

(iii) Conversely, every such triplet is the generating triplet of an infinitely divisible law.

A law is *operator-selfdecomposable* when the contraction in the definition of selfdecomposability is replaced by the linear flow e^{-tB} of a matrix B: the law is its own image under e^{-tB} convolved with a residual, for every $t \ge 0$. The class is Urbanik's [62]. The fact used of it is that, when the flow contracts, $e^{-tB} \to 0$ as $t \to \infty$, the residuals and the law are infinitely divisible. Without that hypothesis the fact fails: for $B = 0$ every law is its own image with the residual δ_0.

Theorem 2.12 (C3: operator-self-decomposable laws with a contracting exponent are infinitely divisible, [30], Thm. 3.3.5, p. 83, with §3.4, pp. 89–90; [62], Prop. 5.2, p. 134)**.** *Let B be a real $d \times d$ matrix with $e^{-tB} \to 0$ as $t \to \infty$, and let μ be a probability measure on $\mathbb{R}^d$ such that for every $t \ge 0$ there is a probability measure ρ_t with $\mu = (e^{-tB})_*\mu * \rho_t$. Then every ρ_t is infinitely divisible, and so is μ.*

Jurek and Mason state their Theorem 3.3.5 for *full* laws, whose support lies in no proper hyperplane ([30], p. 46). The opening paragraph of their §3.4 notes that fullness is used only in the other half of that theorem and drops it, and adds that the law itself is then infinitely divisible. C3 is the last sentence of the theorem read with that paragraph. Urbanik's Proposition 5.2 states the same without fullness. Beyond the letter of the two passages, C3 reads two things. A measure is a probability measure, which is the book's convention. And the passages say that each residual of one family is infinitely divisible, where C3 says it of every probability measure that is a residual at some $t \ge 0$; this is their sentence applied to a family completed around that measure, the hypothesis supplying the residuals at the other t.

The fourth fact is of matrix analysis. The parameter of an aperture in this paper is a positive semidefinite matrix, and an order on these matrices invariant under the general linear group is determined by the relative steps $\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ that it allows (Proposition 4.2). The composite of two steps depends on how their eigenframes sit relative to each other, and C4 bounds its spectrum by the spectra of the two (Lemma C.1(2), Appendix C.1). The eigenvalues of such a product MN are real and positive, being those of $M^{1/2}NM^{1/2}$; write $\lambda^{\downarrow}(A)$ for the eigenvalues of a matrix A with real eigenvalues, in decreasing order and with multiplicity.

Theorem 2.13 (C4: the eigenvalues of a product of positive definite matrices, [4], Cor. III.4.6, (III.21), p. 73)**.** *Let M and N be positive definite $d \times d$ matrices. For $1 \le k \le d$,*

$$\sum_{i\le k} \log \lambda_i^{\downarrow}(MN) \le \sum_{i\le k} \big(\log \lambda_i^{\downarrow}(M) + \log \lambda_i^{\downarrow}(N)\big).$$

The source states (III.21) for complex positive matrices as a majorization, $\log \lambda(AB) \prec \log \lambda^{\downarrow}(A) + \log \lambda^{\downarrow}(B)$: the inequalities of C4 between the partial sums of the k largest entries, with the equality of the totals, which is $\det(MN) = \det M \det N$. The same inequality is stated as Exercise 6.5.3, (6.47), p. 228 of [5], with a pointer to the page above, as an inequality due to Gel'fand, Naimark and Lidskii. The notes to Chapter III of [4] (p. 80) give their papers of 1950.

The fifth fact describes the self-decomposable laws among the infinitely divisible ones by their Lévy measures. In polar coordinates the Lévy measure of such a law has, in each direction, a density with respect to dr/r that does not increase with the radius r.

Theorem 2.14 (C5: the polar criterion, [58], Thm. 15.10, p. 95)**.** *An infinitely divisible μ on $\mathbb{R}^d$ with generating triplet (A, ν, γ) is self-decomposable if and only if $\nu(B) = \int_S \lambda(d\xi) \int_0^\infty \mathbf{1}_B(r\xi)\, k_\xi(r)\, dr/r$ for a finite measure λ on the unit sphere $S = S^{d-1}$ and a nonnegative $k_\xi(r)$, measurable in ξ and nonincreasing in $r > 0$.*

Theorem 2.14 is the source's formula (15.8) with both directions, and the source adds that self-decomposability puts no restriction on A and γ; its Corollary 15.11, on the same page, is the case $d = 1$. Self-decomposability is (2.2), which is the source's Definition 15.1, and the triplet is that of C2. For a rotation-invariant law the isotropic paper reads C5 as the polar form of its class ([16, Prop. 3.11], §2.3).

Each enters with a few statements. C1 makes the increments of a cascade infinitely divisible (Theorem 3.5). C2 enters with the restated classification (Theorem 2.5) and with the laws of the affine orbits and their Lévy measures, from Lemma 4.6 on. C3 enters with the anisotropy budget (Proposition 4.11(1)) and through it with the theorem on the general linear group along the chain of cones (Theorem 4.21). C4 enters with the spectrum of a product (Lemma C.1(2)) and through it with the invariant orders (Theorem 4.3). C5 enters once, with the description of the whole general linear cell along a cone order (Proposition 4.22(2)). Appendix I lists every statement that rests on each.

symbol	meaning	defined in
$\Sigma, \Sigma' \in \mathrm{PSD}(d)$; $\mathrm{PD}(d)$	aperture parameters; the positive definite ones	Def. 3.1
(G, W), $\preceq_W$, P_W	wedge datum: covariance group, order, its pairs	Def. 3.1
$A \cdot \Sigma$, $A_*\mu$	$A\Sigma A^\mathsf{T}$, and the image of a law	Def. 3.1
$\mu_{\Sigma,\Sigma'}$, μ_Σ	kernel of a pair; kernel from scale zero $\mu_{0,\Sigma}$	Def. 3.1
$\mathrm{S}(d)$, $\mathrm{D}_+(d)$, $\mathrm{GL}^+(d)$	similarities, positive diagonal, orientation-preserving group	§4.2
D_s; $\Delta(\Sigma)$	$\mathrm{diag}(s_1, \dots, s_d)$; the diagonal of Σ	§4.3
g	$\widehat{\mu}_\Sigma(\omega) = e^{-g(\langle \omega, \Sigma\omega \rangle)}$ on an affine orbit	Lem. 3.9, Prop. 4.4
M; $\lambda_{\min}(M)$	relative step $\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$; its least eigenvalue	Prop. 4.2
$\mathcal{K}$, $\preceq_\mathcal{K}$	step set in $\mathrm{PD}(d)$, and its relation	Prop. 4.2
C, $\preceq_C$, $\preceq_C^+$	invariant cone in $\mathrm{Sym}(d)$; its order on $\mathrm{PD}(d)$, on $\mathrm{PSD}(d)$	Thm. 4.3
C_α, $0 \le \alpha \le \infty$	the chain of cones from the rays to $\mathrm{PSD}(d)$	Prop. 4.12
$\preceq_\alpha$; $\preceq_\alpha = \preceq_{C_\alpha}^+$	stable order on $\mathrm{PD}(d)$; on $\mathrm{PSD}(d)$	Props. 4.7, 4.20
$\mathcal{K}_\alpha$	stable step cone, the step set of $\preceq_\alpha$	Thm. 3.5
$\mathfrak{k}_\alpha \subseteq \mathfrak{gl}(d)$	exponent cone of $S_\alpha(I)$	Prop. 4.10
$\mathrm{sym}\, B$	$\frac{1}{2}(B + B^\mathsf{T})$, the symmetric part of B	Prop. 4.10
$S_\alpha(H)$, n_H, ψ_H	elliptical stable law, its Lévy density and exponent	Lem. 4.6
k; $R(\lambda)$; $\beta_-(k)$, $\beta(k)$	profile, $k = k_d/\lvert S^{d-1}\rvert$; dilation ratio; lower and growth exponents	Prop. 4.13, Lem. 4.14
k_d, $k_d(0+)$	polar profile and its value at the origin	Def. 2.8
$t(\Sigma)$, $S(\Sigma)$, N, n_i	volume scale, shape, relative shape, its eigenvalues	Lem. 4.1
es, ecc, cond, $\mathcal{M}_s$	exponent spread, eccentricity, condition number, moment matrix	Props. 4.25, H.5
$N(0, C)$	the centred Gaussian law of covariance C	§4.9
$\mathcal{L}_\Sigma$	the operator $-\nabla \cdot \Sigma \nabla$	§4.6

Table 1: Notation of §§3–4.

Some letters keep a second meaning inside a statement, which that statement fixes. They are these.

- a: the Gaussian coefficient a_{V} of (2.1) on the line, and the coefficient a of an origin datum, whose exponent has the term $\frac{1}{2}a|\xi|^2$, so that $a = 2a_{\mathrm{V}}$ at $d = 1$. "Gaussian coefficient 0" in Lemma 2.9 means $a = 0$, and its clause (2) uses $a > 0$ for a dilation factor.

- c: the profile at the origin in Definition 2.8 and Lemma 2.9, and a positive constant elsewhere.

- C: an invariant cone, with $C(\mathcal{K})$ the cone of a step set, and the Gaussian covariance $C(\Sigma)$ of a kernel in Proposition 4.25; a real matrix in Lemma 4.8. C_k is a constant in Lemma 2.9, and C1–C5, upright and without a subscript, are the cited facts.

- g: the function of one variable through which an affine orbit depends on the aperture, $\widehat{\mu}_\Sigma(\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)}$, in every statement that uses it. In Lemma 2.9(2) it is the same function for the orbit of a radial exponent.

- γ: a stable or Matérn index, and the drift of a triplet in C2.

- k: the folded profile of (2.1) on the line, the polar profile k_d of Definition 2.8, and the profile k of Proposition 4.13, taken against $r^{-d}dx$, which is $k_d/|S^{d-1}|$. At $d = 1$ the last is half the folded profile, which counts both directions.

- S: the similarities $\mathrm{S}(d)$, the shape $S(\Sigma)$ of an aperture, the elliptical stable law $S_\alpha(H)$, the relabelling S_λ of the scales in (A8) and (I4), and the unit sphere S^{d-1}, kept apart by their typeface and arguments.

- σ: the surface measure σ_{d-1}, the normalized spectrum $\sigma(\Sigma)$ of Proposition F.3, and, in the prose of §4.8, the slant of a surface.

- t: the volume scale $t(\Sigma)$ of Lemma 4.1, the time of a linear flow in Proposition G.1, and the threshold $t_\alpha(\Sigma)$ of Proposition 4.20(1). The prose writes v and w for variances and e^{sB} for a flow.

Two objects have several names. The stable order of index α is $\preceq_\alpha$ on $\mathrm{PD}(d)$, defined by its step condition in Proposition 4.7(2). Proposition 4.12(1) shows that it equals $\preceq_{C_\alpha}$. On $\mathrm{PSD}(d)$ it is $\leq_\alpha$, which equals $\preceq^+_{C_\alpha}$. Its set of steps is the step cone $\mathcal{K}_\alpha = \exp C_\alpha \subseteq \mathrm{PD}(d)$, a set of relative steps M. The exponent cone $\mathfrak{k}_\alpha \subseteq \mathfrak{gl}(d)$ is a set of exponents, in fraktur as $\mathfrak{gl}(d)$ is, and its symmetric part is C_α (Proposition 4.12(2)).

3 The axiom scheme and the classical affine theory

In dimension $d \geq 2$ an aperture has a shape as well as a size, and the scalar parameter of the line becomes a matrix Σ, proportional to the covariance of the aperture in the Gaussian case. Its size, the scale, is the part that the dilations move. Two structures then act on the parameter. A group G of linear maps of the image moves it by $A \cdot \Sigma = A\Sigma A^{\mathsf{T}}$, and the family is required to be covariant under G. Covariance is what lets a measurement at one shape be carried to another by the image's own change of coordinates, the use of shape adaptation (§4.1). An order W on the matrices says along which pairs the family cascades, so that a coarser record is a further smoothing of a finer one.

The section has two aims. The first is to state the axioms once, for every pair (G, W), so that each case of the classification of §4 is an instance of one definition (§3.1). The second is the theorem at the classical end of that classification: with the general linear group and the Loewner order, one rank-deficient step of the cascade forces the Gaussian (Theorem 3.6, §3.4). That theorem is the reference point from which §4 measures what a smaller group or a smaller order buys. Between the two, two subsections settle what every case shares and what the rest

of the paper uses. The first explains why the positive definite matrices $\mathrm{PD}(d)$ index the kernels. For $\mathrm{GL}(d)$ the axioms supply a rotation-invariant kernel μ_I, and every kernel from scale zero at a positive definite matrix is a linear image of it (§3.2). The second shows that along every order with a path condition each increment is infinitely divisible (Theorem 3.5, §3.3). Every increment then has a generating triplet, a Gaussian covariance, a Lévy measure and a drift, and §4 reads the families through those triplets. The section closes with the prior art and the relation to the line (§3.5).

3.1 The (G, W) scheme, from operators to kernels

This subsection states the axioms for every pair (G, W), on operators and on kernels (Definition 3.1), as the line paper and the isotropic paper state theirs (Definitions 2.1 and 2.4). The operator form is where the axioms have their meaning as measurements, and the kernel form is where the proofs are done. The kernel form for the general linear group with the Loewner order, the classical case, is written out as Definition 3.2. Three statements then serve every pair: a family whose kernels from scale zero have transforms without zeros is determined by them (Lemma 3.3), along rays these kernels are exactly the covariant fields of self-decomposable laws (Proposition 3.4), and on the multiples of I the similarities with the ray order give, up to a change of the scale coordinate, the isotropic theory (Proposition A.3).

The group must contain the dilations, so that zooming the image is always a change of parameter. The order must contain the rays, so that blurring an aperture without changing its shape is always a step of the cascade. With a matrix as parameter, the paper uses two concepts, one from matrix analysis and one from the Lie theory of semigroups. The first is the *Loewner order*: $\Sigma \le \Sigma'$ when $\Sigma' - \Sigma$ is positive semidefinite, that is, when $\langle w, \Sigma w\rangle \le \langle w, \Sigma' w\rangle$ for every $w \in \mathbb{R}^d$ ([24], Ex. 2.2.3, p. 41). For apertures it says that the aperture of Σ' is at least as wide as that of Σ in every direction (Figure 1). Scale-space theory uses it already. The Gaussian kernels form a semigroup over covariance matrices: the kernel of Σ_1 convolved with that of Σ_2 is the kernel of $\Sigma_1 + \Sigma_2$ ([36], (15.34)–(15.35), p. 391). Lindeberg's cascade over covariance matrices requires the difference $\Sigma' - \Sigma$ to be positive semidefinite ([38], fn. 7, p. 594), and the increment of the affine Gaussian scale space from Σ to Σ' is the Gaussian with covariance $2b(\Sigma' - \Sigma)$ (Definition 3.2). The second concept is the *wedge* of Hilgert, Hofmann and Lawson, a closed convex set that contains 0 and is closed under addition ([23], p. 3). For the orders of this paper the steps of W between positive definite matrices are generated by such a cone of symmetric matrices (§4.4), although Definition 3.1 does not require one. The pair (G, W) is called a *wedge datum* after its order.

Definition 3.1 (the (G, W) cascade scheme)**.** Fix an integer $d \ge 1$. Write $\mathrm{PSD}(d)$ for the real symmetric positive semidefinite $d \times d$ matrices and $\mathrm{PD}(d) \subseteq \mathrm{PSD}(d)$ for the positive definite ones. The group $\mathrm{GL}(d)$ acts on $\mathrm{PSD}(d)$ by $A \cdot \Sigma := A\Sigma A^{\mathsf{T}}$, on probability measures on $\mathbb{R}^d$ by pushforward, $A_*\mu := \mu \circ A^{-1}$, and on $L^1(\mathbb{R}^d)$ by $D_A f := |\det A|^{-1} f(A^{-1}\cdot)$; with $\widehat{\mu}(\omega) = \int e^{i\langle \omega, x\rangle}\mu(dx)$ one has $\widehat{A_*\mu}(\omega) = \widehat{\mu}(A^{\mathsf{T}}\omega)$. A *wedge datum in dimension d* is a pair (G, W) of

- a closed subgroup G of $\mathrm{GL}(d)$ containing the dilations λI, $\lambda > 0$, and
- a partial order $\le_W$ on $\mathrm{PSD}(d)$ that is *G-invariant*, $\Sigma \le_W \Sigma'$ implying $A \cdot \Sigma \le_W A \cdot \Sigma'$ for every $A \in G$, and *contains the rays*, $c\,\Sigma \le_W c'\,\Sigma$ whenever $\Sigma \in \mathrm{PSD}(d)$ and $0 \le c \le c'$.

Write P_W for the set of pairs (Σ, Σ') with $\Sigma \le_W \Sigma'$, with the topology of $\mathrm{PSD}(d) \times \mathrm{PSD}(d)$. Since W contains the rays, $0 \le_W \Sigma$ for every Σ, and in particular $(0, I) \in P_W$.

1. *(Operator form.)* A *(G, W)-cascade family of operators* is a family $(\Phi_{\Sigma,\Sigma'})_{(\Sigma,\Sigma')\in P_W}$ of linear operators on the real space $L^1(\mathbb{R}^d)$ such that

   (A1) each $\Phi_{\Sigma,\Sigma'}$ is a bounded operator on $L^1(\mathbb{R}^d)$;

(A2) each $\Phi_{\Sigma,\Sigma'}$ commutes with the translations $T_a f := f(\cdot - a)$, $a \in \mathbb{R}^d$;

(A4) each $\Phi_{\Sigma,\Sigma'}$ maps nonnegative functions to nonnegative functions;

(A5) each $\Phi_{\Sigma,\Sigma'}$ preserves $\int f$ for $f \geq 0$;

(A6) *(cascade)* $\Phi_{\Sigma,\Sigma} = \mathrm{Id}$ and $\Phi_{\Sigma',\Sigma''}\Phi_{\Sigma,\Sigma'} = \Phi_{\Sigma,\Sigma''}$ whenever $\Sigma \leq_W \Sigma' \leq_W \Sigma''$;

(A7) *(continuity)* for every $f \in L^1(\mathbb{R}^d)$ the map $(\Sigma, \Sigma') \mapsto \Phi_{\Sigma,\Sigma'} f$ is continuous from P_W into $L^1(\mathbb{R}^d)$;

(ND) *(nondegeneracy)* $\Phi_{0,I} \neq \mathrm{Id}$;

(Cov) *(covariance)* $D_A \Phi_{\Sigma,\Sigma'} D_A^{-1} = \Phi_{A\cdot\Sigma,\, A\cdot\Sigma'}$ for every $A \in G$ and every $(\Sigma, \Sigma') \in P_W$.

Axioms (A1), (A2), (A4) and (A5) are the single-operator axioms of Definition 2.1 other than the reflection axiom (A3), read in $\mathbb{R}^d$.

2. *(Kernel form.)* A (G, W)*-cascade family of kernels* is a family $(\mu_{\Sigma,\Sigma'})_{(\Sigma,\Sigma')\in P_W}$ of probability measures on $\mathbb{R}^d$ such that

(K1) *(cascade)* $\mu_{\Sigma,\Sigma} = \delta_0$, and $\mu_{\Sigma,\Sigma''} = \mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}$ whenever $\Sigma \leq_W \Sigma' \leq_W \Sigma''$;

(K2) *(continuity)* for every $\omega \in \mathbb{R}^d$ the map $(\Sigma, \Sigma') \mapsto \widehat{\mu}_{\Sigma,\Sigma'}(\omega)$ is continuous on P_W;

(K3) *(G-covariance)* $A_* \mu_{\Sigma,\Sigma'} = \mu_{A\cdot\Sigma,\, A\cdot\Sigma'}$ for every $A \in G$ and every $(\Sigma, \Sigma') \in P_W$;

(K4) *(nondegeneracy)* $\mu_{0,I} \neq \delta_0$.

We write $\mu_\Sigma := \mu_{0,\Sigma}$ for the *kernel from scale zero*.

The *Loewner order*, $\Sigma \leq \Sigma'$ when $\Sigma' - \Sigma \in \mathrm{PSD}(d)$, and the *ray order*, under which Σ precedes Σ' when $\Sigma = 0$ or $\Sigma' = c\,\Sigma$ for some $c \geq 1$, contain the rays and are invariant under every subgroup of $\mathrm{GL}(d)$. The *corners* of the scheme are these four data:

- the similarities $\{cQ : c > 0,\ Q \in \mathrm{O}(d)\}$ with the ray order, read on the multiples sI, $s \geq 0$, of the identity (the isotropic theory);
- $(\mathrm{GL}(d), \text{Loewner})$, the classical case, whose kernel form is Definition 3.2;
- $(\mathrm{GL}(d), \text{rays})$;
- the similarities with the Loewner order.

Four data recur throughout, and the definition ends by listing them as the corners of the scheme. They are the reference points of the landscape. In Table 2 they are the cells of the general linear and the similarity rows in the ray and the Loewner columns. The isotropic theory (§2.3) is the first, read on the multiples of I (below, and Proposition A.3 in Appendix A). The classical theory of this section is the second. The third and the fourth are the two ends of the landscape of §4. The ray order is the smallest order that contains the rays, and $(\mathrm{GL}(d), \text{rays})$ pairs it with the largest group. The similarities with the Loewner order pair the smaller group of the four with the larger order.

The two forms correspond axiom by axiom. The single-operator axioms (A1), (A2), (A4) and (A5) make each $\Phi_{\Sigma,\Sigma'}$ convolution by a probability measure $\mu_{\Sigma,\Sigma'}$, its kernel, as on the line (Lemma 2.2). Then (A6) corresponds to (K1), (A7) to (K2), (Cov) to (K3) and (ND) to (K4). The line's axiom (A8) leaves the relabelling of the scales free; the scheme fixes the action of the group on the parameter, $A \cdot \Sigma$, and for $\mathrm{GL}(d)$ on $\mathrm{PD}(d)$ a relabelling that commutes with that action is a constant multiple (Lemma A.4). The passage from operators to kernels is Lemma A.1, the line's Lemma 2.2 proved again in $\mathbb{R}^d$ with balls for intervals; its clause (5) gives the same passage for the isotropic families. Its converse is Lemma A.2: convolution with the kernels of a family satisfies the operator axioms. Both are stated and proved in Appendix A.

So the two forms describe the same families, and the theory proceeds on kernels from here (§1, "Scope").

Nondegeneracy is required at the one pair $(0, I)$ only. Asking it at every pair would exclude a family of the similarities with the ray order whose kernels at the singular matrices are trivial (Appendix A, after Proposition A.3). For the same corner it also allows the kernel δ_0 at a positive definite aperture. The centred Gaussian kernels with covariance $\phi(\Sigma)\operatorname{tr}(\Sigma)\, I$, $\Sigma \neq 0$, and δ_0 at 0, form such a family when $\phi \geq 0$ is a continuous function of the eigenvalues of Σ divided by its trace, positive at I and zero at some positive definite matrix (Proposition 3.4(2)).

The proofs of §3.4 work with the kernel form at the classical corner, so it is written out once.

Definition 3.2 (affine cascade family)**.** Fix an integer $d \geq 1$, and let $\mathrm{PSD}(d)$, $\mathrm{PD}(d)$, the Loewner order $\Sigma \leq \Sigma'$, the action $A \cdot \Sigma = A\Sigma A^{\mathsf{T}}$ of $\mathrm{GL}(d)$, the pushforward $A_*\mu$ and the transform $\widehat{\mu}$ be as in Definition 3.1. An *affine cascade family in dimension d* is a $(\mathrm{GL}(d), \text{Loewner})$-cascade family of kernels (Definition 3.1(2)), that is, a family $(\mu_{\Sigma,\Sigma'})$ of probability measures on $\mathbb{R}^d$, indexed by the pairs $\Sigma \leq \Sigma'$ in $\mathrm{PSD}(d)$, such that

(K1) *(cascade)* $\mu_{\Sigma,\Sigma} = \delta_0$, and $\mu_{\Sigma,\Sigma''} = \mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}$ whenever $\Sigma \leq \Sigma' \leq \Sigma''$;

(K2) *(continuity)* for every $\omega \in \mathbb{R}^d$ the map $(\Sigma, \Sigma') \mapsto \widehat{\mu}_{\Sigma,\Sigma'}(\omega)$ is continuous on the set of pairs $\Sigma \leq \Sigma'$;

(K3) *(affine covariance)* $A_*\mu_{\Sigma,\Sigma'} = \mu_{A\Sigma A^{\mathsf{T}},\, A\Sigma' A^{\mathsf{T}}}$ for every $A \in \mathrm{GL}(d)$ and every pair;

(K4) *(nondegeneracy)* $\mu_{0,I} \neq \delta_0$.

We write $\mu_\Sigma := \mu_{0,\Sigma}$ for the *kernel from scale zero*. For $b > 0$ the *affine Gaussian scale space with constant b* is the family in which $\mu_{\Sigma,\Sigma'}$ is the centred Gaussian law with covariance matrix $2b(\Sigma' - \Sigma)$, that is

$$\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = e^{-b\langle \omega, (\Sigma' - \Sigma)\omega \rangle}, \tag{3.1}$$

the law being degenerate, carried by the range of $\Sigma' - \Sigma$, when that matrix is singular.

Lindeberg's convention, in which the matrix of a Gaussian aperture is its covariance, is $b = \frac{1}{2}$ ([37], (45)–(46), p. 44). The parameter set includes the origin and the boundary of $\mathrm{PSD}(d)$, so the kernels from scale zero are members of the family. Positivity is the word "probability" in the definition (Remark 3.12).

Every corner of the scheme, and every cell of the landscape of §4, needs the same bookkeeping. A family is determined by its kernels from scale zero once their transforms have no zeros, those kernels are then characterized by four properties, and a family of a larger datum restricts to a smaller one.

Lemma 3.3 (a (G, W) family is its kernels from scale zero)**.** *Let (G, W) be a wedge datum in dimension d (Definition 3.1).*

1. (Restriction.) *Let (G', W') be a wedge datum in dimension d with $G \subseteq G'$ and with $\Sigma \leq_{W'} \Sigma'$ whenever $\Sigma \leq_W \Sigma'$. The restriction to P_W of a (G', W')-cascade family of kernels is a (G, W)-cascade family of kernels, with the same kernels from scale zero.*

2. (The kernels from scale zero.) *The kernels from scale zero $\mu_\Sigma = \mu_{0,\Sigma}$, $\Sigma \in \mathrm{PSD}(d)$, of a (G, W)-cascade family of kernels satisfy*

   (a) $\mu_0 = \delta_0$ and $\mu_I \neq \delta_0$;

   (b) $A_*\mu_\Sigma = \mu_{A\cdot\Sigma}$ for every $A \in G$ and $\Sigma \in \mathrm{PSD}(d)$;

   (c) for every $\omega \in \mathbb{R}^d$ the map $\Sigma \mapsto \widehat{\mu}_\Sigma(\omega)$ is continuous on $\mathrm{PSD}(d)$;

   (d) for every pair $\Sigma \leq_W \Sigma'$ there is a probability measure ρ on $\mathbb{R}^d$ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$, namely $\rho = \mu_{\Sigma,\Sigma'}$.

If no $\widehat{\mu}_\Sigma$ has a zero, $\mu_{\Sigma,\Sigma'}$ is the only ρ in (d), so the family is determined by its kernels from scale zero.

3. (The converse.) *Let $(\mu_\Sigma)_{\Sigma\in\mathrm{PSD}(d)}$ be probability measures on $\mathbb{R}^d$ whose transforms have no zeros and which satisfy (a)–(d). For $\Sigma \le_W \Sigma'$ let $\mu_{\Sigma,\Sigma'}$ be the probability measure ρ of (d). Then $(\mu_{\Sigma,\Sigma'})$ is a (G,W)-cascade family of kernels, its kernels from scale zero are the μ_Σ, and $\widehat{\mu}_{\Sigma,\Sigma'} = \widehat{\mu}_{\Sigma'}/\widehat{\mu}_\Sigma$.*

So a family whose kernels from scale zero have transforms without zeros is the same thing as a continuous G-covariant field of laws, trivial at 0 and not at I, in which every step of W has a probability measure as its increment. For such a family the continuity axiom (K2) on pairs away from scale zero costs nothing beyond continuity of the kernels from scale zero, because an increment is a quotient of two of them. Clause (1) gives the inclusions between the cells of Table 2. The proof is in Appendix A.2.

Every order of the scheme contains the rays, so the restriction of a family to the ray order of its group is a family with the same kernels from scale zero (Lemma 3.3(1)): the families along rays hold the kernels from scale zero of every family. Along rays the scheme requires of them only (a)–(c) of Lemma 3.3(2) and self-decomposability. The group only says how the kernels at different shapes are related, and continuity in Σ is the one link between shapes that the order does not supply.

Proposition 3.4 (every group along rays: covariant fields of self-decomposable laws)**.** *Let (G,W) be a wedge datum in dimension d whose order is the ray order of Definition 3.1: $\Sigma \le_W \Sigma'$ when $\Sigma = 0$ or $\Sigma' = c\,\Sigma$ for some $c \ge 1$.*

1. (Self-decomposable laws.) *If μ is a self-decomposable probability measure on $\mathbb{R}^d$ (2.2), then $\widehat{\mu}$ has no zeros, and $T_*\mu$ is self-decomposable for every real $d\times d$ matrix T.*

2. (The families.) *A map $\Sigma \mapsto \mu_\Sigma$ from $\mathrm{PSD}(d)$ into the probability measures on $\mathbb{R}^d$ is the family of kernels from scale zero of a (G,W)-cascade family of kernels if and only if it satisfies (a), (b) and (c) of Lemma 3.3(2) and every μ_Σ is self-decomposable. The family is then unique, and its increments are given by $\widehat{\mu}_{\Sigma,c\Sigma} = \widehat{\mu}_{c\Sigma}/\widehat{\mu}_\Sigma$.*

3. (The general linear group.) *Let $G = \mathrm{GL}(d)$. The kernels from scale zero of a $(\mathrm{GL}(d),W)$-cascade family of kernels are $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ for every $\Sigma \in \mathrm{PSD}(d)$, where μ_I is rotation invariant, self-decomposable and not δ_0; with $\vartheta(u) := \widehat{\mu}_I(\sqrt{u}\,e_1)$, $\widehat{\mu}_\Sigma(\omega) = \vartheta(\langle\omega,\Sigma\omega\rangle)$. Conversely every such μ_I is the isotropic member of exactly one $(\mathrm{GL}(d),W)$-cascade family.*

Clause (3) is read again without the scheme in §4.5, as Proposition 4.5. The scheme adds the boundary of $\mathrm{PSD}(d)$, by continuity, and the uniqueness of the family. The proof is in Appendix C.2, with the other proofs that the theorem on the general linear row, Theorem 4.21, uses. By the proof of clause (2) the kernels from scale zero of every family of the scheme are self-decomposable, so their transforms have no zeros, and a family is determined by them (Lemma 3.3(2)).

The isotropic corner. The corner of the similarities with the ray order, read on the multiples of I, is the theory of the isotropic paper up to a change of the scale coordinate. Proposition A.3, stated and proved in Appendix A, makes this precise. When G contains $\mathrm{O}(d)$, the restriction of a family to the pairs (sI,tI), its *slice*, is an isotropic cascade family (Definition 2.4), with nondegeneracy at every pair. Conversely every isotropic family is the slice of exactly one $(\mathrm{GL}(d),\text{rays})$-cascade family, once the scale is read in the *variance gauge* $\varphi = \chi^2$ of [15, Rem. 6.7], the coordinate in which the scale of the Gaussian member is proportional

to its variance, with χ the coordinate of [15, Prop. 6.6] in which the dilations act by multiplication. That change of coordinate is unique up to a constant. A dilation by λ acts on the parameter by $\Sigma \mapsto \lambda^2\Sigma$. So for $v \geq 0$ the v in vI is a squared length, while the volume scale $t(\Sigma) := (\det \Sigma)^{1/(2d)}$ of Lemma 4.1 is a length, with $t(vI) = \sqrt{v}$. In the variance gauge the slice has $\widehat{\mu}_{vI,wI}(\omega) = \exp[-(F(\sqrt{w}\,|\omega|) - F(\sqrt{v}\,|\omega|))]$ for $0 \leq v \leq w$, with (χ, F) the pair of Theorem 2.5 for the isotropic family. The volume scale of $\varphi(s)I$ is then $\chi(s)$ for every scale s. The slice determines a family of the general linear group, and for $d \geq 2$ not one of the similarities.

3.2 The parameter space

The parameter of the scheme is a matrix because the group acts on apertures as it acts on covariance matrices, $A \cdot \Sigma = A\Sigma A^{\mathsf{T}}$. This subsection says why the positive definite matrices are the right index for the kernels. The linear images of a law that no proper subspace carries can be indexed by $\mathrm{PD}(d)$, as μ_Σ with $A_*\mu_\Sigma = \mu_{A\Sigma A^{\mathsf{T}}}$, exactly when one of them is rotation invariant. When G contains $\mathrm{O}(d)$, the axioms supply one, the kernel μ_I. The parameter set is $\mathrm{PSD}(d)$, closed so that it carries the origin and the singular matrices, where a step can be rank-deficient. Its interior $\mathrm{PD}(d)$, of dimension $d(d+1)/2$, is what the group makes of I: every positive definite matrix is $A \cdot I = AA^{\mathsf{T}}$ for an invertible A, and A and AQ give the same matrix for every orthogonal Q, so that $\mathrm{PD}(d) = \mathrm{GL}(d)/\mathrm{O}(d)$. For $G = \mathrm{GL}(d)$, (K3) makes every kernel from scale zero at a positive definite matrix a linear image of the one kernel μ_I, $\mu_{AA^{\mathsf{T}}} = A_*\mu_I$: these kernels form one *orbit* of laws under the group. The *stabilizer* of a law is the set of linear maps that leave it unchanged.

Two lemmas, stated and proved in Appendix A, say when an orbit of laws is indexed by $\mathrm{PD}(d)$. Covariance fixes the parametrization up to one constant: a map of $\mathrm{PD}(d)$ into itself that commutes with every $A \in \mathrm{GL}(d)$ is a positive multiple of the identity (Lemma A.4). A law that no proper subspace carries, a *full* law, has a compact stabilizer, and its orbit can be indexed covariantly by $\mathrm{PD}(d)$ exactly when it has a rotation-invariant member (Lemma A.5(1) and (3)). An $A \in G \cap \mathrm{O}(d)$ fixes I, so (K3) makes μ_I invariant under $G \cap \mathrm{O}(d)$; when G contains $\mathrm{O}(d)$ the axioms therefore supply the rotation-invariant member, the kernel μ_I, whose stabilizer is exactly $\mathrm{O}(d)$ (clause (5)). For $G = \mathrm{GL}(d)$ the kernels from scale zero at $\mathrm{PD}(d)$ then correspond one to one to the matrices. For a smaller G containing $\mathrm{O}(d)$ a kernel determines its parameter on $\{AA^{\mathsf{T}} : A \in G\}$ only (clause (5)). Without a rotation-invariant member a full law can have a trivial stabilizer, and its orbit then has dimension d^2 (clause (4)).

3.3 Infinitely divisible increments from the cascade

A cascade builds each increment from smaller ones, and this subsection shows that, with continuity along a monotone path of parameters, this makes every increment *infinitely divisible*: for every n it is the n-fold convolution of one law with itself (§2.2), a blur that can be applied as n equal smaller blurs. The affine Gaussian scale space has this property in plain sight, its increment (3.1) being the n-fold convolution of the Gaussian with covariance $2b(\Sigma' - \Sigma)/n$. Theorem 3.5 gives it to every family whose order has a path condition, and so to every corner. It matters because an infinitely divisible law is described by its generating triplet, a Gaussian covariance, a Lévy measure and a drift (C2, Theorem 2.11), and §4 reads kernels through those triplets (Proposition 4.25).

An increment along a continuous monotone path of parameters is cut into n small steps, continuity makes them a null array (§2.4), and Khintchine's theorem, C1 (Theorem 2.10), makes their convolution, the increment, infinitely divisible.

Theorem 3.5 (a continuous cascade has infinitely divisible increments)**.** *Let $d \geq 1$.*

1. (A cascade along an interval.) *Let $(\mu_{s,t})_{0\le s\le t\le 1}$ be probability measures on $\mathbb{R}^d$ with $\mu_{s,s} = \delta_0$, $\mu_{r,t} = \mu_{s,t} * \mu_{r,s}$ whenever $r \le s \le t$, and $(s,t) \mapsto \widehat{\mu}_{s,t}(\omega)$ continuous on $\{0 \le s \le t \le 1\}$ for every $\omega \in \mathbb{R}^d$. Then every $\mu_{s,t}$ is infinitely divisible.*

2. (The path condition.) *Let $\mathcal{P} \subseteq \mathrm{PSD}(d)$, let $\le_W$ be a partial order on $\mathcal{P}$, and let $(\mu_{\Sigma,\Sigma'})$ be probability measures on $\mathbb{R}^d$, indexed by the pairs $\Sigma \le_W \Sigma'$ in $\mathcal{P}$, satisfying (K1) and (K2) of Definition 3.1 on these pairs. Call a continuous $\zeta : [0,1] \to \mathcal{P}$ with $\zeta(s) \le_W \zeta(t)$ for all $s \le t$ a W-*chain *from $\zeta(0)$ to $\zeta(1)$. If Σ and Σ' are joined by a W-chain, then $\mu_{\Sigma,\Sigma'}$ is infinitely divisible. In particular, for a wedge datum (G, W) satisfying*

   (P) every pair $\Sigma \le_W \Sigma'$ in $\mathrm{PSD}(d)$ is joined by a W-chain,

   every $\mu_{\Sigma,\Sigma'}$ of a (G,W)-cascade family of kernels is infinitely divisible, and so is every kernel from scale zero μ_Σ.

3. (Orders with the path condition.) *The following satisfy the hypothesis of (2) for every pair, the chain being written out. (a) The Loewner order on $\mathrm{PSD}(d)$: $\zeta(s) := \Sigma + s(\Sigma' - \Sigma)$. (b) The ray order on $\mathrm{PSD}(d)$ (Definition 3.1): $\zeta(s) := s\,\Sigma'$ from $\Sigma = 0$, and $\zeta(s) := (1 + s(c-1))\,\Sigma$ from Σ to $\Sigma' = c\,\Sigma$. (c) Let $\mathcal{K} \subseteq \mathrm{PD}(d)$ satisfy $QMQ^{\mathsf{T}} \in \mathcal{K}$ and $M^\tau \in \mathcal{K}$ whenever $M \in \mathcal{K}$, $Q \in \mathrm{O}(d)$ and $\tau \in [0,1]$. On $\mathcal{P} = \mathrm{PD}(d)$ write $\Sigma \le_{\mathcal{K}} \Sigma'$ when $\Sigma^{-1/2}\Sigma'\Sigma^{-1/2} \in \mathcal{K}$, and suppose that this relation is a partial order. Then it satisfies the hypothesis of (2) for every pair, with $\zeta(s) := \Sigma^{1/2} M^s \Sigma^{1/2}$ and $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$. This covers the Loewner order on $\mathrm{PD}(d)$ ($\mathcal{K} = \{M : M \ge I\}$), the ray order on $\mathrm{PD}(d)$ ($\mathcal{K} = \{cI : c \ge 1\}$), and for $0 < \alpha < 2$ the order of the stable step cone, $\mathcal{K}_\alpha := \{M \in \mathrm{PD}(d) : \lambda_{\min}(M)^{d+\alpha} \ge \det M\}$ with $\lambda_{\min}(M)$ the least eigenvalue; $\mathcal{K}_\alpha$ is the set of e^X with X symmetric and $(d+\alpha)\lambda_{\min}(X) \ge \operatorname{tr} X$. These three relations are partial orders.*

4. (Isotropic families.) *Every $\mu_{s,t}$ of an isotropic cascade family in dimension d (Definition 2.4) is infinitely divisible.*

Clause (3)(c) names one order of §4. For the orbit of an isotropic α-stable law, $0 < \alpha < 2$, a step from Σ to Σ' has a probability measure as its increment exactly when the relative step $M = \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ lies in the *stable step cone* $\mathcal{K}_\alpha$ of clause (3)(c) (Proposition 4.7(1)). The relation it defines is the order of that orbit (Proposition 4.7(2)).

The proof is in Appendix A. Clause (1) is Sato's theorem on additive processes, read for a cascade ([58], Thm. 9.7(i)–(ii), p. 51, with Thm. 9.1, p. 47). There the cascade identity and the limits $\mu_{s,t} \to \delta_0$ as $s \uparrow t$ and as $t \downarrow s$ suffice, without the joint continuity of clause (1). The proof needs only (K1), (K2) and the chain, so the theorem holds along every order with the path condition, whatever the group. Every corner has infinitely divisible increments: the classical corner, the ray corners and the similarities with the Loewner order, whose orders have the path condition on all of $\mathrm{PSD}(d)$. The ray chain $\zeta(s) = s\Sigma$ joins 0 to Σ for every order of the scheme, so every kernel from scale zero is infinitely divisible, with or without (P). For the stable orders of Proposition 4.20(1), clause (3)(c) gives chains between positive definite pairs only. With the ray chains from 0 this covers the increments between positive definite parameters and the kernels from scale zero. A pair with a nonzero singular lower end is a step along a ray, which clause (3)(b) covers, so these orders have the path condition on every pair (Proposition 4.20(1)). At the classical corner and $d \ge 2$ the conclusion is weaker than Theorem 3.6 below, which makes every increment Gaussian without the null array. The theorem is about the steps of a continuous cascade. For a single step $\mu_{\Sigma'} = \mu_\Sigma * \rho$ outside one, a computation on a family of kernels that is not a cascade finds an increment that is nonnegative to rounding, and one that is positive would not be infinitely divisible; positivity is not proved (Remark H.3).

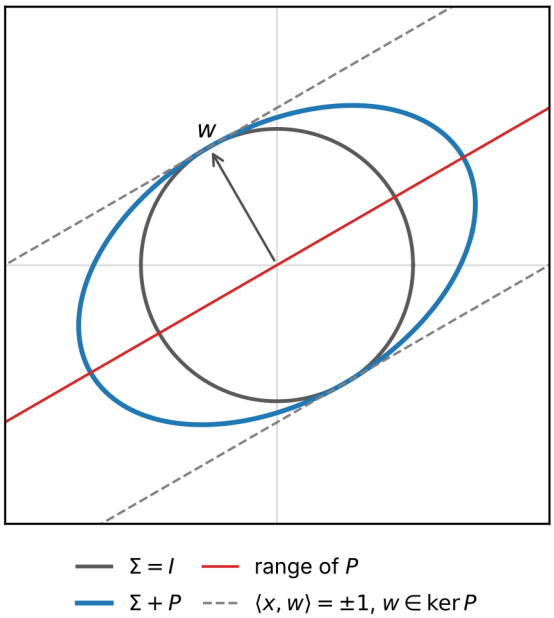


Figure 1: A rank-one Loewner step in the plane, from $\Sigma = I$ to $\Sigma + P$ with $P = \frac{3}{2}uu^{\mathsf{T}}$, $u = (\cos 30°, \sin 30°)$: the apertures $\{x : \langle x, \Sigma^{-1}x\rangle \le 1\}$ of the two matrices. The aperture widens along the range $\mathbb{R}u$ of P. Its extent in a unit direction w of the null space $\ker P$, which is $\langle w, \Sigma w\rangle^{1/2}$, is unchanged, so both ellipses touch the lines $\langle x, w\rangle = \pm 1$. When the kernels depend on Σ through one function g, continuous at 0 with $g(0) = 0$, Theorem 3.7 says that an increment of such a step that is a probability measure is a centred Gaussian law carried by $\mathbb{R}u$, the point mass δ_0 when $g \equiv 0$. Drawn by a script deposited with the paper.

3.4 The single step and rigidity

The theorem of the classical corner is that, for $d \ge 2$, covariance under the whole group $\mathrm{GL}(d)$ with a cascade along the whole Loewner order admits only the affine Gaussian scale space (Theorem 3.6), on kernels and on operators (Theorem 3.13). It derives the classical family from the cascade and covariance, with positivity, continuity and nondegeneracy, without a semigroup or an extremum principle, and Remark 3.15 compares these axioms with Lindeberg's.

The proof turns on one step, the step that shape adaptation takes. The adapted aperture of Lindeberg and Gårding keeps the least eigenvalue of its isotropic first stage, so the step from the first stage to it is a Loewner step from Σ to $\Sigma + P$ with P singular (Proposition 4.23(1)). Such a step widens the aperture along the range of P and leaves the directional variances along the null space $\ker P$ unchanged (Figure 1). Its increment, if it is a probability measure, must then be a blur along the range of P alone. Covariance makes the kernels from scale zero depend on Σ through one function g of $\langle \omega, \Sigma\omega\rangle$ (Lemma 3.9), and for such kernels that requirement is a functional equation whose nonnegative solutions are linear: one such step forces the Gaussian (Theorem 3.7). The rigidity is a corollary of that step.

The rigidity is stated first, then the single step, then the three lemmas on which their proofs rest, which are proved in Appendix A. Positivity enters twice: a probability measure whose transform is 1 on a subspace is carried by its orthogonal complement (Lemma 3.8), and the function g of Lemma 3.9 is nonnegative because a transform has modulus at most 1 (Remark 3.12).

Theorem 3.6 (affine rigidity: the affine Gaussian scale space)**.** *Let $d \ge 2$. A family $(\mu_{\Sigma,\Sigma'})$ is an affine cascade family in dimension d if and only if it is the affine Gaussian scale space with some constant $b > 0$ ((3.1)). In particular the kernel from scale zero at I of an affine cascade*

family in dimension $d \geq 2$ is a centred Gaussian law with covariance a positive multiple of I, and no other probability law on $\mathbb{R}^d$ is the kernel μ_I of one.

Read against Theorem 2.5 and Proposition A.3(1), the theorem says that for $d \geq 2$ no isotropic family with a non-Gaussian kernel is, after a change of gauge, the slice of an affine cascade family, since the kernel μ_I of an affine cascade family is Gaussian. The proof turns on the single step.

Theorem 3.7 (one rank-deficient step forces the Gaussian)**.** *Let $d \geq 2$ and let $g : [0,\infty) \to [0,\infty)$ be continuous at 0 with $g(0) = 0$. Suppose there are $\Sigma \in \mathrm{PD}(d)$, $P \in \mathrm{PSD}(d)$ with $0 < \operatorname{rank} P < d$, and a probability measure ρ on $\mathbb{R}^d$ with*

$$e^{-g(\langle\omega,(\Sigma+P)\omega\rangle)} = e^{-g(\langle\omega,\Sigma\omega\rangle)}\,\widehat{\rho}(\omega) \qquad (\omega \in \mathbb{R}^d). \tag{3.2}$$

Then $g(u) = b\,u$ with $b = g(1) \geq 0$, and ρ is the centred Gaussian law with covariance $2bP$. Conversely, if $g(u) = bu$ with $b \geq 0$, then (3.2) holds with that Gaussian ρ for every $\Sigma, P \in \mathrm{PSD}(d)$.

The hypotheses are about one step alone: ρ is any probability measure, and g is any non-negative function continuous at 0 with $g(0) = 0$, whether or not $e^{-g(\langle\omega,\Sigma\omega\rangle)}$ is a characteristic function. A step of full rank widens the aperture in every direction, and $\ker P = \{0\}$ leaves the subspace argument nothing to act on. Whether such a step can have a probability measure as its increment depends on g: for the Matérn orbit no full-rank step can (Proposition 4.17), and for the isotropic stable orbit the steps of a cone can (Proposition 4.7).

The proofs of the two theorems rest on three lemmas, proved in Appendix A.

Lemma 3.8 (a characteristic function equal to 1 on a subspace)**.** *Let ρ be a probability measure on $\mathbb{R}^d$ and $H \subseteq \mathbb{R}^d$ a linear subspace with $\widehat{\rho}(\eta) = 1$ for every η in a neighbourhood of 0 in H. Then ρ is carried by the orthogonal complement $H^{\perp}$, and $\widehat{\rho}(\omega + \eta) = \widehat{\rho}(\omega)$ for all $\omega \in \mathbb{R}^d$ and $\eta \in H$. In particular, if $\widehat{\rho} = 1$ near 0 then $\rho = \delta_0$.*

The second statement of the lemma also follows from a general fact. For a positive definite function f with $f(0) = 1$, the set G_f where $|f| = 1$ is a group, and $f(x + y) = f(x)f(y)$ for $x \in G_f$ and every y ([57], Cor. 1.4.13, p. 29). For $f = \widehat{\rho}$, every $\eta \in H$ is a sum of points of the neighbourhood where $\widehat{\rho} = 1$, so the product rule gives $\widehat{\rho}(\eta) = 1$. Then $\widehat{\rho}(\omega + \eta) = \widehat{\rho}(\eta)\widehat{\rho}(\omega) = \widehat{\rho}(\omega)$.

The next lemma reduces an affine cascade family to one function of one variable. Covariance makes the kernels from scale zero depend on Σ through $\langle\omega, \Sigma\omega\rangle$. Along the ray of I the cascade (K1) and continuity (K2) exclude a first zero of the transform, so the transform is positive and g is its negative logarithm. Nondegeneracy (K4) makes $g \not\equiv 0$.

Lemma 3.9 (reduction to one function)**.** *Let $(\mu_{\Sigma,\Sigma'})$ be an affine cascade family in dimension d. There is a continuous $g : [0,\infty) \to [0,\infty)$ with $g(0) = 0$ and $g \not\equiv 0$ such that*

$$\widehat{\mu}_{\Sigma}(\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)} \qquad (\Sigma \in \mathrm{PSD}(d),\ \omega \in \mathbb{R}^d), \tag{3.3}$$

and for every pair $\Sigma \leq \Sigma'$

$$\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = \exp\bigl[-g(\langle\omega,\Sigma'\omega\rangle) + g(\langle\omega,\Sigma\omega\rangle)\bigr]. \tag{3.4}$$

A single Loewner step turns (3.4) into a functional equation for g, which the next lemma solves.

Lemma 3.10 (one ratio forces linearity)**.** *Let $g : [0,\infty) \to [0,\infty)$ be continuous at 0 with $g(0) = 0$, and let $a > 1$ be such that*

$$g(au + w) - g(u + w) = g(au) - g(u) \qquad \text{for all } u > 0,\ w \geq 0. \tag{3.5}$$

Then $g(u) = b\,u$ for every $u \geq 0$, with $b = g(1) \geq 0$.

With the three lemmas the two theorems are proved, the single step first.

Proof of Theorem 3.7. Transport to the identity and diagonalize. Put $P' := \Sigma^{-1/2} P \Sigma^{-1/2} \in \mathrm{PSD}(d)$, of rank $r := \operatorname{rank} P$, and choose $Q \in \mathrm{O}(d)$ with $Q^{\mathsf{T}} P' Q = \operatorname{diag}(p_1, \dots, p_r, 0, \dots, 0)$, $p_i > 0$. For $\omega = \Sigma^{-1/2} Q \zeta$ one has $\langle \omega, \Sigma\omega \rangle = |\zeta|^2$ and $\langle \omega, P\omega \rangle = \sum_{i \le r} p_i \zeta_i^2$. Let $C := Q^{\mathsf{T}} \Sigma^{-1/2}$ and $\rho' := C_* \rho$, so that $\widehat{\rho'}(\zeta) = \widehat{\rho}(C^{\mathsf{T}}\zeta) = \widehat{\rho}(\Sigma^{-1/2} Q \zeta)$. Dividing (3.2) by $e^{-g(\langle \omega, \Sigma\omega \rangle)}$, which has no zero,

$$\widehat{\rho'}(\zeta) = \exp\Big[-g\Big(|\zeta|^2 + \sum_{i \le r} p_i \zeta_i^2 \Big) + g\big(|\zeta|^2 \big) \Big].$$

The subspace. On $H := \{\zeta : \zeta_1 = \cdots = \zeta_r = 0\}$, of dimension $d - r \ge 1$, the exponent vanishes and $\widehat{\rho'} = 1$. By Lemma 3.8, $\widehat{\rho'}(\zeta + \eta) = \widehat{\rho'}(\zeta)$ for $\eta \in H$. Take $\zeta = t e_1$ with $t > 0$ and $\eta \in H$ arbitrary, and put $a := 1 + p_1 > 1$, $u := t^2$, $w := |\eta|^2$: the invariance reads

$$g(au + w) - g(u + w) = g(au) - g(u),$$

and as t ranges over $(0, \infty)$ and η over $H \neq \{0\}$, the pair (u, w) ranges over $(0, \infty) \times [0, \infty)$. That is (3.5), and Lemma 3.10 gives $g(u) = bu$ with $b = g(1) \ge 0$.

The increment, and the converse. Now (3.2) reads $\widehat{\rho}(\omega) = e^{-b\langle \omega, P\omega \rangle}$, the characteristic function of the centred Gaussian law with covariance $2bP$ (the law of $(2bP)^{1/2} Z$, Z standard normal on $\mathbb{R}^d$); by uniqueness of the transform ρ is that law. Conversely, for $g(u) = bu$ the exponents in (3.2) are linear in the matrix, and the same Gaussian satisfies it for any $\Sigma, P \in \mathrm{PSD}(d)$. □

Proof of Theorem 3.6. Necessity. By Lemma 3.9 there is a continuous $g \ge 0$ with $g(0) = 0$ and $g \not\equiv 0$ satisfying (3.3) and (3.4). Take $\Sigma = I$ and $P = e_1 e_1^{\mathsf{T}}$, of rank $1 < d$, and $\rho := \mu_{I, I+P}$, a probability measure. Then (3.4) at the pair $I \le I + P$, multiplied by $e^{-g(|\omega|^2)}$, is (3.2). So Theorem 3.7 gives $g(u) = bu$, and $b > 0$ because $g \not\equiv 0$. Now (3.4) is (3.1) for every pair, and by uniqueness of the transform each $\mu_{\Sigma,\Sigma'}$ is the stated Gaussian.

Sufficiency. Let $b > 0$ and let $\mu_{\Sigma,\Sigma'}$ be the centred Gaussian with covariance $2b(\Sigma' - \Sigma)$, which exists for every positive semidefinite covariance as the law of $(2b(\Sigma' - \Sigma))^{1/2} Z$. Its transform is (3.1). The axioms of Definition 3.2 hold. (K1): the transform at $\Sigma = \Sigma'$ is 1, and the exponents of $\mu_{\Sigma',\Sigma''}$ and $\mu_{\Sigma,\Sigma'}$ add to that of $\mu_{\Sigma,\Sigma''}$; by uniqueness of the transform, $\mu_{\Sigma,\Sigma} = \delta_0$ and the convolution identity holds. (K2): the transform is continuous in (Σ, Σ'). (K3): $\widehat{A_* \mu_{\Sigma,\Sigma'}}(\omega) = e^{-b\langle A^{\mathsf{T}}\omega, (\Sigma' - \Sigma) A^{\mathsf{T}} \omega \rangle} = e^{-b\langle \omega, (A\Sigma' A^{\mathsf{T}} - A\Sigma A^{\mathsf{T}})\omega \rangle}$. (K4): $\widehat{\mu}_{0,I}(e_1) = e^{-b} \neq 1$.

The last clause. By necessity, $\widehat{\mu}_I(\omega) = e^{-b|\omega|^2}$ with $b > 0$. □

The proof uses (K1)–(K4) through Lemma 3.9 and then one step of the cascade, $I \le I + e_1 e_1^{\mathsf{T}}$, and of g it needs only continuity at 0 and $g \ge 0$.

Continuity (K2) enters that proof in Lemma 3.9, twice: to exclude a first zero of the transform along the ray of I, and to reach the kernels at the singular apertures. The cascade and the dilations do the first without it, by making μ_I self-decomposable, and the second cannot be done without it.

Proposition 3.11 (continuity is needed only at the boundary of $\mathrm{PSD}(d)$)**.** *Let $d \ge 2$, and let $(\mu_{\Sigma,\Sigma'})$ be probability measures on $\mathbb{R}^d$, indexed by the pairs $\Sigma \le \Sigma'$ in $\mathrm{PSD}(d)$, that satisfy (K1), (K3) and (K4) of Definition 3.2, but not necessarily (K2).*

1. (Between positive definite apertures.) *There is $b > 0$ such that for every $\Sigma \in \mathrm{PD}(d)$ the kernel from scale zero μ_Σ is the centred Gaussian law with covariance $2b\Sigma$, and for every pair $\Sigma \le \Sigma'$ in $\mathrm{PD}(d)$ the increment $\mu_{\Sigma,\Sigma'}$ is the centred Gaussian law with covariance $2b(\Sigma' - \Sigma)$. In particular μ_I is the centred Gaussian law with covariance $2bI$.*

2. (At the boundary.) *Let $b > 0$ and $0 \le \theta_0 < 1$, put $\theta(\Sigma) := 1$ for $\Sigma \in \mathrm{PD}(d)$ and $\theta(\Sigma) := \theta_0$ for singular $\Sigma \in \mathrm{PSD}(d)$, and for $\Sigma \le \Sigma'$ let $\mu_{\Sigma,\Sigma'}$ be the centred Gaussian law with covariance $2b(\theta(\Sigma')\Sigma' - \theta(\Sigma)\Sigma)$. This covariance is positive semidefinite for every pair, and the family satisfies (K1), (K3) and (K4) and agrees with the affine Gaussian scale space with constant b on the pairs in $\mathrm{PD}(d)$. Its kernel from scale zero at a nonzero singular P has covariance $2b\theta_0 P$, so the family is not an affine Gaussian scale space, and it does not satisfy (K2).*

Proof. (1). Let $\vartheta(u) := \widehat{\mu}_I(\sqrt{u}\, e_1)$, $u \ge 0$. As in the first step of the proof of Lemma 3.9, which reads only (K3), μ_I is rotation invariant, $\widehat{\mu}_I(\omega) = \vartheta(|\omega|^2)$, and ϑ is continuous and real with $\vartheta(0) = 1$ and $|\vartheta| \le 1$. For $\Sigma \in \mathrm{PD}(d)$, (K3) with $A = \Sigma^{1/2}$ gives $\widehat{\mu}_\Sigma(\omega) = \vartheta(\langle \omega, \Sigma\omega\rangle)$. For $0 < c < 1$, (K1) at $0 \le c^2 I \le I$ gives $\mu_I = \mu_{c^2 I} * \mu_{c^2 I, I}$, and (K3) with $A = cI$ at the pair $(0, I)$ gives $\mu_{c^2 I} = (cI)_* \mu_I$. So μ_I is self-decomposable (2.2), and its transform has no zero (Proposition 3.4(1)). Being real and continuous with $\vartheta(0) = 1$, ϑ is then positive, and $g := -\log \vartheta$ is continuous, nonnegative and $g(0) = 0$. If $g \equiv 0$, then $\widehat{\mu}_I \equiv 1$ and $\mu_I = \delta_0$ (Lemma 3.8), against (K4). Put $P := e_1 e_1^{\mathsf{T}}$, of rank $1 < d$. (K1) at $0 \le I \le I + P$ gives $\mu_{I+P} = \mu_{I,I+P} * \mu_I$, which is (3.2) with $\Sigma = I$ and $\rho = \mu_{I,I+P}$. So Theorem 3.7 gives $g(u) = bu$ with $b = g(1)$, and $b > 0$ as $g \not\equiv 0$. Hence $\widehat{\mu}_\Sigma(\omega) = e^{-b\langle \omega, \Sigma\omega\rangle}$ for $\Sigma \in \mathrm{PD}(d)$, the transform of the centred Gaussian law with covariance $2b\Sigma$. For $\Sigma \le \Sigma'$ in $\mathrm{PD}(d)$, (K1) at $0 \le \Sigma \le \Sigma'$ gives $\widehat{\mu}_{\Sigma'} = \widehat{\mu}_{\Sigma,\Sigma'}\, \widehat{\mu}_\Sigma$, and as $\widehat{\mu}_\Sigma$ has no zero, $\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = e^{-b\langle \omega, (\Sigma' - \Sigma)\omega\rangle}$. The uniqueness of the transform names each law.

(2), the covariances. Let $\Sigma \le \Sigma'$ and $D := \theta(\Sigma')\Sigma' - \theta(\Sigma)\Sigma$. If Σ is definite, so is $\Sigma' \ge \Sigma$, and $D = \Sigma' - \Sigma \ge 0$. If both are singular, $D = \theta_0(\Sigma' - \Sigma) \ge 0$. If Σ is singular and Σ' definite, $D = \Sigma' - \theta_0\Sigma = (\Sigma' - \Sigma) + (1 - \theta_0)\Sigma \ge 0$. So each $\mu_{\Sigma,\Sigma'}$ exists, with transform $e^{-b\langle \omega, D\omega\rangle}$.

(2), the axioms. (K1): at $\Sigma = \Sigma'$, $D = 0$. For $\Sigma \le \Sigma' \le \Sigma''$ the matrices D of the pairs (Σ, Σ') and (Σ', Σ'') add to that of (Σ, Σ''), so the transforms multiply and the uniqueness of the transform gives the cascade identity. (K3): an $A \in \mathrm{GL}(d)$ preserves rank, so $\theta(A\Sigma A^{\mathsf{T}}) = \theta(\Sigma)$, and $A_* \mu_{\Sigma,\Sigma'}$, the centred Gaussian law with covariance $2bADA^{\mathsf{T}}$, is $\mu_{A\Sigma A^{\mathsf{T}}, A\Sigma' A^{\mathsf{T}}}$. (K4): $\theta(0)\, 0 = 0$, so $\mu_{0,I}$ has covariance $2bI$ and is not δ_0. On a pair in $\mathrm{PD}(d)$, $D = \Sigma' - \Sigma$, which is (3.1). For a nonzero singular P, $\mu_{0,P}$ has covariance $2b\theta_0 P$. An affine Gaussian scale space with this $\mu_{0,I}$ has the constant b, by (3.1) at the pair $(0, I)$, and its kernel at P has covariance $2bP \ne 2b\theta_0 P$, as $\theta_0 < 1$ and $P \ne 0$. Finally, $P + n^{-1} I$ is definite and tends to P, while for ω with $P\omega \ne 0$, $\widehat{\mu}_{0,P+n^{-1}I}(\omega) = e^{-b\langle \omega, (P + n^{-1}I)\omega\rangle} \to e^{-b\langle \omega, P\omega\rangle} \ne e^{-b\theta_0\langle \omega, P\omega\rangle} = \widehat{\mu}_{0,P}(\omega)$. So (K2) fails at the pair $(0, P)$. □

So by Proposition 3.11, on the positive definite apertures the axioms need no continuity, and Iijima's, as reported in Remark 3.15, list none. Continuity fixes only the kernels and increments at the singular apertures.

Remark 3.12 (positivity carries weight). The classification theorems of this paper use positivity: their kernels and increments are probability measures. In the rigidity it enters twice: through Lemma 3.8, which says that a probability measure whose transform is 1 on a subspace is carried by its orthogonal complement, and through the sign of the function g of Lemma 3.9. That function is nonnegative because the transform of a probability law has modulus at most 1, and Lemma 3.10 uses the sign, with continuity at 0, in place of every other regularity in Cauchy's equation. Without positivity the increments need not be signed measures: across shapes the increment of the Matérn orbit of index 1 is not even a finite signed measure (Proposition E.4(3)).

The theorem holds on operators too. The passage from operators to kernels (Lemma A.1) has a converse, from a family of kernels to a family of operators continuous in L^1 (Lemma A.2, both in Appendix A), and the two together carry the rigidity over.

Theorem 3.13 (affine rigidity on operators)**.** *Let $d \ge 2$. A family $(\Phi_{\Sigma,\Sigma'})$ of linear operators on $L^1(\mathbb{R}^d)$, indexed by the pairs $\Sigma \le \Sigma'$ in $\mathrm{PSD}(d)$, is a $(\mathrm{GL}(d), \mathit{Loewner})$-cascade family of*

operators (Definition 3.1(1)) if and only if there is $b > 0$ such that $\Phi_{\Sigma,\Sigma'} f = \mu_{\Sigma,\Sigma'} * f$ for every pair and every $f \in L^1(\mathbb{R}^d)$, where $(\mu_{\Sigma,\Sigma'})$ is the affine Gaussian scale space with constant b (Definition 3.2, (3.1)*). The constant b is then unique.*

Proof. Necessity. By Lemma A.1(4) the measures $\mu_{\Sigma,\Sigma'}$ with $\Phi_{\Sigma,\Sigma'} f = \mu_{\Sigma,\Sigma'} * f$ form an affine cascade family in dimension d, and by Theorem 3.6 it is the affine Gaussian scale space with some $b > 0$.

Sufficiency. By Theorem 3.6 the affine Gaussian scale space with constant $b > 0$ is an affine cascade family, that is a $(\mathrm{GL}(d), \text{Loewner})$-cascade family of kernels, and $(\mathrm{GL}(d), \text{Loewner})$ is a wedge datum (Definition 3.1). By Lemma A.2(2) the operators $f \mapsto \mu_{\Sigma,\Sigma'} * f$ form a $(\mathrm{GL}(d), \text{Loewner})$-cascade family of operators.

Uniqueness of b. If $\Phi_{0,I}$ is convolution by the Gaussian laws of constants b and b', the uniqueness clause of Lemma A.1(1) makes the two laws equal, and their transforms (3.1) at $\Sigma = 0$, $\Sigma' = I$ and $\omega = e_1$ give $e^{-b} = e^{-b'}$. □

3.5 Prior art, and the relation to the line

This subsection places Theorem 3.6 among the axiomatics of scale space and the characterizations of the normal law. The family it selects is classical, and so is its covariance under linear maps. The morphological axiomatics reads "affine" in another sense, and the α-scale spaces admit the stable laws, which the classical corner removes. As a characterization of the normal law its core is Kelker's lemma (Remark 3.14), and as an axiomatic derivation it stands beside Lindeberg's (Remark 3.15). Remark 3.16 reads the scheme on the line.

The affine Gaussian scale space is classical. Lindeberg and Gårding define it in the plane by the Gaussian kernels of the positive semidefinite covariance matrices, and it is closed under linear changes of the image coordinates: an invertible B moves the matrix by $\Sigma \mapsto B\Sigma B^{\mathsf{T}}$ ([40], §§4 and 4.1). The rotationally symmetric linear scale space has that closure only for rotations and uniform rescalings (the same source, §4.1).

The word "affine" has another sense in the morphological axiomatics of Alvarez, Guichard, Lions and Morel. There the scale stays a scalar t, and a linear map A of the image is required to be absorbed by a change of that scalar, $AT_{t'} = T_t A$ for some $t'(t, A)$ ([1], [Projection invariance], p. 206). In the plane, with grey-scale and scale invariance among the hypotheses, this leaves the nonlinear affine morphological scale space $\partial u/\partial t = |Du|\,(t\,\mathrm{curv}(u))^{1/3}$ (Thm. 5(iii), p. 229). Their linear multiscale analyses with isometric invariance are the heat equation, up to a rescaling of t (Thm. 3, p. 214), and its rotationally symmetric kernels are not closed under an anisotropic map. Indexing the kernels by matrices is what lets a linear, positive family be covariant under $\mathrm{GL}(d)$. With the cascade along the whole Loewner order and $d \geq 2$, Theorem 3.6 says that the Gaussian is then the only one.

The other neighbours are the axiomatics that admit the stable laws. For linear, even, recursive and scale-invariant filters on the line, Pauwels, Van Gool, Fiddelaers and Moons obtain the transforms $e^{-a|\omega|^\alpha t}$, $\alpha > 0$, so that these conditions do not single out the Gaussian ([49], p. 694). The kernels have zero crossings for $\alpha > 2$ (pp. 694–695). Duits, Florack, de Graaf and ter Haar Romeny write this class on $\mathbb{R}^d$ as the α scale spaces generated by $-(-\Delta)^\alpha$ ([14], (25), p. 277), whose transforms are $e^{-s|\omega|^{2\alpha}}$. Under positivity the parameter lies in $(0, 1]$, with $\alpha = \frac{1}{2}$ the Poisson and $\alpha = 1$ the Gaussian scale space (p. 267). Pauwels' α is the stable index of this paper, and the α of Duits et al. is half of it. Where they are positive, the kernels of both classes are the isotropic stable laws, which the isotropic theory contains. For $d \geq 2$ the classical corner removes them: by Theorem 3.6 no stable law other than the Gaussian is the kernel μ_I of an affine cascade family. Felsberg and Sommer show that the planar Poisson kernel satisfies Iijima's axioms for a scalar scale ([20], Thm. 1, p. 7), and in an earlier outlook propose a 2×2 scaling matrix in place of that scale ([21], §5, p. 130). The elliptical Poisson orbit is the

stable orbit of index 1, whose positive increments form the cone of Proposition 4.7(1), with no rank-deficient step in it (Theorem 3.7).

Two earlier results bear on Theorem 3.6 directly: Kelker's characterization of the normal law, which is its core (Remark 3.14), and Lindeberg's derivation of the same family from other axioms (Remark 3.15).

Remark 3.14 (Kelker's lemma). A rotation-invariant law with independent coordinates is normal. In the plane this is the Herschel–Maxwell theorem ([6], Thm. 0.0.1, p. 1). In every dimension it is Kelker's lemma, proved for the centred laws whose transform has the form $\psi(\langle \omega, \Sigma\omega \rangle)$ with Σ diagonal, of which the rotation-invariant laws are the case $\Sigma = I$ ([31], Lemma 5, p. 423). Fang, Kotz and Ng restate it with pairwise independence ([18], Thm. 4.11, p. 106). The following sketch shows that it is the core of Theorem 3.6. From the base $I - e_1 e_1^{\mathsf{T}}$ of an affine cascade family, $\mu_I = \mu_{I - e_1 e_1^{\mathsf{T}}} * \mu_{I - e_1 e_1^{\mathsf{T}},\, I}$, and by Lemma 3.8 the first factor is carried by $e_1^{\perp}$ and the second by the e_1-axis. So the first coordinate of μ_I is independent of the others. By rotation invariance every coordinate is, so the transform factors over the coordinates. The coordinates are then independent, and Kelker's lemma makes μ_I normal. The step used has a singular base, which lies outside Theorem 3.7 and inside Definition 3.2. What the scale-space setting adds is the single-step form, Theorem 3.7. It reads one step $\Sigma \to \Sigma + P$, with Σ positive definite and P singular, whose increment is any probability measure. And $e^{-g(\langle \omega, \Sigma\omega \rangle)}$ need not be a characteristic function there. Kelker, Fang, Kotz and Ng, Bryc, and Cambanis, Huang and Simons characterize the normal law among the *elliptically contoured* laws, the linear images of rotation-invariant laws shifted by a vector, by properties of one vector, such as independent coordinates, conditional laws, quadratic forms or marginal densities ([31], Lemma 5, p. 423, Thm. 7 and Lemma 8, pp. 424–426; [7], §5, pp. 379–384). Khatri and Mukerjee characterize the normal law by the law of one quadratic form, and Anderson and Fang by the independence of two complementary quadratic forms of one spherical vector ([32], Thm. 1, p. 187; [2], Thm. 1, p. 309). Kelker's route, and that of Cambanis, Huang and Simons, goes through Cauchy's equation for the generator, as Lemma 3.10 does ([31], p. 423; [7], Thm. 6, pp. 379–380). Where the Gaussian has the whole Loewner order, the non-Gaussian orbits keep cones of full-rank steps (Propositions 4.7 and 4.13). A form without any covariance group, with the Darmois–Skitovich theorem ([60], Thm. 2, p. 190) in place of Kelker's lemma, is open problem 7 of §5.1.

Remark 3.15 (Lindeberg's and Iijima's derivations). Lindeberg [37] assumes a semigroup in a scalar scale s, strong continuity with a Sobolev regularity condition, and non-enhancement of local extrema. From these, his Theorem 5 (p. 42) obtains a local generator of order two with a positive semidefinite matrix Σ_0, and the affine Gaussian scale space is its positive definite, mirror-symmetric case (§4.1.2, p. 45). Positivity, unit mass and closure under affine maps are properties of his solution, not axioms, and non-enhancement is what removes the stable laws (§2, p. 39). Theorem 3.6 takes as axioms positivity, a cascade over Loewner-ordered pairs of matrices, continuity in the pair, nondegeneracy at one pair and $\mathrm{GL}(d)$-covariance, for $d \geq 2$. The two results have different shapes. His classifies all non-enhancing continuous semigroups, one Gaussian ray $\{s\Sigma_0\}$ for each matrix Σ_0 and drift, with the affine group acting on the rays afterwards. This one is a rigidity statement for one two-parameter family with a group action. Neither implies the other from its own hypotheses: his axioms have no positivity and no order on matrices, and this one has no non-enhancement and does not cover $d = 1$. The order appears in his later work as the meaning of a larger matrix scale ([38], fn. 7, p. 594). On the ray from $\Sigma = 0$ to $\Sigma' = s\Sigma_0$, the kernel of (3.1) with $b = \frac{1}{2}$ is his kernel (45), p. 44, with zero drift.

Iijima's axioms of 1962 and 1963, as Weickert, Ishikawa and Imiya report them ([64], p. 241), are closer in form. The scale is a positive definite 2×2 matrix. The transformations are linear, translation invariant and positive, and they are closed under every invertible linear map. Any two of them compose to a third member of the family, with no order on the scales. These axioms give the affine Gaussian scale space. For a scalar scale, Felsberg and Sommer find that Iijima's uniqueness proof assumes implicitly a transform continuously differentiable at the origin, which

that of the Poisson kernel is not ([20], p. 8, and n. 2, p. 24). As hypotheses the two sets are incomparable. Iijima composes every pair of scales and asks that the composite be in the family. Theorem 3.6 indexes its increments by pairs, does not ask that they be members, and asks for them only from a matrix to a larger one in the Loewner order. It adds continuity in the pair, which is needed only at the singular matrices (Proposition 3.11). It holds for every $d \geq 2$.

Remark 3.16 (relation to the line). At $d = 1$, $\mathrm{PSD}(1) = [0, \infty)$, and the Loewner order and the ray order are both the order of the half-line. The similarities are all of $\mathrm{GL}(1)$, so the four corners of Definition 3.1 coincide, and every pair is on the slice. By Proposition A.3(1) and (2) with $d = 1$, the families of the scheme are then the isotropic families in dimension 1, read in the variance gauge, and those are the line's families of Definition 2.1 on kernels [16, Rem. 3.26]. So on the line the classical corner is the line's theory with its gauge fixed, and the kernel μ_1 at scale 1 of such a family can be any symmetric self-decomposable law other than δ_0, and only such a law (Theorem 2.3). On the line the relabelling of the scales is free, and the gauge χ of Theorem 2.3 is part of the answer. For $d \geq 2$ and $G = \mathrm{GL}(d)$ the group acts transitively on $\mathrm{PD}(d)$, and Lemma A.4 forces the parametrization up to one constant. The rigidity belongs to dimension $d \geq 2$: Theorem 3.7 needs a step with $0 < \operatorname{rank} P < d$, which the line does not have.

4 The affine landscape

Theorem 3.6 closes the classical corner of the axiom scheme: for $d \geq 2$, a family covariant under the whole general linear group that cascades along every step of the Loewner order is an affine Gaussian scale space. A user who wants another kernel at elliptical apertures, the Poisson kernel or a Matérn kernel, has to give something up, and there are two things to give up. One is covariance, a smaller group of linear maps of the image. The other is the cascade, an order smaller than the Loewner order: fewer pairs of apertures Σ, Σ' along which the record at Σ' is a further smoothing of the record at Σ. This section maps the families for each choice of the two (Table 2). Every order used here is invariant under all of $\mathrm{GL}(d)$, so it pairs with every group. Along the general linear row the families run from the affine orbits of all rotation-invariant self-decomposable laws, each a round kernel stretched to every elliptical aperture, on the rays (Proposition 4.5), to the Gaussian alone, on the Loewner order. In between, the orbit of the isotropic γ-stable law, $0 < \gamma < 2$, is a family along the stable order of index α exactly when $\gamma \geq \alpha$ (Theorem 4.21(3)), and a Matérn orbit cascades along the rays and along no other step (Proposition 4.17). What turns on the order is how fast an adapted aperture may change its shape between two scales while the record stays a further smoothing (§4.1). A smaller group admits more families in every column.

§4.1 says what an affine scale space is used for, and what each use asks of the axioms. §4.2 lists the groups and reads the Loewner order in volume and shape. §4.3 gives the destination first: the overview table of the families of each group along each order, with forward references. The rest fills it in. §4.4 describes the invariant orders on positive definite matrices through their cones, which index the chain of orders along which the general linear row is read. §4.5 treats the base of the general linear row, the rays, and §4.6 the steps each orbit survives beyond them, up to the stable order on $\mathrm{PSD}(d)$. §4.7 states and proves the theorem on the general linear row. §4.8 reads the practice of shape-adapted smoothing against the families, and §4.9 summarizes the rows of the smaller groups. Appendices A–C hold the proofs that §3 and this section defer, Appendix C the complete proof of the theorem on the general linear row. Appendices D–H hold further results that the table cites.

4.1 What an affine scale space is for

The use of an affine scale space considered here is shape adaptation. An aperture whose shape fits the local image structure measures the same thing in every affine image of a pattern.

The standard form of this is the iteration of Lindeberg and Gårding [39]. An uncommitted, rotationally symmetric first stage, of variance w, fixes a point and an integration scale. The second-moment matrix of the image at that point then fixes the shape of the next aperture. The image is smoothed again with the new covariance, and the step is repeated until the shape no longer changes ([39], p. 396, steps 1–4; [40], §6.2). The iteration is the basis of the affine-invariant features of wide-baseline matching. Baumberg runs a simplified form of it in the warped image with circular kernels at a fixed local scale, and scales the warp to determinant 1 ([3], §4.2, pp. 777–778). Mikolajczyk and Schmid run it in the warped image too, normalizing the warp so that the patch is enlarged and never shrunk ([46], §3.3, p. 72). Each round convolves the original image. The affine scale-space representation is defined by one convolution of the image for each covariance matrix ([39], eq. (15), p. 394), and every round evaluates it at the new covariance. The second of the four steps divides the shape matrix of the round by its least eigenvalue, so the local scale matrix, w times that normalized shape, keeps its least eigenvalue at w. The least amount of smoothing, and with it the suppression of noise, stays that of the first stage ([40], §§6.1–6.2).

Four uses can be read off this practice, and each asks something different of the axioms.

1. *Shape selection.* At a point, choose the shape of the aperture that fits the local structure. The selected shape must transform with the image, which needs the covariance group. Lindeberg and Gårding select at fixed least smoothing. Selection at fixed volume, with $\det \Sigma$ held constant, is the other natural normalization, and they reject it because it overestimates the anisotropy ([40], §6.1). Baumberg's warp of determinant 1 is this normalization. At fixed volume no Loewner step leads from the isotropic first stage to another shape: if $\Sigma \leq \Sigma'$ and $\det \Sigma = \det \Sigma'$, the relative step $\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ has eigenvalues at least 1 with product 1, so $\Sigma' = \Sigma$. Under that normalization even the Gaussian reaches an adapted aperture only by measuring the image afresh.

2. *Following a shape across scale.* Between stages the adapted aperture grows. Along a ray $\Sigma \to \lambda^2\Sigma$ the coarser record is a further smoothing of the finer one only if the rays belong to the order.

3. *Changing shape.* When the shape is re-estimated as the scale grows, the adapted apertures trace a curve in scale and shape, not a ray. The curve is a cascade only if each of its steps lies in the order, and each family limits how fast the shape may change along it.

4. *Singularity theory over the larger group.* A larger covariance group gives more to take the quotient by, and a larger parameter space (x, Σ) in which to study the deep structure of the image. This use is not treated here (§5).

So the question splits into a choice of group and a choice of order. Shape selection, in the sense used here, transports a shape with the image under every linear map, so it needs a group that carries every shape to every other, as $\mathrm{GL}(d)$ does on $\mathrm{PD}(d)$. Among the families covariant under $\mathrm{GL}(d)$, what differs is the order. A family cascades along a step $\Sigma \to \Sigma'$ when its kernel at Σ' is its kernel at Σ convolved with a probability measure, the *increment*. The coarser record is then a further smoothing of the finer one. That the increment is a probability measure carries weight (Remark 3.12). Off the order the axioms ask for no increment. A change of aperture with no increment that is a probability measure is called a *re-measurement* of the original image. The coarser record is then not a further smoothing of the finer one. §4.8 returns to the four uses with the results in hand.

4.2 The groups, and the Loewner order in volume and shape

The landscape has two axes, and this subsection sets out both. Its rows are groups, and the list below says which shapes each group carries to which. Its columns are the rays, the stable

orders and the Loewner order. Each contains the rays and is contained in the Loewner order (Proposition 4.20(1) for the stable orders), so every step of every column is a Loewner step. Read in coordinates of volume and shape, the Loewner order says two things a user of shape adaptation needs (Lemma 4.1). Two distinct apertures of the same volume are never ordered, so a change of shape at fixed volume is never a further smoothing, for the Gaussian either. And one step can change the shape only within a limit set by the growth of the volume. Figure 2 draws that limit in the plane.

A group of the scheme is a closed subgroup of $\mathrm{GL}(d)$ that contains the dilations, acting on $\mathrm{PSD}(d)$ by $A \cdot \Sigma = A\Sigma A^{\mathsf{T}}$ (Definition 3.1). The landscape uses these groups:

- $\mathrm{GL}(d)$ and its orientation-preserving subgroup $\mathrm{GL}^+(d)$, which act transitively on $\mathrm{PD}(d)$, with stabilizers of I the orthogonal group $\mathrm{O}(d)$ and the rotation group $\mathrm{SO}(d)$;
- the similarities $\mathrm{S}(d) = \{cQ : c > 0,\ Q \in \mathrm{O}(d)\}$, whose orbits in $\mathrm{PSD}(d) \setminus \{0\}$ are indexed by the normalized spectrum (Proposition F.3(1));
- the diagonal matrices with positive entries, $\mathrm{D}_+(d)$, whose orbits in $\mathrm{PD}(d)$ are indexed by the correlation matrices (Proposition F.4(1));
- the dilations $\{\lambda I : \lambda > 0\}$, which each of them contains.

The special linear group and $\mathrm{O}(d)$ contain no dilations, so neither is the group of a wedge datum. They enter as the shape part of $\mathrm{GL}(d)$ and as the stabilizer of I. A *one-parameter group* is the set $\{e^{sB} : s \in \mathbb{R}\}$ of the linear maps generated by one real $d \times d$ matrix B, with e^{sB} the matrix exponential, so that $e^{sB}e^{s'B} = e^{(s+s')B}$. For $B = I$ it is the group of dilations $e^s I$, for a skew-symmetric B a group of rotations, and for $B = \mathrm{diag}(1, 0, \dots, 0)$ the stretchings $\mathrm{diag}(e^s, 1, \dots, 1)$ of the first axis. Such a group moves an aperture Σ_0 along the curve $e^{sB}\Sigma_0 e^{sB^{\mathsf{T}}}$, which is not an order on $\mathrm{PSD}(d)$, and its families are treated on their own in Appendix G.

Orientation. For $d \geq 2$ the orientation-preserving group gives no row of its own. An order invariant under $\mathrm{GL}^+(d)$ is invariant under $\mathrm{GL}(d)$, and the $(\mathrm{GL}^+(d), W)$-cascade families of kernels and of operators are the $(\mathrm{GL}(d), W)$-cascade families (Proposition D.1, stated and proved in Appendix D). So for $d \geq 2$ every statement made below for $(\mathrm{GL}(d), W)$ holds verbatim for $(\mathrm{GL}^+(d), W)$.

The space $\mathrm{PD}(d) = \mathrm{GL}(d)/\mathrm{O}(d)$ has natural coordinates of volume and shape, and the Loewner order of Definition 3.2 reads simply in them. The reading tells a vision application which comparisons across scale and shape exist at all.

Lemma 4.1 (the Loewner order in scale–shape coordinates)**.** *For $\Sigma \in \mathrm{PD}(d)$ put $t(\Sigma) := (\det \Sigma)^{1/2d}$, the* volume scale*, and $S(\Sigma) := t(\Sigma)^{-2}\Sigma$, the* shape*, so that $\det S(\Sigma) = 1$. Let $\Sigma_1, \Sigma_2 \in \mathrm{PD}(d)$, $t_k := t(\Sigma_k)$, $S_k := S(\Sigma_k)$, and let $n_1 \leq \cdots \leq n_d$ be the eigenvalues of the* relative shape *$N := S_1^{-1/2} S_2 S_1^{-1/2}$, so that $n_1 \cdots n_d = 1$.*

1. *$\Sigma_1 \leq \Sigma_2$ if and only if $n_1 \geq (t_1/t_2)^2$.*
2. (No order within a scale.) *If $\Sigma_1 \leq \Sigma_2$ then $t_1 \leq t_2$, with equality only if $\Sigma_1 = \Sigma_2$; and $\Sigma_1 \leq \Sigma_2$ with $S_1 = S_2$ holds exactly when $\Sigma_2 = \lambda^2 \Sigma_1$ with $\lambda \geq 1$.*
3. (A speed limit on shape.) *If $\Sigma_1 \leq \Sigma_2$ then $(t_1/t_2)^2 \leq n_1 \leq n_d \leq (t_2/t_1)^{2(d-1)}$, so the anisotropy gained, $\sqrt{n_d/n_1}$, is at most $(t_2/t_1)^d$. Conversely, for every $t_2 \geq t_1$ and every $N \in \mathrm{PD}(d)$ with $\det N = 1$ and least eigenvalue at least $(t_1/t_2)^2$, $\Sigma_2 := t_2^2 S_1^{1/2} N S_1^{1/2}$ satisfies $\Sigma_1 \leq \Sigma_2$, $t(\Sigma_2) = t_2$, and has relative shape N.*
4. (The plane.) *Let $d = 2$ and $\delta(S_1, S_2) := \big((\log n_1)^2 + (\log n_2)^2\big)^{1/2}$. For $t_2 \geq t_1$, the shapes $S(\Sigma_2)$ of the $\Sigma_2 \geq \Sigma_1$ with $t(\Sigma_2) = t_2$ are exactly the shapes S with $\det S = 1$ and $\delta(S_1, S) \leq 2\sqrt{2}\,\log(t_2/t_1)$.*

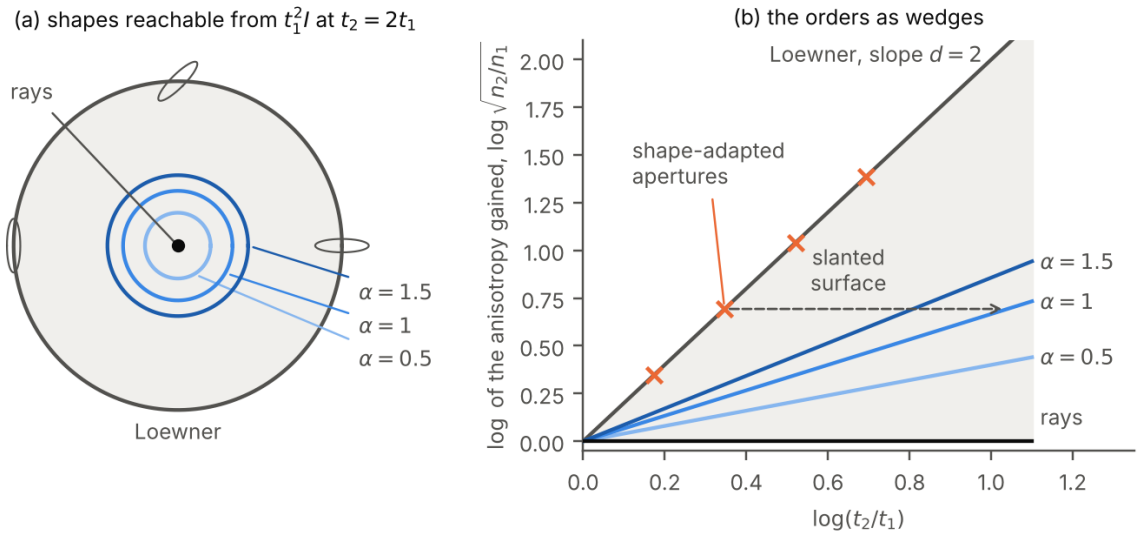


Figure 2: The plane, $d = 2$: one step from the isotropic aperture $\Sigma_1 = t_1^2 I$. (a) The shapes $S(\Sigma_2)$, $\det S = 1$, reachable at volume scale $t_2 = 2t_1$, in a chart whose radius is the distance $\delta(I, S)$ of Lemma 4.1(4) and whose angle is twice the orientation of the major axis. Rotating the aperture through θ turns $\log S$, a point of the plane of trace-free symmetric matrices, through 2θ, so orthogonal orientations lie at opposite points. The Loewner order reaches the disc of radius $2\sqrt{2}\log(t_2/t_1)$ (Lemma 4.1(4)); the stable order of index α the disc of radius $2\sqrt{2}\,\frac{\alpha}{2+\alpha}\log(t_2/t_1)$ (Proposition 4.7(3)); the rays only the centre. The ellipses are apertures on the Loewner edge in three orientations. (b) The same orders in the logarithm of the anisotropy gained, $\sqrt{n_2/n_1}$, against the logarithm of the scale ratio: slopes d, $d\alpha/(d+\alpha)$ and 0. The crosses are the apertures of shape-adapted smoothing at anisotropy $n_2/n_1 = 2$, 4, 8, 16, reached from the isotropic first stage (Proposition 4.23), which lie on the Loewner edge. The dashed arrow is the step for the surface at slant 60° of §4.8, at log-anisotropy $\log 2$. It runs from the aperture of shape-adapted smoothing on the Loewner edge to the edge of the stable order of index $\alpha = 1$, which reaches that shape only at twice the volume scale. Computed, with the discs checked against the two orders on random steps, by a script deposited with the paper.

Clause (2) says that a selection of shape at fixed volume is a re-measurement for every family, the Gaussian included. No two distinct apertures of the same volume are comparable, so no cascade axiom bears on such a selection. Clause (3) is a speed limit on the anisotropy gained per step, and it is the Gaussian's, since the Gaussian cascades along the whole Loewner order. The stable orders of §4.6 have narrower limits of the same form (Proposition 4.7(3)). Figure 2 draws both in the plane: the shapes one step can reach form discs around the starting shape, nested by the order, and the apertures of shape-adapted smoothing lie on the edge of the Loewner disc. The number $\delta(S_1, S_2)$, the norm $(\mathrm{tr}(\log N)^2)^{1/2}$ of the logarithm of the relative shape, is the distance between S_1 and S_2 in the affine-invariant Riemannian metric on the positive definite matrices, the metric invariant under $\Sigma \mapsto A\Sigma A^{\mathsf{T}}$ ([52], §§3.2–3.3, pp. 46–47; [5], Thm. 6.1.6 and (6.13), p. 205). Clauses (1) and (2) rest on textbook facts about the Loewner order. For $\Sigma_1, \Sigma_2 \in \mathrm{PD}(d)$, $\Sigma_1 \le \Sigma_2$ exactly when every eigenvalue of $\Sigma_1^{-1/2}\Sigma_2\Sigma_1^{-1/2}$ is at least 1 ([26], Thm. 7.7.3(a), p. 494). The eigenvalues and the determinant increase along the order (Cor. 7.7.4(c) and (e), p. 495). The proof of the lemma, from the congruence $M = \Sigma_1^{-1/2}\Sigma_2\Sigma_1^{-1/2} = (t_2/t_1)^2 N$, is in Appendix B.

4.3 The landscape at a glance

Table 2 answers, for each choice of covariance group and of the pairs of apertures along which the record must be a further smoothing, which kernels are available. It has the groups of §4.2 as rows and three orders as columns, for $d \geq 2$, and every cell has an entry, citing the statements it summarizes in §§4.4–4.9 or in the appendices. The cells use these terms.

- The *ray order* relates Σ to its multiples $c\,\Sigma$, $c \geq 1$, and 0 to everything.
- The *stable order* $\leq_\alpha$, $0 < \alpha < 2$, relates two definite matrices when the relative step $M = \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ has $\lambda_{\min}(M)^{d+\alpha} \geq \det M$, and adds the origin and the rays (Proposition 4.20(1)). It lies between the rays and the Loewner order. It is the order of the step set (Proposition 4.2, §4.6) of the stable orbit of index α, the linear images $(\Sigma^{1/2})_* S_\alpha(I)$ of the isotropic α-stable law, whose transform is $e^{-|\omega|^\alpha}$ (Lemma 4.6 and Proposition 4.7). At $\alpha = 1$ that law is the kernel of the Poisson scale space at unit scale (§3.5). And it is the order generated by the cone of directions $C_\alpha := \{X \in \mathrm{PSD}(d) : (d+\alpha)\,\lambda_{\min}(X) \geq \operatorname{tr} X\}$ (Proposition 4.12); C_0 gives the rays and $C_\infty := \mathrm{PSD}(d)$ the Loewner order.
- The *frame* is the coordinate frame, the axes of the sampling grid along which the positive diagonal group $\mathrm{D}_+(d)$ rescales. The *frame's shapes* are the positive definite diagonal matrices, the apertures whose axes are those of the frame, and they are the orbit of I under $\mathrm{D}_+(d)$. For $s \in (0,\infty)^d$ write $D_s := \operatorname{diag}(s_1, \ldots, s_d)$, so that the frame's shapes are the D_s^2, and write $\Delta(\Sigma) := \operatorname{diag}(\Sigma_{11}, \ldots, \Sigma_{dd})$ for the diagonal of Σ.
- The *exponent cone* $\mathfrak{k}_\alpha$ of the stable law is the set of matrices B for which $S_\alpha(I)$ is its own image under e^{-tB} convolved with a residual, for every $t > 0$. A law with this property is *B-self-decomposable*, or operator-self-decomposable with exponent B (Proposition 4.10). The symmetric part of B is $\operatorname{sym} B := \frac{1}{2}(B + B^{\mathsf{T}})$.
- *Anisotropy* is measured by the exponent spread es of the non-Gaussian part, the least $K \geq 1$ with $\psi(Q\omega) \leq K\,\psi(\omega)$ for every $Q \in \mathrm{O}(d)$. It is set against the eccentricity $\operatorname{ecc}(\Sigma)$ of the aperture, the ratio of its extreme eigenvalues. Both are defined again where they are used, in Proposition 4.25.
- A non-Gaussian part is *shape-blind* when it is rotation invariant at every Σ, so that it sees the size of the aperture and not its shape. It is shape-blind *at the origin* when its Lévy measure has the same rate near the origin in every direction.
- The profile entries concern a rotation-invariant law with no Gaussian part whose Lévy density at x is $k(|x|)\,|x|^{-d}$, with k nonincreasing, its *profile*, and the law is written ν_k. They use the dilation ratio $R(\lambda) := \inf\{k(r/\lambda)/k(r) : r > 0,\ k(r) > 0\}$, $\lambda \geq 1$, and the lower exponent $\beta_-(k) := \inf_{\lambda>1} \log R(\lambda)/\log\lambda$ (Proposition 4.13 and Lemma 4.14). The condition $R(\lambda) \geq \lambda^\alpha$ for every $\lambda \geq 1$ says that $r^\alpha k(r)$ is nonincreasing: towards the origin the profile grows at least like $r^{-\alpha}$.

The one-parameter row lies outside the scheme, a curve not being an order on $\mathrm{PSD}(d)$, and is listed below the others.

The inclusions. A family of a larger group or a larger order restricts to every smaller datum, with the same kernels from scale zero (Lemma 3.3(1)), so in each column the cells grow from the general linear row to each smaller group's row, and along each row towards the rays. These inclusions are known to be strict:

- along the general linear row, at every step of the chain up to $\alpha = 2$ (Theorem 4.21(3));

G	rays	stable order $\leq_\alpha$, $0 < \alpha < 2$	Loewner order
$\mathrm{GL}(d)$	the orbits $(\Sigma^{1/2})_*\mu_I$ of the rotation-invariant self-decomposable laws $\mu_I \neq \delta_0$, one family for each law (Propositions 4.5 and 3.4(3))	the orbits of exactly the laws $N(0, aI)$, $a > 0$, and $N(0, aI) * \nu_k$, $a \geq 0$, with k right-continuous and $\beta_-(k) \geq \alpha$; among the stable orbits, exactly those of index $\gamma \geq \alpha$ (Theorem 4.21(3), Proposition 4.22, Lemma 4.14(3))	the affine Gaussian scale spaces only (Theorem 3.6)
similarities $\mathrm{S}(d)$	continuous covariant fields of symmetric self-decomposable laws over the shapes, with orbits indexed by the normalized spectrum; the isotropic members are the rotation-invariant self-decomposable laws $\neq \delta_0$, and none fixes its family (Proposition F.3)	every family's kernels are symmetric infinitely divisible with $\mathrm{es}(\psi_\Sigma) \leq 4\,\mathrm{ecc}(\Sigma)^{(d+\alpha)/\alpha}$; contains the restrictions of the $\mathrm{GL}(d)$ cell and of the Loewner cell, and a family in neither (Proposition 4.25)	the isotropic members of the ray cell, each with a shape-blind family (Proposition F.3(3)–(4)); every family's kernels are infinitely divisible with $\mathrm{es}(\psi_\Sigma) \leq 4\,\mathrm{ecc}(\Sigma)$ (Proposition 4.25(2)); a uniform bound when the Lévy measures at the thinnest apertures stay comparable to the isotropic member's, and none in general (Propositions H.5, H.9 and H.6); for a profile bounded at the origin, shape-blind at the origin (Proposition H.8) and need not be beyond it (Proposition H.7(3)); not every family is a Gaussian convolved with a shape-blind part (Proposition H.1(3))
positive diagonal $\mathrm{D}_+(d)$	on the frame's shapes the images $(D_s)_*\mu_I$ of any self-decomposable $\mu_I \neq \delta_0$, not necessarily products; orbits in $\mathrm{PD}(d)$ indexed by the correlation matrices; a rotation-invariant member fixes no family off the frame's shapes (Proposition F.4)	on the frame's shapes $(D_s)_*\mu_I$ with μ_I operator-self-decomposable for every diagonal exponent in $\mathfrak{k}_\alpha$, a necessary condition, and whether every such law is the isotropic member of a family is open; strictly between the restricted Loewner cell and the ray cell, and not the union of the restrictions of the Loewner and $\mathrm{GL}(d)$ cells; off the frame's shapes not described (Proposition F.5)	on the frame's shapes the products $(s_1)_*\nu_1 \otimes \cdots \otimes (s_d)_*\nu_d$ of self-decomposable laws on the line (Propositions F.2 and F.4(3)); off the frame's shapes one extension, $(\Delta(\Sigma)^{1/2})_*(\nu_1 \otimes \cdots \otimes \nu_d)$, the families not described
One linear flow $\{e^{sB}\}$, B with a skew part allowed, along the curve $e^{sB}\Sigma_0 e^{sB^\mathsf{T}}$: the families are the B-self-decomposable laws (Proposition G.1); for a contracting flow a rotation-invariant non-Gaussian member exists exactly when $\mathrm{tr}\, B < (d+2)\lambda_{\min}(\mathrm{sym}\, B)$ (Proposition 4.11).			

Table 2: The affine landscape for $d \geq 2$: the families of each group along each order. The four corners of the scheme (Definition 3.1) are the cells of the first two rows in the ray and the Loewner columns. Each cell cites the statements it summarizes. The cells of the general linear group are complete, and §5 says how far the others go.

- from the $\mathrm{GL}(d)$ cell and the restricted Loewner cell to the cell (similarities, stable order) (Proposition 4.25(3));
- from the restricted Loewner cell to the stable cell, and from there to the ray cell, in the diagonal row (Proposition F.5(4)), and so from the restricted (diagonal, Loewner) cell to the (diagonal, rays) cell (Proposition F.4(4)).

4.4 Invariant wedges

An order of the landscape says which changes of aperture must be further smoothings. This subsection asks which orders there are, and how to describe one by what may happen in an instant. For the Gaussian the answer is familiar. Its aperture may grow in any direction, $\Sigma(t)$ moving with a positive semidefinite velocity $\Sigma'(t)$, and the pairs joined by such a curve are those of the Loewner order (Proposition 4.12(1), at $\alpha = \infty$). The rays allow only a velocity that is a nonnegative multiple of $\Sigma(t)$, a growth without a change of shape. An order in between allows at each aperture a cone of velocities, the *cone field*, and relates two apertures when a curve whose velocity stays in the cones joins them. The main result, Theorem 4.3, says that the closed invariant partial orders on $\mathrm{PD}(d)$ that arise in this way are exactly those whose steps can be taken in fractions, and that every step of such an order is reached along one curve, with no limit of curves needed. The rays, the stable orders and the Loewner order are of this kind, the members C_0, C_α and C_∞ of one chain of cones (Proposition 4.12).

A picture in the plane. For $d = 2$ the cones can be drawn. Write a symmetric matrix as $X = \left(\begin{smallmatrix} a+u & v \\ v & a-u \end{smallmatrix}\right)$, so that $\mathrm{Sym}(2)$ is a space of three dimensions with coordinates (a, u, v), and X has the eigenvalues $a \pm \sqrt{u^2+v^2}$. Read X as the velocity of an aperture at I. Then a is its growth in size, $\sqrt{u^2+v^2}$ its rate of change of shape, and the angle of (u, v) the direction in which it stretches. A rotation of the plane by an angle ϕ rotates (u, v) by 2ϕ and leaves a alone. So a closed convex cone that is invariant under rotations, contains I and contains no line is a round cone about the axis of the multiples of I,

$$\sqrt{u^2+v^2} \le k\, a \qquad \text{for some } k \ge 0.$$

Its opening k bounds how fast the shape may change against how fast the aperture grows (Figure 3). The rays are the axis alone, $k = 0$. The Loewner order allows every positive semidefinite velocity, the cone $k = 1$ of half-angle 45°, and a wider cone allows an aperture to shrink in some direction. The cone C_α of the chain has $k = \alpha/(\alpha+2)$, so the stable orders, $0 < \alpha < 2$, are the cones narrower than $k = \frac{1}{2}$. The cone field carries the cone at I to every aperture by the group, and the order relates two apertures when a curve whose velocity stays in the cones joins them. Lie theory calls a closed convex cone a *wedge* (§3.1), and these wedges, which contain no line, are the tangent objects through which it studies such orders. The rest of the section uses them in two ways. Each affine orbit has its own set of steps along which it cascades, and §§4.5–4.6 find it for the orbits of interest: the rays alone for the Matérn kernels, the order of C_γ for the stable kernel of index γ. And in the plane Theorem 4.21 reads off the opening. A cone with $k < \frac{1}{2}$ carries non-Gaussian families, a cone with $\frac{1}{2} \le k \le 1$ carries only the affine Gaussian scale spaces, and a cone with $k > 1$ carries none.

The rest of the subsection is linear algebra, in the language of matrix analysis and of the Lie theory of semigroups, where the orders of a homogeneous space are studied through their cones ("Prior art" at the end of the subsection). An order is described by its set of relative steps $M = \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ (Proposition 4.2), and an $\mathrm{O}(d)$-invariant closed convex cone C of symmetric matrices plays the part of the allowed velocities, through the cone field of Theorem 4.3. A single step is tested by its logarithm: Theorem 4.3(1) shows that a curve of that cone field joins Σ to Σ' exactly when $\log M \in C$. Steps do not compose by multiplication. The composite of two

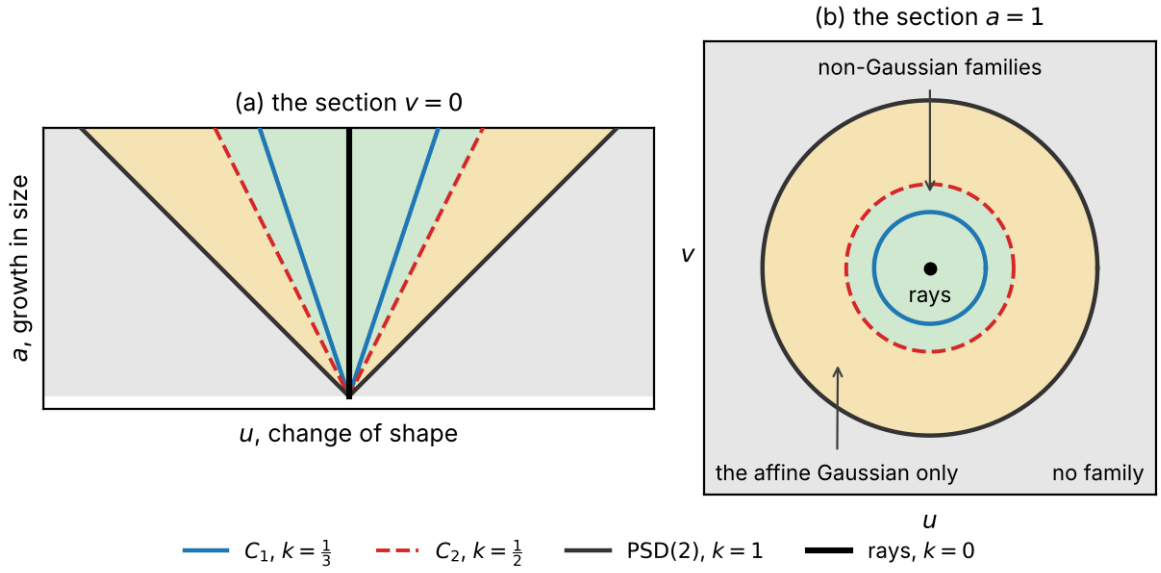


Figure 3: The rotation-invariant cones of velocities in the plane, in the coordinates (a, u, v) of $X = \left(\begin{smallmatrix} a+u & v \\ v & a-u \end{smallmatrix}\right)$: each is a round cone $\sqrt{u^2+v^2} \le k\,a$ about the axis of the multiples of I. (a) The section $v = 0$, where each cone is a sector. (b) The section $a = 1$, where each cone is a disc of radius k. Drawn are the rays ($k = 0$), the cone C_1 ($k = \frac{1}{3}$), along whose order the orbit of the Poisson kernel cascades, the cone C_2 ($k = \frac{1}{2}$) and the Loewner cone PSD(2) ($k = 1$). The shading reads Theorem 4.21 in the plane: a cone with $k < \frac{1}{2}$ (green) carries non-Gaussian families, one with $\frac{1}{2} \le k \le 1$ (amber) only the affine Gaussian scale spaces, and one with $k > 1$ (grey) none. Drawn by a script deposited with the paper.

steps M and N is $M^{1/2}N'M^{1/2}$ with N' a rotation of N (Proposition 4.2(3)), and its logarithm is not $\log M + \log N'$ when M and N' do not commute. So it is not obvious that the test is transitive. A comparison of spectra by partial sums, majorization, bounds the spectrum of the composite whatever the relative position of the eigenframes, with one cited inequality (C4, Theorem 2.13). It gives both the transitivity and the fact that a curve of the cone field ends at a step in $\exp C$. Proposition 4.2 sets up the dictionary between invariant orders and sets of steps in which the theorem is stated.

Proposition 4.2 (invariant relations on $\mathrm{PD}(d)$ and their steps)**.** *Let $d \ge 1$ and write $\mathrm{Sym}(d)$ for the real symmetric $d \times d$ matrices. Call a set $\mathcal{K} \subseteq \mathrm{PD}(d)$ a* step set *if $QMQ^{\mathsf{T}} \in \mathcal{K}$ whenever $M \in \mathcal{K}$ and $Q \in \mathrm{O}(d)$. For a step set write $\Sigma \le_{\mathcal{K}} \Sigma'$, for $\Sigma, \Sigma' \in \mathrm{PD}(d)$, when the* relative step *$\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ lies in $\mathcal{K}$, and put $\mathcal{S}_{\mathcal{K}} := \{A \in \mathrm{GL}(d) : AA^{\mathsf{T}} \in \mathcal{K}\}$.*

1. (Invariant relations are step sets.) *For $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and every $A \in \mathrm{GL}(d)$ with $AA^{\mathsf{T}} = \Sigma$, $\Sigma \le_{\mathcal{K}} \Sigma'$ if and only if $A^{-1}\Sigma' A^{-\mathsf{T}} \in \mathcal{K}$. The relation $\le_{\mathcal{K}}$ is $\mathrm{GL}(d)$-invariant: $\Sigma \le_{\mathcal{K}} \Sigma'$ if and only if $B\Sigma B^{\mathsf{T}} \le_{\mathcal{K}} B\Sigma' B^{\mathsf{T}}$, for every $B \in \mathrm{GL}(d)$. Every $\mathrm{GL}(d)$-invariant relation R on $\mathrm{PD}(d)$ is $\le_{\mathcal{K}}$ for exactly one step set, $\mathcal{K} = \{M \in \mathrm{PD}(d) : I\,R\,M\}$.*

2. (Order properties.) *$\le_{\mathcal{K}}$ is reflexive if and only if $I \in \mathcal{K}$; it is antisymmetric if and only if no $M \ne I$ has both M and M^{-1} in $\mathcal{K}$; and it is closed in $\mathrm{PD}(d) \times \mathrm{PD}(d)$ if and only if $\mathcal{K}$ is closed in $\mathrm{PD}(d)$.*

3. (How steps compose.) *If $\Sigma \le_{\mathcal{K}} \Sigma' \le_{\mathcal{K}} \Sigma''$, M is the relative step of (Σ, Σ') and N that of (Σ', Σ''), then the relative step of (Σ, Σ'') is $M^{1/2}N'M^{1/2}$ with $N' = Q^{\mathsf{T}}NQ$ for some $Q \in \mathrm{O}(d)$, and its eigenvalues are those of MN'. Conversely, for $M, N \in \mathcal{K}$ the triple I, M, $M^{1/2}NM^{1/2}$ is such a chain. So $\le_{\mathcal{K}}$ is transitive if and only if $M^{1/2}NM^{1/2} \in \mathcal{K}$ for*

all $M, N \in \mathcal{K}$, if and only if $\mathcal{S}_{\mathcal{K}}$ is closed under products. The spectrum of the composite is not determined by the spectra of the two steps: for $d = 2$ the step $\mathrm{diag}(4, 1)$ *followed by* $\mathrm{diag}(4, 1)$ *gives* $\mathrm{diag}(16, 1)$*, and followed by* $\mathrm{diag}(1, 4)$ *gives* $4I$*.*

4. (The infinitesimal cone.) *The set $C(\mathcal{K}) := \{X \in \mathrm{Sym}(d) : e^{tX} \in \mathcal{K} \text{ for every } t \geq 0\}$ is invariant under $X \mapsto QXQ^{\mathsf{T}}$, $Q \in \mathrm{O}(d)$, and under $X \mapsto cX$, $c \geq 0$. It is closed when $\mathcal{K}$ is closed in* $\mathrm{PD}(d)$*, and if $\leq_{\mathcal{K}}$ is antisymmetric then $X \in C(\mathcal{K})$ and $-X \in C(\mathcal{K})$ only for $X = 0$.*

Clause (1) is the dictionary between the orders of the scheme and sets of matrices: a $\mathrm{GL}(d)$-invariant order on $\mathrm{PD}(d) = \mathrm{GL}(d)/\mathrm{O}(d)$ is its set of relative steps at I. The proof is in Appendix C.

The invariant cones are read through their eigenvalues. For $X \in \mathrm{Sym}(d)$ write $\lambda^{\uparrow}(X)$ for its eigenvalues in increasing order, and for a cone $C \subseteq \mathrm{Sym}(d)$ put $\exp C := \{e^X : X \in C\}$.

An invariant cone generates an order along curves that move in the cone at each point. The theorem says which orders arise in this way, and its clause (4) gives the orders on $\mathrm{PSD}(d)$ along which the general linear row is read in §4.7.

Theorem 4.3 (invariant orders generated by cone fields)**.** *Let $d \geq 1$ and let $C \subseteq \mathrm{Sym}(d)$ be a closed convex cone invariant under $X \mapsto QXQ^{\mathsf{T}}$, $Q \in \mathrm{O}(d)$. Its* cone field *assigns to $\Sigma \in \mathrm{PD}(d)$ the cone $\Sigma^{1/2} C \Sigma^{1/2}$ of tangent vectors, which equals ACA^{T} for every A with $AA^{\mathsf{T}} = \Sigma$, so that $B(\Sigma^{1/2} C \Sigma^{1/2})B^{\mathsf{T}}$ is the cone at $B\Sigma B^{\mathsf{T}}$ for $B \in \mathrm{GL}(d)$. An absolutely continuous $\gamma : [0, 1] \to \mathrm{PD}(d)$ is C-*causal *if $\gamma(t)^{-1/2} \gamma'(t) \gamma(t)^{-1/2} \in C$ for almost every t; write $\Sigma \preceq_C \Sigma'$ when a C-causal curve runs from Σ to Σ'. Use the notation of Proposition 4.2 (step sets $\mathcal{K}$, their relations $\leq_{\mathcal{K}}$, the sets $\mathcal{S}_{\mathcal{K}}$ and the cones $C(\mathcal{K})$), write $\exp C := \{e^X : X \in C\}$, and write $\lambda_1^{\uparrow}(M)$ for the least eigenvalue of M.*

1. (The order of a cone field.) *$\Sigma \preceq_C \Sigma'$ if and only if the relative step $M = \Sigma^{-1/2} \Sigma' \Sigma^{-1/2}$ lies in $\exp C$, and then $\gamma(t) := \Sigma^{1/2} M^t \Sigma^{1/2}$ is C-causal. So $\preceq_C$ is $\leq_{\exp C}$: a closed $\mathrm{GL}(d)$-invariant preorder, which is a partial order if and only if $C \cap (-C) = \{0\}$; and $C(\exp C) = C$.*

2. (Closed invariant orders.) *Let $\mathcal{K}$ be a closed step set for which $\leq_{\mathcal{K}}$ is a partial order. Then $\{B \in \mathfrak{gl}(d) : e^{tB} \in \mathcal{S}_{\mathcal{K}} \text{ for every } t \geq 0\} = \{B : \frac{1}{2}(B + B^{\mathsf{T}}) \in C(\mathcal{K})\}$, and this is a closed convex cone containing every skew-symmetric matrix; $C(\mathcal{K})$ is an $\mathrm{O}(d)$-invariant closed convex cone with $C(\mathcal{K}) \cap (-C(\mathcal{K})) = \{0\}$; and $\preceq_{C(\mathcal{K})}$ is contained in $\leq_{\mathcal{K}}$. The following are equivalent: (a) $\leq_{\mathcal{K}}$ is $\preceq_{C'}$ for some cone C' as above; (b) $\mathcal{K} = \exp C(\mathcal{K})$; (c) $M^{\tau} \in \mathcal{K}$ whenever $M \in \mathcal{K}$ and $0 \leq \tau \leq 1$. When they hold, $C' = C(\mathcal{K})$.*

3. (A closed order without a cone.) *$\mathcal{K} := \{I\} \cup \{M \in \mathrm{PD}(d) : \lambda_1^{\uparrow}(M) \geq 2\}$ is a closed step set, $\leq_{\mathcal{K}}$ is a partial order, and $C(\mathcal{K}) = \{0\}$, so that $\preceq_{C(\mathcal{K})}$ is equality; hence $\leq_{\mathcal{K}}$ is $\preceq_{C'}$ for no cone C'.*

4. (On $\mathrm{PSD}(d)$.) *Suppose $I \in C$ and $C \cap (-C) = \{0\}$. For $\Sigma, \Sigma' \in \mathrm{PSD}(d)$ let $\Sigma \preceq_C^+ \Sigma'$ when $\Sigma = 0$, or $\Sigma' = c\,\Sigma$ for some $c \geq 1$, or $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and $\Sigma \preceq_C \Sigma'$. Then $\preceq_C^+$ is a $\mathrm{GL}(d)$-invariant partial order on $\mathrm{PSD}(d)$ that contains the rays, every pair $\Sigma \preceq_C^+ \Sigma'$ is joined by a chain as in Theorem 3.5(2), and $(G, \preceq_C^+)$ is a wedge datum with the path condition (P) for every closed subgroup G of $\mathrm{GL}(d)$ containing the dilations. It is contained in the Loewner order if and only if $C \subseteq \mathrm{PSD}(d)$. For $C = \{cI : c \geq 0\}$ it is the ray order of Definition 3.1.*

So an $\mathrm{O}(d)$-invariant closed convex cone generates an invariant preorder whose steps are exactly the exponentials of the cone, and a partial order when $C \cap (-C) = \{0\}$. The steps are reached along the geodesics $\Sigma^{1/2} M^t \Sigma^{1/2}$, and no closure is needed. A closed invariant order

comes from a cone exactly when its steps are closed under fractional powers, and clause (3) shows that the restriction is needed: a closed invariant order can have no infinitesimal part at all. At $C = \mathrm{PSD}(d)$, clause (4) adds to the Loewner order of $\mathrm{PD}(d)$ only the origin and the rays. It is therefore smaller than the Loewner order of $\mathrm{PSD}(d)$, which also has steps from nonzero singular matrices to definite ones. The proof, in Appendix C, runs through one family of functionals, which Lemma C.1 makes superadditive along any three points.

A vector $x \in \mathbb{R}^d$ is *majorized* by y when, for every k, the sum of the k smallest entries of x is at least that of y, with equal totals: x is the less spread of the two. The source of C4 states its inequality in the equivalent form of partial sums of the largest entries (§2.4). Two matrix inequalities carry the correspondence. The eigenvalues of a sum $X + Y$ are majorized by the sum of the two spectra, and the logarithmic spectrum of a product $M^{1/2}NM^{1/2}$ by the sum of the logarithmic spectra of M and N. Clause (3) of Proposition 4.2 is the reason majorization is the tool here: the spectrum of a composite is not a function of the two spectra, but majorization bounds it uniformly over the rotation between the eigenframes. Lemma C.1 states the two inequalities, proves the first, and derives the second from the cited fact C4 and a determinant, with the description of the $\mathrm{O}(d)$-invariant closed convex cones by symmetric cones of eigenvalues and the transitivity of $\leq_{\exp C}$. It is in Appendix C, with the proofs it serves.

The cones that matter here form one chain, from the rays to the Loewner order: $C_\alpha := \{X \in \mathrm{PSD}(d) : (d + \alpha)\,\lambda_{\min}(X) \geq \operatorname{tr} X\}$ for $0 \leq \alpha < \infty$, and $C_\infty := \mathrm{PSD}(d)$. Its members of index $0 < \alpha < 2$ are the symmetric parts of the exponent cones of the stable laws, and Proposition 4.12 in §4.6 shows that their orders are the stable orders.

Prior art. The correspondence between invariant cone fields, Lie wedges, semigroups and closed orders on a homogeneous space G/H is Lawson's ([34], Thm. 4.9, p. 290; Props. 6.3–6.4, pp. 295–296; Thms. 7.4–7.5, pp. 300–302), on the Lie theory of semigroups of Hilgert, Hofmann and Lawson ([23], Prop. V.1.7, pp. 375–376). Read on $\mathrm{GL}(d)$ with the involution $A \mapsto A^{-\mathsf{T}}$, their Cor. V.4.58 (p. 440) gives the transitivity of Theorem 4.3(1) for pointed cones, and their abelian example (p. 376) is the model of Theorem 4.3(3). Hilgert and Ólafsson state the bijection between invariant regular cones and invariant regular causal structures ([24], Thm. 2.2.2, p. 40), and the dual description of a closed convex cone is in Faraut and Korányi ([19], Thm. I.1.1, p. 2). Here the correspondence is read on $\mathrm{PD}(d)$ through step sets and their spectra and proved by majorization. The semigroup is in Lawson's paper on polar decompositions, which allows cones that are not pointed and derives the closedness that Cor. V.4.58 asks for: $\mathcal{S}_{\exp C} = (\exp C)\mathrm{O}(d)$ is a closed semigroup with subtangent wedge $C \oplus \mathfrak{so}(d)$, by his Thm. 3.1 ([35], pp. 198–199), as $\operatorname{ad} X$ has real spectrum for symmetric X (his Lem. 4.1 and Example 4.2, p. 203) and the centre of $\mathfrak{gl}(d)$ is $\mathbb{R}I$, which gives his condition (4). By Proposition 4.2(3) that is the transitivity of $\leq_{\exp C}$. His paper has no order on the quotient. The characterization by powers is the form on $\mathrm{PD}(d)$ of their theorem that a closed subsemigroup of a connected Lie group is divisible, with roots of every order, exactly when it is the exponential image of its tangent wedge, the set of generators of the one-parameter semigroups it contains ([23], Thm. V.6.5, p. 460). Their divisibility asks for roots of the elements of $\mathcal{S}_K$, while the powers in (c) are those of the positive definite steps. The two agree only on the symmetric elements. The proof here is by majorization. As orders, the stable orders are the Loewner order in other coordinates (Proposition 4.12(3)), and this paper reads them as the orders of the stable orbits (Proposition 4.7).

4.5 The base: the general linear group along rays

The smallest order of the scheme is the ray order. It requires only that the record at $\lambda^2\Sigma$, $\lambda > 1$, be a further smoothing of the record at Σ: a zoom out without a change of shape. This subsection asks which kernels can then be carried to every elliptical aperture covariantly under the whole general linear group, and the answer is that this costs nothing beyond what the isotropic theory requires. The kernel μ_I at the round aperture fixes the family, which is its

affine orbit: the kernel at Σ is $(\Sigma^{1/2})_*\mu_I$, the round kernel stretched by $\Sigma^{1/2}$, as the Gaussian of covariance $2b\Sigma$ is the round Gaussian of covariance $2bI$ stretched. The orbit is a family along the rays exactly when μ_I is rotation invariant and self-decomposable (2.2) (Proposition 4.5). The rotation-invariant self-decomposable laws other than δ_0 are the kernels from scale zero of the isotropic families (Theorem 2.5 and Lemma 2.6(1)), so every isotropic scale space, the stable and the Matérn ones among them, has an affine orbit along the rays. What a non-Gaussian orbit cannot do is cascade along a step that widens the aperture in some directions and not in others, a step of rank below d (Proposition 4.4(3)). The Matérn orbit of Proposition 4.4(4) is the example to keep in mind.

Proposition 4.4 (affine orbits of isotropic laws)**.** *Let $d \geq 1$ and let $g : [0,\infty) \to [0,\infty)$ be continuous with $g(0) = 0$ and such that $\omega \mapsto e^{-g(|\omega|^2)}$ is the characteristic function of a probability measure μ_I on $\mathbb{R}^d$. For $\Sigma \in \mathrm{PD}(d)$ put $\mu_\Sigma := (\Sigma^{1/2})_*\mu_I$, the* affine orbit *of μ_I.*

1. (Covariance.) *$\widehat{\mu}_\Sigma(\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)}$, and $A_*\mu_\Sigma = \mu_{A\Sigma A^\mathsf{T}}$ for every $A \in \mathrm{GL}(d)$ and $\Sigma \in \mathrm{PD}(d)$.*

2. (Cascade along rays.) *If for some $\lambda > 1$ there is a probability measure ν_λ on $\mathbb{R}^d$ with $e^{-g(\lambda^2|\omega|^2)} = e^{-g(|\omega|^2)}\,\widehat{\nu}_\lambda(\omega)$, then $\mu_{\lambda^2\Sigma} = \mu_\Sigma * (\Sigma^{1/2})_*\nu_\lambda$ for every $\Sigma \in \mathrm{PD}(d)$.*

3. (No cascade across shapes.) *If $d \geq 2$ and g is not linear, then for no $\Sigma \in \mathrm{PD}(d)$ and no $P \in \mathrm{PSD}(d)$ with $0 < \operatorname{rank} P < d$ is $\mu_{\Sigma+P} = \mu_\Sigma * \rho$ with ρ a probability measure. In particular the orbit is not the family of kernels from scale zero of any affine cascade family.*

4. (The Matérn orbits.) *For $\gamma > 0$, $g_\gamma(u) := \gamma\log(1+u)$ satisfies the hypothesis in every dimension: $(1+|\omega|^2)^{-\gamma}$ is the characteristic function of the law of $\sqrt{2V}\,Z$, with V Gamma distributed of shape γ and scale 1 and Z standard normal on $\mathbb{R}^d$, independent. Its orbit has $\widehat{\mu}_\Sigma(\omega) = (1+\langle\omega,\Sigma\omega\rangle)^{-\gamma}$; clause (3) applies to it, g_γ not being linear; and the hypothesis of clause (2) holds for every $\gamma > 0$ and every $\lambda > 1$, with ν_λ the law of $\sqrt{2Y}\,Z$, where Z is standard normal on $\mathbb{R}^d$ and $Y \geq 0$, independent of Z, is compound Poisson with Lévy measure $\gamma\,(e^{-x/\lambda^2} - e^{-x})\,x^{-1}dx$ on $(0,\infty)$, of total mass $2\gamma\log\lambda$.*

Clause (3) is Theorem 3.7 read on the orbit. The stable orbit $e^{-\langle\omega,\Sigma\omega\rangle^{\alpha/2}}$, $0 < \alpha < 2$, is another instance of the proposition (Lemma 4.6), so by clause (3) it is not a family along the Loewner order. For $\gamma > d/2$ the family of clause (4) is the Matérn covariance of spatial statistics ([44], (2.4.5)–(2.4.7), pp. 17–18; [66], §9, pp. 447–448; [41], §2.1, eqs. (1)–(2), p. 426), with geometric anisotropy ([41], §3.4, p. 437), in which the matrix is a parameter of the covariance. Clause (3) excludes the rank-deficient steps between two values of it, and Proposition 4.17 shows that no step across shapes has a positive increment. The proof is in Appendix B.

The converse holds: a family covariant under the general linear group that cascades along the ray of I is such an orbit, and the law must be self-decomposable.

Proposition 4.5 (the ray families are the orbits of the self-decomposable isotropic laws)**.** *Let $d \geq 1$ and let $(\mu_\Sigma)_{\Sigma\in\mathrm{PD}(d)}$ be probability measures on $\mathbb{R}^d$ with*

(R1) (affine covariance) *$A_*\mu_\Sigma = \mu_{A\Sigma A^\mathsf{T}}$ for every $A \in \mathrm{GL}(d)$ and $\Sigma \in \mathrm{PD}(d)$;*

(R2) (a cascade along the ray of I) *for every $\lambda > 1$ there is a probability measure ρ_λ on $\mathbb{R}^d$ with $\mu_{\lambda^2 I} = \mu_I * \rho_\lambda$.*

Then μ_I is rotation invariant and self-decomposable (2.2), $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ for every Σ, and the family cascades along every ray: $\mu_{\lambda^2\Sigma} = \mu_\Sigma * (\Sigma^{1/2})_*\rho_\lambda$ for every $\Sigma \in \mathrm{PD}(d)$ and $\lambda > 1$. Conversely, for every rotation-invariant self-decomposable law μ_I on $\mathbb{R}^d$ the family $\mu_\Sigma := (\Sigma^{1/2})_*\mu_I$ satisfies (R1) and (R2).

With Theorem 3.6 this places the obstruction for $d \geq 2$: along the rays every rotation-invariant self-decomposable law keeps its orbit, and across every step of the Loewner order only the Gaussian does. Which steps across shapes an orbit survives depends on the law, and that is the next subsection.

Proof. (R1) is (K3) of Definition 3.2 for the kernels from scale zero, and (R2) is (K1) along the ray of I alone.

Covariance determines the orbit. (R1) with $A = Q \in \mathrm{O}(d)$ and $\Sigma = I$ gives $Q_*\mu_I = \mu_{QQ^\mathsf{T}} = \mu_I$, and with $A = \Sigma^{1/2}$ it gives $(\Sigma^{1/2})_*\mu_I = \mu_\Sigma$, as for the orbits of Proposition 4.4.

Self-decomposability. By (R1) with $A = \lambda I$, $\mu_{\lambda^2 I} = (\lambda I)_*\mu_I$, so (R2) reads $(\lambda I)_*\mu_I = \mu_I * \rho_\lambda$. Applying $(\lambda^{-1}I)_*$ gives $\mu_I = (cI)_*\mu_I * (cI)_*\rho_\lambda$ with $c := \lambda^{-1}$, which is (2.2) with the residual $(cI)_*\rho_\lambda$, and every $c \in (0, 1)$ arises.

Every ray. $\mu_{\lambda^2\Sigma} = (\Sigma^{1/2})_*\mu_{\lambda^2 I} = (\Sigma^{1/2})_*(\mu_I * \rho_\lambda) = \mu_\Sigma * (\Sigma^{1/2})_*\rho_\lambda$.

Converse. For (R1), let $B := A\Sigma^{1/2}$ and $\Sigma' := A\Sigma A^\mathsf{T} = BB^\mathsf{T}$. Then $Q := \Sigma'^{-1/2}B$ is orthogonal, so $A_*\mu_\Sigma = B_*\mu_I = (\Sigma'^{1/2})_*Q_*\mu_I = \mu_{\Sigma'}$. For (R2), self-decomposability at $c := \lambda^{-1}$ gives $\mu_I = (cI)_*\mu_I * \rho_c$, and applying $(\lambda I)_*$ gives $\mu_{\lambda^2 I} = \mu_I * (\lambda I)_*\rho_c$. □

4.6 The step set of an orbit

Shape adaptation changes the shape of the aperture as the scale grows (§4.1, the third use). For the Gaussian every Loewner step is a further smoothing, so the shape may change as fast as Lemma 4.1(3) allows: in the plane, a step that doubles the volume scale reaches an anisotropy gained of 4 and no more. This subsection asks the same of the other affine orbits. Along which steps $\Sigma \to \Sigma'$ across shapes is the kernel at Σ' the kernel at Σ convolved with a probability measure? Along those steps the coarser record is a further smoothing of the finer one, and along every other step a change of shape is a re-measurement of the image.

To state the question for every orbit, relate two apertures $\Sigma, \Sigma' \in \mathrm{PD}(d)$ when $\mu_{\Sigma'} = \mu_\Sigma * \rho$ for a positive finite measure ρ. The orbit is covariant under $\mathrm{GL}(d)$ (Proposition 4.4(1)), so this relation is $\mathrm{GL}(d)$-invariant, and by Proposition 4.2(1) it is $\leq_\mathcal{K}$ for exactly one step set $\mathcal{K}$: the set of M with $\mu_M = \mu_I * \rho$. Call $\mathcal{K}$ the *step set* of the orbit. It contains the rays exactly when μ_I is self-decomposable (Proposition 4.5). For a non-Gaussian orbit it contains no step $\Sigma \to \Sigma + P$ with $0 < \operatorname{rank} P < d$ (Proposition 4.4(3)), so the question is about steps of full rank.

The answer depends on the kernel. The orbit of the isotropic α-stable law, $0 < \alpha < 2$, cascades on full-rank steps of every orientation, under a speed limit of the Gaussian's form and narrower: with t_1, t_2 the volume scales of the two apertures, the anisotropy gained is at most $(t_2/t_1)^{d\alpha/(d+\alpha)}$, against $(t_2/t_1)^d$ for the Gaussian (Proposition 4.7(3) and Lemma 4.1(3)). For the Poisson kernel, $\alpha = 1$ (§3.5), a step in the plane that doubles the volume scale reaches an anisotropy gained of $2^{2/3} \approx 1.59$ and no more, and Figure 2 draws both limits. At the other end, the Matérn orbits cascade along the rays and along no other step (Proposition 4.17). No orbit whose transform falls off like a power of the frequency, in the sense of Proposition 4.15, has a positive increment on a step off the rays, and the orbits of decaying rational symbols are among them (Proposition 4.18). For the orbit of a general profile, one function of the profile, its dilation ratio, gives a set of steps on which the orbit cascades (Proposition 4.13); whether it cascades on more steps is open. The subsection ends with what the theorem on the general linear row (§4.7) takes from the orbits: the exponents of the isotropic stable law, the bound they put on a non-Gaussian law, and the stable order extended to $\mathrm{PSD}(d)$.

The answers are read through the Lévy–Khintchine representation (C2, Theorem 2.11), and from here on the subsection uses its vocabulary. The kernels of the stable orbits and of the orbits of Proposition 4.13 are *infinitely divisible* laws with no Gaussian part. Each is the law of a sum of independent displacements, and its *Lévy measure* gives the expected number of displacements of each size and direction. Here that measure has a density $n_\Sigma(x)$, the rate of the displacements per unit volume, not the density of the kernel. A step $\Sigma \to \Sigma'$ is a further

smoothing when the kernel at Σ' has at least the rate of displacements of the kernel at Σ at every size and in every direction, $n_\Sigma \le n_{\Sigma'}$ (Proposition 4.13(2)), and for the stable orbits only then (Proposition 4.7(1)).

The stable rung

The chain of cones C_α of §4.4 is a ladder from the rays ($\alpha = 0$) to the Loewner order ($\alpha = \infty$). Call the *rung* of an orbit the largest index α whose stable order its step set contains. The elliptical stable laws are built from their Lévy densities.

Lemma 4.6 (the elliptical stable laws)**.** *Let $d \ge 1$ and $0 < \alpha < 2$, and put*

$$c_{d,\alpha} := \int_{\mathbb{R}^d} (1 - \cos y_1)\, |y|^{-d-\alpha}\, dy.$$

Then $0 < c_{d,\alpha} < \infty$. For $H \in \mathrm{PD}(d)$ and $x \in \mathbb{R}^d \setminus \{0\}$ put

$$n_H(x) := c_{d,\alpha}^{-1} (\det H)^{-1/2} \langle x, H^{-1}x\rangle^{-(d+\alpha)/2}, \qquad \psi_H(\omega) := \langle \omega, H\omega\rangle^{\alpha/2}. \tag{4.1}$$

1. *For every $H \in \mathrm{PD}(d)$ and $\omega \in \mathbb{R}^d$, $\psi_H(\omega) = \int_{\mathbb{R}^d\setminus\{0\}} (1 - \cos\langle\omega, x\rangle)\, n_H(x)\, dx$, and $\int_{\mathbb{R}^d\setminus\{0\}} (|x|^2 \wedge 1)\, n_H(x)\, dx < \infty$.*

2. *For every $H \in \mathrm{PD}(d)$ there is a unique probability measure $S_\alpha(H)$ on $\mathbb{R}^d$ with $\widehat{S_\alpha(H)} = e^{-\psi_H}$. It is infinitely divisible with generating triplet $(0, n_H(x)\, dx, 0)$, and $A_* S_\alpha(H) = S_\alpha(AHA^{\mathsf{T}})$ for every $A \in \mathrm{GL}(d)$.*

3. *$n_{AHA^{\mathsf{T}}}(x) = |\det A|^{-1}\, n_H(A^{-1}x)$ for $A \in \mathrm{GL}(d)$, $H \in \mathrm{PD}(d)$ and $x \ne 0$.*

In particular $g(u) := u^{\alpha/2}$ satisfies the hypothesis of Proposition 4.4, with $\mu_I = S_\alpha(I)$, and the affine orbit of $S_\alpha(I)$ is $\mu_\Sigma = S_\alpha(\Sigma)$, the stable orbit*.*

The law $S_\alpha(H)$ is the rotation-invariant α-stable law ([58], Thm. 14.14, p. 86) transported by $H^{1/2}$, with its constant normalized to 1. In the terms of Samorodnitsky and Taqqu it is a sub-Gaussian stable law. The existence in (2) is their Proposition 2.5.2 with $R = 2H$, and their Proposition 2.5.7 makes every such law a linear image of $S_\alpha(I)$ ([55], pp. 78 and 81). At $\alpha = 1$, $S_1(I)$ is the kernel of the Poisson scale space at unit scale, with transform $e^{-|\omega|}$ (§3.5). The proof is in Appendix B.

The stable orbit cascades on a cone of full-rank steps, the steps whose Lévy densities (4.1) are ordered.

Proposition 4.7 (the stable orbit cascades on a cone of full-rank steps)**.** *Let $d \ge 1$, $0 < \alpha < 2$, and let $\mu_\Sigma = S_\alpha(\Sigma)$, $\Sigma \in \mathrm{PD}(d)$, be the stable orbit (Lemma 4.6). For $\Sigma, \Sigma' \in \mathrm{PD}(d)$ put $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ and let $\lambda_{\min}(M)$ be its least eigenvalue.*

1. (The step cone.) *The following are equivalent: (a) there is a positive finite measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$; (b) $n_\Sigma(x) \le n_{\Sigma'}(x)$ for every $x \ne 0$; (c) $\lambda_{\min}(M)^{d+\alpha} \ge \det M$. When they hold, $\Sigma \le \Sigma'$, and ρ is unique: it is the infinitely divisible probability measure $\rho_{\Sigma,\Sigma'}$ with generating triplet $(0, (n_{\Sigma'} - n_\Sigma)(x)\, dx, 0)$.*

2. (The order.) *Write $\Sigma \preceq_\alpha \Sigma'$ when (c) holds. This is a partial order on $\mathrm{PD}(d)$, contained in the Loewner order, and $\Sigma \preceq_\alpha \Sigma'$ if and only if $A\Sigma A^{\mathsf{T}} \preceq_\alpha A\Sigma' A^{\mathsf{T}}$, for every $A \in \mathrm{GL}(d)$. The increments satisfy $\rho_{\Sigma,\Sigma} = \delta_0$, $\rho_{\Sigma,\Sigma''} = \rho_{\Sigma',\Sigma''} * \rho_{\Sigma,\Sigma'}$ whenever $\Sigma \preceq_\alpha \Sigma' \preceq_\alpha \Sigma''$, and $A_*\rho_{\Sigma,\Sigma'} = \rho_{A\Sigma A^{\mathsf{T}}, A\Sigma' A^{\mathsf{T}}}$.*

3. (Which steps.) *Every ray, $\Sigma' = \lambda^2\Sigma$ with $\lambda \geq 1$, satisfies $\Sigma \preceq_\alpha \Sigma'$, and no step $\Sigma' = \Sigma + P$ with $P \in \mathrm{PSD}(d)$ and $0 < \operatorname{rank} P < d$ does. In the coordinates of Lemma 4.1 (volume scales t_1, t_2, the eigenvalues $n_1 \leq \cdots \leq n_d$ of the relative shape N),*

$$\Sigma_1 \preceq_\alpha \Sigma_2 \iff n_1 \geq (t_1/t_2)^{2\alpha/(d+\alpha)},$$

so the anisotropy gained, $\sqrt{n_d/n_1}$, is at most $(t_2/t_1)^{d\alpha/(d+\alpha)}$; and for every $t_2 \geq t_1$ and every $N \in \mathrm{PD}(d)$ with $\det N = 1$ and least eigenvalue at least $(t_1/t_2)^{2\alpha/(d+\alpha)}$, the matrix $\Sigma_2 := t_2^2 S_1^{1/2} N S_1^{1/2}$ satisfies $\Sigma_1 \preceq_\alpha \Sigma_2$. In particular, for $d \geq 2$ and $t_2 > t_1$ the relative shape of a step in the cone may be any $N \neq I$ in that range, with any eigenvectors.

So the stable orbits, which are not Gaussian, cascade across shapes, on full-rank steps of every orientation. What limits a step is its relative anisotropy, the gap between $\lambda_{\min}(M)$ and the geometric mean $(\det M)^{1/d}$, and not its orientation. The cone lies between the rays and the Loewner order: the Loewner order is $n_1 \geq (t_1/t_2)^2$ (Lemma 4.1(1)). Clause (3) excludes the steps of rank below d by condition (c), as Theorem 3.7 does for $g(u) = u^{\alpha/2}$. By Proposition 4.12(1) the order is that of the cone field C_α.

The proof of (a)⇒(b) rests on a lemma that reads the Lévy densities off the tail of an increment. Positivity is tested against functions supported at a large scale, so only a weak form of the tail is needed, and the homogeneity of the exponent gives it.

Lemma 4.8 (the tail of an increment is the difference of the Lévy densities)**.** *Let $d \geq 1$, $0 < \alpha < 2$, $H, H' \in \mathrm{PD}(d)$, and C a real $d \times d$ matrix. Suppose ρ is a positive finite measure on $\mathbb{R}^d$ with*

$$\widehat{\rho}(\omega) = \exp\bigl[-\langle\omega, C\omega\rangle - \psi_{H'}(\omega) + \psi_H(\omega)\bigr] \qquad (\omega \in \mathbb{R}^d), \tag{4.2}$$

with ψ_H and n_H as in (4.1). Then ρ is a probability measure, and for every smooth ϕ with compact support in $\mathbb{R}^d \setminus \{0\}$

$$\lim_{R\to\infty} R^\alpha \int \phi(x/R)\,\rho(dx) = \int \phi(x)\,\bigl(n_{H'}(x) - n_H(x)\bigr)\,dx. \tag{4.3}$$

Consequently $n_H(x) \leq n_{H'}(x)$ for every $x \neq 0$.

Since $n_{H'} - n_H$ is homogeneous of degree $-d-\alpha$, the right-hand side of (4.3) equals $R^\alpha \int \phi(x/R)(n_{H'} - n_H)(x)\,dx$ for every R: at large scales the increment's mass is that of the signed difference of Lévy densities. The limit compares masses at large scales, not densities. The Gaussian factor $\langle\omega, C\omega\rangle$ is there for Proposition 4.25(3).

The proof writes $\int \phi(x/R)\,\rho(dx)$ by Fourier inversion as an integral of $\widehat{\phi}$ against $e^{-E(\cdot/R)} - 1$, with $\widehat{\rho} = e^{-E}$; the term 1 costs nothing because ϕ vanishes at the origin. As the non-Gaussian part of the exponent is homogeneous of degree α and its Gaussian part of degree $2 > \alpha$, the factor R^α turns this into $\int \phi\,(n_{H'} - n_H)\,dx$ in the limit, and a nonnegative bump supported where $n_{H'} < n_H$ would make it negative. The complete proof is in Appendix C.

Proof. (b)⇔(c). Substitute $x = \Sigma^{1/2} y$. Then $\langle x, \Sigma^{-1}x\rangle = |y|^2$, $\langle x, \Sigma'^{-1}x\rangle = \langle y, M^{-1}y\rangle$ and $\det\Sigma'/\det\Sigma = \det M$, so (b) says $\langle y, M^{-1}y\rangle/|y|^2 \leq (\det M)^{-1/(d+\alpha)}$ for every $y \neq 0$. The supremum of the left-hand side is $1/\lambda_{\min}(M)$, so (b) is (c).

(c) gives $\Sigma \leq \Sigma'$. $\det M \geq \lambda_{\min}(M)^d$, so (c) gives $\lambda_{\min}(M) \geq 1$, hence $M \geq I$ and $\Sigma' - \Sigma = \Sigma^{1/2}(M - I)\Sigma^{1/2} \geq 0$.

(a)⇒(b). $e^{-\psi_\Sigma}$ has no zero, so $\widehat{\rho} = e^{-\psi_{\Sigma'} + \psi_\Sigma}$, which is (4.2) with $C = 0$, and Lemma 4.8 gives (b). The same identity shows that ρ is unique.

(b)⇒(a). $f := n_{\Sigma'} - n_\Sigma$ is nonnegative, even and at most $n_{\Sigma'}$, so by Lemma 4.6(1) and C2(iii) there is an infinitely divisible ρ with triplet $(0, f(x)\,dx, 0)$ and $\widehat{\rho} = e^{-\psi_{\Sigma'} + \psi_\Sigma}$. So $\mu_\Sigma * \rho$ and $\mu_{\Sigma'}$ have the same transform.

Clause (2). Transitivity is that of (b), antisymmetry that of the Loewner order, and by Lemma 4.6(3) condition (b) for $(A\Sigma A^{\mathsf{T}}, A\Sigma' A^{\mathsf{T}})$ is (b) for (Σ, Σ') at $A^{-1}x$. The transforms $e^{-\psi_{\Sigma'}+\psi_\Sigma}$ telescope, which gives the cascade, and $\psi_\Sigma(A^{\mathsf{T}}\omega) = \psi_{A\Sigma A^{\mathsf{T}}}(\omega)$ gives the covariance.

Clause (3). A ray has $M = \lambda^2 I$, and (c) reads $\lambda^{2(d+\alpha)} \geq \lambda^{2d}$. For $0 < \operatorname{rank} P < d$, $M = I + \Sigma^{-1/2} P \Sigma^{-1/2}$ has the eigenvalue 1 and one above 1, so $\lambda_{\min}(M) = 1 < \det M$. In the coordinates of Lemma 4.1, $M = (t_2/t_1)^2 N$, so (c) is $n_1^{d+\alpha} \geq (t_1/t_2)^{2\alpha}$, and $n_d/n_1 \leq n_1^{-d} \leq (t_2/t_1)^{2d\alpha/(d+\alpha)}$. The converse is Lemma 4.1(3). For the last sentence take $N = Q \operatorname{diag}(a, a^{-1/(d-1)}, \ldots, a^{-1/(d-1)}) Q^{\mathsf{T}}$ with Q arbitrary and $(t_1/t_2)^{2\alpha/(d+\alpha)} \leq a < 1$. □

Remark 4.9 (a second route to (a)⇒(b)). The implication (a)⇒(b) of Proposition 4.7(1) also follows from the uniqueness of the generating triplet. The increment ρ has the transform e^{-E} with $E := \psi_{\Sigma'} - \psi_\Sigma$, which is homogeneous of degree α. So the image of ρ under $x \mapsto n^{-1/\alpha}x$ is an n-th convolution root of ρ for every n, and ρ is infinitely divisible. The generating triplets of μ_Σ and ρ then add to one of $\mu_{\Sigma'}$. By the uniqueness of the triplet ([58], Thm. 8.1(ii), pp. 37–38) the Lévy measure of ρ is $(n_{\Sigma'} - n_\Sigma)(x)\,dx$. So $n_{\Sigma'} - n_\Sigma \geq 0$ almost everywhere, and by continuity at every $x \neq 0$.

The theorem on the general linear row needs a bound on how anisotropic the exponents of a rotation-invariant non-Gaussian law can be. A law is *operator-self-decomposable* with exponent B when it is its own image under e^{-tB} convolved with a residual, for every $t > 0$ (§2.4). Applying $(e^{tB})_*$, this says that the law stretched by e^{tB} is the law convolved with a probability measure, a further smoothing. For $B = I$ the stretch is a zoom, and the condition is self-decomposability (2.2). On an affine orbit the kernel at I stretched by e^{tB} is the kernel at $e^{tB}e^{tB^{\mathsf{T}}}$ (Proposition 4.4(1)), so an exponent of μ_I is a direction in which the orbit leaves the round aperture along further smoothings. The stable law gives the bound, and for a single linear flow e^{tB} the same computation gives the set of its exponents.

Proposition 4.10 (the exponent cone of the isotropic stable law)**.** *Let $d \geq 1$ and B a real $d \times d$ matrix, and write* $\operatorname{sym} B := \frac{1}{2}(B + B^{\mathsf{T}})$. *Call a probability measure μ on $\mathbb{R}^d$ B-*self-decomposable *if for every $t > 0$ there is a probability measure ρ_t with $\mu = (e^{-tB})_*\mu * \rho_t$. For $0 < \alpha < 2$, $S_\alpha(I)$ is B-self-decomposable if and only if*

$$(d + \alpha)\,\lambda_{\min}(\operatorname{sym} B) \geq \operatorname{tr} B. \tag{4.4}$$

The set $\mathfrak{k}_\alpha$ of such B is a closed convex cone, invariant under $B \mapsto QBQ^{\mathsf{T}}$ for $Q \in \mathrm{O}(d)$; it contains every skew-symmetric matrix and I, and every $B \in \mathfrak{k}_\alpha$ has $\operatorname{sym} B \geq 0$, *with $\lambda_{\min}(\operatorname{sym} B) > 0$ unless B is skew-symmetric.*

Only $\operatorname{sym} B$ enters, so the skew part of B, the turning of the flow, is free; what is limited is the anisotropy of $\operatorname{sym} B$. $B = I$ is ordinary self-decomposability. The cone grows with α. The closure of the union of the $\mathfrak{k}_\alpha$ is the cone that (4.4) defines at $\alpha = 2$, and for $d \geq 2$ it is smaller than the Gaussian's cone $\{\operatorname{sym} B \geq 0\}$. The proof is in Appendix B.

The exponent cones of the stable laws bound the contracting exponents of every rotation-invariant non-Gaussian law: such an exponent lies in $\mathfrak{k}_\alpha$ for some $\alpha < 2$. The proof of Theorem 4.21 uses this bound.

Proposition 4.11 (the anisotropy budget of a rotation-invariant law)**.** *Let $d \geq 1$, let B be a real $d \times d$ matrix, and write $\lambda := \lambda_{\min}(\operatorname{sym} B)$ (Proposition 4.10).*

1. *If $e^{-tB} \to 0$ as $t \to \infty$ and μ is a rotation-invariant B-self-decomposable law on $\mathbb{R}^d$ that is not Gaussian (a point mass counting as Gaussian), then $\lambda > 0$ and*

$$\operatorname{tr} B < (d + 2)\,\lambda.$$

2. *Conversely, if $\operatorname{tr} B < (d+2)\lambda$, then $\lambda > 0$, $e^{-tB} \to 0$, and $S_\alpha(I)$ is rotation invariant, not Gaussian, and B-self-decomposable for every $\alpha \in (0, 2)$ with $\alpha \geq \operatorname{tr} B/\lambda - d$.*

The ratio $\operatorname{tr} B/\lambda$ is at least d, with equality for a multiple of I plus a skew part. For a contracting flow, a rotation-invariant non-Gaussian B-self-decomposable law exists exactly when the ratio stays below $d+2$: an anisotropy budget of 2, the ceiling of the stable index. The bound is strict, unlike (4.4), because the Lévy measure must integrate $|x|^2$ near the origin. Clause (1) is proved in Appendix C, clause (2) in Appendix B.

Prior art. The families of one linear flow are Urbanik's operator-self-decomposable laws [62]. Their exponents are Jurek's U-exponents with the sign reversed. Jurek proves that the set of them is a closed convex cone and a Lie wedge, a cone of matrices closed under the conjugations $e^X Y e^{-X}$ by its own edge, and computes the Gaussian's ([29], Prop. 1.2, Cor. 1.3 and the Example, pp. 1055–1057); for the standard Gaussian it is $\{\operatorname{sym} B \geq 0\}$, the closure of the union of the cones that (4.4) defines for $0 < \alpha < \infty$. The book of Jurek and Mason restates this frame, and in its terms $\mathfrak{k}_\alpha$ is the set of exponents of $S_\alpha(I)$ ([30], p. 85, Prop. 3.3.6, p. 86, and Prop. 3.3.10(a), p. 88). Their Proposition 3.4.1 characterizes a contracting exponent B of a law by its Lévy measure ν, with a matching condition on the Gaussian part ([30], Prop. 3.4.1, p. 90). The condition on ν is $\nu(U) \geq \nu(e^{tB}U)$ for every Borel set U and every $t \geq 0$. For contracting B that reduces membership in $\mathfrak{k}_\alpha$ to a computation on the density n_I. The step set of an orbit corresponds to Urbanik's decomposability semigroup of μ_I, transported to the orbit ([30], Lemma 2.3.3, p. 54). Jurek and Mason compute that semigroup for Gaussian laws, where it gives the whole Loewner order (Prop. 2.3.8, p. 58). The qualitative fact behind Proposition 4.11 is Jurek's: a rotation-invariant law with such an exponent outside the skew-symmetric matrices is self-decomposable ([29], Prop. 1.6, p. 1056). Operator-stable laws, the limits of sums of independent copies normalized by matrices, have exponents of their own, a different set ([45], Thm. 7.2.11, p. 262). Hudson and Mason compute them in the plane, and Holmes, Hudson and Mason in every dimension, for the laws whose symmetry group is conjugate to the orthogonal group ([27], Thm. 3, p. 483; [25], Thm. 3 and Cor. 3, p. 603, and p. 606). For $S_\alpha(I)$ they are $\alpha^{-1}I$ plus the skew-symmetric matrices. By Jurek's (1.7), read in the sign of this paper, the stability exponents of an infinitely divisible law are exponents of decomposability ([29], p. 1057), so these matrices lie in $\mathfrak{k}_\alpha$, and on them $\operatorname{sym} B$ is a multiple of I. None of these sources computes the exponents of self-decomposability of a non-Gaussian law, bounds $\operatorname{tr} B$ by $\lambda_{\min}(\operatorname{sym} B)$, or compares two members of the stable orbit. So the cone $\mathfrak{k}_\alpha$, the quantitative budget of Proposition 4.11 with its sharpness through $S_\alpha(I)$, and the step cone of Proposition 4.7 are new against the sources read. The Lévy measure of the elliptical stable law, which the step cone compares, is classical ([27], Thm. 3, p. 483, in the plane; [25], proof of Thm. 3, pp. 605–606, in every dimension).

The step cone and the exponent cone are one object read in two ways. The chain of cones C_α of §4.4 generates, in the sense of Theorem 4.3, exactly the stable orders, and its members of index $0 < \alpha < 2$ are the symmetric parts of the exponent cones. The link is the *tangent cone* of a closed semigroup S of matrices, which the Lie theory of semigroups calls its tangent wedge: the set of B with $e^{tB} \in S$ for every $t \geq 0$, the generators of the one-parameter semigroups that S contains ([23], Prop. V.1.7, pp. 375–376).

Proposition 4.12 (the stable step cone is the order of the cone field C_α)**.** *Let $d \geq 1$. For $0 \leq \alpha < \infty$ put*

$$C_\alpha := \{X \in \operatorname{Sym}(d) : X \text{ positive semidefinite},\ (d+\alpha)\,\lambda_{\min}(X) \geq \operatorname{tr} X\}, \qquad C_\infty := \operatorname{PSD}(d),$$

and write $\preceq_{C_\alpha}$ for the order of the cone field C_α (Theorem 4.3), $\lambda_{\min}$ the least eigenvalue. Each C_α is an $\mathrm{O}(d)$-invariant closed convex cone containing I, with $C_\alpha \cap (-C_\alpha) = \{0\}$. For $\Sigma, \Sigma' \in \operatorname{PD}(d)$ let $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$.

1. (Three orders, each generated by its cone field.) *$C_0 = \{cI : c \geq 0\}$, and $\Sigma \preceq_{C_0} \Sigma'$ if and only if $\Sigma' = c\,\Sigma$ with $c \geq 1$, the ray order on $\operatorname{PD}(d)$. For $0 < \alpha < \infty$ the condition $X \geq 0$*

in C_α is implied by the inequality, and $\Sigma \preceq_{C_\alpha} \Sigma'$ if and only if $\lambda_{\min}(M)^{d+\alpha} \geq \det M$; for $0 < \alpha < 2$ this is the order $\preceq_\alpha$ of Proposition 4.7(2). $\Sigma \preceq_{C_\infty} \Sigma'$ if and only if $\Sigma \leq \Sigma'$, the Loewner order. In each case the related pairs are exactly the pairs joined by a C_α-causal curve, and $\Sigma^{1/2} M^t \Sigma^{1/2}$ is one.

2. (The exponent cone is the tangent cone.) *For $0 < \alpha < 2$ the set $\mathfrak{k}_\alpha$ of (4.4) equals $\{B \in \mathfrak{gl}(d) : e^{tB} e^{tB^\mathsf{T}} \in \exp C_\alpha$ for every $t \geq 0\}$, the tangent cone of the semigroup $\{A : I \preceq_{C_\alpha} AA^\mathsf{T}\}$, the set of B with e^{tB} in it for every $t \geq 0$, and $\mathfrak{k}_\alpha \cap \mathrm{Sym}(d) = C_\alpha$.*

3. (The Loewner order in other coordinates.) *For $0 < \alpha < \infty$ let $\Phi_\alpha(\Sigma) := (\det \Sigma)^{-1/(d+\alpha)}\, \Sigma$ on $\mathrm{PD}(d)$. Then Φ_α is a diffeomorphism of $\mathrm{PD}(d)$ onto itself with inverse $T \mapsto (\det T)^{1/\alpha}\, T$; $\Sigma \preceq_{C_\alpha} \Sigma'$ if and only if $\Phi_\alpha(\Sigma) \leq \Phi_\alpha(\Sigma')$; the derivative of Φ_α at Σ maps the cone $\Sigma^{1/2} C_\alpha \Sigma^{1/2}$ onto $\mathrm{PSD}(d)$; and $\Phi_\alpha(A\Sigma A^\mathsf{T}) = \psi_\alpha(A)\, \Phi_\alpha(\Sigma)\, \psi_\alpha(A)^\mathsf{T}$ for $A \in \mathrm{GL}(d)$, where $\psi_\alpha(A) := |\det A|^{-1/(d+\alpha)} A$ is an automorphism of the group $\mathrm{GL}(d)$. In particular ψ_α maps the semigroup $\{A : I \preceq_{C_\alpha} AA^\mathsf{T}\}$ onto the semigroup $\{A : AA^\mathsf{T} \geq I\}$ of the Loewner order.*

So up to a change of coordinates the stable orders are the Loewner order: $\preceq_\alpha$ is the Loewner order read through Φ_α. The parameter interpolates between the rays, as $\alpha \to 0$, where Φ_α degenerates to the shape map $\Sigma \mapsto (\det \Sigma)^{-1/d}\Sigma$, and the Loewner order at $\alpha = \infty$. The map Φ_α acts on the parameters only. The proof is in Appendix C.

The cones form a chain, $C_0 \subsetneq C_\alpha \subsetneq C_\beta \subsetneq C_2 \subsetneq C_\infty$ for $0 < \alpha < \beta < 2$. For $\alpha < \infty$ every nonzero member of C_α is positive definite, while C_∞ contains the singular directions. Which general linear families each cone carries is Theorem 4.21, stated and proved in §4.7.

Every profile: the dilation ratio

A rotation-invariant law with no Gaussian part and a Lévy density is described by one function of the radius, its *profile* k: the Lévy density at x is $k(|x|)\,|x|^{-d}$, so that the displacements with radius between r_1 and r_2 occur at the rate $|S^{d-1}| \int_{r_1}^{r_2} k(r)\, dr/r$. It is the polar profile of Definition 2.8 divided by the area of the sphere, $k = k_d/|S^{d-1}|$. The stable profile is $k(r) = c_{d,\alpha}^{-1} r^{-\alpha}$ (Lemma 4.6). For a general nonincreasing profile the ordering of Lévy densities is read off one function, the *dilation ratio* $R(\lambda)$, the least factor by which the profile, the rate of displacements per unit of $\log r$ in each direction, grows when the radius is divided by λ. It is λ^α for the stable profile and 1 for a profile bounded at the origin (Proposition 4.13(3)).

Proposition 4.13 (the ordered-density step set of an affine orbit)**.** *Let $d \geq 1$ and let $k : (0,\infty) \to [0,\infty)$ be nonincreasing, not identically 0, and put $m(r) := k(r)\, r^{-d}$, with $\int_{\mathbb{R}^d} (|x|^2 \wedge 1)\, m(|x|)\, dx < \infty$. For $\lambda \geq 1$ put*

$$R(\lambda) := \inf\{k(r/\lambda)/k(r) : r > 0,\ k(r) > 0\} \in [1, \infty).$$

1. (The orbit.) *There is a unique probability measure μ_I on $\mathbb{R}^d$ with $\widehat{\mu}_I(\omega) = e^{-\psi_I(\omega)}$, $\psi_I(\omega) := \int (1 - \cos\langle \omega, x\rangle)\, m(|x|)\, dx$; it is rotation invariant and infinitely divisible with generating triplet $(0, m(|x|)\, dx, 0)$. For $\Sigma \in \mathrm{PD}(d)$ its orbit $\mu_\Sigma := (\Sigma^{1/2})_* \mu_I$ has $\widehat{\mu}_\Sigma(\omega) = \exp[-\int (1 - \cos\langle \omega, x\rangle)\, n_\Sigma(x)\, dx]$ with $n_\Sigma(x) := (\det \Sigma)^{-1/2} m(|\Sigma^{-1/2} x|)$.*

2. (The steps.) *For $\Sigma, \Sigma' \in \mathrm{PD}(d)$ put $M := \Sigma^{-1/2} \Sigma' \Sigma^{-1/2}$. Then $n_\Sigma \leq n_{\Sigma'}$ everywhere if and only if $\lambda_{\min}(M) \geq 1$ and*

$$\det M \leq \lambda_{\min}(M)^d\, R\big(\lambda_{\min}(M)^{1/2}\big)^2;$$

and then $\Sigma \leq \Sigma'$, and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ the infinitely divisible probability measure with generating triplet $(0, (n_{\Sigma'} - n_\Sigma)(x)\, dx, 0)$.

3. (Which steps.) *The relation of (2) is a partial order on* $\mathrm{PD}(d)$*, invariant under* $\Sigma \mapsto A\Sigma A^{\mathsf{T}}$ *for* $A \in \mathrm{GL}(d)$*; it contains every ray* $\Sigma \le \lambda^2\Sigma$*,* $\lambda \ge 1$*, and no step* $\Sigma \to \Sigma + P$ *with* $0 < \operatorname{rank} P < d$*. If* $k(r) = c\, r^{-\alpha}$*,* $0 < \alpha < 2$*, then* $R(\lambda) = \lambda^{\alpha}$ *and its step set is the cone of Proposition 4.7. If* $k(0+) := \lim_{r \downarrow 0} k(r)$ *is finite, then* $R \equiv 1$ *and its step set is the rays.*

The least eigenvalue of M fixes how far a step may widen its narrowest direction, and R how much volume that buys. By the polar criterion of self-decomposability, C5 (§2.4; §2.3), the laws of clause (1) are the rotation-invariant self-decomposable laws with no Gaussian part whose Lévy measure has a density, so clause (2) applies to the orbit of every such law. The proposition gives sufficiency. That a positive increment forces ordered densities is proved for the stable profiles (Lemma 4.8) and for the profiles bounded at the origin (Proposition 4.19). For a general profile it is open: the positivity of an increment alone does not exclude a signed Lévy measure. Inside a cascade family along a stable order or a cone order it holds for every profile, since there the increments are infinitely divisible (Proposition 4.22). The class I_0 does not close the gap. By Cuppens's criterion, a law of clause (1) whose Lévy density is continuous and positive off the origin lies outside the class I_0 of laws whose factors are all infinitely divisible ([11], Thm. 7.2.1, p. 157). The stable and Matérn laws are among them (for the stable laws on the line, [42], Thm. 6.7.2 and Ex. 1, p. 218).

The proof substitutes $x = \Sigma^{1/2}y$, which turns the ordering of densities into a comparison of m at $|y|$ and at $\langle y, M^{-1}y\rangle^{1/2}$. On each sphere the comparison is hardest along an eigenvector for $\lambda_{\min}(M)$, where it reads $k(r/\lambda) \ge (\det M)^{1/2}\lambda^{-d}k(r)$ with $\lambda := \lambda_{\min}(M)^{1/2}$, and the definition of R makes this the inequality of (2). The complete proof is in Appendix C.

A step set, not a cone. The relative steps M allowed by clause (2) form a set $\mathcal{K}_R$ invariant under orthogonal conjugation, a step set in the sense of Proposition 4.2. For the stable profiles and the profiles bounded at the origin it is $\exp C$ for a cone C (clause (3)), and in general it need not be. The dyadic profile of Lemma 4.14(5) below has $R = 1$ on $[1, 2)$ and $R(2) = 2$. For $d \ge 2$, the matrix $M := \operatorname{diag}(4, \dots, 4, 16)$ has $\det M = 4^d \cdot 4 = \lambda_{\min}(M)^d R(2)^2$, so $M \in \mathcal{K}_R$. Its square root $M^{1/2} = \operatorname{diag}(2, \dots, 2, 4)$ has $\det M^{1/2} = 2^d \cdot 2$, above $\lambda_{\min}(M^{1/2})^d R(\sqrt{2})^2 = 2^d$, so $M^{1/2} \notin \mathcal{K}_R$. A set $\exp C$ contains $e^{X/2}$ with e^X, so this $\mathcal{K}_R$ is $\exp C$ for no cone C.

The dilation ratio is monotone and supermultiplicative, and it has a growth exponent. The growth exponent bounds R from above, so the rung that ordered densities give a profile is its lower exponent instead.

Lemma 4.14 (the dilation ratio: monotonicity and two exponents)**.** *Let* $d \ge 1$*, let* k *be a profile as in Proposition 4.13, and let* R *be its dilation ratio,* $R(\lambda) = \inf\{k(r/\lambda)/k(r) : r > 0,\ k(r) > 0\}$ *for* $\lambda \ge 1$*.*

1. (Monotone and supermultiplicative.) $R(1) = 1$*,* R *is nondecreasing on* $[1, \infty)$*, and* $R(\lambda\nu) \ge R(\lambda)\, R(\nu)$ *for all* $\lambda, \nu \ge 1$*.*

2. (Below the square.) $R(\lambda) < \lambda^2$ *for every* $\lambda > 1$*.*

3. (Two exponents.) *Put* $\beta_-(k) := \inf_{\lambda > 1} \log R(\lambda)/\log\lambda$*. The limit* $\beta(k) := \lim_{\lambda \to \infty} \log R(\lambda)/\log\lambda$ *exists and equals* $\sup_{\lambda > 1} \log R(\lambda)/\log\lambda$*. They satisfy* $0 \le \beta_-(k) \le \beta(k) \le 2$ *and* $\beta_-(k) < 2$*, and*
$$\lambda^{\beta_-(k)} \le R(\lambda) \le \lambda^{\beta(k)} \qquad (\lambda \ge 1).$$
For $\alpha \ge 0$*,* $R(\lambda) \ge \lambda^{\alpha}$ *for every* $\lambda \ge 1$ *if and only if* $\alpha \le \beta_-(k)$*.*

4. (The rung.) *Let* $d \ge 2$ *and* $0 < \alpha < 2$*, and let* $\preceq_\alpha$ *be the order of Proposition 4.7(2). Every pair* $\Sigma \preceq_\alpha \Sigma'$ *satisfies* $n_\Sigma \le n_{\Sigma'}$ *everywhere, in the notation of Proposition 4.13(1), if and only if* $\alpha \le \beta_-(k)$*.*

5. (Values.) *If $k(r) = c\,r^{-\alpha}$ with $c > 0$ and $0 < \alpha < 2$, then $\beta_-(k) = \beta(k) = \alpha$. If $k(0+)$ is finite, then $\beta_-(k) = \beta(k) = 0$. The profile $k(r) := 2^n$ for $2^{-n-1} < r \le 2^{-n}$, $n \in \mathbb{Z}$, satisfies the hypotheses of Proposition 4.13 and has $\beta_-(k) = 0$ and $\beta(k) = 1$.*

So the condition $R(\lambda) \ge \lambda^\alpha$ for every $\lambda \ge 1$, which appears in Theorem 4.21(3) and Proposition 4.20(2), is $\alpha \le \beta_-(k)$, and for $d \ge 2$ it is exactly the condition that the ordered-density step set of k contain the stable order of index α. The two exponents agree for the stable profiles, for the profiles bounded at the origin and for the tempered profiles below, and the dyadic profile of clause (5) shows that they need not agree in general. The value $\beta(k) = 2$ occurs, for $k(r) = r^{-2}(1 + \log(1/r))^{-2}$ on $(0, 1]$ and r^{-2} beyond. The lower exponent also decides which stable orders the orbit's family survives, for every profile and with any Gaussian part (Proposition 4.22). Whether a single positive increment outside a family orders the densities is the open converse of Proposition 4.13. The proof is in Appendix B.

Profiles stable at the origin. A profile $k(r) = r^{-\alpha} j(r)$, with j nonincreasing and $j(0+)$ finite and positive, has the dilation ratio λ^α of the stable profile. So its orbit's ordered-density step set is the stable cone of index α, however the profile is tempered at large radii. With an exponential tempering the orbit has exponential moments, as the Matérn law does and the stable law does not. One such family has the transforms $\exp[-((1 + \langle\omega, \Sigma\omega\rangle)^{\alpha/2} - 1)]$. With its exponent normalized to $(2\gamma/\alpha)((1 + u)^{\alpha/2} - 1)$, its limit as $\alpha \downarrow 0$ is the Matérn orbit of index γ, whose profile is bounded at the origin. These are Proposition E.1, in Appendix E.1.

Index 0: the rays

At the bottom of the ladder, rung 0, are the orbits whose transform falls off like a power of the frequency: $e^{-g(cu)}/e^{-g(u)} \to c^{-\kappa}$ as $u \to \infty$, for every $c > 0$. The Matérn orbit of index γ is one, with $\kappa = \gamma$. For them one positive increment between two apertures forces a ray. The statement is about a single pair of apertures of the orbit, so it holds along every order.

Proposition 4.15 (orbits of logarithmic exponent cascade along rays only)**.** *Let $d \ge 1$, let g satisfy the hypothesis of Proposition 4.4, and let (μ_Σ) be its affine orbit, $\widehat{\mu}_\Sigma(\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)}$. Suppose there is $\kappa > 0$ with*

$$g(cu) - g(u) \to \kappa \log c \qquad (u \to \infty), \text{ for every } c > 0.$$

If $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ for a positive finite measure ρ, then $\Sigma' = c\,\Sigma$ for some $c > 0$. In particular, for every $\gamma > 0$, no step of the Matérn orbit of index γ, $\widehat{\mu}_\Sigma(\omega) = (1 + \langle\omega, \Sigma\omega\rangle)^{-\gamma}$ (Proposition 4.4(4)), off the rays has a positive increment.

At small scales the increment looks like the distribution whose transform is $(\langle\omega, \Sigma\omega\rangle/\langle\omega, \Sigma'\omega\rangle)^\kappa$, a function homogeneous of degree 0. A positive measure that is invariant under dilations and finite on compact sets is a point mass, so that transform must be constant. Which rays carry a positive increment is a separate question: for a general orbit, exactly those of a self-decomposable μ_I (Proposition 4.5). The proof is in Appendix B.

Remark 4.16 (the analytic step in the literature). The proof in Appendix B shows that the limit $h(\omega) := \lim_{t\to\infty} \widehat{\rho}(t\omega)$, which is $(\langle\omega, \Sigma\omega\rangle/\langle\omega, \Sigma'\omega\rangle)^\kappa$ for $\omega \ne 0$, is constant off 0. That step has a second, published route. The limit h of $\widehat{\rho}(t\,\cdot)$, with $h(0) = 1$, is a pointwise limit of continuous functions, so it is measurable. As a pointwise limit of characteristic functions it is positive definite ([57], Thm. 1.4.10, p. 28). A measurable positive definite function on $\mathbb{R}^d$ equals a continuous one f_c almost everywhere ([10], Theorem, p. 548; [57], Thm. 2.11.7, p. 130). As h is continuous off 0, $h = f_c$ off 0. So f_c is homogeneous of degree 0 off 0 and continuous at 0, hence constant, and h is that constant off 0.

For the Matérn orbit of every index the two halves combine: the rays carry positive increments, and nothing else does.

Proposition 4.17 (the Matérn orbit of every index: the positive steps are the rays)**.** *Let $d \geq 1$, $\gamma > 0$, and let $(\mu_\Sigma)_{\Sigma \in \mathrm{PD}(d)}$ be the Matérn orbit of index γ, $\widehat{\mu}_\Sigma(\omega) = (1 + \langle \omega, \Sigma\omega \rangle)^{-\gamma}$ (Proposition 4.4(4)). Let $\Sigma, \Sigma' \in \mathrm{PD}(d)$ with $\Sigma' \neq \Sigma$. Then there is a positive finite measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$ if and only if $\Sigma' = \lambda^2 \Sigma$ for some $\lambda > 1$; and then ρ is unique, and it is the probability measure $(\Sigma^{1/2})_* \nu_\lambda$, with ν_λ the law of $\sqrt{2Y}\, Z$ of Proposition 4.4(4).*

No hypothesis $\Sigma \leq \Sigma'$ is needed: no pair of distinct apertures off a ray is joined by a positive increment, comparable or not.

Proof. $\widehat{\mu}_\Sigma$ has no zero, so an increment ρ, if there is one, has the transform $\widehat{\rho}(\omega) = \big((1 + \langle \omega, \Sigma\omega \rangle)/(1 + \langle \omega, \Sigma'\omega \rangle)\big)^\gamma$, which determines it. *Only if.* By Proposition 4.15, $\Sigma' = c\,\Sigma$ for some $c > 0$, and $c \neq 1$ as $\Sigma' \neq \Sigma$. As $\widehat{\rho}(0) = 1$, ρ is a probability measure and $|\widehat{\rho}| \leq 1$; at $\omega \neq 0$, with $u := \langle \omega, \Sigma\omega \rangle > 0$, $\widehat{\rho}(\omega) = ((1+u)/(1+cu))^\gamma$, which exceeds 1 if $c < 1$. So $c > 1$, and $\lambda := c^{1/2}$. *If.* By Proposition 4.4(4), the hypothesis of Proposition 4.4(2) holds for λ with the stated ν_λ, and Proposition 4.4(2) gives $\mu_{\lambda^2\Sigma} = \mu_\Sigma * (\Sigma^{1/2})_* \nu_\lambda$. □

A filter computed by finitely many elliptic solves in the shape's own operator has a rational symbol, and a decaying rational symbol has a logarithmic exponent.

Proposition 4.18 (rational symbols cascade along rays only)**.** *Let $d \geq 1$, and let P and Q be real polynomials with $\deg P > \deg Q$ and $P(0) = Q(0)$, neither of which vanishes on $[0, \infty)$. Suppose that $\omega \mapsto Q(|\omega|^2)/P(|\omega|^2)$ is the characteristic function of a probability measure μ_I on $\mathbb{R}^d$, and let $\mu_\Sigma := (\Sigma^{1/2})_* \mu_I$ for $\Sigma \in \mathrm{PD}(d)$, so that $\widehat{\mu}_\Sigma(\omega) = Q(\langle \omega, \Sigma\omega \rangle)/P(\langle \omega, \Sigma\omega \rangle)$. Then $g(u) := \log(P(u)/Q(u))$ satisfies the hypothesis of Proposition 4.4, and that of Proposition 4.15 with $\kappa = \deg P - \deg Q$. So if $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ for a positive finite measure ρ, then $\Sigma' = c\,\Sigma$ for some $c \geq 1$. This applies, with $n \geq 1$ and $a_1, \ldots, a_n > 0$, to the Matérn orbits of integer index γ ($Q = 1$, $P = (1+u)^\gamma$, $\kappa = \gamma$), to the products $\prod_{i \leq n} (1 + a_i u)^{-1}$ ($\kappa = n$), and to the mixtures $\sum_{i \leq n} w_i (1 + a_i u)^{-1}$ with $w_i \geq 0$ and $\sum_i w_i = 1$ ($\kappa = 1$).*

Write $\mathcal{L}_\Sigma := -\nabla \cdot \Sigma \nabla$, the operator with symbol $u = \langle \omega, \Sigma\omega \rangle$. Finitely many solves of shifted equations $(\mathcal{L}_\Sigma - z)v = w$, $z \in \mathbb{C} \setminus [0, \infty)$, and finitely many applications of $\mathcal{L}_\Sigma$ multiply the transform by a rational function of u. Conversely $Q(u)/P(u)$, with P factored over $\mathbb{C}$, is computed by $\deg P$ such solves and one application of $Q(\mathcal{L}_\Sigma)$. A decaying rational symbol behaves like $u^{-\kappa}$ at infinity with κ a positive integer. So every affine orbit of a law that such solves implement exactly, with a transform that does not vanish and decays, changes shape only by re-measurement. The converse fails: the Matérn orbits of every index and the orbits of profiles bounded at the origin cascade along rays only too (Propositions 4.17 and 4.19), and the tempered orbits of Proposition E.1(3), whose symbols are not rational, have at least the stable cone. Symbols that do not decay, with an atom at the origin, are not covered. The proof is in Appendix B.

The profiles bounded at the origin close the converse of Proposition 4.13(2): by Lemma 2.9(2) such a profile gives a logarithmic exponent.

Proposition 4.19 (orbits of a profile bounded at the origin cascade along rays only)**.** *Let $d \geq 1$, let k satisfy the hypotheses of Proposition 4.13 with $k(0+) := \lim_{r \downarrow 0} k(r)$ finite, and let (μ_Σ) be the orbit of Proposition 4.13(1), with Lévy densities n_Σ. For a unit vector e the function $g(u) := \psi_I(\sqrt{u}\, e)$ satisfies the hypothesis of Proposition 4.4, and that of Proposition 4.15 with $\kappa = \frac{1}{2}|S^{d-1}|\, k(0+) > 0$. Consequently, for $\Sigma, \Sigma' \in \mathrm{PD}(d)$ the following are equivalent:*

1. *$\mu_{\Sigma'} = \mu_\Sigma * \rho$ for a positive finite measure ρ;*
2. *$n_\Sigma \leq n_{\Sigma'}$ everywhere;*

3. $\Sigma' = c\,\Sigma$ *for some* $c \geq 1$.

So for every profile bounded at the origin a positive increment, ordered Lévy densities and a ray are the same condition. The class contains the Matérn orbits (Proposition E.1(4), with $\kappa = \gamma$), every subordinated isotropic datum with $a = 0$ (Definition 2.8) whose delay profile is nonzero and bounded at the origin, and the unit-step profiles $k = \mathbf{1}_{(0,\tau)}$. The equivalence holds on both sides of the boundedness threshold of the isotropic paper [16, Thm. 3.13], the value $k_d(0+) = d$ of the polar profile at the origin above which a kernel without Gaussian part is bounded at its centre and at or below which it is not: every bounded profile has $R \equiv 1$, whatever its value at the origin. For a profile unbounded at the origin the three conditions part: for the stable profiles (1) and (2) are equivalent and (3) fails (Proposition 4.7), and for the others it is open whether (1) implies (2). The proof is in Appendix B.

The size of the failure. For the Matérn orbit of index 1, $d \geq 2$, and a step from $\Sigma \in \mathrm{PD}(d)$ to $\Sigma' = \Sigma + P$ with $0 \neq P \in \mathrm{PSD}(d)$, the would-be increment has the transform $1 - \langle \omega, P\omega \rangle\, \widehat{\mu}_{\Sigma'}(\omega)$. Off the origin it is the function $q = \nabla \cdot P \nabla K_{\Sigma'}$, where the density $K_{\Sigma'}$ of $\mu_{\Sigma'}$ is a linear image of the Green's function of $1 - \Delta$. Near the origin, in the coordinates $y = \Sigma'^{-1/2} x$ and with $P' := \Sigma'^{-1/2} P \Sigma'^{-1/2}$, q is $(\det \Sigma')^{-1/2} |S^{d-1}|^{-1} |y|^{-d} (d\langle \widehat{y}, P'\widehat{y} \rangle - \operatorname{tr} P')$ to leading order. Unless Σ' is a multiple of Σ, the bracket is negative around a least eigenvector of P'. The negative mass on $\{a < |y| < 1\}$ then grows like $\log(1/a)$, so the increment is not even a finite signed measure (Proposition E.2 with Remark E.3, and Proposition E.4, with a computation on a grid in Remark E.5; all four in Appendix E.2).

The stable order on the positive semidefinite matrices

The theorem on the general linear row (§4.7) reads the families along the stable order, extended to $\mathrm{PSD}(d)$ by the origin and the rays. The next proposition gives that order and the general linear cell along it, on which the theorem rests. The same order for the similarities is the subject of Proposition 4.25 in §4.9.

Proposition 4.20 (the stable order on $\mathrm{PSD}(d)$ and the general linear row along it)**.** *Let* $d \geq 2$ *and* $0 < \alpha < 2$. *Write* $\preceq_\alpha$ *for the order of Proposition 4.7(2) on* $\mathrm{PD}(d)$, *and* $\mathrm{ecc}(\Sigma) := \lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma)$ *for* $\Sigma \in \mathrm{PD}(d)$.

1. (The stable order on $\mathrm{PSD}(d)$.) *For* $\Sigma, \Sigma' \in \mathrm{PSD}(d)$ *put* $\Sigma \leq_\alpha \Sigma'$ *when* $\Sigma = 0$, *or* $\Sigma' = c\,\Sigma$ *for some* $c \geq 1$, *or* $\Sigma, \Sigma' \in \mathrm{PD}(d)$ *and* $\Sigma \preceq_\alpha \Sigma'$. *This is a partial order on* $\mathrm{PSD}(d)$, *invariant under* $\mathrm{GL}(d)$, *containing the rays and contained in the Loewner order, and every pair* $\Sigma \leq_\alpha \Sigma'$ *is joined by a chain as in Theorem 3.5(2); so* $(G, \leq_\alpha)$ *is a wedge datum with the path condition (P) for every closed subgroup* G *of* $\mathrm{GL}(d)$ *containing the dilations. If* $\alpha \leq \beta < 2$, *then* $\Sigma \leq_\alpha \Sigma'$ *implies* $\Sigma \leq_\beta \Sigma'$. *For* $\Sigma \in \mathrm{PD}(d)$ *with least and largest eigenvalues* $\lambda_1 \leq \lambda_d$, *and* $\varepsilon, t \geq 0$,

$$\varepsilon I \leq_\alpha \Sigma \Leftrightarrow \varepsilon \leq \varepsilon_\alpha(\Sigma) := \Big(\frac{\lambda_1^{d+\alpha}}{\det \Sigma}\Big)^{1/\alpha}, \qquad \Sigma \leq_\alpha tI \Leftrightarrow t \geq t_\alpha(\Sigma) := \Big(\frac{\lambda_d^{d+\alpha}}{\det \Sigma}\Big)^{1/\alpha},$$

and $t_\alpha(\Sigma)/\varepsilon_\alpha(\Sigma) = \mathrm{ecc}(\Sigma)^{(d+\alpha)/\alpha}$.

2. (The general linear row.) *Let* k *be a profile as in Proposition 4.13 whose dilation ratio has* $R(\lambda) \geq \lambda^\alpha$ *for every* $\lambda \geq 1$, *and* μ_I *the law of that proposition's clause (1). Then* $\Sigma \mapsto (\Sigma^{1/2})_* \mu_I$, $\Sigma \in \mathrm{PSD}(d)$, *is the family of kernels from scale zero of a* $(\mathrm{GL}(d), \leq_\alpha)$*-cascade family of kernels. This applies to the stable law* $S_\beta(I)$ *of every index* $\beta \in [\alpha, 2)$ *(the profile* $k(r) = c_{d,\beta}^{-1} r^{-\beta}$ *of Lemma 4.6), and to the profiles of Proposition E.1 of every index* $\beta \in [\alpha, 2)$.

The order $\leq_\alpha$ of clause (1) is $\preceq^+_{C_\alpha}$ of Theorem 4.3(4), and the general linear row along it is Theorem 4.21(3). The thresholds of clause (1) place every definite Σ between two multiples of I in the order, the sandwich on which the bounds of Proposition 4.25 rest. The proof is in Appendix C.

4.7 The general linear row along the chain

A practitioner who keeps covariance under the whole general linear group and wants a cascade across shapes with a kernel other than the Gaussian has to give up part of the Loewner order, and the part kept decides how fast the shape may change from one stage to the next. Along the Loewner order the anisotropy gained in a step from volume scale t_1 to t_2 is at most $(t_2/t_1)^d$ (Lemma 4.1(3)); along the stable order of index α it is at most $(t_2/t_1)^{d\alpha/(d+\alpha)}$ (Proposition 4.7(3)); along the rays the shape does not change. So the question is how large an order a non-Gaussian kernel can keep, and which kernels keep it. Theorem 4.21 answers it for the orders generated by the invariant cones of Theorem 4.3(4). A cone not contained in $\mathrm{PSD}(d)$ carries no family, and a cone in $\mathrm{PSD}(d)$ that contains a nonzero singular matrix, the direction of a rank-deficient step, leaves the affine Gaussian scale spaces alone. For a cone whose nonzero members are positive definite, one number $\beta(C)$ decides: a family whose kernel at I is not Gaussian exists exactly when $\beta(C) < 2$, and for $0 < \gamma < 2$ the orbit of the isotropic γ-stable law is a family exactly when $\gamma \geq \beta(C)$ (clause (2)). The stable kernels are the familiar non-Gaussian ones, those that the α-scale spaces of Pauwels et al. and of Duits et al. admit where their kernels are positive ([49], p. 694; [14], p. 277). Along the chain C_α the orbit of each keeps a cascade up to the order of its own index, and from $\alpha = 2$ on no non-Gaussian kernel keeps one (clause (3)).

The theorem assembles the general linear cells of §§4.4–4.6 along the chain of cones C_α. For an $\mathrm{O}(d)$-invariant closed convex cone $C \subseteq \mathrm{Sym}(d)$ with $I \in C$ and $C \cap (-C) = \{0\}$, the order $\preceq^+_C$ on $\mathrm{PSD}(d)$ relates two definite matrices when a curve whose relative velocity lies in C at almost every time joins them, and adds the origin and the rays (Theorem 4.3(4)). The cones $C_\alpha := \{X \in \mathrm{PSD}(d) : (d+\alpha)\,\lambda_{\min}(X) \geq \operatorname{tr} X\}$, $0 \leq \alpha < \infty$, and $C_\infty := \mathrm{PSD}(d)$ form a chain from the rays ($\alpha = 0$) through the stable orders ($0 < \alpha < 2$) and the cones C_α, $2 \leq \alpha < \infty$, to the Loewner order ($\alpha = \infty$), and $\preceq_\alpha$ is the order of C_α on $\mathrm{PD}(d)$ (Proposition 4.12). For $0 < \alpha < 2$ it is the stable order, and on $\mathrm{PSD}(d)$ the order $\preceq^+_{C_\alpha}$ is the $\leq_\alpha$ of Proposition 4.20(1) (Table 1). Here $S_\gamma(I)$ is the rotation-invariant γ-stable law, with transform $e^{-|\omega|^\gamma}$ (Lemma 4.6).

Non-Gaussian families stop at $\alpha = 2$, below the Loewner order, for two reasons, one for the cones that reach the boundary of $\mathrm{PSD}(d)$ and one for those that do not. When a cone reaches the boundary, whose nonzero points are the nonzero singular matrices, a single rank-deficient step forces the Gaussian (Theorem 3.7), and no C_α with $\alpha < \infty$ reaches that boundary. Between C_2 and the boundary the anisotropy budget of Proposition 4.11 excludes every non-Gaussian law, at $\alpha = 2$ itself included, since the stable cones fill only the open part of C_2 (Theorem 4.21(1)).

Theorem 4.21 (the general linear row along the chain of cones)**.** *Let $d \geq 2$, and let C_α and $\preceq_{C_\alpha}$ be as in Proposition 4.12 and $\preceq^+_C$ as in Theorem 4.3(4).*

1. (The chain.) *For $0 \leq \alpha < \beta \leq \infty$, $C_\alpha \subsetneq C_\beta$ and $\preceq_{C_\alpha} \subsetneq \preceq_{C_\beta}$. For $\alpha < \infty$ every nonzero member of C_α is positive definite, while $C_\infty = \mathrm{PSD}(d)$ contains every nonzero singular positive semidefinite matrix. The union of the C_α, $0 \leq \alpha < 2$, is $\{0\} \cup \{X \in \mathrm{Sym}(d) : (d+2)\lambda_{\min}(X) > \operatorname{tr} X\}$, a proper subset of C_2.*

2. (Which cones carry general linear families.) *Let $C \subseteq \mathrm{Sym}(d)$ be an $\mathrm{O}(d)$-invariant closed convex cone with $I \in C$ and $C \cap (-C) = \{0\}$, and consider the $(\mathrm{GL}(d), \preceq^+_C)$-cascade families of kernels (Definition 3.1(2)).*

   (a) *If $C \not\subseteq \mathrm{PSD}(d)$, there is none.*

(b) *If $C \subseteq \mathrm{PSD}(d)$ and C contains a nonzero singular matrix, they are exactly the restrictions to the pairs $\Sigma \preceq_C^+ \Sigma'$ of the affine Gaussian scale spaces* (3.1)*, $b > 0$.*

(c) *Otherwise every nonzero member of C is positive definite, the largest value $\beta(C)$ of $\operatorname{tr} X/\lambda_{\min}(X) - d$ over $X \in C \setminus \{0\}$ is attained, and $\beta(C) \in [0, \infty)$ is the least α with $C \subseteq C_\alpha$. For $0 < \gamma < 2$, $\Sigma \mapsto (\Sigma^{1/2})_* S_\gamma(I)$ is the family of kernels from scale zero of a $(\mathrm{GL}(d), \preceq_C^+)$-cascade family if and only if $\gamma \geq \beta(C)$. A $(\mathrm{GL}(d), \preceq_C^+)$-cascade family whose μ_I is not Gaussian exists if and only if $\beta(C) < 2$; if $\beta(C) \geq 2$, the families are, as in (b), exactly the restrictions to the pairs $\Sigma \preceq_C^+ \Sigma'$ of the affine Gaussian scale spaces* (3.1)*, $b > 0$.*

3. (The row along the chain.) *For $0 \leq \alpha \leq \infty$ let $\mathfrak{F}_\alpha$ be the set of laws μ_I of $(\mathrm{GL}(d), \preceq_{C_\alpha}^+)$-cascade families of kernels.*

(a) *$\mathfrak{F}_0$ is the set of rotation-invariant self-decomposable laws other than δ_0.*

(b) *$\mathfrak{F}_\beta \subseteq \mathfrak{F}_\alpha$ for $\alpha \leq \beta$, strictly when $\alpha < \beta \leq 2$.*

(c) *For $0 < \alpha < 2$, $\mathfrak{F}_\alpha$ contains $S_\gamma(I)$, $0 < \gamma < 2$, if and only if $\gamma \geq \alpha$, and contains the law μ_I of Proposition 4.13(1) for every profile whose dilation ratio has $R(\lambda) \geq \lambda^\alpha$ for $\lambda \geq 1$.*

(d) *For $2 \leq \alpha \leq \infty$, $\mathfrak{F}_\alpha$ is the set of centred Gaussian laws with covariance $2bI$, $b > 0$.*

So the general linear row shrinks from the orbits of all rotation-invariant self-decomposable laws other than δ_0 to the Gaussian, and for the stable orbits the stable index decides exactly where on the chain they stop. For the orbit of every profile, with any Gaussian part, the converse of the profile clause of (3)(c) holds as well (Proposition 4.22 below).

The proof reads clauses (2) and (3) off the steps $I \preceq_C^+ e^X$, $X \in C$, and the increment that a family gives each of them. A negative eigenvalue of X leaves no family, and a singular $X \neq 0$ makes the step rank-deficient, which forces the Gaussian (Theorem 3.7). For a cone of definite matrices the stable cone of Proposition 4.7(1) and the anisotropy budget of Proposition 4.11(1) give the threshold $\beta(C)$, and clause (3) is clause (2) for the cones C_α, which have $\beta(C_\alpha) = \alpha$. The complete proof, with complete proofs of the statements it spends, is in Appendix C.

The increments of a family along these orders are infinitely divisible (Theorem 3.5(2)), and an infinitely divisible increment orders the Lévy densities (Proposition H.2(1)). So for families the ordered-density condition of Proposition 4.13 is necessary, and the stable rung of a profile, its lower exponent, decides where its orbit stops on the chain. With C5 this describes the cells for every finite $\alpha > 0$ completely. From $\alpha = 2$ on no profile is left, which gives the laws of Theorem 4.21(3)(d) for finite α without C3.

Proposition 4.22 (the general linear row along the chain: the profiles, and the whole cell)**.** *Let $d \geq 2$, and let C_α, $\preceq_C^+$, the number $\beta(C)$ and $\mathfrak{F}_\alpha$ be as in Theorem 4.21. For a cone C as in Theorem 4.21(2) with $C \subseteq \mathrm{PSD}(d)$ and no nonzero singular member, the case (2)(c) there, write $\mathfrak{F}(C)$ for the set of laws μ_I of $(\mathrm{GL}(d), \preceq_C^+)$-cascade families of kernels, so that $\mathfrak{F}_\alpha = \mathfrak{F}(C_\alpha)$ for $0 \leq \alpha < \infty$. Let k be a profile as in Proposition 4.13, with dilation ratio R and lower exponent $\beta_-(k)$ (Lemma 4.14(3)), let ν_I be its law (Proposition 4.13(1)), and for $a \geq 0$ write $N(0, aI)$ for the centred Gaussian law with covariance aI.*

1. (The profiles.) *Let C be such a cone, with $\beta(C) > 0$. Then $N(0, aI) * \nu_I \in \mathfrak{F}(C)$ if and only if $\beta(C) \leq \beta_-(k)$. In particular, for $0 < \alpha < 2$, $N(0, aI) * \nu_I \in \mathfrak{F}_\alpha$ if and only if $R(\lambda) \geq \lambda^\alpha$ for every $\lambda \geq 1$, and then $\Sigma \mapsto N(0, a\Sigma) * (\Sigma^{1/2})_* \nu_I$, $\Sigma \in \mathrm{PSD}(d)$, is the family of kernels from scale zero of a $(\mathrm{GL}(d), \preceq_{C_\alpha}^+)$-cascade family of kernels.*

2. (The whole cell.) *For $0 < \alpha < \infty$, $\mathfrak{F}_\alpha$ consists of the laws $N(0, aI)$, $a > 0$, and the laws $N(0, aI) * \nu_I$, $a \geq 0$, of the right-continuous profiles k with $\beta_-(k) \geq \alpha$. From $\alpha = 2$ on there is no such profile, so $\mathfrak{F}_\alpha$ consists of the laws $N(0, aI)$, $a > 0$. For a cone C as above, $\mathfrak{F}(C) = \mathfrak{F}_{\beta(C)}$: the laws μ_I of the families along $\preceq_C^+$ depend on C only through $\beta(C)$.*

Proof. The orders. For $0 < \gamma < 2$, $\preceq^+_{C_\gamma}$ is the order $\leq_\gamma$ of Proposition 4.20(1): on $\mathrm{PD}(d)$ both are $\preceq_\gamma$ (Proposition 4.12(1)), and both add the origin and the rays. Let C be a cone as in the statement. By Theorem 4.21(2)(c) the largest value $\beta(C)$ of $\operatorname{tr} X/\lambda_{\min}(X) - d$ over $X \in C \setminus \{0\}$ is attained and $C \subseteq C_{\beta(C)}$. So $\preceq^+_C$ is contained in $\preceq^+_{C_{\beta(C)}}$, a C-causal curve being $C_{\beta(C)}$-causal, and a $(\mathrm{GL}(d), \preceq^+_{C_{\beta(C)}})$-cascade family restricts to a $(\mathrm{GL}(d), \preceq^+_C)$-cascade family with the same μ_I (the proof of Theorem 4.21, the orders). For $X \in C$ and $t \geq 0$ the curve $\zeta(s) := e^{stX}$, $s \in [0,1]$, is a chain from I to e^{tX} for $\preceq^+_C$ as in Theorem 3.5(2). It is continuous, and for $s \leq s'$, $I \preceq^+_C e^{(s'-s)tX}$ as $(s'-s)tX \in C$ (the proof of Theorem 4.21, the orders). The matrix $A := e^{stX/2}$ carries I to $\zeta(s)$ and $e^{(s'-s)tX}$ to $\zeta(s')$, so $\zeta(s) \preceq^+_C \zeta(s')$ by the $\mathrm{GL}(d)$-invariance of the order.

(1), sufficiency. Let $\beta(C) \leq \beta_-(k)$ and put $\gamma := \beta(C)$. Then $0 < \gamma < 2$, as $\beta_-(k) < 2$, and $R(\lambda) \geq \lambda^\gamma$ for every $\lambda \geq 1$ (Lemma 4.14(3)). By Proposition 4.20(2) the laws $\nu_\Sigma := (\Sigma^{1/2})_* \nu_I$, $\Sigma \in \mathrm{PSD}(d)$, are the kernels from scale zero of a $(\mathrm{GL}(d), \leq_\gamma)$-cascade family $(\nu_{\Sigma,\Sigma'})$. Put $\mu_\Sigma := N(0, a\Sigma) * \nu_\Sigma$. Its transform $e^{-a\langle\omega,\Sigma\omega\rangle/2}\, \widehat{\nu}_\Sigma(\omega)$ has no zero, as $\widehat{\nu}_\Sigma(\omega) = e^{-\psi_I(\Sigma^{1/2}\omega)}$ (Proposition 4.13(1)). The conditions (a)–(d) of Lemma 3.3(2) hold for $G = \mathrm{GL}(d)$ and the order $\leq_\gamma$, as they hold for each factor.

- (a): $\mu_0 = \delta_0$, and $\mu_I \neq \delta_0$, as $\widehat{\mu}_I = e^{-a|\omega|^2/2 - \psi_I}$ and ψ_I is not identically 0, k being positive near 0.
- (b): $A_* N(0, a\Sigma) = N(0, aA\Sigma A^{\mathsf{T}})$ and $A_* \nu_\Sigma = \nu_{A\cdot\Sigma}$ for $A \in \mathrm{GL}(d)$.
- (c): both transforms are continuous in Σ.
- (d): for $\Sigma \leq_\gamma \Sigma'$, $\Sigma' - \Sigma \geq 0$ (Proposition 4.20(1)), so $\rho := N(0, a(\Sigma' - \Sigma)) * \nu_{\Sigma,\Sigma'}$ is a probability measure, and $\mu_{\Sigma'} = \mu_\Sigma * \rho$.

By Lemma 3.3(3) the μ_Σ are the kernels from scale zero of a $(\mathrm{GL}(d), \leq_\gamma)$-cascade family, whose law at I is $N(0, aI) * \nu_I$, and it restricts to $\preceq^+_C$.

(1), the right-continuous modification. Put $\tilde{k}(r) := k(r+) = \lim_{s \downarrow r} k(s)$ for $r > 0$. It is nonincreasing and right-continuous, and $\tilde{k} \leq k$, so $\tilde{k}$ satisfies the integrability conditions that k does. If $k(r_0) > 0$, then $\tilde{k} \geq k(r_0)$ on $(0, r_0)$, so $\tilde{k}$ is a profile as in Proposition 4.13. A monotone function has at most countably many discontinuities, so $\tilde{k} = k$ off a countable subset of $(0, \infty)$, and the densities $k(|x|)|x|^{-d}$ and $\tilde{k}(|x|)|x|^{-d}$ agree off the set of the x with $|x|$ in that countable set, a null set of $\mathbb{R}^d$. So the two profiles have the same ψ_I and the same law ν_I. They also have the same lower exponent. Let $\tilde{R}$ be the dilation ratio of $\tilde{k}$ and $\lambda \geq 1$. Every $s > 0$ has $k(s/\lambda) \geq R(\lambda) k(s)$, by the definition of R when $k(s) > 0$ and trivially otherwise. Letting s decrease to r gives $\tilde{k}(r/\lambda) \geq R(\lambda)\tilde{k}(r)$, so $\tilde{R}(\lambda) \geq R(\lambda)$. Conversely, let $\alpha \geq 0$ with $\tilde{R}(\tau) \geq \tau^\alpha$ for every $\tau \geq 1$, let $\lambda > 1$, and let $k(r) > 0$. For $r/\lambda < s < r$, $\tilde{k}(s) \geq k(r) > 0$, as $k \geq k(r)$ on (s, r), and $\tau := s\lambda/r > 1$. So $k(r/\lambda) \geq \tilde{k}(r/\lambda) = \tilde{k}(s/\tau) \geq \tau^\alpha \tilde{k}(s) \geq (s\lambda/r)^\alpha k(r)$, and letting s increase to r gives $k(r/\lambda) \geq \lambda^\alpha k(r)$. So $R(\lambda) \geq \lambda^\alpha$, which holds at $\lambda = 1$ as well. By Lemma 4.14(3), applied to k and to $\tilde{k}$, the numbers $\alpha \geq 0$ with $\alpha \leq \beta_-(k)$ are those with $\alpha \leq \beta_-(\tilde{k})$, and $\beta_-(k) = \beta_-(\tilde{k})$. Both sides of the equivalence in (1) are therefore the same for k and $\tilde{k}$. In the necessity k may be replaced by $\tilde{k}$, and so taken right-continuous.

(1), necessity. By the modification we may take k right-continuous. Let $\mu_I := N(0, aI) * \nu_I$ be the law at I of a $(\mathrm{GL}(d), \preceq^+_C)$-cascade family $(\mu_{\Sigma,\Sigma'})$. For $\Sigma \in \mathrm{PD}(d)$, (K3) with $A = \Sigma^{1/2}$ at the pair $(0, I)$ gives $\mu_\Sigma = (\Sigma^{1/2})_* \mu_I = N(0, a\Sigma) * (\Sigma^{1/2})_* \nu_I$. Let $X \in C \setminus \{0\}$ attain $\beta(C)$, so that $\operatorname{tr} X = (d + \beta(C))\lambda$ with $\lambda := \lambda_{\min}(X) > 0$, X being positive definite. Let $t > 0$ and $M := e^{tX}$. By the chain of the first paragraph and Theorem 3.5(2) the increment $\rho := \mu_{I,M}$ is infinitely divisible, and (K1) at $0 \preceq^+_C I \preceq^+_C M$ gives $\mu_M = \mu_I * \rho$. So Proposition H.2(1), with h the identity and the Gaussian covariance $\Sigma \mapsto a\Sigma$, puts the step $I \to M$ in the step set of Proposition 4.13(2): $\det M \leq \lambda_{\min}(M)^d R(\lambda_{\min}(M)^{1/2})^2$. Here $\lambda_{\min}(M) = e^{t\lambda}$ and $\det M = e^{t \operatorname{tr} X} = e^{t(d+\beta(C))\lambda}$, so

with $u := e^{t\lambda/2}$ the inequality reads $u^{2(d+\beta(C))} \le u^{2d} R(u)^2$, that is $R(u) \ge u^{\beta(C)}$. As t ranges over $(0,\infty)$, u ranges over $(1,\infty)$, and $R(1) = 1$. So $\beta(C) \le \beta_-(k)$ by Lemma 4.14(3). For $C = C_\alpha$ one may take $X = \mathrm{diag}(1,\dots,1,1+\alpha)$, and M is the step of the proof of Lemma 4.14(4).

(1), the stable orders. For $0 < \alpha < 2$, C_α is a cone as in the statement with $\beta(C_\alpha) = \alpha$ (the proof of Theorem 4.21, (3)), and $\mathfrak{F}_\alpha = \mathfrak{F}(C_\alpha)$. By Lemma 4.14(3), $\alpha \le \beta_-(k)$ if and only if $R(\lambda) \ge \lambda^\alpha$ for every $\lambda \ge 1$, and the family is the one built in the sufficiency with $\gamma = \alpha$, along $\preceq_\alpha = \preceq^+_{C_\alpha}$.

(2), the form of μ_I. Let C be a cone as in the statement and $\mu_I \in \mathfrak{F}(C)$. The ray order is contained in $\preceq^+_C$, so the family restricts to a $(\mathrm{GL}(d), \text{rays})$-cascade family with the same μ_I, and by Proposition 3.4(3) and (1), μ_I is rotation invariant, self-decomposable and not δ_0, and $\widehat{\mu}_I$ has no zero. The pair $0 \preceq^+_C I$ is joined by the chain $s \mapsto sI$, so $\mu_I = \mu_{0,I}$ is infinitely divisible (Theorem 3.5(2)). Its transform is radial, and real as μ_I is invariant under $-I$. Having no zero, it is $\widehat{\mu}_I(\omega) = e^{-F(|\omega|)}$ with $F(r) := -\log \widehat{\mu}_I(re_1) \ge 0$. By C2(i), μ_I has a generating triplet (A,ν,γ). The real part of $e^{i\theta} - 1 - i\theta \mathbf{1}_{\{|x|\le 1\}}$ is $\cos\theta - 1$, so taking moduli in the representation,

$$F(|\omega|) = \tfrac{1}{2}\langle \omega, A\omega\rangle + \int_{\mathbb{R}^d} \big(1 - \cos\langle \omega, x\rangle\big)\, \nu(dx), \qquad \omega \in \mathbb{R}^d.$$

As μ_I is self-decomposable, the polar criterion C5 gives $\nu(B) = \int_S \lambda(d\xi) \int_0^\infty \mathbf{1}_B(r\xi)\, k_\xi(r)\, dr/r$ for Borel B, with λ a finite measure on $S = S^{d-1}$ and $k_\xi(r) \ge 0$ measurable in ξ and nonincreasing in r. A nonincreasing k_ξ differs from its right limit $k_\xi(r+) = \sup\{k_\xi(q) : q \in \mathbb{Q},\ q > r\}$ at countably many r only, so replacing $k_\xi(r)$ by $k_\xi(r+)$ changes no inner integral, and it makes the kernel jointly measurable in (ξ, r), as a countable supremum of measurable functions. Let σ_{d-1} be the surface measure of S, with total $|S^{d-1}| = \sigma_{d-1}(S)$, and put

$$k(r) := \frac{1}{|S^{d-1}|} \int_S k_\xi(r+)\, \lambda(d\xi) \in [0,\infty], \qquad r > 0.$$

It is nonincreasing. Exchanging the integrals (the integrands are nonnegative) and passing to polar coordinates $x = r\eta$, $dx = r^{d-1}\, dr\, \sigma_{d-1}(d\eta)$,

$$\int_{\mathbb{R}^d} (|x|^2 \wedge 1)\, \nu(dx) = |S^{d-1}| \int_0^\infty (r^2 \wedge 1)\, k(r)\, \frac{dr}{r} = \int_{\mathbb{R}^d} (|x|^2 \wedge 1)\, k(|x|)\, |x|^{-d}\, dx,$$

which is finite by C2(i). So k is finite: a nonincreasing function infinite at some r_0 is infinite on $(0, r_0]$, and the integral would diverge. The left side of the display for F depends on ω only through $|\omega|$, so it equals the mean of the right side at the points $|\omega|\theta$, $\theta \in S$, against $\sigma_{d-1}/|S^{d-1}|$. The Gaussian term has the mean $\frac{1}{2}a|\omega|^2$ with $a := \mathrm{tr}\, A/d \ge 0$, as $\int_S \theta_i\theta_j\, \sigma_{d-1}(d\theta) = \delta_{ij}|S^{d-1}|/d$. For the other term let $c(s) := |S^{d-1}|^{-1} \int_S (1 - \cos(s\langle\theta,\xi\rangle))\, \sigma_{d-1}(d\theta)$ for $s \ge 0$ and a unit vector ξ. It does not depend on ξ, since σ_{d-1} is invariant under a rotation carrying one unit vector to another. Exchanging the integrals as before and then passing to polar coordinates, with $c(r|\omega|) = |S^{d-1}|^{-1} \int_S (1 - \cos\langle\omega, r\eta\rangle)\, \sigma_{d-1}(d\eta)$ (take $\xi = \omega/|\omega|$ when $\omega \ne 0$),

$$F(|\omega|) = \tfrac{1}{2}a|\omega|^2 + |S^{d-1}| \int_0^\infty c(r|\omega|)\, k(r)\, \frac{dr}{r} = \tfrac{1}{2}a|\omega|^2 + \int_{\mathbb{R}^d} \big(1 - \cos\langle\omega, x\rangle\big)\, k(|x|)\, |x|^{-d}\, dx, \qquad \omega \in \mathbb{R}^d.$$

If $k \equiv 0$, then $\widehat{\mu}_I(\omega) = e^{-a|\omega|^2/2}$ and $\mu_I = N(0, aI)$, with $a > 0$ as $\mu_I \ne \delta_0$. Otherwise k is a profile as in Proposition 4.13, and so is its right-continuous modification $\tilde{k}$, with the same exponent (the paragraph on the modification). So $\widehat{\mu}_I = e^{-a|\omega|^2/2}\, \widehat{\nu}_I$ for the law ν_I of $\tilde{k}$, and $\mu_I = N(0, aI) * \nu_I$ by the uniqueness of the transform.

(2), the equality. If $\beta(C) = 0$, then $C \subseteq C_0 = \{cI : c \ge 0\}$ (Proposition 4.12(1)), and $C_0 \subseteq C$ as $I \in C$. So $C = C_0$, and there is nothing to prove. Let $\beta(C) > 0$. By the first paragraph, $\mathfrak{F}_{\beta(C)} \subseteq \mathfrak{F}(C)$. Conversely let $\mu_I \in \mathfrak{F}(C)$, of one of the two forms above. A law $N(0, aI)$, $a > 0$, is the law at I of the affine Gaussian scale space with constant $a/2$, an affine

cascade family (Theorem 3.6), and so of its restriction to the pairs of $\preceq^{+}_{C_{\beta(C)}}$, which are Loewner pairs, $C_{\beta(C)}$ being contained in PSD(d) (the proof of Theorem 4.21, the orders). So it lies in $\mathfrak{F}_{\beta(C)}$. A law $N(0,aI) * \nu_I$ has $\beta(C) \le \beta_-(k)$ by (1) for C, and so lies in $\mathfrak{F}(C_{\beta(C)}) = \mathfrak{F}_{\beta(C)}$ by (1) for $C_{\beta(C)}$, whose number is $\beta(C)$. Hence $\mathfrak{F}(C) = \mathfrak{F}_{\beta(C)}$. For $C = C_\alpha$, $0 < \alpha < \infty$, a cone as in the statement with $\beta(C_\alpha) = \alpha$ (the proof of Theorem 4.21, (3)), the same two cases describe $\mathfrak{F}_\alpha$: each member has one of the two forms, each $N(0,aI)$ with $a > 0$ lies in it, and $N(0,aI) * \nu_I$ lies in it exactly when $\beta_-(k) \ge \alpha$, by (1). For $\alpha \ge 2$ the second case does not occur, as $\beta_-(k) < 2$ for every profile (Lemma 4.14(3)). □

The single-step converse is not used, and it stays open. A positive increment that is not known to be infinitely divisible may fail to order the densities, as the computation of Remark H.3 suggests.

4.8 Shape-adapted smoothing across the rungs

Shape adaptation is the use this paper has for an affine scale space, and the place where the choice of kernel shows. The iteration of Lindeberg and Gårding, which the affine-invariant features of Baumberg and of Mikolajczyk and Schmid run in simplified forms (§4.1), measures the image at each point through an aperture fitted to the second-moment matrix there. A reader who uses it can ask whether the adapted record is a further smoothing of the isotropic first stage, computed from it by one more convolution with a probability kernel, or a new measurement of the image, and whether the answer depends on the kernel. It depends on the kernel and on the normalization. Holding the least smoothing fixed, as Lindeberg and Gårding do, makes the step from the first stage to the adapted aperture rank-deficient, and among the affine orbits only the Gaussian follows such a step by an increment that is a probability measure (Proposition 4.23). Holding the volume fixed, as Baumberg's warp of determinant 1 does, makes the adapted aperture a re-measurement for every family, the Gaussian included (Lemma 4.1(2)). The first three uses of §4.1 are read in turn below, with the results of §§4.5–4.7 and the smaller groups of §4.9; the fourth, singularity theory, is not treated (§5).

Shape selection at fixed least smoothing. The practice keeps the least eigenvalue of the local scale matrix at the variance w of the isotropic first stage. The step from the first stage to the adapted aperture is then rank-deficient, and among the affine orbits only the Gaussian has a positive increment along it.

Proposition 4.23 (the apertures of shape-adapted smoothing are rank-deficient steps)**.** *Let $d \ge 2$, $w > 0$, and $M \in \mathrm{PD}(d)$ with least eigenvalue 1, and put $\Sigma_w := wM$, so that the least eigenvalue of Σ_w is w, and $P := \Sigma_w - wI$.*

1. (The step.) *$P = w(M-I)$ is positive semidefinite. Its null space $\ker P$ is the eigenspace of M for the eigenvalue 1, and its range is spanned by the eigenvectors of M with eigenvalues above 1. So $\operatorname{rank} P \le d-1$, with $\operatorname{rank} P = 0$ exactly when $M = I$; for $d = 2$ and $M \ne I$, $\operatorname{rank} P = 1$.*

2. (Only the Gaussian follows the apertures.) *Let g, μ_I and the affine orbit $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$, $\Sigma \in \mathrm{PD}(d)$, be as in Proposition 4.4, and let $M \ne I$. There is a probability measure ρ on $\mathbb{R}^d$ with $\mu_{\Sigma_w} = \mu_{wI} * \rho$ if and only if $g(u) = bu$ for some $b \ge 0$, that is, μ_I is the centred Gaussian law with covariance $2bI$ (δ_0 when $b = 0$). Then ρ is the centred Gaussian law with covariance $2bP$, carried by the range of P.*

3. (In the scheme.) *Let $(\mathrm{GL}(d), W)$ be a wedge datum (Definition 3.1) with $wI \le_W wM$ for some $w > 0$ and some $M \ne I$ as above. Then for every $(\mathrm{GL}(d), W)$-cascade family of kernels there is a $b > 0$ with $\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = e^{-b\langle\omega,(\Sigma'-\Sigma)\omega\rangle}$ for every pair $\Sigma \le_W \Sigma'$.*

The apertures of the practice are the $\Sigma_w = wM$ of Proposition 4.23, with M the shape matrix divided by its least eigenvalue ([39], p. 396, step 2; [40], §6.2). In the Gaussian, every adapted aperture is therefore a further smoothing of the first stage, whether or not it is computed that way, by the degenerate Gaussian along the widened directions; in the plane it is the first stage blurred once more along the major axis. In every other affine orbit the adapted aperture is a re-measurement of the image, related to the first stage by no increment that is a probability measure, though the orbit may cascade along other steps of full rank (Proposition 4.7). Clause (3) reads this in the scheme: an order invariant under the general linear group that contains one such step leaves only Gaussian increments, as the Loewner order does.

Proof. (1). The least eigenvalue of M is $1 = \min_{|x|=1}\langle x, Mx\rangle$, and it is attained at a unit eigenvector v. So $\langle x, (M-I)x\rangle \geq 0$ for every x, that is, $M - I$ is positive semidefinite, and $Mv = v$. Then $P = w(M-I)$ is positive semidefinite, $\lambda_{\min}(wM) = w\lambda_{\min}(M) = w$, and $Px = 0$ exactly when $Mx = x$, as $w > 0$: the null space $\ker P$ is the eigenspace E_1 of M for 1. For the range: if $Mx = cx$ with $c > 1$ then $x = P\big((w(c-1))^{-1}x\big)$, so every eigenspace E_c, $c > 1$, lies in the range, and so does their span K. The range is orthogonal to E_1: P is symmetric, so $\langle z, Px\rangle = \langle Pz, x\rangle = 0$ for $z \in E_1$. The eigenspaces of the symmetric matrix M span $\mathbb{R}^d$, and $E_c = 0$ for $c < 1$, since $0 \leq \langle x, (M-I)x\rangle = (c-1)|x|^2$ on E_c. So every y is $a + k$ with $a \in E_1$ and $k \in K$. If y is in the range, then $a = y - k$ is in the range as well as in E_1, so $\langle a, a\rangle = 0$, and $y = k \in K$: the range is K. By rank–nullity the rank of P is d minus the dimension of its null space, which contains $v \neq 0$, so $\operatorname{rank} P \leq d-1$. It is 0 exactly when $P = 0$, that is $M = I$. For $d = 2$ and $M \neq I$ it is therefore 1.

(2). By Proposition 4.4(1), $\widehat{\mu}_\Sigma(\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)}$. If $\mu_{\Sigma_w} = \mu_{wI} * \rho$, taking transforms gives $e^{-g(\langle\omega,(wI+P)\omega\rangle)} = e^{-g(\langle\omega,wI\omega\rangle)}\,\widehat{\rho}(\omega)$, which is (3.2) with $\Sigma = wI \in \mathrm{PD}(d)$ and P positive semidefinite with $0 < \operatorname{rank} P < d$ by (1), and g is continuous with $g(0) = 0$ and $g \geq 0$. Theorem 3.7 gives $g(u) = bu$ with $b = g(1) \geq 0$ and ρ the centred Gaussian law with covariance $2bP$, which is carried by the range of P, the law of $(2bP)^{1/2}Z$ with Z standard normal. Then $\widehat{\mu}_I(\omega) = e^{-b|\omega|^2}$, the transform of the centred Gaussian law with covariance $2bI$. Conversely, if $g(u) = bu$ with $b \geq 0$, the converse half of Theorem 3.7 gives (3.2) for $\Sigma = wI$ and this P with that Gaussian ρ, and the transforms of μ_{Σ_w} and $\mu_{wI} * \rho$ agree.

(3). Let $(\mu_{\Sigma,\Sigma'})$ be a $(\mathrm{GL}(d), W)$-cascade family of kernels, with kernels from scale zero μ_Σ. W contains the rays, so the ray order is contained in it, and Lemma 3.3(1) restricts the family to $(\mathrm{GL}(d), \text{rays})$ with the same kernels from scale zero. By Proposition 3.4(3), $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ with μ_I rotation invariant, self-decomposable and not δ_0, and $\widehat{\mu}_\Sigma(\omega) = F(\langle\omega, \Sigma\omega\rangle)$ with $F(u) := \widehat{\mu}_I(\sqrt{u}\,e_1)$ continuous and $F(0) = 1$. By Proposition 3.4(1), F has no zero. It is real, as μ_I is invariant under $-I$, and $|F| \leq 1$, so $F > 0$, and $g := -\log F$ is continuous, nonnegative, with $g(0) = 0$ and $e^{-g(|\omega|^2)} = \widehat{\mu}_I(\omega)$: the hypotheses of Proposition 4.4 hold, and the μ_Σ, $\Sigma \in \mathrm{PD}(d)$, are the orbit of μ_I. (K1) of Definition 3.1 at $0 \leq_W wI \leq_W wM$ gives $\mu_{wM} = \mu_{wI} * \mu_{wI,wM}$, so (2) gives $g(u) = bu$, and $b > 0$ because $\mu_I \neq \delta_0$. Then $\widehat{\mu}_\Sigma(\omega) = e^{-b\langle\omega,\Sigma\omega\rangle}$ for every $\Sigma \in \mathrm{PSD}(d)$, by the form above. These transforms have no zero, and (K1) at $0 \leq_W \Sigma \leq_W \Sigma'$ gives $\widehat{\mu}_{\Sigma'} = \widehat{\mu}_{\Sigma,\Sigma'}\,\widehat{\mu}_\Sigma$, so $\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = e^{-b\langle\omega,(\Sigma'-\Sigma)\omega\rangle}$. □

So the normalization of the practice places the adaptation on the boundary of the Loewner cone (Figure 2(b)), the part that Theorem 3.7 closes to every non-Gaussian orbit. A stable orbit can follow the practice's shapes by moving up the scale axis to the boundary of its own step set. An orbit whose transform decays like a power of the frequency, such as a Matérn orbit, cannot follow them at all: its step set lies inside the rays (Propositions 4.15 and 4.17). For the stable orbit of index α, write wI for the first stage, as in Proposition 4.23, and sS for an aperture of shape S, with $\det S = 1$ and least eigenvalue n_1; w and s are scale parameters in the variance gauge, squared lengths rather than volume scales. Condition (c) of Proposition 4.7(1) at $\Sigma = wI$ says that sS is a further smoothing of wI exactly when $s \geq w\,n_1^{-(d+\alpha)/\alpha}$. At the boundary the scale parameter along the minor axis, $s\,n_1$, is $n_1^{-d/\alpha}$ times that of the first stage.

family	steps across shapes that are further smoothings	anisotropy gained per step
Gaussian	exactly the Loewner steps	at most $(t_2/t_1)^d$, in any orientation (Lemma 4.1(3))
isotropic α-stable orbit	exactly the steps with $\lambda_{\min}(M)^{d+\alpha} \geq \det M$	at most $(t_2/t_1)^{d\alpha/(d+\alpha)}$, in any orientation (Proposition 4.7(3))
stable profiles of index α under a nonincreasing tempering	at least the stable cone of index α; whether more is open	at least the stable orbit's freedom (Proposition E.1(1))
orbit of a self-decomposable profile k	at least the steps read off R; exactly the rays when k is bounded at the origin	read off R for ordered densities; not known in general (Propositions 4.13 and 4.19)
Matérn orbit of every index; every orbit of logarithmic exponent	at most the rays; exactly the rays for the Matérn orbits and whenever the law is self-decomposable	none: every change of shape is a re-measurement (Propositions 4.15, 4.17 and 4.5)
frame families	the coordinatewise order of one frame	the Gaussian's bound along the frame's axes only; no turning (Propositions F.1 and F.2)
one linear flow	the curve $e^{sB}\Sigma_0 e^{sB^\mathsf{T}}$	drift locked to scale; for a rotation-invariant non-Gaussian law with a contracting flow the anisotropy of $\operatorname{sym} B$ is budgeted (Propositions G.1 and 4.11)

Table 3: The speed limit on shape drift, family by family, for the affine orbits and the families of §4.9; $t_1 < t_2$ are volume scales and M is the relative step.

A planar pattern seen at slant σ, the angle between the line of sight and the normal of the surface, is foreshortened by the factor $\cos\sigma$ across the tilt, so the shape that fits it has the eigenvalue ratio $1/\cos^2\sigma$. At $\sigma = 60°$ that ratio is 4, so $n_1 = 1/2$ in the plane. The factor $n_1^{-d/\alpha}$ by which the minor axis must grow is then 4 at $\alpha = 1$, about 2.52 at $\alpha = 1.5$ and about 2.07 at $\alpha = 1.9$. For the Gaussian it is 1: the minor axis keeps the scale parameter of the first stage. The selection at fixed volume, the normalization Lindeberg and Gårding reject ([40], §6.1), is a re-measurement for every family by Lemma 4.1(2), so no cascade axiom bears on it.

Following a shape across scale. Every family covariant under $\mathrm{GL}(d)$ that cascades along rays is the orbit of a rotation-invariant self-decomposable law (Proposition 4.5). Those laws are the kernels from scale zero of the isotropic families of the isotropic paper (Theorem 2.5 and Lemma 2.6(1), restated in §2.3), so each of them supports this use: within a fixed shape, a coarser record is the isotropic increment transported by $\Sigma^{1/2}$.

Shape change: the speed limit. When the adapted shape is re-estimated as the scale grows, the apertures trace a curve in scale and shape, and it is a cascade only if each step lies in the family's step set. With $t_1 < t_2$ the volume scales of two apertures, Table 3 gives each family's limit on the anisotropy gained by further smoothing.

Beyond its step set every family re-measures, the Gaussian included. The stable orbits turn too, in any orientation. Among the general linear families the Gaussian alone has the Loewner order's gain per step. The affine orbits whose symbol in the shape's operator is a decaying rational function without zeros on $[0, \infty)$, the same for every shape, cascade at most along rays (Proposition 4.18), and along all of them when the law is self-decomposable. For them a different shape is a re-measurement of the original image rather than a further smoothing of the current one.

Remark 4.24 (relation to the isotropic theory and to the line). At $d = 1$ every step is a ray, so every order of this section is the order of $[0, \infty)$, and the general linear row is the line's theory: Proposition 4.5 is then the definition of self-decomposability, and Proposition 4.17 the self-decomposability of the law of $\sqrt{2V}Z$. Even there the group matters: under $\mathrm{GL}^+(1)$ the axioms admit point masses that $\mathrm{GL}(1)$ excludes (Proposition D.1(3)). For $d \geq 2$ the base of the general linear row, the rays, carries exactly the kernels from scale zero of the isotropic families of the isotropic paper, transported by $\Sigma^{1/2}$ (Proposition 4.5, with Theorem 2.5). What dimension adds is the room to change shape. Steps of rank below d exist only for $d \geq 2$, and they are what forces the Gaussian (Theorem 3.7). Steps of full rank across shapes separate the families: the stable index bounds the anisotropy per step, a lower bound on the step set of an orbit is read off the dilation ratio of its profile, and the orbit of a profile bounded at the origin cascades along no step across shapes.

4.9 Smaller groups

A smaller group serves a use that transports fewer shapes. A comparison of an image with its rotated and rescaled copies, but not with its sheared ones, needs covariance under the similarities only. A filter applied in the fixed axes of the sampling grid, with a scale for each axis, needs covariance under the positive diagonal matrices only. Neither group carries every shape to every other, so neither supports shape selection in the sense of §4.1. Following a shape across scale and changing it remain, and a smaller group admits at least as many families for them in every column of Table 2, strictly more in the cells listed after it. The main gain is in the Loewner column: with the similarities in place of the general linear group, the classical order keeps non-Gaussian families. For $0 < \alpha < 2$, $b \geq 0$ and $\kappa > 0$ with $\alpha\kappa^2 - (d+1)\kappa - 1 \geq 0$, one is an affine Gaussian, trivial when $b = 0$, convolved with a stable kernel whose shape is pulled towards the isotropic one,

$$\mu_\Sigma = N(0, 2b\Sigma) * S_\alpha(\Sigma + \kappa \operatorname{tr}(\Sigma) I),$$

with $N(0, C)$ the centred Gaussian law of covariance C. It cascades along every Loewner step and is covariant under the similarities, and its stable part is rotation invariant only at the multiples of I (Proposition H.1(3)). The statements behind the rows of the two smaller groups are in Appendices F–H, except Proposition 4.25, which ends this subsection.

Frames. The affine image of a product of d cascades on the line cascades along the axes of its own frame. Only the Gaussian loses its frame, in the sense that its kernel depends on the aperture only through $A\Sigma A^{\mathsf{T}}$ and not on the frame A (Proposition F.1). Conversely, a family covariant under the positive diagonal matrices that cascades along each axis is a product of self-decomposable laws on the line, one per axis (Proposition F.2). Such a family gains anisotropy along its axes as fast per step as the Gaussian does and cannot turn, while the stable orbits turn with a smaller gain per step. Turning alone therefore does not exclude non-Gaussian kernels.

The ray column. Along rays a family of any group is a continuous covariant field of self-decomposable laws (Proposition 3.4(2)). For the similarities the orbits of the shapes form a simplex of dimension $d - 1$, and for the positive diagonal matrices they are indexed by the correlation matrices. In both rows the isotropic member no longer determines the family (Propositions F.3 and F.4), and the similarity row keeps every rotation-invariant self-decomposable law as an isotropic member in every column, the Loewner column included.

One linear flow. The families covariant under one group $\{e^{sB}\}$ that cascade along its curve $e^{sB}\Sigma_0 e^{sB^{\mathsf{T}}}$ lie outside the scheme, the curve not being an order on $\mathrm{PSD}(d)$. They are the operator-self-decomposable laws with exponent B (Proposition G.1, Appendix G),

and for a contracting flow a rotation-invariant non-Gaussian one exists exactly when $\operatorname{tr} B < (d+2)\,\lambda_{\min}(\operatorname{sym} B)$ (Proposition 4.11).

The similarities with the Loewner order. This is the classical order with the smaller group, and here non-Gaussian families survive. Theorem 3.6 needs the whole general linear group, since its proof transports one rank-deficient step to every other. A kernel is infinitely divisible, a Gaussian part convolved with a non-Gaussian part of exponent ψ_Σ (Proposition H.9). What the order still controls is how far that part is from rotation invariance, measured by its exponent spread against the eccentricity of the aperture, both defined in Proposition 4.25 below: every family has $\operatorname{es}(\psi_\Sigma) \le 4\operatorname{ecc}(\Sigma)$ (clause (2) of that proposition). The spread is bounded uniformly in Σ when the rank-one limit of the Lévy measures of the part (Proposition H.4(4)) stays comparable to the Lévy measure of the isotropic member (Proposition H.5), and without such a condition it can grow like a power of the eccentricity (Proposition H.6). When the profile of the isotropic member is bounded at the origin, as the Matérn profile is, the Lévy measure of every part has the same logarithmic rate at the origin in every direction (Propositions H.2 and H.8); away from the origin it need not be rotation invariant (Proposition H.7).

The stable order. For the diagonal group the isotropic member must be operator-self-decomposable for every diagonal exponent of the cone $\mathfrak{k}_\alpha$, and on the frame's shapes the cell lies strictly between its neighbours in the diagonal row (Proposition F.5). For the similarities the stable order $\le_\alpha$ of Proposition 4.20 bounds how far a kernel is from rotation invariance by the *eccentricity* $\operatorname{ecc}(\Sigma)$ of the aperture, the ratio of its extreme eigenvalues. The main measure is the *exponent spread* of the exponent of the non-Gaussian part, which is 1 exactly for a rotation-invariant exponent. The others are condition numbers of the Gaussian part and of moment matrices of the Lévy measure. All are defined in the statement, and Appendix H studies them further. The bounds come from a sandwich: every definite Σ lies between two multiples of I in the order (Proposition 4.20(1)). They are relative to the shape of Σ, as they must be, since the stable orbit belongs to the cell.

Proposition 4.25 (similarity covariance along the stable order: the families and the shape of their kernels)**.** *Let $d \ge 2$ and $0 < \alpha < 2$, and let $\le_\alpha$ be the stable order on $\mathrm{PSD}(d)$ and $\varepsilon_\alpha(\Sigma)$ and $t_\alpha(\Sigma)$ its thresholds, as in Proposition 4.20(1). Write $\mathrm{S}(d)$ for the similarities cQ, $c > 0$, $Q \in \mathrm{O}(d)$, and $\operatorname{ecc}(\Sigma) := \lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma)$ for $\Sigma \in \mathrm{PD}(d)$. For a measure L on $\mathbb{R}^d \setminus \{0\}$ and $s > 0$ put*

$$\psi_L(\omega) := \int (1 - \cos\langle \omega, x\rangle)\, L(dx), \qquad \mathcal{M}_s(L) := \int (|x|^2 \wedge s^2)\, \frac{xx^\mathsf{T}}{|x|^2}\, L(dx);$$

for a function $\psi \ge 0$ on $\mathbb{R}^d$ let the exponent spread $\operatorname{es}(\psi)$ *be the infimum of the $K \ge 1$ with $\psi(Q\omega) \le K\,\psi(\omega)$ for every $Q \in \mathrm{O}(d)$ and $\omega \in \mathbb{R}^d$, the infimum of the empty set being ∞; and put* $\operatorname{cond} M := \lambda_{\max}(M)/\lambda_{\min}(M) \in [1, \infty]$ *for $M \in \mathrm{PSD}(d) \setminus \{0\}$. These are the functionals of Proposition H.5.*

1. (The kernels of the similarity row.) *Let G be a closed subgroup of $\mathrm{GL}(d)$ with $\mathrm{S}(d) \subseteq G$, and $(\mu_{\Sigma,\Sigma'})$ a $(G, \le_\alpha)$-cascade family of kernels. Every $\mu_{\Sigma,\Sigma'}$ and every kernel from scale zero μ_Σ is infinitely divisible; for $\Sigma \in \mathrm{PD}(d)$, μ_Σ is a symmetric law with generating triplet $(C(\Sigma), L_\Sigma, 0)$; the family $(\mu_\Sigma)_{\Sigma \in \mathrm{PD}(d)}$ is covariant under the similarities, $(\sqrt{c}\,Q)_*\mu_\Sigma = \mu_{cQ\Sigma Q^\mathsf{T}}$ for $c > 0$, $Q \in \mathrm{O}(d)$ and $\Sigma \in \mathrm{PD}(d)$ (so it satisfies the hypotheses of Proposition H.4); and every step $\Sigma \le_\alpha \Sigma'$ in $\mathrm{PD}(d)$ has the infinitely divisible increment $\mu_{\Sigma,\Sigma'}$. For $G = \mathrm{S}(d)$ the isotropic members μ_I of these families are exactly the rotation-invariant self-decomposable laws other than δ_0.*

2. (The sandwich, and the shape.) *In the setting of (1) let* $\Sigma \in \mathrm{PD}(d)$, $\varepsilon := \varepsilon_\alpha(\Sigma)$ *and* $t := t_\alpha(\Sigma)$, *and write* $\psi_\Sigma := \psi_{L_\Sigma}$. *Then* $C(I) = c_0 I$ *with* $c_0 \geq 0$, *and*

$$\varepsilon c_0 I \leq C(\Sigma) \leq t c_0 I, \qquad (\sqrt{\varepsilon}\, I)_* L_I \leq L_\Sigma \leq (\sqrt{t}\, I)_* L_I, \qquad \psi_I(\sqrt{\varepsilon}\,\omega) \leq \psi_\Sigma(\omega) \leq \psi_I(\sqrt{t}\,\omega).$$

Consequently $\operatorname{cond} C(\Sigma) \leq \operatorname{ecc}(\Sigma)^{(d+\alpha)/\alpha}$ *if* $c_0 > 0$, $\operatorname{cond} \mathcal{M}_s(L_\Sigma) \leq \operatorname{ecc}(\Sigma)^{(d+\alpha)/\alpha}$ *for every* $s > 0$ *if* $L_I \neq 0$, *and* $\operatorname{es}(\psi_\Sigma) \leq 4 \operatorname{ecc}(\Sigma)^{(d+\alpha)/\alpha}$. *For a* $(G, \textit{Loewner})$*-cascade family of kernels the same holds with* $\varepsilon := \lambda_{\min}(\Sigma)$, $t := \lambda_{\max}(\Sigma)$ *and* $\operatorname{ecc}(\Sigma)$ *in place of* $\operatorname{ecc}(\Sigma)^{(d+\alpha)/\alpha}$.

3. (The cell is not the union of its neighbours.) *Let* $b > 0$ *and* $\kappa > 0$, *and for* $\Sigma \in \mathrm{PSD}(d)$ *put* $S_\alpha(\Sigma) := (\Sigma^{1/2})_* S_\alpha(I)$ *and*

$$\mu_\Sigma := N\big(0,\, 2b(\Sigma + \kappa \operatorname{tr}(\Sigma)\, I)\big) * S_\alpha(\Sigma).$$

Then (μ_Σ) *is the family of kernels from scale zero of an* $(\mathrm{S}(d), \leq_\alpha)$*-cascade family of kernels, and it is not covariant under* $\mathrm{GL}(d)$, *hence the kernels of no* $(\mathrm{GL}(d), \leq_\alpha)$*-family (Lemma 3.3(2)). For* $P \in \mathrm{PSD}(d)$ *with* $0 < \operatorname{rank} P < d$ *there is no positive finite measure* ρ *with* $\mu_{I+P} = \mu_I * \rho$; *so* (μ_Σ) *is the family of kernels from scale zero of no* $(\mathrm{S}(d), \textit{Loewner})$*-cascade family.*

A uniform bound on the anisotropy is impossible in this cell, since the stable orbit belongs to it by Proposition 4.20(2) with exponent spread $\operatorname{ecc}(\Sigma)^{\alpha/2}$. The last sentence of clause (2), the bound $4 \operatorname{ecc}(\Sigma)$, holds for every (similarities, Loewner)-cascade family, with no further hypothesis. The cell contains the restrictions of its two neighbours, the general linear cell of Proposition 4.20(2) and the (similarities, Loewner) cell (Lemma 3.3(1)), and clause (3) gives a family in neither. Whether the exponent $(d+\alpha)/\alpha$ is sharp is open; the stable orbit has $\alpha/2$. The proof is in Appendix B.

5 Conclusion

This paper asked which positive scale spaces indexed by covariance matrices exist, for a covariance group G and an order W, and what each can do for shape adaptation. For the general linear group the answer runs from the rays, where every rotation-invariant self-decomposable law keeps its affine orbit (Proposition 4.5), to the Loewner order, where for $d \geq 2$ the Gaussian is alone and one step of shape-adapted smoothing already forces it (Theorem 3.6 and Proposition 4.23(3)). Between the two a stable law keeps a cascade at a price in speed of shape change (Theorem 4.21(3)), and the orbits whose transform decays like a power of the frequency, the Matérn orbits among them, cascade at most along rays (Propositions 4.15 and 4.17). A smaller group admits more families. With the similarities, non-Gaussian families survive along the Loewner order, with an exponent spread of the non-Gaussian part at most four times the eccentricity of the aperture (Propositions H.1(3) and 4.25(2)).

What this means for shape adaptation. The normalization of Lindeberg and Gårding, which holds the least smoothing fixed ([40], §§6.1–6.2), is what places the adapted apertures where the rigidity acts: on the boundary of the Loewner cone of the first stage, where one step leaves the Gaussian alone. For the Gaussian the adapted record is a further smoothing of the first stage, whether or not it is computed that way. For every other kernel it is a new measurement of the image, which the iteration makes anyway, since each round convolves the original image (§4.1). So, as far as the apertures are concerned, a non-Gaussian kernel loses nothing the practice uses, and it loses the cascade between its stages. If the cascade is wanted, the order must shrink, and Theorem 4.21 prices the shrinking: a stable kernel keeps a cascade only at a lower speed of shape change (Table 3), and from $\alpha = 2$ on no non-Gaussian

kernel keeps one. The affine orbits whose symbol in the shape's operator is a decaying rational function without zeros on $[0,\infty)$, the same for every shape, keep none across shapes, although finitely many elliptic solves in that operator compute them: they cascade at most along rays (Proposition 4.18). A smaller group admits more families, at the price of shape selection that transforms with the image, which needs a group that carries every shape to every other.

The paper has four limitations. Its results concern families of kernels, and through Lemma A.2 the operators they define, with exact cascades, and it makes no statement about images. It treats the apertures a family offers, not how a shape is chosen. In shape-adapted smoothing the shape at a point is selected through a second-moment matrix at an integration scale (§4.1), Proposition 4.23 reads only the apertures that this selection produces, and how the selection and its iteration behave for the non-Gaussian orbits is not treated. It does not treat the fourth use of §4.1, singularity theory over the larger group. Along shape directions at fixed volume there is no order (Lemma 4.1(2)), so the singularities met there are bifurcations of a parametrized family; which monotonicity statements hold along the step set of a family in dimension d, and how extrema behave along the shape directions, are not stated here. And, apart from the measurement of Remark E.5, it does not compute the families on a grid. What it keeps of computation is Proposition 4.18; the computation by recursive filters and in a pyramid is left to a further paper by the author, on scale-space pyramids, in preparation. Within these limits the three cells of the general linear group in Table 2 are complete. Along the stable orders the orbit of a profile, with any Gaussian part, is a family exactly when the lower exponent of the profile is at least the index of the order (Proposition 4.22). Three cells stop short of a classification. The similarities along the stable and Loewner orders are described through bounds on the shape of their kernels. For the diagonal group along the stable orders the condition on the frame's shapes is necessary, and whether it is sufficient for a family is open (problems 4 and 5 below). Three more are classified on part of their parameters: the ray cells of the two smaller groups up to which kernels extend to a continuous field over the shapes, and the diagonal group with the Loewner order on the frame's shapes (problem 4).

5.1 Open problems

Each problem names the statement it would complete.

1. *The single-step converse of the ordered-density steps.* Proposition 4.13 gives a positive increment for every step whose Lévy densities are ordered, $n_\Sigma \le n_{\Sigma'}$. Inside a family along a stable order or a cone order the converse holds for every profile, since there the increments are infinitely divisible (Proposition 4.22). For a single step, that a positive increment forces ordered densities is proved for the stable profiles (Proposition 4.7) and for the profiles bounded at the origin (Proposition 4.19). For the other profiles it is open, the tempered profiles $r^{-\alpha}j$ of Proposition E.1 included. The positivity of an increment does not by itself exclude a signed Lévy measure. The converse would make exact the rows of Table 3 that read "at least".

2. *The step set of a general orbit.* The step set of an affine orbit is a step set in the sense of Proposition 4.2 (§4.6). Whether its relation is a closed partial order whose steps are closed under the fractional powers $M \mapsto M^\tau$, $0 \le \tau \le 1$, so that by Theorem 4.3(2) it is generated by a cone, is open for a general orbit. Closedness asks that a limit of increments be an increment, and Lévy's continuity theorem on $\mathbb{R}^d$, in its form for a pointwise limit of transforms continuous at the origin, is the natural tool for it. The fractional powers have no such route.

3. *Orders that no cone generates.* Theorem 4.21 describes the general linear row along the orders of invariant cones. Along a closed invariant order with no cone, of the kind of Theorem 4.3(3), the row is not described. Along the orders of the cones of Theorem 4.21(2)(c)

the laws μ_I of the families depend on the cone only through the number $\beta(C)$ of Theorem 4.21(2)(c) (Proposition 4.22(2)).

4. *Continuous fields over the shapes.* A family of the similarities with the ray order is determined by its kernels at the diagonal matrices with nonincreasing nonnegative entries and trace 1, one for each orbit of the similarities (Proposition F.3(1)–(2)). Which choices of those kernels extend to a continuous field on $\mathrm{PSD}(d)$ is open. So is the same question over the correlation matrices for the positive diagonal group (Proposition F.4). Along the stable order the families of the diagonal group off the frame's shapes are not described, and whether every law of Proposition F.5(2) is the isotropic member of a family is open. The groups that contain $\mathrm{SO}(d)$ but not $\mathrm{O}(d)$ and are not transitive on $\mathrm{PD}(d)$, such as the orientation-preserving similarities, are open as well (Appendix D).

5. *The shape of the similarity families.* Four questions remain. (a) For the similarities along the stable order of index α, the exponent spread of the non-Gaussian part of a kernel is at most $4\,\mathrm{ecc}(\Sigma)^{(d+\alpha)/\alpha}$ (Proposition 4.25(2)). The stable orbit has the spread $\mathrm{ecc}(\Sigma)^{\alpha/2}$, and the sharp exponent is open. (b) The comparability hypotheses of Proposition H.5 are sufficient and not necessary, and a necessary condition for a bounded shape is open. (c) Hypothesis (D) of Proposition H.8(3), for a general part whose polar profiles are bounded at the origin, asks for the second term of the exponent's expansion, and it is open. Clause (1) of that proposition gives the index at the origin direction by direction for infinitely divisible increments. Whether it holds when the increments are only known to be positive is not known; clause (2) does hold then, when there is no Gaussian part. (d) For families of kernels that are not cascades, with a Gaussian part and positive increments only, the infinite divisibility that Proposition H.2(1)–(3) asks for need not hold: the computation of Remark H.3 finds a positive increment that is not infinitely divisible, a numerical observation. Whether the shape of such families is bounded is open.

6. *How much the Matérn orbit fails.* The mechanism of the failure off the rays, and its negative mass diverging like a logarithm at the origin, are proved for the Matérn orbit of index 1 (Remark E.3 and Proposition E.4). The growth that the computation of Remark E.5 measures on a grid, linear in the logarithm of the aperture, is not proved on the lattice. At index 2, the other index computed, the grid follows within 4 % the rate that the same heuristic reads off the degree-0 limit of the increment's transform, as at index 1 (Remark E.5). That rate is computed and not proved.

7. *A form of the rigidity without covariance.* The core of Theorem 3.6 that Remark 3.14 isolates suggests a form without any covariance group, which is not proved. Its hypotheses are these. A kernel is reached from scale zero along two decompositions into rank-one steps; the step directions of each decomposition form a basis of $\mathbb{R}^d$; the two bases differ; and each increment is a law carried by the line of its step. Each decomposition writes the kernel as the law of a sum of independent one-dimensional components along its basis. These are the hypotheses of the theorem of Darmois and Skitovich and of its matrix form in Comon, with one step missing: the two decompositions must be read as one vector written twice as a linear image of independent components, and which pairs of bases count as mixing must be settled. The expected conclusion is that the Darmois–Skitovich theorem makes Gaussian every non-degenerate component that the two bases mix: in Skitovich's form ([60], Thm. 2, p. 190, with Thm. 1, p. 185, for the case in which every component is mixed), and in Darmois's as Comon states it ([9], Thm. 19, p. 306), with its matrix form, Thm. 10, p. 294. It would extend Theorem 3.7 to families with no group.

A Between operators and kernels, and the axiom scheme

The two lemmas of the first subsection pass between the operator form of the scheme of §3.1 and its kernel form, in both directions. The other subsections prove the statements of §§3.1–3.4 whose proofs the main text defers, in their order. Two of them also state what the main text summarizes in a paragraph: the isotropic corner, and the two lemmas on the parameter space.

A.1 Between operators and kernels

The axioms of the scheme are stated on operators, in Definition 3.1(1), as the line paper states its own (Definition 2.1), and the paper works on kernels. The two lemmas of this subsection show that nothing is lost. Lemma A.1 passes from a family of operators to its kernels, as Lemma 2.2 does on the line, and Lemma A.2 passes back. Clause (1) of the first is, on the real space $L^1(\mathbb{R}^d)$, Wendel's theorem on left centralizers ([65], Thm. 1, p. 252; for operators commuting with the translations, [33], Thm. 0.1.1, p. 2), with positivity and mass making the measure a probability measure.

Lemma A.1 (from operators to kernels in dimension d)**.** *Fix $d \geq 1$. On the real space $L^1(\mathbb{R}^d)$ write $T_a f := f(\cdot - a)$ for $a \in \mathbb{R}^d$, $D_A f := |\det A|^{-1} f(A^{-1}\,\cdot)$ for $A \in \mathrm{GL}(d)$, and, for a probability measure μ on $\mathbb{R}^d$, $(\mu * f)(x) := \int f(x-y)\,\mu(dy)$. Let $\varphi(x) := (2\pi)^{-d/2} e^{-|x|^2/2}$ be the standard Gaussian density.*

1. (Representation.) *Let $\Phi : L^1(\mathbb{R}^d) \to L^1(\mathbb{R}^d)$ be linear and satisfy the single-operator axioms of Definition 2.1 other than the reflection axiom (A3), read in $\mathbb{R}^d$: (A1) Φ is bounded; (A2) $\Phi T_a = T_a \Phi$ for every $a \in \mathbb{R}^d$; (A4) $\Phi f \geq 0$ whenever $f \geq 0$; (A5) $\int \Phi f = \int f$ whenever $f \geq 0$. Then there is a unique probability measure μ on $\mathbb{R}^d$ with*

$$\Phi f = \mu * f, \qquad f \in L^1(\mathbb{R}^d).$$

   Uniqueness holds in the strong form: if μ and ν are probability measures on $\mathbb{R}^d$ with $\mu * \varphi = \nu * \varphi$, then $\mu = \nu$. Conversely, for every probability measure μ on $\mathbb{R}^d$ the operator $f \mapsto \mu * f$ satisfies (A1), (A2), (A4) and (A5).

2. (Transport.) *For every probability measure μ on $\mathbb{R}^d$, every $A \in \mathrm{GL}(d)$ and every $f \in L^1(\mathbb{R}^d)$, $D_A(\mu * f) = (A_*\mu) * D_A f$. Consequently, for probability measures μ and ν, the operators satisfy $D_A\,(\mu * \cdot)\,D_A^{-1} = \nu * \cdot$ if and only if $A_*\mu = \nu$.*

3. (Continuity.) *If μ_n and μ are probability measures on $\mathbb{R}^d$ with $\mu_n * \varphi \to \mu * \varphi$ in $L^1(\mathbb{R}^d)$, then $\widehat{\mu}_n(\omega) \to \widehat{\mu}(\omega)$ for every $\omega \in \mathbb{R}^d$.*

4. (The (G,W) families, and the affine family.) *Let (G,W) be a wedge datum in dimension d and $(\Phi_{\Sigma,\Sigma'})_{(\Sigma,\Sigma') \in P_W}$ a (G,W)-cascade family of operators (Definition 3.1(1)). Then the measures $\mu_{\Sigma,\Sigma'}$ that (1) gives, $\Phi_{\Sigma,\Sigma'} f = \mu_{\Sigma,\Sigma'} * f$, form a (G,W)-cascade family of kernels (Definition 3.1(2)). In particular, let $(\Phi_{\Sigma,\Sigma'})$ be linear operators on $L^1(\mathbb{R}^d)$, indexed by the pairs $\Sigma \leq \Sigma'$ in $\mathrm{PSD}(d)$, with the axioms of the case $(\mathrm{GL}(d), \textit{Loewner})$: each $\Phi_{\Sigma,\Sigma'}$ satisfies (A1), (A2), (A4) and (A5); (A6) $\Phi_{\Sigma,\Sigma} = \mathrm{Id}$ and $\Phi_{\Sigma',\Sigma''}\Phi_{\Sigma,\Sigma'} = \Phi_{\Sigma,\Sigma''}$ whenever $\Sigma \leq \Sigma' \leq \Sigma''$; (A7) for every $f \in L^1(\mathbb{R}^d)$ the map $(\Sigma,\Sigma') \mapsto \Phi_{\Sigma,\Sigma'} f$ is continuous from the set of pairs into $L^1(\mathbb{R}^d)$; $D_A \Phi_{\Sigma,\Sigma'} D_A^{-1} = \Phi_{A\Sigma A^{\mathsf{T}},\, A\Sigma' A^{\mathsf{T}}}$ for every $A \in \mathrm{GL}(d)$ and every pair; and $\Phi_{0,I} \neq \mathrm{Id}$. Then the measures form an affine cascade family in dimension d (Definition 3.2).*

5. (The isotropic family.) *Let $(\Phi_{s,t})_{0 \leq s \leq t}$ be linear operators on $L^1(\mathbb{R}^d)$ with these axioms: each $\Phi_{s,t}$ satisfies (A1), (A2), (A4) and (A5); (A6) $\Phi_{t,t} = \mathrm{Id}$ and $\Phi_{s,t}\Phi_{r,s} = \Phi_{r,t}$ for $r \leq s \leq t$; (A7) for every $f \in L^1(\mathbb{R}^d)$ the map $(s,t) \mapsto \Phi_{s,t} f$ is continuous from $\{0 \leq s \leq t\}$ into $L^1(\mathbb{R}^d)$; $\Phi_{s,t} R_Q = R_Q \Phi_{s,t}$ for every $Q \in \mathrm{O}(d)$, with $R_Q f := f(Q^{-1}\,\cdot)$; (A8)*

there are increasing bijections S_λ of $[0,\infty)$, $\lambda > 0$, with $D_\lambda \Phi_{s,t} = \Phi_{S_\lambda s,\, S_\lambda t} D_\lambda$, where $D_\lambda f := \lambda^{-d} f(\cdot/\lambda)$; and (ND) $\Phi_{s,t} \neq \mathrm{Id}$ whenever $s < t$. Then the measures $\mu_{s,t}$ that (1) gives form an isotropic cascade family in dimension d (Definition 2.4).

Proof. Two facts are used throughout. For a probability measure μ and $f \in L^1(\mathbb{R}^d)$, $\mu * f$ is the $L^1(\mathbb{R}^d)$-valued Bochner integral $\int T_y f \, \mu(dy)$: the map $y \mapsto T_y f$ is continuous because translation acts continuously on $L^1(\mathbb{R}^d)$, and it is bounded by $\|f\|_1$. And with $\widehat{h}(\omega) := \int e^{i\langle \omega, x\rangle} h(x)\, dx$ for $h \in L^1(\mathbb{R}^d)$, Fubini gives the pairing

$$\widehat{\mu * f}(\omega) = \widehat{\mu}(\omega)\, \widehat{f}(\omega), \qquad |\widehat{h}(\omega)| \le \|h\|_1,$$

and $\widehat{\varphi}(\omega) = e^{-|\omega|^2/2} > 0$.

(1), the interchange identity. For $f, g \in L^1(\mathbb{R}^d)$, $f * g = \int f(y)\, T_y g \, dy$ as a Bochner integral, dominated by $|f(y)|\, \|g\|_1$. A bounded operator commutes with an $L^1(\mathbb{R}^d)$-valued integral, so (A1) and (A2) give $\Phi(f * g) = \int f(y)\, T_y \Phi g \, dy = f * \Phi g$.

The approximants. Let $B_\varepsilon := \{|x| < \varepsilon\}$, $|B_\varepsilon|$ its volume, $\rho_\varepsilon := |B_\varepsilon|^{-1} \mathbf{1}_{B_\varepsilon}$ and $h_\varepsilon := \Phi \rho_\varepsilon$. By (A4) and (A5), $h_\varepsilon \ge 0$ with $\int h_\varepsilon = 1$, so $\nu_\varepsilon := h_\varepsilon\, dx$ is a probability measure on $\mathbb{R}^d$. For every f, $\|f * \rho_\varepsilon - f\|_1 \le \sup_{|y|<\varepsilon} \|T_y f - f\|_1 \to 0$ as $\varepsilon \downarrow 0$.

Tightness. By the interchange identity $\rho_1 * h_\varepsilon = \Phi(\rho_1 * \rho_\varepsilon)$, which tends to $\Phi \rho_1$ in $L^1(\mathbb{R}^d)$ by (A1). Along $\varepsilon_n := 1/(n+1)$ the functions $\rho_1 * h_{\varepsilon_n}$ therefore have uniformly small tails: given $\eta > 0$ there is R with $\int_{|x|>R} (\rho_1 * h_{\varepsilon_n}) < \eta$ for every n (choose N with $\|\rho_1 * h_{\varepsilon_n} - \Phi\rho_1\|_1 < \eta/2$ for $n \ge N$, then R for $\Phi\rho_1$ and the first N terms). Convolution with a nonnegative unit-mass density carried by the unit ball moves mass by at most 1: for $|y| > R+1$ the translate $\rho_1(\cdot - y)$ is carried by $\{|x| > R\}$, so by Tonelli

$$\int_{|x|>R} (\rho_1 * h_\varepsilon)(x)\, dx = \int h_\varepsilon(y) \int_{|x|>R} \rho_1(x-y)\, dx\, dy \ge \nu_\varepsilon\big(\{|y| > R+1\}\big).$$

So the sequence (ν_{ε_n}) is tight, with the closed balls of radius $R+1$ as compacts.

The limit. By Prokhorov's theorem a subsequence $\nu_{\varepsilon_{n_k}}$ converges weakly to a probability measure μ on $\mathbb{R}^d$.

Identification. Let $\Psi \in L^\infty(\mathbb{R}^d)$, acting on $L^1(\mathbb{R}^d)$ by $\Psi(h) := \int \Psi h$, and $f \in L^1(\mathbb{R}^d)$. The map $y \mapsto \Psi(T_y f)$ is continuous and bounded by $\|\Psi\|_\infty \|f\|_1$, hence a test function for weak convergence. Since $f * h_\varepsilon = \int h_\varepsilon(y)\, T_y f \, dy$ and Ψ commutes with the integral,

$$\Psi(f * h_{\varepsilon_{n_k}}) = \int \Psi(T_y f)\, \nu_{\varepsilon_{n_k}}(dy) \xrightarrow{k\to\infty} \int \Psi(T_y f)\, \mu(dy) = \Psi(\mu * f).$$

On the other hand $f * h_\varepsilon = \Phi(f * \rho_\varepsilon) \to \Phi f$ in $L^1(\mathbb{R}^d)$, by the interchange identity and (A1). Hence $\Psi(\Phi f - \mu * f) = 0$ for every Ψ; taking Ψ the sign of $\Phi f - \mu * f$ gives $\Phi f = \mu * f$.

Uniqueness. If $\mu * \varphi = \nu * \varphi$, the pairing gives $\widehat{\mu}(\omega)\, \widehat{\varphi}(\omega) = \widehat{\nu}(\omega)\, \widehat{\varphi}(\omega)$, so $\widehat{\mu} = \widehat{\nu}$ because $\widehat{\varphi} > 0$, and $\mu = \nu$ because a finite measure on $\mathbb{R}^d$ is determined by its Fourier transform. In particular two measures representing the same operator agree.

Converse. For a probability measure μ, $\|\mu * f\|_1 \le \|f\|_1$ by Minkowski's integral inequality (A1); $T_a(\mu * f) = \mu * T_a f$, both being $x \mapsto \int f(x - a - y)\, \mu(dy)$ (A2); $\mu * f \ge 0$ when $f \ge 0$ (A4); and $\int \mu * f = \int f$ for $f \ge 0$ by Tonelli (A5).

(2). For $x \in \mathbb{R}^d$, substituting $z = Ay$,

$$\big((A_*\mu) * D_A f\big)(x) = \int |\det A|^{-1} f\big(A^{-1}(x-z)\big)\, (A_*\mu)(dz) = |\det A|^{-1} \int f(A^{-1}x - y)\, \mu(dy) = D_A(\mu * f)(x).$$

D_A is invertible on $L^1(\mathbb{R}^d)$ with inverse $D_{A^{-1}}$, so applying the identity to $D_A^{-1} g$ gives $D_A\,(\mu * D_A^{-1} g) = (A_*\mu) * g$ for every g: the operator $D_A\,(\mu * \cdot\,)\, D_A^{-1}$ is convolution by $A_*\mu$. It is convolution by ν if and only if $A_*\mu = \nu$, by the uniqueness clause of (1).

(3). By the pairing, $|\widehat{\mu}_n(\omega) - \widehat{\mu}(\omega)|\,\widehat{\varphi}(\omega) \le \|\mu_n * \varphi - \mu * \varphi\|_1 \to 0$; divide by $\widehat{\varphi}(\omega) > 0$. The test density must have a zero-free transform, which is why the indicator of a ball would not do.

(4). By (1), each $\Phi_{\Sigma,\Sigma'}$ is convolution by a unique probability measure $\mu_{\Sigma,\Sigma'}$. The four axioms of the kernel form follow.

- (K1): $\Phi_{\Sigma,\Sigma} = \mathrm{Id}$ is convolution by δ_0, so $\mu_{\Sigma,\Sigma} = \delta_0$ by uniqueness. By Fubini $\mu_{\Sigma',\Sigma''} * (\mu_{\Sigma,\Sigma'} * f) = (\mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}) * f$, so the cascade says that $\mu_{\Sigma,\Sigma''}$ and $\mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}$ represent the same operator, and they are equal by uniqueness.

- (K2): P_W is a subset of a finite-dimensional space, so continuity on it is sequential. If $(\Sigma_n, \Sigma_n') \to (\Sigma, \Sigma')$ in P_W, continuity at $f = \varphi$ gives $\mu_{\Sigma_n,\Sigma_n'} * \varphi \to \mu_{\Sigma,\Sigma'} * \varphi$ in $L^1(\mathbb{R}^d)$, and (3) gives $\widehat{\mu}_{\Sigma_n,\Sigma_n'}(\omega) \to \widehat{\mu}_{\Sigma,\Sigma'}(\omega)$.

- (K3): for $A \in G$, $A \cdot \Sigma \le_W A \cdot \Sigma'$ because W is G-invariant, so the right-hand side of the covariance axiom is a member of the family, and (2) with $\mu = \mu_{\Sigma,\Sigma'}$ and $\nu = \mu_{A\cdot\Sigma, A\cdot\Sigma'}$ turns the axiom into $A_*\mu_{\Sigma,\Sigma'} = \mu_{A\cdot\Sigma,\, A\cdot\Sigma'}$.

- (K4): if $\mu_{0,I} = \delta_0$ then $\Phi_{0,I} f = \delta_0 * f = f$ for every f, against $\Phi_{0,I} \neq \mathrm{Id}$.

The pair $(\mathrm{GL}(d), \text{Loewner})$ is a wedge datum: $\mathrm{GL}(d)$ is closed in itself and contains the dilations, and the Loewner order is $\mathrm{GL}(d)$-invariant, since $A(\Sigma' - \Sigma)A^{\mathsf{T}}$ is positive semidefinite with $\Sigma' - \Sigma$, and contains the rays, since $c'\Sigma - c\Sigma = (c' - c)\Sigma$; its kernel form is Definition 3.2.

(5). The same argument on the parameter set $\{0 \le s \le t\}$ gives (I1) from (A6) and (I2) from (A7). (I3): $|\det Q| = 1$, so $R_Q = D_Q$, and the commutation reads $D_Q \Phi_{s,t} D_Q^{-1} = \Phi_{s,t}$; (2) with $\nu = \mu = \mu_{s,t}$ gives $Q_*\mu_{s,t} = \mu_{s,t}$. (I4): $D_\lambda = D_{\lambda I}$, and (A8) reads $D_{\lambda I}\Phi_{s,t}D_{\lambda I}^{-1} = \Phi_{S_\lambda s,\, S_\lambda t}$; (2) gives $(\lambda I)_*\mu_{s,t} = \mu_{S_\lambda s,\, S_\lambda t}$. (ND): if $\mu_{s,t} = \delta_0$ with $s < t$ then $\Phi_{s,t} = \mathrm{Id}$, against (ND). □

The converses of (4) and (5) hold as well, and the next lemma states them: convolution with the kernels of a (G, W)-cascade family is a (G, W)-cascade family of operators, and convolution with an isotropic cascade family satisfies the operator axioms of (5). The step that is not formal is continuity in L^1, which (K2) gives only pointwise in the frequency. Clause (1) of the lemma supplies it by smoothing with a Gaussian, without weak convergence.

Lemma A.2 (from kernels to operators in dimension d)**.** *Fix $d \ge 1$, and use the notation of Lemma A.1.*

1. (Continuity.) *If μ_n and μ are probability measures on $\mathbb{R}^d$ with $\widehat{\mu}_n(\omega) \to \widehat{\mu}(\omega)$ for every $\omega \in \mathbb{R}^d$, then $\mu_n * f \to \mu * f$ in $L^1(\mathbb{R}^d)$ for every $f \in L^1(\mathbb{R}^d)$.*

2. (The (G, W) families.) *Let (G, W) be a wedge datum in dimension d and $(\mu_{\Sigma,\Sigma'})_{(\Sigma,\Sigma')\in P_W}$ a (G, W)-cascade family of kernels (Definition 3.1(2)). Then the operators $\Phi_{\Sigma,\Sigma'} f := \mu_{\Sigma,\Sigma'} * f$ on $L^1(\mathbb{R}^d)$ form a (G, W)-cascade family of operators (Definition 3.1(1)), and the measures that Lemma A.1(1) gives for them are the $\mu_{\Sigma,\Sigma'}$. So, for every wedge datum, convolution is a one-to-one correspondence between the (G, W)-cascade families of kernels and those of operators, and its inverse is the passage of Lemma A.1(4).*

3. (The isotropic family.) *Let $(\mu_{s,t})_{0\le s\le t}$ be an isotropic cascade family in dimension d (Definition 2.4). Then the operators $\Phi_{s,t} f := \mu_{s,t} * f$ on $L^1(\mathbb{R}^d)$ satisfy the axioms of Lemma A.1(5), with the bijections S_λ of (I4) in (A8), and the measures that Lemma A.1(1) gives for them are the $\mu_{s,t}$.*

Proof. For $\sigma > 0$ write $\varphi_\sigma(x) := \sigma^{-d}\varphi(x/\sigma)$, the centred Gaussian density with covariance $\sigma^2 I$. For a probability measure ν and $f, g \in L^1(\mathbb{R}^d)$ three facts are used, each by Fubini's theorem: $\|\nu * f\|_1 \le \|f\|_1$; $\|g * f\|_1 \le \|g\|_1\|f\|_1$; and $\nu * (g * f) = (\nu * g) * f$, where $\nu * g$ is the integrable function $x \mapsto \int g(x - y)\,\nu(dy)$.

(1), the smoothed measures. The Gaussian integral gives, for every $z \in \mathbb{R}^d$,

$$\varphi_\sigma(z) = (2\pi)^{-d} \int e^{-i\langle \omega, z\rangle} e^{-\sigma^2|\omega|^2/2}\, d\omega.$$

Put $z = x - y$ and integrate in $\nu(dy)$. The integrand is bounded by $e^{-\sigma^2|\omega|^2/2}$, which is integrable for $d\omega \otimes \nu$, so Fubini's theorem gives, for every probability measure ν and every $x \in \mathbb{R}^d$,

$$(\nu * \varphi_\sigma)(x) = (2\pi)^{-d} \int e^{-i\langle \omega, x\rangle}\, \widehat{\nu}(\omega)\, e^{-\sigma^2|\omega|^2/2}\, d\omega.$$

Put $h_n := \mu_n * \varphi_\sigma$ and $h := \mu * \varphi_\sigma$. Since $|\widehat{\mu}_n| \le 1$ and $\widehat{\mu}_n \to \widehat{\mu}$ pointwise, dominated convergence gives $h_n(x) \to h(x)$ for every x. The h_n and h are nonnegative with integral 1 (Tonelli), so $\int (h_n - h) = 0$ and

$$\|h_n - h\|_1 = 2 \int (h - h_n)^+ \xrightarrow{n\to\infty} 0,$$

by dominated convergence again, since $0 \le (h - h_n)^+ \le h$ and $(h - h_n)^+ \to 0$ pointwise.

(1), the approximation. By Minkowski's integral inequality and the substitution $y = \sigma z$,

$$\|\varphi_\sigma * f - f\|_1 \le \int \varphi_\sigma(y)\, \|T_y f - f\|_1\, dy = \int \varphi(z)\, \|T_{\sigma z} f - f\|_1\, dz,$$

which tends to 0 as $\sigma \downarrow 0$ by dominated convergence: the integrand is bounded by $2\|f\|_1 \varphi(z)$ and tends to 0 pointwise, translation being continuous on $L^1(\mathbb{R}^d)$. Now split $f = (f - \varphi_\sigma * f) + \varphi_\sigma * f$. By the three facts, $\mu_n * (\varphi_\sigma * f) = h_n * f$ and $\mu * (\varphi_\sigma * f) = h * f$, so

$$\|\mu_n * f - \mu * f\|_1 \le 2\, \|f - \varphi_\sigma * f\|_1 + \|(h_n - h) * f\|_1 \le 2\, \|f - \varphi_\sigma * f\|_1 + \|h_n - h\|_1\, \|f\|_1.$$

Given $\varepsilon > 0$, choose σ so that the first term is below ε, and then n_0 so that the second is below ε for $n \ge n_0$.

(2). Each $\Phi_{\Sigma,\Sigma'}$ is linear and satisfies (A1), (A2), (A4) and (A5) by the converse in Lemma A.1(1). *Cascade (A6).* $\Phi_{\Sigma,\Sigma} = \delta_0 * \cdot = \mathrm{Id}$ by (K1), and for $\Sigma \le_W \Sigma' \le_W \Sigma''$, by the associativity of convolution (Fubini) and (K1), $\Phi_{\Sigma',\Sigma''}\Phi_{\Sigma,\Sigma'} f = \mu_{\Sigma',\Sigma''} * (\mu_{\Sigma,\Sigma'} * f) = (\mu_{\Sigma',\Sigma''} * \mu_{\Sigma,\Sigma'}) * f = \Phi_{\Sigma,\Sigma''} f$. *Continuity (A7).* P_W is a subset of the finite-dimensional space of pairs of symmetric matrices, so continuity on it is sequential. If $(\Sigma_n, \Sigma_n') \to (\Sigma, \Sigma')$ in P_W, (K2) gives $\widehat{\mu}_{\Sigma_n,\Sigma_n'}(\omega) \to \widehat{\mu}_{\Sigma,\Sigma'}(\omega)$ for every ω, and (1) gives $\Phi_{\Sigma_n,\Sigma_n'} f \to \Phi_{\Sigma,\Sigma'} f$ in $L^1(\mathbb{R}^d)$ for every f. *(ND).* If $\Phi_{0,I} = \mathrm{Id}$, then $\mu_{0,I}$ and δ_0 represent the same operator, so $\mu_{0,I} = \delta_0$ by the uniqueness clause of Lemma A.1(1), against (K4). *Covariance.* For $A \in G$ and $(\Sigma, \Sigma') \in P_W$, the pair $(A \cdot \Sigma, A \cdot \Sigma')$ lies in P_W because W is G-invariant, and (K3) says $A_* \mu_{\Sigma,\Sigma'} = \mu_{A\cdot\Sigma,\, A\cdot\Sigma'}$; by Lemma A.1(2) this is $D_A \Phi_{\Sigma,\Sigma'} D_A^{-1} = \Phi_{A\cdot\Sigma,\, A\cdot\Sigma'}$. *The measures.* $\Phi_{\Sigma,\Sigma'}$ is convolution by $\mu_{\Sigma,\Sigma'}$, so by the uniqueness clause of Lemma A.1(1) the measure that clause gives is $\mu_{\Sigma,\Sigma'}$. *The correspondence.* Kernels to operators to kernels returns the kernels, by the last step. Operators to kernels to operators returns the operators, since the measures of Lemma A.1(4) satisfy $\Phi_{\Sigma,\Sigma'} f = \mu_{\Sigma,\Sigma'} * f$.

(3). The same argument on the parameter set $\{0 \le s \le t\}$, a subset of $\mathbb{R}^2$, gives the four single-operator axioms, (A6) from (I1), (A7) from (I2) and (1), and (ND) at every $s < t$ from (ND). For $Q \in \mathrm{O}(d)$, $|\det Q| = 1$, so $D_Q = R_Q$, and (I3) with Lemma A.1(2) gives $R_Q \Phi_{s,t} R_Q^{-1} = \Phi_{s,t}$, that is $\Phi_{s,t} R_Q = R_Q \Phi_{s,t}$. For $\lambda > 0$, $D_\lambda = D_{\lambda I}$, and (I4) with Lemma A.1(2) gives $D_\lambda \Phi_{s,t} D_\lambda^{-1} = \Phi_{S_\lambda s,\, S_\lambda t}$, which is (A8). The measures are the $\mu_{s,t}$, as in (2). □

The sufficiency statements of the paper on (G, W)-cascade families and on isotropic families are stated on kernels, and by Lemma A.2 each holds on operators as well. The rigidity on operators is Theorem 3.13. The families of one linear flow (Proposition G.1) lie outside the scheme and are stated on kernels only.

A.2 A family is its kernels from scale zero

Proof of Lemma 3.3. (1). $P_W \subseteq P_{W'}$, and a triple $\Sigma \le_W \Sigma' \le_W \Sigma''$ is a triple for W', so (K1) holds on P_W; (K2) is the restriction of a continuous map to the subspace P_W; (K3) holds for every $A \in G \subseteq G'$, and $(A\cdot\Sigma, A\cdot\Sigma') \in P_W$ by the G-invariance of W; and $(0, I) \in P_W$, so (K4) is the same condition.

(2). (a) is (K1) at the pair $(0,0)$ and (K4). (b) is (K3) at the pair $(0,\Sigma)$, as $A\cdot 0 = 0$. (c): $0 \le_W \Sigma$ for every Σ, since W contains the rays, and $\Sigma \mapsto (0,\Sigma)$ is continuous from $\mathrm{PSD}(d)$ into P_W; compose with (K2). (d) is (K1) at $0 \le_W \Sigma \le_W \Sigma'$. If $\mu_{\Sigma'} = \mu_\Sigma * \rho$, then $\widehat{\rho} = \widehat{\mu}_{\Sigma'}/\widehat{\mu}_\Sigma$ wherever $\widehat{\mu}_\Sigma \neq 0$, which is everywhere when $\widehat{\mu}_\Sigma$ has no zero; so ρ is determined by its transform.

(3). By the last sentence, $\mu_{\Sigma,\Sigma'}$ is well defined and its transform is the quotient $q_{\Sigma,\Sigma'} := \widehat{\mu}_{\Sigma'}/\widehat{\mu}_\Sigma$. At $\Sigma = 0$ the quotient is $\widehat{\mu}_{\Sigma'}$, by (a), so $\mu_{0,\Sigma'} = \mu_{\Sigma'}$. (K1): $q_{\Sigma,\Sigma} = 1$, and $q_{\Sigma',\Sigma''}\, q_{\Sigma,\Sigma'} = q_{\Sigma,\Sigma''}$ whenever $\Sigma \le_W \Sigma' \le_W \Sigma''$; uniqueness of the transform turns these into $\mu_{\Sigma,\Sigma} = \delta_0$ and the cascade identity. (K2): by (c) the numerator and the denominator of $q_{\Sigma,\Sigma'}(\omega)$ are continuous on $\mathrm{PSD}(d)\times\mathrm{PSD}(d)$ and the denominator has no zero, so the quotient is continuous there, and in particular on P_W. (K3): for $A \in G$, $(A\cdot\Sigma, A\cdot\Sigma') \in P_W$ by G-invariance, and by (b) the transform of $A_*\mu_{\Sigma,\Sigma'}$ at ω is $q_{\Sigma,\Sigma'}(A^{\mathsf{T}}\omega) = \widehat{\mu}_{A\cdot\Sigma'}(\omega)/\widehat{\mu}_{A\cdot\Sigma}(\omega) = q_{A\cdot\Sigma,A\cdot\Sigma'}(\omega)$. (K4): $\mu_{0,I} = \mu_I \neq \delta_0$. □

A.3 The isotropic corner

The isotropic paper states the isotropic theory with its own definition, Definition 2.4. That definition differs from the scheme's isotropic corner, read on the multiples of I, in two ways. It leaves the relabelling S_λ of the scales free, where the scheme lets the dilation λI act on the multiples of I by $s \mapsto \lambda^2 s$. And it requires nondegeneracy at every pair, where the scheme requires it at $(0, I)$ alone. The next proposition shows that, when G contains $\mathrm{O}(d)$, neither difference matters on the slice: the two theories agree up to a change of the scale coordinate, which is unique up to a constant. The new coordinate is $\varphi = \chi^2$, where χ is the coordinate of [15, Prop. 6.6] in which the dilations act by multiplication, $\chi(S_\lambda t) = \lambda\,\chi(t)$. In [15, Rem. 6.7] φ is the *variance gauge*: in it the Gaussian member is a one-parameter semigroup, and its scale is proportional to the variance. The proposition calls the restriction of a family to the pairs (sI, tI) its *slice*. Its proof, at the end of this subsection, reads an isotropic family through its marginal family on a line (Lemma 2.7) and builds the families it needs from the ray order (Proposition 3.4).

Proposition A.3 (the slice of the isotropic corner is the isotropic theory)**.** *Let $d \ge 1$, and let $\mathrm{S}(d) := \{cQ : c > 0,\ Q \in \mathrm{O}(d)\}$ be the similarities. Let (G, W) be a wedge datum in dimension d (Definition 3.1) and $(\nu_{\Sigma,\Sigma'})$ a (G,W)-cascade family of kernels, with kernels from scale zero $\nu_\Sigma := \nu_{0,\Sigma}$. Since W contains the rays, $(sI, tI) \in P_W$ whenever $0 \le s \le t$; the family $(\nu_{sI,tI})_{0\le s\le t}$ is the* slice *of $(\nu_{\Sigma,\Sigma'})$.*

1. (The slice is an isotropic family.) *If $\mathrm{O}(d) \subseteq G$, the slice of every (G,W)-cascade family of kernels is an isotropic cascade family in dimension d (Definition 2.4) with the dilation action $S_\lambda s = \lambda^2 s$. In particular $\nu_{sI,tI} \neq \delta_0$ whenever $s < t$.*

2. (Every isotropic family is a slice, in the variance gauge.) *Let $(\mu_{s,t})$ be an isotropic cascade family in dimension d with dilation action (S_λ). There is a gauge χ with $\chi(S_\lambda t) = \lambda\,\chi(t)$ for every $\lambda > 0$ and $t \ge 0$; put $\varphi := \chi^2$, a gauge. Then there is exactly one $(\mathrm{GL}(d), \text{rays})$-cascade family of kernels $(\nu_{\Sigma,\Sigma'})$ with*

$$\mu_{s,t} = \nu_{\varphi(s)I,\,\varphi(t)I}, \qquad 0 \le s \le t,$$

and its restriction to the datum $(\mathrm{S}(d), \text{rays})$ is an $(\mathrm{S}(d), \text{rays})$-cascade family of kernels with the same property.

3. (The change of gauge is unique up to a constant.) *In the setting of (2), let ψ be a gauge, (G, W) a wedge datum, and $(\nu'_{\Sigma,\Sigma'})$ a (G, W)-cascade family of kernels with $\mu_{s,t} = \nu'_{\psi(s)I,\,\psi(t)I}$ for $0 \le s \le t$. Then $\psi = c\,\varphi$ for some $c > 0$. Conversely, for every $c > 0$ the family $(\nu_{c^{-1}\Sigma,\,c^{-1}\Sigma'})$ is a $(\mathrm{GL}(d), \text{rays})$-cascade family of kernels with this property for $\psi = c\,\varphi$.*

4. (What the slice determines.) *A $(\mathrm{GL}(d), \text{rays})$-cascade family of kernels is determined by its slice. For $d \ge 2$ an $(\mathrm{S}(d), \text{rays})$-cascade family of kernels is not: the slice of every such family is the slice of two such families whose kernels from scale zero differ at every $\Sigma \in \mathrm{PD}(d)$ that is not a multiple of I.*

Clause (1) says that, when G contains $\mathrm{O}(d)$, nondegeneracy at the one pair $(0, I)$ gives it at every pair of the slice. Off the slice it does not follow. For $d \ge 2$ the similarities with the ray order admit a family whose kernel from scale zero at Σ is $((\det\Sigma)^{1/(2d)}I)_*m$, for a rotation-invariant self-decomposable law $m \ne \delta_0$ (Proposition F.3(3)). Its kernels at the singular Σ are all δ_0, so its increments along the ray of such a Σ are trivial, and (ND) at every pair would exclude it. Clauses (2) and (3) say that the free gauge costs nothing: the change of coordinate is the variance gauge $\varphi = \chi^2$, with χ taken from the marginal family on a line (Lemma 2.7). Clause (4) says what the corner adds to its slice: nothing for $\mathrm{GL}(d)$, the shapes of the apertures for the similarities.

What the proof takes from the ray order is this. By Proposition 3.4(3), a $(\mathrm{GL}(d), \text{rays})$-cascade family is determined by its kernel at I, and for every rotation-invariant self-decomposable law $m \ne \delta_0$ there is one with kernel m at I, whose kernels from scale zero are $\nu_\Sigma = (\Sigma^{1/2})_*m$. By Proposition 3.4(2), for a wedge datum with the ray order the kernels from scale zero of a family are self-decomposable, and every map $\Sigma \mapsto \nu_\Sigma$ into self-decomposable laws with $\nu_0 = \delta_0$, $\nu_I \ne \delta_0$, covariance under the group and transforms continuous in Σ is the family of kernels from scale zero of exactly one family. For the similarities $\Sigma \mapsto ((\det\Sigma)^{1/(2d)}I)_*m$ is such a map, and it gives the second family of clause (4). By Lemma A.5(2), a rotation-invariant law other than δ_0 has the orthogonal group as its stabilizer, which is what separates the two families off the multiples of I.

The proof also uses three elementary facts. The transform of a self-decomposable law has no zero, and a linear image $A_*\mu$ of a self-decomposable law is self-decomposable, with the residuals $A_*\rho_c$ (Proposition 3.4(1)). When $\widehat{\nu}_\Sigma$ has no zero, $\nu_{\Sigma'} = \nu_\Sigma * \rho$ determines $\widehat{\rho} = \widehat{\nu}_{\Sigma'}/\widehat{\nu}_\Sigma$, so the increment $\nu_{\Sigma,\Sigma'}$ is the only such law ρ (Lemma 3.3(2)). And a family of the general linear group with the ray order is also one of the similarities, with the same kernels, since (K1), (K2) and (K4) do not involve the group and (K3) for the similarities is part of (K3) for $\mathrm{GL}(d)$; the similarities with the ray order are a wedge datum, as the proof shows.

Proof of Proposition A.3 (sketch). A fact used twice is the argument of [15, Lem. 6.3]: *if a law m on $\mathbb{R}^d$ satisfies $(aI)_*m = m$ for one $a > 0$ with $a \ne 1$, then $m = \delta_0$.* Indeed $\widehat{m}(\omega) = \widehat{m}(a^n\omega)$ for every $n \in \mathbb{Z}$, and letting $a^n \to 0$ gives $\widehat{m}(\omega) = 1$.

(1). Since W contains the rays and $Q \cdot (sI) = sI$ for $Q \in \mathrm{O}(d)$, (K1), (K2) and (K3) of Definition 3.1 give (I1), (I2) and (I3) of Definition 2.4. For $\lambda > 0$, $(\lambda I) \cdot (sI) = \lambda^2 s\, I$, so (K3) at λI is (I4) with $S_\lambda s = \lambda^2 s$. For (ND), (K3) with $A = \sqrt{t}\, I$ gives $\nu_{tI} = (\sqrt{t}\, I)_*\nu_I$. If $\nu_{sI,tI} = \delta_0$ with $s < t$, (K1) gives $\nu_{tI} = \nu_{sI}$. For $s = 0$ this makes $\nu_I = \delta_0$. For $s > 0$ it makes ν_I invariant under $(\sqrt{s/t}\, I)_*$, and the fact gives $\nu_I = \delta_0$. Both contradict (K4).

(2). The marginal family of Lemma 2.7, the images $\kappa_{s,t}$ of the $\mu_{s,t}$ on a line through the origin acting by $f \mapsto \kappa_{s,t} * f$, satisfies (A1)–(A8) and (ND) with the same (S_λ). [15, Prop. 6.6], restated in the part used: *for such a family, $\ell(\lambda) := S_\lambda 1$ is a continuous strictly increasing bijection of $(0, \infty)$ with $\ell(1) = 1$, and $\chi := \ell^{-1}$, with $\chi(0) := 0$, is an increasing bijection of $[0, \infty)$ with $\chi(S_\lambda t) = \lambda\,\chi(t)$.* With $\theta := \varphi^{-1}$ this gives $S_\lambda\theta(u) = \theta(\lambda^2 u)$. Put $m := \mu_{0,\theta(1)}$. By (I3), (I4) and (ND), m is rotation invariant, not δ_0, and $\mu_{0,\theta(s)} = (\sqrt{s}\, I)_*m$ for $s \ge 0$. By (I1) it

is self-decomposable, (2.2), with the residuals $\mu_{\theta(c^2),\theta(1)}$. Let $(\nu_{\Sigma,\Sigma'})$ be the $(\mathrm{GL}(d), \text{rays})$-cascade family with kernel m at I. Then $\nu_{uI} = \mu_{0,\theta(u)}$, and since the transforms of its kernels from scale zero have no zero, (I1) and the uniqueness of increments (Lemma 3.3(2)) give $\nu_{uI,vI} = \mu_{\theta(u),\theta(v)}$ for $u \le v$. With $u = \varphi(s)$ and $v = \varphi(t)$ this is the claim. A second such family has the same kernel m at I, hence is the same family. For every $d \ge 1$ the similarities $\mathrm{S}(d)$ form a subgroup containing the dilations, closed in $\mathrm{GL}(d)$, as an invertible limit A of similarities has $AA^{\mathsf{T}} = kI$ with $k > 0$; so $(\mathrm{S}(d), \text{rays})$ is a wedge datum, the ray order being a partial order that contains the rays and is invariant under $\mathrm{GL}(d)$. As $\mathrm{S}(d) \subseteq \mathrm{GL}(d)$, (K1), (K2) and (K4) do not involve the group, and (K3) for $\mathrm{S}(d)$ is part of (K3) for $\mathrm{GL}(d)$, the family is also an $(\mathrm{S}(d), \text{rays})$-cascade family, with the same kernels.

(3). Put $m' := \nu'_I$, which is not δ_0 by (K4). As in (1), $\nu'_{uI} = (\sqrt{u}\, I)_* m'$. At the pairs $(0,t)$ the two families give $(\sqrt{\psi(t)}\, I)_* m' = \mu_{0,t} = (\sqrt{\varphi(t)}\, I)_* m$, and at $t_0 := \theta(1)$ this reads $m = (\sqrt{\psi(t_0)}\, I)_* m'$. So m' is invariant under $((\varphi(t)\psi(t_0)/\psi(t))^{1/2} I)_*$ for every $t > 0$, and the fact gives $\psi = \psi(t_0)\,\varphi$. The converse is a check of (K1)–(K4) for $\nu_{c^{-1}\Sigma,\, c^{-1}\Sigma'}$, the ray order being invariant under $\Sigma \mapsto c^{-1}\Sigma$.

(4). Two $(\mathrm{GL}(d), \text{rays})$-cascade families with the same slice have the same kernel at I, and are equal. Let $d \ge 2$, and let $(\nu_{\Sigma,\Sigma'})$ be an $(\mathrm{S}(d), \text{rays})$-cascade family and $m := \nu_I$. It is rotation invariant, self-decomposable and not δ_0, and $\nu_{sI} = (\sqrt{s}\, I)_* m$. Proposition 3.4 gives two families with the kernels from scale zero $(\Sigma^{1/2})_* m$ and $((\det \Sigma)^{1/(2d)} I)_* m$. They have the kernel $(\sqrt{s}\, I)_* m$ at sI as well, and by the uniqueness of increments the three slices agree. If the two kernels agree at $\Sigma \in \mathrm{PD}(d)$, then $v^{-1}\Sigma^{1/2}$, with $v := (\det \Sigma)^{1/(2d)}$, fixes m, so it is orthogonal by Lemma A.5(2), and $\Sigma = v^2 I$. □

A.4 The parameter space

The two lemmas of §3.2 say why the kernels of the scheme at the positive definite matrices are indexed by $\mathrm{PD}(d)$, and what it costs. The first says that covariance fixes the parametrization up to one constant. Their proofs follow them.

Lemma A.4 (the affine gauge is a constant)**.** *Let $h : \mathrm{PD}(d) \to \mathrm{PD}(d)$ satisfy $h(A\Sigma A^{\mathsf{T}}) = A\, h(\Sigma)\, A^{\mathsf{T}}$ for every $A \in \mathrm{GL}(d)$ and $\Sigma \in \mathrm{PD}(d)$. Then there is a $c > 0$ with $h(\Sigma) = c\,\Sigma$ for every Σ. On the isotropic slice $\{sI : s > 0\}$ the dilation $x \mapsto \lambda x$, $\lambda > 0$, which is $A = \lambda I$, acts by $s \mapsto \lambda^2 s$.*

The group acts transitively on $\mathrm{PD}(d)$, and the parametrization there is forced up to one constant. On the multiples of I, Proposition A.3(2) and (3) identify the parameter with the variance gauge, up to that constant. In the Gaussian family (3.1) the parameter is the covariance matrix up to the factor $2b$.

The second lemma says why $\mathrm{PD}(d)$ at all. A law that no proper subspace carries has a compact stabilizer, and a rotation-invariant law other than δ_0 has exactly $\mathrm{O}(d)$. An orbit of $\mathrm{GL}(d)$ through a full law can therefore be indexed covariantly by $\mathrm{PD}(d) = \mathrm{GL}(d)/\mathrm{O}(d)$ exactly when it has a rotation-invariant member. Clause (1) is a classical consequence of the convergence of types ([45], Thm. 2.3.10, p. 33, where the symmetries may carry a shift), and (3)(b) has the form of their Thm. 2.3.11, p. 34, on the compact subgroups of $\mathrm{GL}(d)$.

Lemma A.5 (the parameter space: compact stabilizers and the isotropic member)**.** *Let $d \ge 1$. For a probability measure μ on $\mathbb{R}^d$ write* $\mathrm{Stab}(\mu) := \{A \in \mathrm{GL}(d) : A_*\mu = \mu\}$ *and* $\mathrm{Orb}(\mu) := \{A_*\mu : A \in \mathrm{GL}(d)\}$*, and call μ* full *if $\mu(V) < 1$ for every linear subspace $V \ne \mathbb{R}^d$.*

1. (Compact stabilizers.) $\mathrm{Stab}(\mu)$ *is a closed subgroup of* $\mathrm{GL}(d)$*. If μ is full,* $\mathrm{Stab}(\mu)$ *is compact. So a law whose stabilizer is not compact is carried by a proper linear subspace; for $d \ge 2$ it need not be δ_0: the stabilizer of δ_{e_1} contains every A with $Ae_1 = e_1$.*

2. (Rotation-invariant laws.) *If $Q_*\mu = \mu$ for every $Q \in \mathrm{O}(d)$ and $\mu \neq \delta_0$, then μ is full and* $\mathrm{Stab}(\mu) = \mathrm{O}(d)$.

3. (The isotropic member is the axiom of the parameter space $\mathrm{PD}(d)$.) *Let μ_0 be full. The following are equivalent: (a)* $\mathrm{Orb}(\mu_0)$ *has a rotation-invariant member; (b)* $\mathrm{Stab}(\mu_0) = C\,\mathrm{O}(d)\,C^{-1}$ *for some $C \in \mathrm{GL}(d)$; (c) there is a map $\Sigma \mapsto \mu_\Sigma$ from* $\mathrm{PD}(d)$ *into* $\mathrm{Orb}(\mu_0)$ *with $A_*\mu_\Sigma = \mu_{A\Sigma A^\mathsf{T}}$ for every $A \in \mathrm{GL}(d)$ and $\Sigma \in \mathrm{PD}(d)$. When they hold, every map as in (c) is a bijection of* $\mathrm{PD}(d)$ *onto* $\mathrm{Orb}(\mu_0)$*, and for any two such maps μ and μ' there is a $c > 0$ with $\mu'_\Sigma = \mu_{c\Sigma}$ for every $\Sigma \in \mathrm{PD}(d)$.*

4. (An orbit without an isotropic member.) *Let $p_1, \dots, p_d > 0$ be distinct with $p_1 + \cdots + p_d = 1$, and $\mu_0 := \sum_j p_j\,\delta_{e_j}$. Then μ_0 is full,* $\mathrm{Stab}(\mu_0) = \{I\}$*, the map $A \mapsto A_*\mu_0$ is a bijection of* $\mathrm{GL}(d)$ *onto* $\mathrm{Orb}(\mu_0)$*, and no member of* $\mathrm{Orb}(\mu_0)$ *is rotation invariant.*

5. (In the scheme.) *Let (G, W) be a wedge datum with $\mathrm{O}(d) \subseteq G$ and $(\mu_{\Sigma,\Sigma'})$ a (G, W)-cascade family of kernels (Definition 3.1(2)). Then μ_I is rotation invariant and full,* $\mathrm{Stab}(\mu_I) = \mathrm{O}(d)$*, and $\Sigma \mapsto \mu_\Sigma$ is injective on $\{AA^\mathsf{T} : A \in G\}$, which is* $\mathrm{PD}(d)$ *when $G = \mathrm{GL}(d)$. Without $\mathrm{O}(d) \subseteq G$, (K4) need not make μ_I full: for $d \geq 2$, G the invertible diagonal matrices and W the Loewner order, the centred Gaussian laws $\mu_{\Sigma,\Sigma'}$ with $\widehat{\mu}_{\Sigma,\Sigma'}(\omega) = \exp[-\frac{1}{2}(\Sigma' - \Sigma)_{11}\,\omega_1^2]$ form a (G, W)-cascade family of kernels whose μ_I is carried by the first axis and has a stabilizer that is not compact.*

For $G = \mathrm{GL}(d)$, clause (5) makes the kernels from scale zero at $\mathrm{PD}(d)$ correspond one to one to $\mathrm{GL}(d)/\mathrm{O}(d) = \mathrm{PD}(d)$. For a smaller group containing $\mathrm{O}(d)$ it gives injectivity on $\{AA^\mathsf{T} : A \in G\}$ only. Without an isotropic member, a full law can have a trivial stabilizer, and its orbit is then in bijection with $\mathrm{GL}(d)$ itself, of dimension d^2 against $d(d+1)/2$ (clause (4)).

Proof of Lemma A.4. For $Q \in \mathrm{O}(d)$, $QQ^\mathsf{T} = I$ gives $Q\,h(I)\,Q^\mathsf{T} = h(I)$: the symmetric matrix $H := h(I)$ commutes with every orthogonal matrix. Taking Q the reflection $I - 2e_ie_i^\mathsf{T}$, the entry $(QHQ^\mathsf{T})_{ik}$ is $-H_{ik}$ for $k \neq i$, so $H_{ik} = 0$ and H is diagonal; taking Q the permutation matrix exchanging e_i and e_k gives $H_{ii} = H_{kk}$. So $H = cI$, and $c > 0$ because $H \in \mathrm{PD}(d)$. For $\Sigma \in \mathrm{PD}(d)$ put $A = \Sigma^{1/2}$: then $h(\Sigma) = h(AIA^\mathsf{T}) = A(cI)A^\mathsf{T} = c\,\Sigma$. Finally $(\lambda I)(sI)(\lambda I)^\mathsf{T} = \lambda^2 s\,I$, which is how λI moves the parameter of Definition 3.2. □

Proof of Lemma A.5 (sketch). (1). By uniqueness of the transform, $A_*\mu = \mu$ exactly when $\widehat{\mu}(A^\mathsf{T}\omega) = \widehat{\mu}(\omega)$ for every ω, so $\mathrm{Stab}(\mu)$ is a subgroup, closed by the continuity of $\widehat{\mu}$. Let μ be full. The function $f(B) := \int \min(1, |Bx|)\,\mu(dx)$ is continuous, by dominated convergence, and has a minimum c on the compact sphere $\{\|B\| = 1\}$; $c > 0$, since $f(B) = 0$ would make μ carried by the proper subspace $\ker B$. For A in the stabilizer and $t := \|A\|$, since $A_*\mu = \mu$,

$$c \leq f(A/t) = \int \min\bigl(1, |y|/t\bigr)\,\mu(dy),$$

which tends to 0 as $t \to \infty$. The stabilizer is therefore bounded, and as it contains the inverses, $|\det A|$ is bounded away from 0 on it. A limit of its elements is then invertible, and in it, as it is closed in $\mathrm{GL}(d)$; so it is compact. For δ_{e_1} the matrices $I + t\,e_1e_2^\mathsf{T}$, $t \in \mathbb{R}$, fix e_1 and are unbounded.

(2). If a proper subspace carried μ, so would a hyperplane $u^\perp$, hence by rotation invariance every hyperplane, and so would the intersection $\{0\}$ of the coordinate hyperplanes. So μ is full, and by (1) its stabilizer K is compact and contains $\mathrm{O}(d)$. For $A \in K$ with polar decomposition $A = PQ$, the factor $P = AQ^{-1} \in \mathrm{PD}(d)$ lies in K, and so do its powers P^n, $n \in \mathbb{Z}$. An eigenvalue p of P then has p^n bounded, so $p = 1$ and $A = Q$.

(3). For $B \in \mathrm{GL}(d)$, $B_*\mu$ is full with μ, and $\mathrm{Stab}(B_*\mu) = B\,\mathrm{Stab}(\mu)B^{-1}$. So (a) gives (b) by (2). From (b), $\nu := (C^{-1})_*\mu_0$ is rotation invariant, and $\mu_\Sigma := (\Sigma^{1/2})_*\nu$ is as in (c):

for $\Sigma' := A\Sigma A^{\mathsf{T}}$ the matrix $\Sigma'^{-1/2} A \Sigma^{1/2}$ is orthogonal. From (c), μ_I is a rotation-invariant member. For a map as in (c), covariance with $A = \Sigma^{1/2}$ gives $\mu_\Sigma = (\Sigma^{1/2})_* \mu_I$, which is onto the orbit and, by (2), one to one. For a second map μ', $h := \mu^{-1} \circ \mu'$ satisfies $h(A\Sigma A^{\mathsf{T}}) = A\, h(\Sigma) A^{\mathsf{T}}$, and Lemma A.4 gives $h(\Sigma) = c\,\Sigma$.

(4). A proper subspace misses some e_j, so it has μ_0-measure at most $1 - p_j$. An A that fixes μ_0 fixes each atom, the masses being distinct, so $A = I$. A member $A_*\mu_0$ gives positive mass only to the d points Ae_j, and mass p_1 to $x := Ae_1 \neq 0$. If it were rotation invariant, every point of norm $r := |x|$, being Qx for an orthogonal Q, would carry that mass and be one of those d points; but the $2d$ points $\pm r\, e_j$ have norm r.

(5). For $Q \in \mathrm{O}(d) \subseteq G$, $Q \cdot 0 = 0$ and $Q \cdot I = I$, so (K3) and (K4) of Definition 3.1 make μ_I rotation invariant and not δ_0, and (2) applies. (K3) gives $\mu_{AA^{\mathsf{T}}} = A_* \mu_I$ for $A \in G$, and $A_*\mu_I = B_*\mu_I$ forces $B^{-1}A \in \mathrm{O}(d)$, hence $AA^{\mathsf{T}} = BB^{\mathsf{T}}$. In the example the exponents are additive in $\Sigma' - \Sigma$ and continuous, a diagonal A multiplies them by a_1^2, and $\widehat{\mu}_{0,I}(e_1) = e^{-1/2} \neq 1$. The stabilizer of its μ_I contains the unbounded matrices $\mathrm{diag}(1, t, 1, \ldots, 1)$, $t > 0$. □

A.5 Infinitely divisible increments

Proof of Theorem 3.5. Small increments. Under the hypotheses of (1), for every $\varepsilon > 0$ and $\eta > 0$ there is $\delta > 0$ with $\mu_{a,b}(\{|x| > \varepsilon\}) < \eta$ whenever $0 \le a \le b \le 1$ and $b - a < \delta$. Otherwise there are pairs $0 \le a_n \le b_n \le 1$ with $b_n - a_n \to 0$ and $\mu_{a_n,b_n}(\{|x| > \varepsilon\}) \ge \eta$. For each ω the function $(a, b) \mapsto \widehat{\mu}_{a,b}(\omega)$ is uniformly continuous on the compact triangle $\{0 \le a \le b \le 1\}$ and equal to 1 on its diagonal, so $\widehat{\mu}_{a_n,b_n}(\omega) \to 1$. By Lévy's continuity theorem on $\mathbb{R}^d$, $\mu_{a_n,b_n} \to \delta_0$ weakly, and the portmanteau inequality for the closed set $\{|x| \ge \varepsilon\}$ bounds the upper limit of its measures by 0, a contradiction.

(1). Fix $s \le t$, and for $n \ge 1$ put $t_{n,k} := s + k(t - s)/(n + 1)$ and $\mu_{n,k} := \mu_{t_{n,k-1}, t_{n,k}}$, $1 \le k \le n + 1$. The cascade identity and the commutativity of convolution give, by induction on the number of steps,

$$\mu_{s,t} = \mu_{n,1} * \mu_{n,2} * \cdots * \mu_{n,n+1} \qquad \text{for every } n.$$

The rows are infinitesimal. Given $\varepsilon > 0$ and $\eta > 0$, take δ from the small increments: once $(t - s)/(n + 1) < \delta$, every $\mu_{n,k}(\{|x| > \varepsilon\})$ is below η. *Khintchine.* The $\mu_{n,k}$ form a null array with $r_n = n + 1$ entries in row n, whose row products all equal $\mu_{s,t}$, so C1 with $b_n = 0$ makes $\mu_{s,t}$ infinitely divisible.

(2). For a W-chain ζ from Σ to Σ', the laws $\nu_{u,v} := \mu_{\zeta(u),\zeta(v)}$, $0 \le u \le v \le 1$, satisfy the hypotheses of (1) by (K1) and (K2) of Definition 3.1, since $(u, v) \mapsto (\zeta(u), \zeta(v))$ is continuous. And $\nu_{0,1} = \mu_{\Sigma,\Sigma'}$. The kernel μ_Σ is the increment of the pair $0 \le_W \Sigma$.

(3). (a) $\zeta(t) - \zeta(s) = (t - s)(\Sigma' - \Sigma)$ is positive semidefinite. (b) Along each chain a later value is a multiple of an earlier one by a factor at least 1. (c) By the spectral theorem M^s is positive definite, continuous in s, and $M^a M^b = M^{a+b}$. For $s \le t$ put $B := \Sigma^{1/2} M^{s/2}$, so that $\zeta(s) = BB^{\mathsf{T}}$ and $\zeta(t) = BM^{t-s}B^{\mathsf{T}}$. Then $Q := \zeta(s)^{-1/2} B$ is orthogonal, and $\zeta(s)^{-1/2}\zeta(t)\zeta(s)^{-1/2} = QM^{t-s}Q^{\mathsf{T}}$ lies in $\mathcal{K}$ by the two closure properties. The three sets named are closed under orthogonal conjugation, and $M \mapsto M^\tau$ raises every eigenvalue, the least one and the determinant to the power $\tau \ge 0$, which preserves each defining inequality. For $M = e^X$, $\lambda_{\min}(M) = e^{\lambda_{\min}(X)}$ and $\det M = e^{\mathrm{tr}\, X}$, which gives the description of $\mathcal{K}_\alpha$. The Loewner and the ray order are partial orders directly. That the relation of $\mathcal{K}_\alpha$ is one is the first sentence of Proposition 4.7(2), which rests on no cited fact.

(4). For $s \le t$ the laws $\nu_{u,v} := \mu_{s+u(t-s),\, s+v(t-s)}$ satisfy the hypotheses of (1) by (I1) and (I2) of Definition 2.4, and $\nu_{0,1} = \mu_{s,t}$. □

A.6 The single step

Proof of Lemma 3.8. Let $U := \{\eta \in H : |\eta| < r\}$, with $r > 0$ so small that $\widehat{\rho} = 1$ on U. For $\eta \in U$, $0 = \mathrm{Re}\big(1 - \widehat{\rho}(\eta)\big) = \int \big(1 - \cos\langle \eta, x\rangle\big)\rho(dx)$ with a nonnegative integrand, so $\langle \eta, x\rangle \in 2\pi\mathbb{Z}$ for every x outside a ρ-null set N_η. Let $D \subseteq U$ be countable and dense in U, and $N := \bigcup_{\eta\in D} N_\eta$, a ρ-null set. For $x \notin N$ the linear map $\ell_x : H \to \mathbb{R}$, $\eta \mapsto \langle \eta, x\rangle$, is continuous and maps D into the closed set $2\pi\mathbb{Z}$, hence maps U, which lies in the closure of D, into $2\pi\mathbb{Z}$. As U is connected, $\ell_x(U)$ is a connected subset of $2\pi\mathbb{Z}$, a single point, and it contains $\ell_x(0) = 0$; so $\ell_x = 0$ on U, hence on H, which is the union of the dilates of U. That is $x \in H^\perp$. Thus $\rho(\mathbb{R}^d \setminus H^\perp) \le \rho(N) = 0$, and for $\eta \in H$, $\widehat{\rho}(\omega + \eta) = \int_{H^\perp} e^{i\langle \omega, x\rangle} e^{i\langle \eta, x\rangle}\rho(dx) = \widehat{\rho}(\omega)$. For $H = \mathbb{R}^d$ the complement is $\{0\}$, so $\rho = \delta_0$. □

Proof of Lemma 3.9. The kernel from scale zero at I is radial and real. For $Q \in \mathrm{O}(d)$, (K3) of Definition 3.2 with $\Sigma = 0$, $\Sigma' = I$ gives $Q_*\mu_I = \mu_{QQ^\mathsf{T}} = \mu_I$, so $\widehat{\mu}_I(Q^\mathsf{T}\omega) = \widehat{\mu}_I(\omega)$ and $\widehat{\mu}_I$ is constant on spheres. Put $\vartheta(u) := \widehat{\mu}_I(\sqrt{u}\, e_1)$ for $u \ge 0$; then $\widehat{\mu}_I(\omega) = \vartheta(|\omega|^2)$, ϑ is continuous because a characteristic function is, and $\vartheta(0) = 1$. With $Q = -I$, $\widehat{\mu}_I(\omega) = \widehat{\mu}_I(-\omega) = \overline{\widehat{\mu}_I(\omega)}$, so ϑ is real, and $|\vartheta| \le 1$.

Every kernel from scale zero. For $\Sigma \in \mathrm{PD}(d)$, (K3) with $A = \Sigma^{1/2}$ gives $\mu_\Sigma = A_*\mu_I$, so $\widehat{\mu}_\Sigma(\omega) = \widehat{\mu}_I(\Sigma^{1/2}\omega) = \vartheta(\langle \omega, \Sigma\omega\rangle)$. For $\Sigma \in \mathrm{PSD}(d)$ singular, $\Sigma_n := \Sigma + n^{-1}I \in \mathrm{PD}(d)$ tends to Σ, and (K2) along the pairs $(0, \Sigma_n) \to (0, \Sigma)$ with the continuity of ϑ gives $\widehat{\mu}_\Sigma(\omega) = \lim_n \vartheta(\langle \omega, \Sigma_n\omega\rangle) = \vartheta(\langle \omega, \Sigma\omega\rangle)$.

No zeros. Fix ω and put $f(s,t) := \widehat{\mu}_{sI,tI}(\omega)$ for $0 \le s \le t$. By (K2) f is continuous; by (K1), and because the transform of a convolution is the product of the transforms, $f(t,t) = 1$ and $f(r,t) = f(s,t)\, f(r,s)$ for $r \le s \le t$. Suppose $f(0,t) = 0$ for some t, and let t^* be the infimum of such t. As $f(0,\cdot)$ is continuous with $f(0,0) = 1$, $t^* > 0$ and $f(0,t^*) = 0$. By continuity at (t^*, t^*), where $f = 1$, there is $s \in (0, t^*)$ with $f(s,t^*) \ne 0$, and $f(0,s) \ne 0$ because $s < t^*$; then $f(0,t^*) = f(s,t^*)\, f(0,s) \ne 0$, a contradiction. Hence $\vartheta(t|\omega|^2) = \widehat{\mu}_{tI}(\omega) \ne 0$ for all $t \ge 0$ and ω, so ϑ has no zero on $[0,\infty)$. Being real and continuous with $\vartheta(0) = 1$, it is positive, and $g := -\log\vartheta$ is continuous, nonnegative because $\vartheta \le 1$, and $g(0) = 0$. This is (3.3).

Increments. For $\Sigma \le \Sigma'$, (K1) gives $\mu_{\Sigma'} = \mu_{\Sigma,\Sigma'} * \mu_\Sigma$, so $\widehat{\mu}_{\Sigma'} = \widehat{\mu}_{\Sigma,\Sigma'}\,\widehat{\mu}_\Sigma$, and $\widehat{\mu}_\Sigma > 0$ gives (3.4).

Nondegeneracy. If $g \equiv 0$ then $\widehat{\mu}_I \equiv 1$ and Lemma 3.8 with $H = \mathbb{R}^d$ gives $\mu_I = \delta_0$, against (K4). □

Proof of Lemma 3.10. A constant difference on a half-line. Fix $\delta > 0$ and put $u_\delta := \delta/(a-1)$, so that $au_\delta = u_\delta + \delta$. For $x \ge u_\delta$, (3.5) at $u = u_\delta$ and $w = x - u_\delta \ge 0$ reads $g(x+\delta) - g(x) = g(u_\delta + \delta) - g(u_\delta) =: \kappa(\delta)$, independent of x on $[u_\delta, \infty)$.

A constant difference everywhere. Fix $\delta > 0$ and $y > 0$. For every integer $n \ge \delta/((a-1)y)$ the points $y + j\delta/n$, $0 \le j < n$, are at least $y \ge u_{\delta/n}$, so the previous step with δ/n in place of δ gives

$$g(y+\delta) - g(y) = \sum_{j=0}^{n-1} \Big[g\big(y + \tfrac{(j+1)\delta}{n}\big) - g\big(y + \tfrac{j\delta}{n}\big) \Big] = n\,\kappa(\delta/n).$$

Given $y_1, y_2 > 0$, an n admissible for both shows $g(y_1+\delta) - g(y_1) = g(y_2+\delta) - g(y_2)$. So $K(\delta) := g(y+\delta) - g(y)$ does not depend on $y > 0$, and $K(2\delta) = [g(y+2\delta) - g(y+\delta)] + [g(y+\delta) - g(y)] = 2K(\delta)$.

Additivity, from continuity at 0 alone. Put $h(\delta) := g(\delta) - K(\delta)$ for $\delta > 0$. With $y = \delta$ the previous step gives $g(2\delta) = g(\delta) + K(\delta)$, so $h(2\delta) = g(\delta) + K(\delta) - 2K(\delta) = h(\delta)$. By induction, for every $n \ge 0$,

$$h(\delta) = h(\delta 2^{-n}), \qquad K(\delta) = 2^n K(\delta 2^{-n}), \qquad \text{hence} \qquad h(\delta) = g(\delta 2^{-n}) - 2^{-n}K(\delta).$$

Letting $n \to \infty$, continuity at 0 and $g(0) = 0$ give $h(\delta) = 0$, that is $g(\delta) = K(\delta)$. Hence $g(y+\delta) = g(y) + g(\delta)$ for all $y, \delta > 0$.

Cauchy's equation. By induction $g(n\delta) = n\,g(\delta)$ for integers $n \geq 1$, so $q\,g(p/q) = g(p) = p\,g(1)$ and $g(r) = r\,g(1)$ for every positive rational r. As $g \geq 0$, $g(y+\delta) \geq g(y)$: g is nondecreasing on $(0,\infty)$. For $x > 0$ and rationals $0 < r < x < r'$ this gives $r\,g(1) \leq g(x) \leq r'\,g(1)$, and letting $r, r' \to x$ gives $g(x) = x\,g(1)$. With $g(0) = 0$ that is the claim. □

B Deferred proofs for the affine landscape

This appendix holds the routine proofs of §4. The statements are in that section, each followed by a pointer here, and the subsections follow their order. The statements whose proofs carry the argument of Theorem 4.21, the invariant orders of §4.4 among them, are proved completely in Appendix C. For one of them, Proposition 4.11, this appendix proves the remaining clause. It also gives the complete proofs of Propositions 4.15 and 4.19, the rays at index 0, and the sketch of the proof of Proposition 4.25.

B.1 The groups and the Loewner order

Proof of Lemma 4.1. Clause (1). Congruence by an invertible matrix preserves positive semidefiniteness in both directions, so $\Sigma_1 \leq \Sigma_2$ exactly when $M := \Sigma_1^{-1/2}\Sigma_2\Sigma_1^{-1/2} \geq I$ (the Loewner order of Definition 3.2), that is, exactly when every eigenvalue of M is at least 1. Since $\Sigma_k^{\pm 1/2} = t_k^{\pm 1} S_k^{\pm 1/2}$, $M = (t_2/t_1)^2 N$, whose least eigenvalue is $(t_2/t_1)^2 n_1$.

Clause (2). $\det M = \det\Sigma_2 / \det\Sigma_1 = (t_2/t_1)^{2d}$ is the product of the eigenvalues of M. Under $M \geq I$ each is at least 1, so $t_2 \geq t_1$, with equality only if all are 1, that is, $M = I$ and $\Sigma_2 = \Sigma_1$. If $S_1 = S_2$ then $N = I$ and $\Sigma_2 = (t_2/t_1)^2\Sigma_1$. Conversely $\lambda^2\Sigma_1 - \Sigma_1 \geq 0$ for $\lambda \geq 1$, and $S(\lambda^2\Sigma_1) = S_1$.

Clause (3). The lower bound on n_1 is clause (1). Then $n_d = \left(\prod_{i<d} n_i\right)^{-1} \leq (t_2/t_1)^{2(d-1)}$, and $n_d/n_1 \leq (t_2/t_1)^{2(d-1)}(t_2/t_1)^2$. For the converse, $\det\Sigma_2 = t_2^{2d}\det S_1 \det N = t_2^{2d}$, so $t(\Sigma_2) = t_2$ and $S(\Sigma_2) = S_1^{1/2} N S_1^{1/2}$, whose relative shape is N, and clause (1) gives $\Sigma_1 \leq \Sigma_2$.

Clause (4). At $d = 2$, $n_1 n_2 = 1$ and $n_1 \leq 1$, so $\delta(S_1, S_2) = \sqrt{2}\,\log(1/n_1)$, and $n_1 \geq (t_1/t_2)^2$ is $\delta(S_1, S_2) \leq 2\sqrt{2}\log(t_2/t_1)$. By clause (1) every $S(\Sigma_2)$ with $\Sigma_2 \geq \Sigma_1$ and $t(\Sigma_2) = t_2$ lies in the set. By clause (3) every S in the set, whose relative shape $N = S_1^{-1/2} S S_1^{-1/2}$ has determinant 1 and least eigenvalue at least $(t_1/t_2)^2$, is the shape of $t_2^2 S = t_2^2 S_1^{1/2} N S_1^{1/2}$. □

B.2 The base, and the step set of an orbit

Proof of Proposition 4.4 (sketch). (1) and *(2)* are transport by $\Sigma^{1/2}$: $\widehat{\mu}_\Sigma(\omega) = \widehat{\mu}_I(\Sigma^{1/2}\omega) = e^{-g(\langle\omega,\Sigma\omega\rangle)}$, and the transform of $(\Sigma^{1/2})_*\nu_\lambda$ times $\widehat{\mu}_\Sigma$ is $e^{-g(\lambda^2\langle\omega,\Sigma\omega\rangle)}$.

(3). Taking transforms in $\mu_{\Sigma+P} = \mu_\Sigma * \rho$ gives the single-step identity (3.2), and Theorem 3.7 makes g linear. For the last sentence, (K1) of an affine cascade family would give $\mu_{I+e_1e_1^\mathsf{T}} = \mu_{I,I+e_1e_1^\mathsf{T}} * \mu_I$, which the first sentence excludes; equivalently, Theorem 3.6 would make μ_I Gaussian.

(4). The Gamma integral gives $(1+u)^{-\gamma} = \Gamma(\gamma)^{-1}\int_0^\infty v^{\gamma-1}e^{-v}e^{-vu}\,dv$, and conditioning on V gives the law of $\sqrt{2V}Z$. For the ray increments, *Frullani's integral* is

$$\int_0^\infty \left(e^{-ax} - e^{-bx}\right)\frac{dx}{x} = \log\frac{b}{a}, \qquad 0 < a < b :$$

on (ε,∞) the substitutions $y = ax$ and $y = bx$ turn the left-hand side into $\int_{a\varepsilon}^{b\varepsilon} e^{-y}y^{-1}dy$, which lies between $e^{-b\varepsilon}\log(b/a)$ and $\log(b/a)$. Put $n(x) := (e^{-x/\lambda^2} - e^{-x})x^{-1}$, of total mass $\log\lambda^2$ by Frullani's integral at $(\lambda^{-2}, 1)$. The same integral at $(u+\lambda^{-2}, u+1)$ gives

$$\int_0^\infty (1 - e^{-ux})\,n(x)\,dx = \log\frac{1+\lambda^2 u}{1+u},$$

so the compound Poisson variable Y with Lévy measure $\gamma\, n(x)\, dx$ has $\mathbb{E}\, e^{-uY} = ((1+u)/(1+\lambda^2 u))^\gamma$. Conditioning on Y, the law of $\sqrt{2Y}Z$ has transform $e^{-g_\gamma(\lambda^2|\omega|^2)}/e^{-g_\gamma(|\omega|^2)}$. □

Proof of Lemma 4.6. The constant. Since $1-\cos y_1 \le \min(2, y_1^2/2) \le 2(|y|^2 \wedge 1)$, the integrand is at most $2(|y|^2\wedge 1)|y|^{-d-\alpha}$, whose integral is finite because $0<\alpha<2$. The integrand is positive off the null set $\{y_1 \in 2\pi\mathbb{Z}\}$, so $c_{d,\alpha}>0$.

One integral. Put $J(\omega) := \int (1-\cos\langle\omega, y\rangle)|y|^{-d-\alpha}dy$. The substitutions $y = Qz$, $Q\in \mathrm{O}(d)$, and $y = z/s$ give $J(Q\omega) = J(\omega)$ and $J(s\omega) = s^\alpha J(\omega)$, so $J(\omega) = c_{d,\alpha}|\omega|^\alpha$.

Clause (1). Substitute $x = H^{1/2}y$, so that $dx = (\det H)^{1/2}dy$ and $\langle x, H^{-1}x\rangle = |y|^2$: the integral is $c_{d,\alpha}^{-1} J(H^{1/2}\omega) = \psi_H(\omega)$. The same substitution bounds the second integral by a multiple of $\int(|y|^2\wedge 1)|y|^{-d-\alpha}dy$.

Clause (3) is the pair of identities

$$\det(AHA^{\mathsf{T}})^{-1/2} = |\det A|^{-1}(\det H)^{-1/2}, \qquad \langle x, (AHA^{\mathsf{T}})^{-1}x\rangle = \langle A^{-1}x, H^{-1}A^{-1}x\rangle.$$

Clause (2). By clause (1), $\nu := n_H(x)\,dx$ satisfies the conditions of C2, and C2(iii) gives an infinitely divisible law with triplet $(0,\nu,0)$. As n_H is even, the imaginary part of the integrand of C2(i) is odd and integrable and integrates to 0, and clause (1) makes the transform $e^{-\psi_H}$. The transform determines the law, and $e^{-\psi_H(A^{\mathsf{T}}\omega)} = e^{-\psi_{AHA^{\mathsf{T}}}(\omega)}$ gives the covariance.

The orbit. $g(u) = u^{\alpha/2}$ is continuous with $g(0)=0$, and $e^{-|\omega|^\alpha}$ is the transform of $S_\alpha(I)$; so the hypothesis of Proposition 4.4 holds, and clause (2) gives $(\Sigma^{1/2})_* S_\alpha(I) = S_\alpha(\Sigma)$. □

Proof of Proposition 4.10. By Lemma 4.6(2), $(e^{-tB})_* S_\alpha(I) = S_\alpha(\Sigma_t)$ with $\Sigma_t := e^{-tB}e^{-tB^{\mathsf{T}}}$. So $S_\alpha(I)$ is B-self-decomposable exactly when $\Sigma_t \preceq_\alpha I$ for every $t>0$, by Proposition 4.7(1). Here $M_t = e^{tB^{\mathsf{T}}}e^{tB}$, whose least eigenvalue is $\min_{|x|=1}|e^{tB}x|^2$ and whose determinant is $e^{2t\,\mathrm{tr}\,B}$, as $\det e^{tB} = e^{t\,\mathrm{tr}\,B}$. So the condition is $|e^{tB}x| \ge e^{t\,\mathrm{tr}\,B/(d+\alpha)}$ for all $t>0$ and all unit x. Write $\lambda := \lambda_{\min}(\mathrm{sym}\,B)$. If (4.4) holds, then $\frac{d}{dt}|e^{tB}x|^2 = 2\langle e^{tB}x, (\mathrm{sym}\,B)e^{tB}x\rangle \ge 2\lambda|e^{tB}x|^2$, so $|e^{tB}x| \ge e^{t\lambda} \ge e^{t\,\mathrm{tr}\,B/(d+\alpha)}$. Conversely, for a unit eigenvector x of $\mathrm{sym}\,B$ with eigenvalue λ, $f(t) := |e^{tB}x|^2$ has $f(0)=1$ and $f'(0)=2\lambda$, and $f(t) \ge e^{2t\,\mathrm{tr}\,B/(d+\alpha)}$ with equality at $t=0$ gives $2\lambda \ge 2\,\mathrm{tr}\,B/(d+\alpha)$.

As $\lambda_{\min}(\mathrm{sym}\,B) = \min_{|x|=1}\langle x, Bx\rangle$, $\mathfrak{k}_\alpha$ is the intersection over unit vectors x of the closed half-spaces $\{B : \mathrm{tr}\,B \le (d+\alpha)\langle x, Bx\rangle\}$, each bounded by a hyperplane through 0; so $\mathfrak{k}_\alpha$ is a closed convex cone, and conjugation by Q changes neither side. For skew B both sides of (4.4) vanish, and for $B=I$ it reads $d+\alpha \ge d$. Finally $\mathrm{tr}\,B \ge d\lambda$, so $B\in\mathfrak{k}_\alpha$ gives $\lambda\ge 0$, and if $\lambda = 0$ then $\mathrm{sym}\,B \ge 0$ has trace at most 0, so $\mathrm{sym}\,B = 0$. □

Proof of Proposition 4.11, clause (2). Clause (2). $\mathrm{tr}\,B \ge d\lambda$, so $\mathrm{tr}\,B < (d+2)\lambda$ forces $\lambda>0$ and $e^{-tB}\to 0$ (for $y := e^{-tB}x$, $\frac{d}{dt}|y|^2 \le -2\lambda|y|^2$, so $|e^{-tB}x| \le e^{-t\lambda}|x|$), and $(d+\alpha)\lambda \ge \mathrm{tr}\,B$ for the stated α, so Proposition 4.10 applies. $S_\alpha(I)$ is rotation invariant by Lemma 4.6(2), and not Gaussian: its marginal along e_1 has the transform $e^{-|t|^\alpha}$, and a Gaussian marginal has one of modulus $e^{-vt^2/2}$; equality at $t=1$ and $t=2$ would give $2^\alpha = 4$, so $\alpha = 2$.

□

Proof of Lemma 4.14 (sketch). (1). $R(1)=1$, monotonicity is that of k, and

$$\frac{k(r/(\lambda\nu))}{k(r)} = \frac{k((r/\nu)/\lambda)}{k(r/\nu)}\cdot\frac{k(r/\nu)}{k(r)} \ge R(\lambda)\,R(\nu).$$

(2). In polar coordinates the integrability of the Lévy density is $\int_0^1 r\,k(r)\,dr < \infty$. Suppose $R(\lambda_0) \ge \lambda_0^2$ for some $\lambda_0 > 1$, and pick r_0 with $k(r_0)>0$. Then $k(r_0\lambda_0^{-n}) \ge \lambda_0^{2n}k(r_0)$ for every $n\ge 0$. For $0<r\le r_0$ choose n with $\lambda_0^n \le r_0/r < \lambda_0^{n+1}$; as k is nonincreasing,

$$k(r) \ge k(r_0\lambda_0^{-n}) \ge \lambda_0^{2n}k(r_0) \ge \frac{r_0^2\,k(r_0)}{\lambda_0^2\,r^2},$$

so $\int_0^{\min(r_0,1)} r\,k(r)\,dr = \infty$, a contradiction.

(3). $\phi(s) := \log R(e^s)$ is nonnegative and superadditive by (1), and below $2s$ by (2). Let $L := \sup_{s>0} \phi(s)/s$. For $s_0 > 0$ and $s \geq s_0$ let n be the integer with $ns_0 \leq s < (n+1)s_0$. Superadditivity and $\phi \geq 0$ give $\phi(s) \geq n\,\phi(s_0)$, so

$$L \geq \frac{\phi(s)}{s} \geq \Big(1 - \frac{s_0}{s}\Big)\frac{\phi(s_0)}{s_0}.$$

As s_0 is arbitrary, $\phi(s)/s \to L$ (Fekete's argument for superadditive functions), so $\beta(k)$ exists and equals L. The displayed bounds are the definitions of infimum and supremum.

(4). If $\alpha \leq \beta_-(k)$ and $\Sigma \preceq_\alpha \Sigma'$, then with $m := \lambda_{\min}(M) \geq 1$, $\det M \leq m^{d+\alpha} \leq m^d R(m^{1/2})^2$, which puts the step in the step set of Proposition 4.13(2). Conversely, for $\lambda > 1$ the step $M := \operatorname{diag}(\lambda^2, \dots, \lambda^2, \lambda^{2(1+\alpha)})$ lies in $\preceq_\alpha$ by condition (c) of Proposition 4.7(1), because $d \geq 2$ leaves at least one entry λ^2, and ordered densities for it give $R(\lambda) \geq \lambda^\alpha$.

(5). The stable and the bounded profiles are Proposition 4.13(3). For the dyadic profile, $k(r) \leq 1/r$ gives the integrability; $r = 1$ in the definition of R gives $R(\lambda) \leq k(1/\lambda) \leq \lambda$, with $R = 1$ on $(1, 2)$ by (1), while a dilation by 2 moves every point one shell inward, so $R(2) = 2$ and the supremum of the quotients is 1. □

Proof of Proposition 4.15. Test functions are the smooth real functions with compact support; for one of them, ϕ, write $\Phi(\xi) := \int e^{-i\langle \xi, x\rangle}\phi(x)\,dx$, a Schwartz function with $(2\pi)^{-d}\int \Phi = \phi(0)$ and $\phi(x) = (2\pi)^{-d}\int \Phi(\xi) e^{i\langle \xi, x\rangle} d\xi$.

The increment's transform. $\widehat{\mu}_\Sigma$ has no zero, so $\widehat{\rho}(\omega) = \exp[-(g(\langle \omega, \Sigma'\omega\rangle) - g(\langle \omega, \Sigma\omega\rangle))]$; $\widehat{\rho}(0) = 1$, so ρ is a probability measure, and $|\widehat{\rho}| \leq 1$.

The blow-up. For $\omega \neq 0$ and $t > 0$ put $u := t^2\langle \omega, \Sigma\omega\rangle$ and $c := \langle \omega, \Sigma'\omega\rangle / \langle \omega, \Sigma\omega\rangle$; as $t \to \infty$, $u \to \infty$ and

$$\widehat{\rho}(t\omega) = e^{-(g(cu) - g(u))} \to c^{-\kappa} = h(\omega) := \Big(\frac{\langle \omega, \Sigma\omega\rangle}{\langle \omega, \Sigma'\omega\rangle}\Big)^\kappa,$$

a function that is even, bounded (by $(\lambda_{\max}(\Sigma)/\lambda_{\min}(\Sigma'))^\kappa$), homogeneous of degree 0 and continuous off 0. By Fubini and Fourier inversion, $\int \phi(x/R)\,\rho(dx) = (2\pi)^{-d}\int \Phi(\xi)\,\widehat{\rho}(\xi/R)\,d\xi$ for $R > 0$, and dominated convergence ($|\Phi\,\widehat{\rho}| \leq |\Phi|$) gives, as $R \downarrow 0$,

$$\int \phi(x/R)\,\rho(dx) \to \Lambda(\phi) := (2\pi)^{-d}\operatorname{Re}\int \Phi(\xi)\,h(\xi)\,d\xi.$$

So Λ is a limit of the positive linear functionals $\phi \mapsto \int \phi(x/R)\,\rho(dx)$: it is linear, $\Lambda(\phi) \geq 0$ for $\phi \geq 0$, hence monotone, and $\Lambda(\phi(\cdot/\lambda)) = \Lambda(\phi)$ for $\lambda > 0$, as $\int \phi(x/(\lambda R))\,\rho(dx) \to \Lambda(\phi)$ too.

Λ *vanishes on test functions vanishing near* 0. Let $\phi \geq 0$ vanish on $\{|x| < r\}$ and on $\{|x| > R\}$, $0 < r \leq R$, with $\phi \leq M$, and take $\lambda > R/r$. The dilates $\phi(\lambda^n x)$, $n \geq 0$, have disjoint supports (a point x with $\phi(\lambda^n x) \neq 0$ has $r \leq \lambda^n|x| \leq R$, and $\lambda^{n+1}|x| \geq \lambda r > R$), so $\sum_{n<N} \phi(\lambda^n x) \leq M\chi(x)$, with χ a smooth bump equal to 1 on $\{|x| \leq R\}$ and $0 \leq \chi \leq 1$. By monotonicity and dilation invariance, $N\Lambda(\phi) \leq M\Lambda(\chi)$ for every N, so $\Lambda(\phi) = 0$. A general ϕ vanishing near 0 is $(\phi + M\chi') - M\chi'$, with $\chi' \geq 0$ a smooth bump equal to 1 on $\{r \leq |x| \leq R\}$ and vanishing near 0, both terms nonnegative and vanishing near 0; so $\Lambda(\phi) = 0$.

Λ *is a multiple of the point mass.* Let χ_0 be a smooth bump with $\chi_0 = 1$ on $\{|x| \leq 1\}$, $0 \leq \chi_0 \leq 1$, and put $c_0 := \Lambda(\chi_0) \geq 0$. For a test function ϕ, $\psi := \phi - \phi(0)\chi_0$ has $\psi(0) = 0$. Given $\eta > 0$, take $\delta \leq 1/2$ with $|\psi| \leq \eta$ on $\{|x| < 2\delta\}$ and a smooth bump b with $0 \leq b \leq 1$, equal to 1 on $\{|x| \leq \delta\}$ and vanishing off $\{|x| < 2\delta\}$. Then $\psi(1-b)$ vanishes near 0, so $\Lambda(\psi(1-b)) = 0$, and $|\psi b| \leq \eta\chi_0$, so $|\Lambda(\psi b)| \leq \eta c_0$ by monotonicity. Hence $|\Lambda(\psi)| \leq \eta c_0$ for every $\eta > 0$, $\Lambda(\psi) = 0$, and $\Lambda(\phi) = c_0\phi(0)$.

h is constant. Fix $a \neq 0$ and, for $N \geq 1$, the test function $\phi_N(x) := \chi_0(x/N)\cos\langle a, x\rangle$, which has $\phi_N(0) = 1$, so $\Lambda(\phi_N) = c_0$. Writing Φ_0 for the transform of χ_0, the transform of ϕ_N

is $\frac{N^d}{2}\big(\Phi_0(N(\xi-a))+\Phi_0(N(\xi+a))\big)$, and the substitution $\xi=\pm a+\eta/N$ gives

$$c_0=\Lambda(\phi_N)=(2\pi)^{-d}\,\mathrm{Re}\Big[\tfrac12\int\Phi_0(\eta)\,h(a+\eta/N)\,d\eta+\tfrac12\int\Phi_0(\eta)\,h(-a+\eta/N)\,d\eta\Big].$$

As $N\to\infty$, dominated convergence (h bounded, continuous at $\pm a$, Φ_0 integrable) gives the limit $\frac12(h(a)+h(-a))(2\pi)^{-d}\int\Phi_0=h(a)\chi_0(0)=h(a)$. So $h(a)=c_0$ for every $a\neq0$.

The step is a ray. $h(\omega)=c_0$ for every $\omega\neq0$ forces $c_0>0$ and $\langle\omega,\Sigma'\omega\rangle=c_0^{-1/\kappa}\langle\omega,\Sigma\omega\rangle$ for every ω; a symmetric matrix is determined by its quadratic form, so $\Sigma'=c_0^{-1/\kappa}\Sigma$. For the Matérn orbit, $g(u)=\gamma\log(1+u)$ and $g(cu)-g(u)=\gamma\log((1+cu)/(1+u))\to\gamma\log c$, so $\kappa=\gamma$. □

Proof of Proposition 4.18 (sketch). P and Q have one sign on $[0,\infty)$ and agree at 0, so $g=\log(P/Q)$ is continuous, vanishes at 0, and is nonnegative because a characteristic function has modulus at most 1: the hypothesis of Proposition 4.4 holds. With $p:=\deg P$ and $q:=\deg Q$, $g(cu)-g(u)\to(p-q)\log c$, which is the hypothesis of Proposition 4.15 with $\kappa=p-q\geq1$. That proposition gives $\Sigma'=c\,\Sigma$, and $\widehat\rho(s\omega)\to c^{-\kappa}\leq1$ as $s\to\infty$ gives $c\geq1$. The examples are products and mixtures of the index-1 Matérn transforms of Proposition 4.4(4), and the coefficient of u^{n-1} in the mixture's Q is positive, and $Q>0$ on $[0,\infty)$ with $Q(0)=\sum_i w_i=1=P(0)$. □

Proof of Proposition 4.19. The datum. Put $k_d:=|S^{d-1}|\,k$. It is nonincreasing, and $\int_{\mathbb{R}^d}(|x|^2\wedge1)\,m(|x|)\,dx=\int_0^1 k_d(r)\,r\,dr+\int_1^\infty k_d(r)\,dr/r$ in polar coordinates, finite by hypothesis; so $(0,k_d)$ is an isotropic origin datum in dimension d (Definition 2.8), whose Lévy measure has the radial density $k_d(r)\big(|S^{d-1}|r^d\big)^{-1}=m(r)$ and whose exponent is therefore ψ_I. Its profile at the origin is $|S^{d-1}|k(0+)$, finite by hypothesis and positive because k is nonincreasing and not identically 0.

The hypotheses on g. ψ_I is rotation invariant, so $e^{-g(|\omega|^2)}=e^{-\psi_I(\omega)}=\widehat\mu_I(\omega)$, the characteristic function of a probability measure; $g(0)=0$ and $g\geq0$, as ψ_I is; and g is continuous on $[0,\infty)$, being $-\log$ of the continuous positive function $u\mapsto\widehat\mu_I(\sqrt u\,e)$. So g satisfies the hypothesis of Proposition 4.4, and the orbit $(\Sigma^{1/2})_*\mu_I$ of that proposition is the orbit here. By Lemma 2.9(2), $g(au)-g(u)\to\frac12|S^{d-1}|k(0+)\log a=\kappa\log a$ as $u\to\infty$, for every $a>0$.

The equivalence. (2)⇒(1) is Proposition 4.13(2), with ρ the infinitely divisible law named there. (3)⇒(2): $c=\lambda^2$ with $\lambda\geq1$, and Proposition 4.13(3) puts every ray $\Sigma\leq\lambda^2\Sigma$ in the relation of its clause (2). (1)⇒(3): Proposition 4.15 gives $\Sigma'=c\,\Sigma$ with $c>0$. Its proof shows that ρ is a probability measure, so $|\widehat\rho|\leq1$; and for $\omega\neq0$ and $s\to\infty$, with $u_s:=s^2\langle\omega,\Sigma\omega\rangle\to\infty$, $\widehat\rho(s\omega)=\widehat\mu_{\Sigma'}(s\omega)/\widehat\mu_\Sigma(s\omega)=e^{-(g(cu_s)-g(u_s))}\to c^{-\kappa}$. So $c^{-\kappa}\leq1$, and $c\geq1$. □

B.3 Smaller groups

Proof of Proposition 4.25 (sketch). (1). By Proposition 4.20(1) and Theorem 3.5(2) every increment and kernel is infinitely divisible, and the rest is the proof of Proposition H.9(1), with $\leq_\alpha$ and Proposition 4.20(1) in place of the Loewner order and Theorem 3.5(3)(a). The last sentence is Proposition F.3(4).

(2). Proposition H.4(1) with $c=1$ and $\Sigma=I$ gives $C(I)=Q\,C(I)\,Q^{\mathsf T}$ for every $Q\in\mathrm{O}(d)$, so $C(I)=c_0I$ by the argument of the proof of Lemma A.4, and $c_0\geq0$. With $Q=I$ and $c=s>0$ it gives $C(sI)=sc_0I$, $L_{sI}=(\sqrt s\,I)_*L_I$ and $\psi_{sI}(\omega)=\psi_I(\sqrt s\,\omega)$, and L_I and ψ_I are rotation invariant. By Proposition 4.20(1), $\varepsilon I\leq_\alpha\Sigma\leq_\alpha tI$, and by (1) both steps have infinitely divisible increments, so Proposition H.4(2) gives the display. The three bounds follow from it.

- *The Gaussian part.* The eigenvalues of $C(\Sigma)$ lie in $[\varepsilon c_0,tc_0]$, so $\operatorname{cond}C(\Sigma)\leq t/\varepsilon$.

- *The moment matrices.* $\mathcal{M}_s$ is monotone in the Lévy measure, $\mathcal{M}_s((aI)_*L) = a^2\mathcal{M}_{s/a}(L)$, and $\mathcal{M}_r(L_I) = d^{-1}T_r(L_I)\,I$ with $T_r(L) := \operatorname{tr}\mathcal{M}_r(L) = \int(|x|^2 \wedge r^2)\,L(dx)$. Hence

$$\varepsilon\, d^{-1}T_{s/\sqrt{\varepsilon}}(L_I)\,I \le \mathcal{M}_s(L_\Sigma) \le t\, d^{-1}T_{s/\sqrt{t}}(L_I)\,I,$$

  and $T_r(L_I)$ is nondecreasing in r and positive when $L_I \neq 0$, so $\operatorname{cond}\mathcal{M}_s(L_\Sigma) \le t/\varepsilon$.

- *The exponent spread.* Put $g(u) := \psi_I(\sqrt{u}\,e_1)$. It is nondecreasing, the ray steps having increments. As $1-\cos 2\tau \le 4(1-\cos\tau)$, integrating at $\tau = \langle\omega, x\rangle$ against $L_I(dx)$ gives $\psi_I(2\omega) \le 4\psi_I(\omega)$, that is $g(4u) \le 4g(u)$, and so $g(cu) \le 4c\,g(u)$ for $c \ge 1$. For $R \in \mathrm{O}(d)$,

$$\psi_\Sigma(R\omega) \le \psi_I(\sqrt{t}\,R\omega) = g(t|\omega|^2) \le 4\frac{t}{\varepsilon}\,g(\varepsilon|\omega|^2) = 4\frac{t}{\varepsilon}\,\psi_I(\sqrt{\varepsilon}\,\omega) \le 4\frac{t}{\varepsilon}\,\psi_\Sigma(\omega).$$

By Proposition 4.20(1), $t/\varepsilon = \operatorname{ecc}(\Sigma)^{(d+\alpha)/\alpha}$. For the Loewner order, $\lambda_{\min}(\Sigma)I \le \Sigma \le \lambda_{\max}(\Sigma)I$, and Proposition H.9(1) supplies the increments, with $t/\varepsilon = \operatorname{ecc}(\Sigma)$.

(3). The kernels have positive transforms, satisfy (a)–(c) of Lemma 3.3(2), and get increments from Proposition 4.20(2) and from $P + \kappa\operatorname{tr}(P)I \ge 0$. At $A = (I + e_1e_1^{\mathsf{T}})^{1/2}$ and $\omega = e_1$ the trace term breaks $\mathrm{GL}(d)$-covariance. A rank-deficient step from I would give, by Lemma 4.8, ordered stable densities, which Proposition 4.7(3) excludes. □

C The complete proof of the general linear row

Section 4.7 states Theorem 4.21 and the idea of its proof. This appendix proves it completely, together with every statement the proof uses whose proof the main text does not give in full, so that every step can be checked here. The proofs are in the order of their dependence.

- §C.1, invariant orders and their cones: Proposition 4.2, Lemma C.1, Theorem 4.3 and Proposition 4.12, which give the chain of cones and its orders.
- §C.2, the ray column: Proposition 3.4, the base of the row.
- §C.3, the step set of an orbit: Lemma 4.8, Proposition 4.13, Proposition 4.20 and clause (1) of Proposition 4.11, which decide which laws give families along each cone.
- §C.4, the theorem.

The other statements it uses are proved completely where they stand: Theorems 3.7 and 3.6 (§3.4), Proposition 4.7 (§4.6), Lemma 3.3, Theorem 3.5 and the three lemmas of the single step, Lemmas 3.8, 3.9 and 3.10 (Appendix A), Lemma 4.6 and Proposition 4.10 (Appendix B), and clause (1) of Proposition E.1, whose short proof is in Appendix E.

C.1 Invariant orders and their cones

This subsection proves that the invariant orders on $\mathrm{PD}(d)$ are read off their step sets, that the orders generated by cone fields are those of the closed invariant cones, and that the chain C_α generates the stable orders and the Loewner order.

Proof of Proposition 4.2. (1). Let $AA^{\mathsf{T}} = \Sigma$ and $Q := \Sigma^{-1/2}A$; then $QQ^{\mathsf{T}} = \Sigma^{-1/2}AA^{\mathsf{T}}\Sigma^{-1/2} = I$, so $Q \in \mathrm{O}(d)$ and $A^{-1}\Sigma' A^{-\mathsf{T}} = Q^{\mathsf{T}}\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}Q$, which lies in $\mathcal{K}$ exactly when the relative step does, $\mathcal{K}$ being a step set. For $B \in \mathrm{GL}(d)$, BA satisfies $(BA)(BA)^{\mathsf{T}} = B\Sigma B^{\mathsf{T}}$ and $(BA)^{-1}B\Sigma' B^{\mathsf{T}}(BA)^{-\mathsf{T}} = A^{-1}\Sigma' A^{-\mathsf{T}}$; with what was just shown, this is the invariance. Let R be a $\mathrm{GL}(d)$-invariant relation and $\mathcal{K} := \{M : I\,R\,M\}$. For $Q \in \mathrm{O}(d)$, $QIQ^{\mathsf{T}} = I$, so $I\,R\,M$ gives $I\,R\,QMQ^{\mathsf{T}}$: $\mathcal{K}$ is a step set. By invariance under $B = \Sigma^{-1/2}$, $\Sigma\,R\,\Sigma'$ if and only if

$I\,R\,\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$, that is, $\Sigma \le_{\mathcal{K}} \Sigma'$. A step set is recovered from its relation as $\{M : I \le_{\mathcal{K}} M\}$, which gives uniqueness.

(2). The relative step of (Σ, Σ) is I. For antisymmetry let M be the relative step of (Σ, Σ'). The matrix $A := \Sigma^{1/2}M^{1/2}$ has $AA^{\mathsf{T}} = \Sigma^{1/2}M\Sigma^{1/2} = \Sigma'$ and $A^{-1}\Sigma A^{-\mathsf{T}} = M^{-1/2}(\Sigma^{-1/2}\Sigma\Sigma^{-1/2})M^{-1/2} = M^{-1}$, so by (1) $\Sigma' \le_{\mathcal{K}} \Sigma$ if and only if $M^{-1} \in \mathcal{K}$. The relative step M is I exactly when $\Sigma = \Sigma'$, and every matrix of $\mathrm{PD}(d)$ is the relative step of a pair, N that of (I, N). So $\le_{\mathcal{K}}$ is antisymmetric exactly when no $M \ne I$ has M and M^{-1} in $\mathcal{K}$. The map $\Sigma \mapsto \Sigma^{1/2}$ is continuous on $\mathrm{PSD}(d)$: if $\Sigma_n \to \Sigma$, the entries of $\Sigma_n^{1/2}$ are bounded, $(\Sigma_n^{1/2})_{ij}^2 \le (\Sigma_n)_{ii}$, and every limit point of $(\Sigma_n^{1/2})$ is positive semidefinite with square Σ, hence is $\Sigma^{1/2}$ by the uniqueness of the positive square root. So $(\Sigma, \Sigma') \mapsto \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ is continuous on $\mathrm{PD}(d) \times \mathrm{PD}(d)$, and a closed $\mathcal{K}$ gives a closed relation; conversely $\mathcal{K}$ is the slice $\{M : (I, M) \in \le_{\mathcal{K}}\}$ of the relation.

(3). Put $A := \Sigma^{1/2}M^{1/2}$, so that $AA^{\mathsf{T}} = \Sigma^{1/2}M\Sigma^{1/2} = \Sigma'$, and, as in (1), $A = \Sigma'^{1/2}Q$ with $Q \in \mathrm{O}(d)$. Then $\Sigma'' = \Sigma'^{1/2}N\Sigma'^{1/2} = AQ^{\mathsf{T}}NQA^{\mathsf{T}} = \Sigma^{1/2}M^{1/2}N'M^{1/2}\Sigma^{1/2}$, so the relative step of (Σ, Σ'') is $M^{1/2}N'M^{1/2} = M^{-1/2}(MN')M^{1/2}$, which has the eigenvalues of MN'. The converse chain has relative steps M and $M^{-1/2}(M^{1/2}NM^{1/2})M^{-1/2} = N$. So transitivity is the stated closure property, $N' \in \mathcal{K}$ ranging over $\mathcal{K}$ as N does. If it holds and $A_1, A_2 \in \mathcal{S}_{\mathcal{K}}$, put $M_1 := A_1A_1^{\mathsf{T}}$ and write $A_1 = M_1^{1/2}Q_1$ with $Q_1 \in \mathrm{O}(d)$. Then $(A_1A_2)(A_1A_2)^{\mathsf{T}} = M_1^{1/2}(Q_1A_2A_2^{\mathsf{T}}Q_1^{\mathsf{T}})M_1^{1/2}$ with $Q_1A_2A_2^{\mathsf{T}}Q_1^{\mathsf{T}}$ in $\mathcal{K}$, so $A_1A_2 \in \mathcal{S}_{\mathcal{K}}$. Conversely, if $\mathcal{S}_{\mathcal{K}}$ is closed under products and $M, N \in \mathcal{K}$, then $M^{1/2}, N^{1/2} \in \mathcal{S}_{\mathcal{K}}$, and so is $M^{1/2}N^{1/2}$, that is, $M^{1/2}NM^{1/2} \in \mathcal{K}$. For the example, $\mathrm{diag}(2,1)\,\mathrm{diag}(4,1)\,\mathrm{diag}(2,1) = \mathrm{diag}(16,1)$ and $\mathrm{diag}(2,1)\,\mathrm{diag}(1,4)\,\mathrm{diag}(2,1) = \mathrm{diag}(4,4)$.

(4). $e^{tQXQ^{\mathsf{T}}} = Qe^{tX}Q^{\mathsf{T}}$ and $e^{t(cX)} = e^{(tc)X}$ give the two invariances. If $X_n \to X$ with $X_n \in C(\mathcal{K})$, then $e^{tX_n} \to e^{tX}$ in $\mathrm{PD}(d)$ for each t, so a closed $\mathcal{K}$ contains e^{tX}. If X and $-X$ lie in $C(\mathcal{K})$, then e^X and its inverse e^{-X} lie in $\mathcal{K}$, so antisymmetry gives $e^X = I$, hence $X = 0$, the exponential being injective on $\mathrm{Sym}(d)$. □

The invariant cones are read through their eigenvalues. The lemma below gives the two matrix inequalities the correspondence needs: a minimum principle for the partial sums of the eigenvalues of a sum, and the majorization of the logarithmic spectrum of a product, which rests on the cited fact C4.

Lemma C.1 (invariant cones, and the spectra of sums and products)**.** *Let $d \ge 1$. For $X \in \mathrm{Sym}(d)$ write $\lambda^{\uparrow}(X) = (\lambda_1^{\uparrow}(X), \dots, \lambda_d^{\uparrow}(X))$ for its eigenvalues in increasing order, and for $M \in \mathrm{PD}(d)$ put $\ell(M) := \lambda^{\uparrow}(\log M) = (\log \lambda_1^{\uparrow}(M), \dots, \log \lambda_d^{\uparrow}(M))$. For $x \in \mathbb{R}^d$ let $x^{\uparrow}$ and $x^{\downarrow}$ be its increasing and decreasing rearrangements, $s_k(x) := x_1^{\uparrow} + \cdots + x_k^{\uparrow}$, and write $x \prec y$ when $s_k(x) \ge s_k(y)$ for $1 \le k < d$ and $s_d(x) = s_d(y)$. A set $D \subseteq \mathbb{R}^d$ is* symmetric *if it is invariant under permutations of the coordinates, and $D^{\#} := \{w \in \mathbb{R}^d : \langle w, x\rangle \ge 0$ for every $x \in D\}$.*

1. (Sums.) $s_k(\lambda^{\uparrow}(X)) = \min\{\mathrm{tr}(U^{\mathsf{T}}XU) : U \in \mathbb{R}^{d\times k},\ U^{\mathsf{T}}U = I\}$. *Consequently $|s_k(\lambda^{\uparrow}(X)) - s_k(\lambda^{\uparrow}(Y))| \le k\|X - Y\|$, with $\|\cdot\|$ the operator norm, and $\lambda^{\uparrow}(X + Y) \prec \lambda^{\uparrow}(X) + \lambda^{\uparrow}(Y)$.*

2. (Products.) *For $M, N \in \mathrm{PD}(d)$, $\ell(M^{1/2}NM^{1/2}) \prec \ell(M) + \ell(N)$.*

3. (Symmetric cones are closed under majorization.) *Let $D \subseteq \mathbb{R}^d$ be a symmetric closed convex cone. Then $x \in D$ if and only if $\sum_i w_i^{\downarrow}x_i^{\uparrow} \ge 0$ for every $w \in D^{\#}$. Consequently $y \in D$ and $x \prec y$ imply $x \in D$.*

4. (Cones in $\mathrm{Sym}(d)$ and in eigenvalue space.) *For a symmetric closed convex cone $D \subseteq \mathbb{R}^d$, $C_D := \{X \in \mathrm{Sym}(d) : \lambda^{\uparrow}(X) \in D\}$ is a closed convex cone invariant under $X \mapsto QXQ^{\mathsf{T}}$, $Q \in \mathrm{O}(d)$. Every closed convex cone $C \subseteq \mathrm{Sym}(d)$ with that invariance is C_D for exactly one such D, namely $D = \{x : \mathrm{diag}(x) \in C\}$, and $C \cap (-C) = \{0\}$ if and only if $D \cap (-D) =$*

$\{0\}$. *If moreover* $C \neq \{0\}$ *and* $C \cap (-C) = \{0\}$, *then exactly one of* I *and* $-I$ *lies in* C, *and* $\operatorname{tr} X$ *has that sign for every* $X \in C \setminus \{0\}$.

5. (Exponentials.) *For* $C = C_D$ *as in (4),* $\exp C := \{e^X : X \in C\}$ *equals* $\{M \in \mathrm{PD}(d) : \ell(M) \in D\}$; *it is a step set (Proposition 4.2), closed in* $\mathrm{PD}(d)$, *and* $M^{1/2} N M^{1/2} \in \exp C$ *whenever* $M, N \in \exp C$.

Clause (5) is the transitivity of every order whose steps are the exponentials of an invariant closed convex cone (Proposition 4.2(3)). Clause (4) reads such a cone through the permutation-symmetric cone of its eigenvalues. Its last sentence says that every invariant closed convex cone $C \neq \{0\}$ with $C \cap (-C) = \{0\}$ contains I or $-I$, the directions of $\mathrm{Sym}(d)$ fixed by $\mathrm{O}(d)$. Clause (1) is Ky Fan's extremal principle with his majorization of the eigenvalues of a sum ([43], 20.A.2, p. 785, and 9.G.1, p. 329). Clause (2) is the log-majorization of the eigenvalues of a product (9.H.1.a, pp. 338–339). The correspondence in (4) follows from Davis's theorem on orthogonally invariant convex functions ([12], Cors. 1–2, p. 277).

Proof of Lemma C.1. (1). Write $X = \sum_j \lambda_j v_j v_j^{\mathsf{T}}$ with $\lambda_j = \lambda_j^{\uparrow}(X)$ and (v_j) an orthonormal basis. For U with $U^{\mathsf{T}}U = I$, $\operatorname{tr}(U^{\mathsf{T}}XU) = \sum_j \lambda_j p_j$ with $p_j := |U^{\mathsf{T}} v_j|^2$; since UU^{T} is an orthogonal projection, $0 \le p_j \le 1$, and $\sum_j p_j = \operatorname{tr}(U^{\mathsf{T}}U) = k$. Then

$$\sum_j \lambda_j p_j - \sum_{j \le k} \lambda_j = \sum_{j \le k} \lambda_j (p_j - 1) + \sum_{j > k} \lambda_j p_j \ge \lambda_k \sum_{j \le k} (p_j - 1) + \lambda_k \sum_{j > k} p_j = 0,$$

using $p_j - 1 \le 0$ and $\lambda_j \le \lambda_k$ for $j \le k$, and $\lambda_j \ge \lambda_k$ for $j > k$; equality holds for $U = (v_1, \dots, v_k)$. Each $X \mapsto \operatorname{tr}(U^{\mathsf{T}}XU)$ is linear and $k\|\cdot\|$-Lipschitz, so their minimum is $k\|\cdot\|$-Lipschitz, and the minimum of a sum is at least the sum of the minima: $s_k(\lambda^{\uparrow}(X+Y)) \ge s_k(\lambda^{\uparrow}(X)) + s_k(\lambda^{\uparrow}(Y))$, with equality at $k = d$, where both sides are traces. The vector $\lambda^{\uparrow}(X) + \lambda^{\uparrow}(Y)$ is increasing, so its s_k is the right-hand side.

(2). Let $P := M^{1/2} N M^{1/2}$. The products $MN = M^{1/2}(M^{1/2}N)$ and $(M^{1/2}N)M^{1/2} = P$ have the same characteristic polynomial, so the eigenvalues of MN, with multiplicity, are those of P. By the inequality for the eigenvalues of a product in its weak form (C4), for $1 \le k \le d$

$$\sum_{i \le k} \log \lambda_i^{\downarrow}(P) \le \sum_{i \le k} \big(\log \lambda_i^{\downarrow}(M) + \log \lambda_i^{\downarrow}(N) \big),$$

with $\lambda^{\downarrow}$ the eigenvalues in decreasing order: the sum of the k largest entries of $\ell(P)$ is at most that of $\ell(M) + \ell(N)$, both vectors being increasing. The totals are equal, $\sum_i \ell_i(P) = \log\det P = \log(\det M \det N) = \sum_i (\ell_i(M) + \ell_i(N))$. Subtracting from the totals, $s_j(\ell(P)) \ge s_j(\ell(M) + \ell(N))$ for $j = d - k$, $1 \le j < d$, with equality at $j = d$, which is the claim.

(3). The dual description. If $x \in D$ and $w \in D^{\#}$, every rearrangement of x lies in D, so $\langle w, \sigma x \rangle \ge 0$ for every permutation σ; the number $\sum_i w_i^{\downarrow} x_i^{\uparrow}$ is one of these, the largest entries of w meeting the smallest of x, and it is the least of them (the rearrangement inequality). Conversely let $x \notin D$. As D is closed and convex, a linear functional f separates x from D strictly: $f(x) < u < f(z)$ for a number u and every $z \in D$ (Hahn–Banach). Write $f(z) = \langle w, z \rangle$. With $z = 0$, $u < 0$; and if $\langle w, z \rangle < 0$ for some $z \in D$, then $cz \in D$ with $c := u / \langle w, z \rangle > 0$ gives $u < \langle w, cz \rangle = u$, which is absurd. So $w \in D^{\#}$ and $\langle w, x \rangle < u < 0$, and by the rearrangement inequality $\sum_i w_i^{\downarrow} x_i^{\uparrow} \le \langle w, x \rangle < 0$. *Majorization.* By Abel summation, for $w \in \mathbb{R}^d$ and $x \in \mathbb{R}^d$,

$$\sum_{i=1}^{d} w_i^{\downarrow} x_i^{\uparrow} = \sum_{k=1}^{d-1} \big(w_k^{\downarrow} - w_{k+1}^{\downarrow} \big) s_k(x) + w_d^{\downarrow}\, s_d(x),$$

with every coefficient $w_k^{\downarrow} - w_{k+1}^{\downarrow} \ge 0$. If $x \prec y$, each term for x is at least the corresponding term for y (the last ones are equal), so $\sum_i w_i^{\downarrow} x_i^{\uparrow} \ge \sum_i w_i^{\downarrow} y_i^{\uparrow} \ge 0$ for $w \in D^{\#}$ when $y \in D$, and $x \in D$ by the dual description.

(4). $\lambda^\uparrow(QXQ^\mathsf{T}) = \lambda^\uparrow(X)$ and $\lambda^\uparrow(cX) = c\lambda^\uparrow(X)$ for $c \ge 0$. Each $\lambda^\uparrow_k = s_k - s_{k-1}$ is continuous by (1), and D is closed, so C_D is closed. If $X, Y \in C_D$, then $\lambda^\uparrow(X) + \lambda^\uparrow(Y) \in D$, D being a convex cone, and (1) with (3) gives $\lambda^\uparrow(X+Y) \in D$. So C_D is a convex cone. Given C, the set $D := \{x : \operatorname{diag}(x) \in C\}$ is a closed convex cone, symmetric because permutation matrices are orthogonal. Writing $X = Q \operatorname{diag}(\lambda^\uparrow(X)) Q^\mathsf{T}$, $X \in C$ if and only if $\lambda^\uparrow(X) \in D$, so $C = C_D$. And D is determined by C_D as $\{x : \operatorname{diag}(x) \in C_D\}$. If $0 \ne x \in D \cap (-D)$ then $\operatorname{diag}(x) \in C \cap (-C)$; if $0 \ne X \in C \cap (-C)$ then $\lambda^\uparrow(X)$ and $\lambda^\uparrow(-X)$, a rearrangement of $-\lambda^\uparrow(X)$, lie in D, so $0 \ne \lambda^\uparrow(X) \in D \cap (-D)$. For the last sentence let $0 \ne X \in C$ and $x := \lambda^\uparrow(X) \ne 0$. For $m = 0, \dots, d-1$ let $\sigma_m x$ be the cyclic shift $(\sigma_m x)_i := x_{i+m}$, indices modulo d. Each $\sigma_m x$ lies in D, and their sum is $(\operatorname{tr} X)\mathbf{1}$, with $\mathbf{1} = (1, \dots, 1)$, since every coordinate of it is $\sum_j x_j$; so $(\operatorname{tr} X)\mathbf{1} \in D$. If $\operatorname{tr} X = 0$ this sum is 0, so $x = \sigma_0 x = -\sum_{m \ne 0} \sigma_m x \in -D$ and $x \in D \cap (-D)$, which is excluded. So $\operatorname{tr} X \ne 0$, and $(\operatorname{sign} \operatorname{tr} X)\mathbf{1} \in D$, that is, $(\operatorname{sign} \operatorname{tr} X)\, I \in C$. As I and $-I$ cannot both lie in C, the sign is the same for every nonzero $X \in C$, which is the claim.

(5). For $X \in \operatorname{Sym}(d)$, e^X has the eigenvalues $e^{\lambda_i(X)}$, and log is the inverse of exp from $\operatorname{Sym}(d)$ onto $\operatorname{PD}(d)$; so $M \in \exp C$ if and only if $\log M \in C$, that is, $\ell(M) \in D$. Orthogonal conjugation commutes with exp, so $\exp C$ is a step set. On $\operatorname{PD}(d)$, $\ell(M) = (\log \lambda^\uparrow_i(M))_i$ is continuous, each $\lambda^\uparrow_i$ being continuous by (1) and positive there, and D is closed, so $\exp C = \{M : \ell(M) \in D\}$ is closed in $\operatorname{PD}(d)$. For $M, N \in \exp C$, $\ell(M) + \ell(N) \in D$, and (2) with (3) gives $\ell(M^{1/2} N M^{1/2}) \in D$. □

Proof of Theorem 4.3. Write $C = C_D$ (Lemma C.1(4)) and, for $w \in D^\#$ and $\Sigma, \Sigma' \in \operatorname{PD}(d)$,

$$F_w(\Sigma, \Sigma') := \sum_{i=1}^{d} w^\downarrow_i \, \ell_i\big(\Sigma^{-1/2} \Sigma' \Sigma^{-1/2}\big) = \sum_{k=1}^{d-1} \big(w^\downarrow_k - w^\downarrow_{k+1}\big) s_k(\ell) + w^\downarrow_d \, s_d(\ell),$$

ℓ the logarithmic spectrum of the relative step, increasing. By Lemma C.1(5) and (3), $\Sigma \le_{\exp C} \Sigma'$ if and only if $F_w(\Sigma, \Sigma') \ge 0$ for every $w \in D^\#$. *Superadditivity.* For any $\Sigma, \Sigma', \Sigma'' \in \operatorname{PD}(d)$, with M, N the relative steps of (Σ, Σ'), (Σ', Σ''), the relative step of (Σ, Σ'') is $M^{1/2} N' M^{1/2}$ with N' orthogonally conjugate to N (the computation of Proposition 4.2(3), which does not use the order); by Lemma C.1(2) its s_k is at least $s_k(\ell(M)) + s_k(\ell(N))$ for $k < d$, with equality at $k = d$. The coefficients $w^\downarrow_k - w^\downarrow_{k+1}$ are nonnegative, so $F_w(\Sigma, \Sigma'') \ge F_w(\Sigma, \Sigma') + F_w(\Sigma', \Sigma'')$.

(1), the geodesic. Let $M \in \exp C$, $X := \log M \in C$ and $\gamma(t) = \Sigma^{1/2} e^{tX} \Sigma^{1/2}$. With $A := \Sigma^{1/2} e^{tX/2}$, $\gamma(t) = AA^\mathsf{T}$ and $\gamma'(t) = \Sigma^{1/2} X e^{tX} \Sigma^{1/2} = AXA^\mathsf{T}$, so $\gamma(t)^{-1/2} \gamma'(t) \gamma(t)^{-1/2} = QXQ^\mathsf{T}$ with $Q := \gamma(t)^{-1/2} A$, and $QQ^\mathsf{T} = \gamma(t)^{-1/2} AA^\mathsf{T} \gamma(t)^{-1/2} = I$; so it lies in C. The entries of γ are, in an eigenbasis of X, finite sums of exponentials e^{tx_k}, so γ is smooth. *(1), causal curves end in the order.* Let γ be C-causal and $w \in D^\#$, and put $g(t) := F_w(\gamma(0), \gamma(t))$, so $g(0) = 0$. On $[0,1]$, $g(t) = \sum_i w^\downarrow_i \log \lambda^\uparrow_i(N(t))$ with $N(t) := \gamma(0)^{-1/2} \gamma(t) \gamma(0)^{-1/2}$. The least eigenvalue of $N(t)$ is continuous in t (Lemma C.1(1)) and positive, so it is at least some $m > 0$ on $[0,1]$, and log is $(1/m)$-Lipschitz on $[m, \infty)$. Each $\lambda^\uparrow_i$ is Lipschitz in the operator norm by Lemma C.1(1), and the operator norm of $N(s) - N(t)$ is at most $\|\gamma(0)^{-1/2}\|^2 \sum_{a,b} |\gamma_{ab}(s) - \gamma_{ab}(t)|$. So $|g(s) - g(t)|$ is bounded by a constant times $\sum_{a,b} |\gamma_{ab}(s) - \gamma_{ab}(t)|$, and g is absolutely continuous because the entries of γ are. Let $t \in (0,1)$ be a point at which the entries of γ are differentiable and $Y := \gamma(t)^{-1/2} \gamma'(t) \gamma(t)^{-1/2} \in C$; almost every t is such. For $h > 0$, $N_h := \gamma(t)^{-1/2} \gamma(t+h) \gamma(t)^{-1/2} = I + hY + o(h)$, and as $\lambda^\uparrow(I + hY) = \mathbf{1} + h\lambda^\uparrow(Y)$, the Lipschitz bound of Lemma C.1(1) gives $x_i(h) := \lambda^\uparrow_i(N_h) = 1 + h\lambda^\uparrow_i(Y) + o(h)$. From $(x-1)/x \le \log x \le x - 1$ for $x > 0$, $\log x_i(h) = h\lambda^\uparrow_i(Y) + o(h)$. Superadditivity gives $g(t+h) - g(t) \ge F_w(\gamma(t), \gamma(t+h))$, hence, where g is differentiable at t,

$$g'(t) \ge \lim_{h \downarrow 0} \frac{1}{h} F_w(\gamma(t), \gamma(t+h)) = \sum_i w^\downarrow_i \lambda^\uparrow_i(Y) \ge 0,$$

by Lemma C.1(3), as $\lambda^\uparrow(Y) \in D$. An absolutely continuous function is the integral of its derivative, which exists almost everywhere; so $g(1) = \int_0^1 g' \ge 0$, and as w was arbitrary,

$\gamma(0) \leq_{\exp C} \gamma(1)$. *(1), the rest.* $\exp C$ is a closed step set containing I (Lemma C.1(5)), so $\leq_{\exp C}$ is closed, reflexive and, by Lemma C.1(5) and Proposition 4.2(3), transitive; M and M^{-1} lie in $\exp C$ exactly when $\log M \in C \cap (-C)$, which with Proposition 4.2(2) is the criterion for antisymmetry; and $e^{tX} \in \exp C$ if and only if $tX \in C$, log being injective, so $C(\exp C) = C$.

(2). Let $\mathcal{L}$ be the set of B named. $\mathcal{S}_\mathcal{K}$ is closed in $\mathrm{GL}(d)$, as the preimage of the closed set $\mathcal{K}$ under $A \mapsto AA^\mathsf{T}$; it contains $\mathrm{O}(d)$, as $I \in \mathcal{K}$; and it is closed under products by Proposition 4.2(3). $\mathcal{L}$ is closed under positive multiples and, $\mathcal{S}_\mathcal{K}$ being closed, under limits. It is closed under sums by the Lie product formula $(e^{B/n}e^{B'/n})^n \to e^{B+B'}$. Put $P_n := e^{B/n}e^{B'/n}$, $Q_n := e^{(B+B')/n}$ and $a := \|B\| + \|B'\|$. The curve $\varphi(s) := e^{sB}e^{sB'} - e^{s(B+B')}$ has $\varphi(0) = 0$ and $\varphi'(0) = B + B' - (B + B') = 0$, so $n\|P_n - Q_n\| = n\|\varphi(1/n)\| \to 0$; and $\|P_n\|, \|Q_n\| \leq e^{a/n}$, as $\|e^Z\| \leq e^{\|Z\|}$. From $P^n - Q^n = P(P^{n-1} - Q^{n-1}) + (P - Q)Q^{n-1}$, by induction,

$$\|P_n^n - Q_n^n\| \leq n\,(e^{a/n})^n\,\|P_n - Q_n\| = e^a\,n\|P_n - Q_n\| \to 0,$$

and $Q_n^n = e^{B+B'}$. Applied to tB and tB', the products $(e^{tB/n}e^{tB'/n})^n$ lie in $\mathcal{S}_\mathcal{K}$ and tend to $e^{t(B+B')}$. Three cases then describe $\mathcal{L}$.

- For symmetric X, $e^{tX}(e^{tX})^\mathsf{T} = e^{2tX}$, so $X \in \mathcal{L}$ exactly when $X \in C(\mathcal{K})$.
- For skew K, $e^{tK} \in \mathrm{O}(d) \subseteq \mathcal{S}_\mathcal{K}$, so $\pm K \in \mathcal{L}$.
- Adding the skew part of B or its negative, $B \in \mathcal{L}$ if and only if $\frac{1}{2}(B + B^\mathsf{T}) \in \mathcal{L}$.

So $\mathcal{L}$ is as stated, and $C(\mathcal{K}) = \mathcal{L} \cap \mathrm{Sym}(d)$ is a convex cone; it is closed, invariant and pointed by Proposition 4.2(4). By (1), $\preceq_{C(\mathcal{K})}$ is $\leq_{\exp C(\mathcal{K})}$, and $\exp C(\mathcal{K}) \subseteq \mathcal{K}$ by definition. (c)⇒(b): let $M \in \mathcal{K}$ and $X := \log M$. Then $e^{tX} = M^t$ lies in $\mathcal{K}$ for $t \in [0, 1]$ by (c). For $t = j + \tau$ with $j \geq 1$ an integer and $0 \leq \tau < 1$, induction on j gives it: $M^{j+\tau} = M^{1/2}M^{j-1+\tau}M^{1/2}$ is the composite of the steps M and $M^{j-1+\tau}$ of $\mathcal{K}$, which lies in $\mathcal{K}$ by transitivity (Proposition 4.2(3)). So $X \in C(\mathcal{K})$ and $\mathcal{K} \subseteq \exp C(\mathcal{K})$. (b)⇒(a) is (1) with $C' = C(\mathcal{K})$. (a)⇒(c): by (1), $\mathcal{K} = \exp C'$, closed under $M \mapsto M^\tau$ as C' is a cone, and $C' = C(\exp C') = C(\mathcal{K})$.

(3). For M, N with least eigenvalues at least 2 and a unit x, $\langle x, M^{1/2}NM^{1/2}x\rangle \geq 2|M^{1/2}x|^2 \geq 4$, so composites stay in $\mathcal{K}$, and composites with I trivially do: $\leq_\mathcal{K}$ is transitive (Proposition 4.2(3)), and reflexive. $\mathcal{K}$ is a step set, and closed since $\lambda_1^\uparrow$ is continuous. If $M \neq I$ and $M \in \mathcal{K}$, then $\lambda_1^\uparrow(M^{-1}) = 1/\lambda_d^\uparrow(M) \leq 1/2$, so $M^{-1} \notin \mathcal{K}$: antisymmetry. For $X \neq 0$ and small $t > 0$, $e^{tX} \neq I$ and its least eigenvalue $e^{t\lambda_1^\uparrow(X)}$ is below 2, so $X \notin C(\mathcal{K})$. A $\{0\}$-causal curve has derivative 0 almost everywhere, and an absolutely continuous curve with derivative 0 almost everywhere is constant, so $\preceq_{\{0\}}$ is equality. If $\leq_\mathcal{K}$ were $\preceq_{C'}$, then for $X \in C'$ and $t \geq 0$ the geodesic of (1) from I to e^{tX} would be C'-causal, so $I \leq_\mathcal{K} e^{tX}$, that is $e^{tX} \in \mathcal{K}$: $C' \subseteq C(\mathcal{K}) = \{0\}$. Then $\leq_\mathcal{K}$ would be equality, while $I \leq_\mathcal{K} 2I$.

(4). By Lemma C.1(4) every nonzero $X \in C$ has $\operatorname{tr} X > 0$. On $\mathrm{PD}(d)$ the second case is contained in the third, since $\Sigma' = c\Sigma$ has relative step $cI = e^{(\log c)I}$ with $(\log c)I \in C$. If Σ is definite, so is every Σ' above it. If $\Sigma \neq 0$ is singular, the Σ' above it are its multiples $c\Sigma$, $c \geq 1$. The claims follow.

- Reflexivity is $c = 1$.
- Transitivity: from 0 trivially, from a nonzero singular matrix along its multiples, and on $\mathrm{PD}(d)$ by that of $\preceq_C$.
- Antisymmetry: on $\mathrm{PD}(d)$ it is (1). $\Sigma' \preceq_C^+ 0$ forces $\Sigma' = 0$. And $c\Sigma = \Sigma$ with $\Sigma = c'(c\Sigma)$, $c, c' \geq 1$, forces $c = 1$.
- Invariance: $A \cdot 0 = 0$, $A \cdot (c\Sigma) = c\,A \cdot \Sigma$, and $\preceq_C$ is invariant by (1). G-invariance follows from $\mathrm{GL}(d)$-invariance.

- The rays: $c\Sigma \preceq_C^+ c'\Sigma$ for $0 \le c \le c'$, by the first case if $c = 0$ and the second otherwise.
- Chains: $s\Sigma'$ from 0 and $(1 + s(c-1))\Sigma$ along a ray are chains by Theorem 3.5(3)(b). On $\mathrm{PD}(d)$ the geodesic γ of (1) is one, by Theorem 3.5(3)(c) applied to $\mathcal{K} = \exp C$, a step set whose order is a partial order by (1) and which is closed under $M \mapsto M^\tau$, $0 \le \tau \le 1$, as C is a cone; and $\le_{\exp C}$ is $\preceq_C$ by (1). So (P) holds.
- The Loewner order: if $C \subseteq \mathrm{PSD}(d)$, then $e^X \ge I$ for $X \in C$, so every pair of $\preceq_C^+$ is a Loewner pair. If $X \in C$ is not positive semidefinite, $I \preceq_C^+ e^X$ while $e^X - I$ is not positive semidefinite.
- The ray order: for $C = \{cI : c \ge 0\}$, $\exp C = \{cI : c \ge 1\}$, so on $\mathrm{PD}(d)$ the third case is the second, and $\preceq_C^+$ is the ray order.

□

Proof of Proposition 4.12. The cones. The conditions defining C_α are invariant under orthogonal conjugation and positive scaling; $\lambda_{\min}$ is concave and continuous and tr is linear, so C_α is a closed convex cone; I satisfies $d + \alpha \ge d$; and $C_\alpha \subseteq \mathrm{PSD}(d)$ meets $-\mathrm{PSD}(d)$ only in 0.

Three facts about curves. (a) Let $\eta : [0,1] \to \mathrm{Sym}(d)$ have absolutely continuous entries. If $\eta'(t)$ is positive semidefinite for almost every t, then for each x the function $\langle x, \eta(t)x\rangle$ is absolutely continuous with derivative $\langle x, \eta'(t)x\rangle \ge 0$ almost everywhere, so it is the integral of its derivative and $\eta(0) \le \eta(1)$. If $\eta' = 0$ almost everywhere, each entry is constant, and $\eta(0) = \eta(1)$. (b) For $s \in \mathbb{R}$ and a C-causal γ (C any cone), the entries of $(\det\gamma)^s\gamma$ are absolutely continuous on $[0,1]$: $\det\gamma$ is a polynomial in the entries of γ, positive and continuous, hence between two positive numbers on $[0,1]$, where $x \mapsto x^s$ is Lipschitz. At almost every t the entries of γ are differentiable; there, writing $\gamma(u) = \gamma(t)\cdot\gamma(t)^{-1}\gamma(u)$ and expanding the determinant of $\gamma(t)^{-1}\gamma(u)$, which is I at $u = t$, gives Jacobi's formula $(\det\gamma)'(t) = \det\gamma(t)\ \mathrm{tr}(\gamma(t)^{-1}\gamma'(t))$. With $Y := \gamma^{-1/2}\gamma'\gamma^{-1/2}$, so that $\gamma' = \gamma^{1/2}Y\gamma^{1/2}$ and $\mathrm{tr}(\gamma^{-1}\gamma') = \mathrm{tr}\,Y$,

$$\big((\det\gamma)^s\gamma\big)' = (\det\gamma)^s\big(\gamma' + s(\mathrm{tr}\,Y)\gamma\big) = (\det\gamma)^s\,\gamma^{1/2}\big(Y + s(\mathrm{tr}\,Y)I\big)\gamma^{1/2}.$$

(c) If C is invariant under orthogonal conjugation, $X \in C$ and $M = e^X$ is the relative step of (Σ, Σ'), then $\gamma(t) := \Sigma^{1/2}M^t\Sigma^{1/2} = \Sigma^{1/2}e^{tX}\Sigma^{1/2}$ is a C-causal curve from Σ to Σ': its entries are smooth, and with $A := \Sigma^{1/2}e^{tX/2}$, $\gamma(t) = AA^\mathsf{T}$ and $\gamma'(t) = AXA^\mathsf{T}$, so $\gamma(t)^{-1/2}\gamma'(t)\gamma(t)^{-1/2} = QXQ^\mathsf{T}$ with $Q := \gamma(t)^{-1/2}A$ orthogonal.

(1). $X \in C_0$ means $X \ge 0$ and $d\lambda_{\min}(X) \ge \mathrm{tr}\,X$, that is, every eigenvalue equals the least one and is nonnegative: $X = cI$, $c \ge 0$. Let γ be C_0-causal from Σ to Σ'. Almost everywhere $Y = cI$ for some $c \ge 0$, so $Y + s(\mathrm{tr}\,Y)I = 0$ for $s = -1/d$, and by (b) and (a) $(\det\gamma)^{-1/d}\gamma$ is constant: $\Sigma' = c\,\Sigma$ with $c = (\det\Sigma'/\det\Sigma)^{1/d}$. As $C_0 \subseteq \mathrm{PSD}(d)$, $\gamma' = \gamma^{1/2}Y\gamma^{1/2}$ is positive semidefinite almost everywhere, so $\Sigma \le \Sigma'$ by (a), and $(c-1)\Sigma \ge 0$ forces $c \ge 1$. Conversely the relative step of $(\Sigma, c\Sigma)$ is $cI = e^{(\log c)I}$ with $(\log c)I \in C_0$ for $c \ge 1$, and (c) gives the curve. For $0 < \alpha < \infty$, $\mathrm{tr}\,X \ge d\lambda_{\min}(X)$, so $(d+\alpha)\lambda_{\min}(X) \ge \mathrm{tr}\,X$ gives $\alpha\lambda_{\min}(X) \ge 0$, hence $X \ge 0$. With $M = e^X$, $\lambda_{\min}(M) = e^{\lambda_{\min}(X)}$ and $\det M = e^{\mathrm{tr}\,X}$, so $M \in \exp C_\alpha$ if and only if $\lambda_{\min}(M)^{d+\alpha} \ge \det M$, which is condition (c) of Proposition 4.7(1) when $0 < \alpha < 2$. Let γ be C_α-causal and $s := -1/(d+\alpha)$. Almost everywhere $Y \in C_\alpha$, so $Y + s(\mathrm{tr}\,Y)I = Y - \frac{\mathrm{tr}\,Y}{d+\alpha}I \ge 0$, and by (b) and (a) $\Phi_\alpha(\Sigma) \le \Phi_\alpha(\Sigma')$. Dividing by $(\det\Sigma)^s > 0$ and congruence by $\Sigma^{-1/2}$, $\Phi_\alpha(\Sigma) \le \Phi_\alpha(\Sigma')$ is $I \le (\det M)^s M$, that is $\lambda_{\min}(M)(\det M)^s \ge 1$, which is $\lambda_{\min}(M)^{d+\alpha} \ge \det M$. Conversely, if that holds, $M \in \exp C_\alpha$ and (c) gives the curve. Finally $e^X \ge I$ if and only if $X \ge 0$, so $\exp C_\infty = \{M : M \ge I\}$, and $M \ge I$ is $\Sigma \le \Sigma'$, congruence preserving positive semidefiniteness. A $\mathrm{PSD}(d)$-causal curve has $\gamma' = \gamma^{1/2}Y\gamma^{1/2} \ge 0$ almost everywhere, so $\Sigma \le \Sigma'$ by (a); conversely $\Sigma \le \Sigma'$ puts M in $\exp C_\infty$, and (c) gives the curve.

(2). Let $s := -1/(d+\alpha)$ and, for a matrix B, $B' := B + s(\mathrm{tr}\,B)I$. As $|\det e^{tB}|^s = e^{st\,\mathrm{tr}\,B}$ and I commutes with B, $\psi_\alpha(e^{tB}) = e^{tB'}$. By (1) and (3), $e^{tB}e^{tB^\mathsf{T}} \in \exp C_\alpha$ if and only if

$I \preceq_{C_\alpha} e^{tB}(e^{tB})^{\mathsf T}$, if and only if $I \le \Phi_\alpha(e^{tB}(e^{tB})^{\mathsf T}) = \psi_\alpha(e^{tB})\psi_\alpha(e^{tB})^{\mathsf T} = e^{tB'}e^{tB'^{\mathsf T}}$. For $x \in \mathbb{R}^d$ put $f(t) := |e^{tB'^{\mathsf T}}x|^2 = \langle x, e^{tB'}e^{tB'^{\mathsf T}}x\rangle$, whose derivative is $\langle y, (B'+B'^{\mathsf T})y\rangle$ with $y = e^{tB'^{\mathsf T}}x$. If $\lambda_{\min}(\frac12(B'+B'^{\mathsf T})) \ge 0$ then $f(t) \ge f(0) = |x|^2$ for $t \ge 0$; conversely $f(t) \ge f(0)$ for $t \ge 0$ gives $f'(0) = \langle x, (B'+B'^{\mathsf T})x\rangle \ge 0$. So $e^{tB}e^{tB^{\mathsf T}} \in \exp C_\alpha$ for every $t \ge 0$ if and only if $\lambda_{\min}(\frac12(B+B^{\mathsf T}))+s\operatorname{tr}B \ge 0$, that is $(d+\alpha)\lambda_{\min}(\frac12(B+B^{\mathsf T})) \ge \operatorname{tr}B$, which is (4.4). A symmetric X has $\frac12(X+X^{\mathsf T}) = X$, and the condition $X \ge 0$ is implied for $\alpha > 0$, so $\mathfrak{k}_\alpha \cap \operatorname{Sym}(d) = C_\alpha$.

(3). Put $s := -1/(d+\alpha)$, so $1+ds = \alpha/(d+\alpha) > 0$. For $T := \Phi_\alpha(\Sigma)$, $\det T = (\det\Sigma)^{1+ds}$, so $(\det T)^{1/\alpha} = (\det\Sigma)^{1/(d+\alpha)}$ and $(\det T)^{1/\alpha}T = \Sigma$; in the same way $\Phi_\alpha((\det T)^{1/\alpha}T) = T$. Both maps are smooth. By (1), $\Sigma \preceq_{C_\alpha} \Sigma'$ if and only if $\Phi_\alpha(\Sigma) \le \Phi_\alpha(\Sigma')$. By Jacobi's formula of (b) applied to $\varepsilon \mapsto \Sigma + \varepsilon V$, the derivative of Φ_α at Σ in the direction V is

$$D\Phi_\alpha(\Sigma)V = (\det\Sigma)^s\big(V + s\operatorname{tr}(\Sigma^{-1}V)\,\Sigma\big), \qquad D\Phi_\alpha(\Sigma)\big(\Sigma^{1/2}Y\Sigma^{1/2}\big) = (\det\Sigma)^s\,\Sigma^{1/2}\big(Y + s(\operatorname{tr}Y)I\big)\Sigma^{1/2}.$$

The second is positive semidefinite when $\lambda_{\min}(Y) \ge (\operatorname{tr}Y)/(d+\alpha)$, in particular for $Y \in C_\alpha$. Conversely, for $P \ge 0$ put $Z := (\det\Sigma)^{-s}\Sigma^{-1/2}P\Sigma^{-1/2} \ge 0$ and $Y := Z - \frac{s}{1+ds}(\operatorname{tr}Z)I$; then $\operatorname{tr}Y = \operatorname{tr}Z/(1+ds)$ and $Y + s(\operatorname{tr}Y)I = Z \ge 0$, so $Y \in C_\alpha$, and $D\Phi_\alpha(\Sigma)(\Sigma^{1/2}Y\Sigma^{1/2}) = P$: every positive semidefinite direction is attained. For $A \in \mathrm{GL}(d)$, $\det(A\Sigma A^{\mathsf T}) = (\det A)^2\det\Sigma$ gives the covariance. Finally $\psi_\alpha(AB) = \psi_\alpha(A)\psi_\alpha(B)$, $|\det A|^s$ being a character and scalars central, and with $s' := -s/(1+ds)$ the map $A \mapsto |\det A|^{s'}A$ is its inverse, since $|\det(|\det A|^{s'}A)|^s|\det A|^{s'} = |\det A|^{s'(1+ds)+s} = 1$, and symmetrically for the other composition. As $\Phi_\alpha(I) = I$, $I \preceq_{C_\alpha} AA^{\mathsf T}$ if and only if $I \le \Phi_\alpha(AA^{\mathsf T}) = \psi_\alpha(A)\psi_\alpha(A)^{\mathsf T}$, which is the last sentence. □

C.2 The ray column

The base of the general linear row is the ray order. Its families are the orbits of the rotation-invariant self-decomposable laws, and the proof below shows it from the axioms of the scheme.

Proof of Proposition 3.4. (1), images. If $\widehat\mu(\omega) = \widehat\mu(c\omega)\,\widehat\rho_c(\omega)$, then at $T^{\mathsf T}\omega$, $\widehat{T_*\mu}(\omega) = \widehat\mu(T^{\mathsf T}\omega) = \widehat{T_*\mu}(c\omega)\,\widehat{T_*\rho_c}(\omega)$, which is (2.2) for $T_*\mu$ with the residual $T_*\rho_c$.

(1), no zeros. First an inequality: for a probability measure ρ on $\mathbb{R}^d$ and $\omega \in \mathbb{R}^d$,

$$1 - |\widehat\rho(2\omega)|^2 \le 4\big(1 - |\widehat\rho(\omega)|^2\big).$$

Indeed $|\widehat\rho(\omega)|^2 = \widehat\rho(\omega)\widehat\rho(-\omega)$ is the transform of $\kappa := \rho * (-I)_*\rho$, a symmetric probability measure, so $1 - |\widehat\rho(\omega)|^2 = \int(1-\cos\langle\omega,x\rangle)\,\kappa(dx)$. As $1-\cos 2\tau = 2(1-\cos\tau)(1+\cos\tau) \le 4(1-\cos\tau)$, integrating at $\tau = \langle\omega,x\rangle$ against κ gives the inequality. Now suppose $\widehat\mu(\omega_1) = 0$ for some ω_1, and let $t_* := \inf\{t > 0 : \widehat\mu(t\omega_1) = 0\}$. As $\widehat\mu$ is continuous with $\widehat\mu(0) = 1$, $t_* > 0$, $\omega_0 := t_*\omega_1$ has $\widehat\mu(\omega_0) = 0$, and $\widehat\mu(t\omega_0) \ne 0$ for $0 \le t < 1$. For $c \in (0,1)$, (2.2) at ω_0 reads $0 = \widehat\mu(c\omega_0)\,\widehat\rho_c(\omega_0)$ with $\widehat\mu(c\omega_0) \ne 0$, so $\widehat\rho_c(\omega_0) = 0$; at $\omega_0/2$ it gives $\widehat\rho_c(\omega_0/2) = \widehat\mu(\omega_0/2)/\widehat\mu(c\omega_0/2)$, which tends to 1 as $c \uparrow 1$ by continuity. The inequality at $\omega = \omega_0/2$ gives $1 \le 4(1 - |\widehat\rho_c(\omega_0/2)|^2) \to 0$, a contradiction.

(2), necessity. By Lemma 3.3(2) the kernels from scale zero satisfy (a)–(c). Let $\Sigma \in \mathrm{PSD}(d)$ and $\lambda > 1$. As $\Sigma \le_W \lambda^2\Sigma$, the cascade gives $\mu_{\lambda^2\Sigma} = \mu_\Sigma * \rho$ with $\rho := \mu_{\Sigma,\lambda^2\Sigma}$, and (b) with $A = \lambda I \in G$ gives $\mu_{\lambda^2\Sigma} = (\lambda I)_*\mu_\Sigma$. Applying $(\lambda^{-1}I)_*$, $\mu_\Sigma = (\lambda^{-1}I)_*\mu_\Sigma * (\lambda^{-1}I)_*\rho$, which is (2.2) with $c = \lambda^{-1}$; every $c \in (0,1)$ arises. The argument reads only that W contains the rays and G the dilations, so the kernels from scale zero of a cascade family are self-decomposable for every wedge datum, not only for the ray order.

(2), sufficiency. By (1) no $\widehat\mu_\Sigma$ has a zero, so by Lemma 3.3(3) it suffices to give each pair $\Sigma \le_W \Sigma'$ a probability measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$. For $\Sigma = 0$ take $\rho := \mu_{\Sigma'}$, using $\mu_0 = \delta_0$. For $\Sigma' = c\,\Sigma$ with $c = 1$ take δ_0; with $c > 1$, the self-decomposability of μ_Σ at $c^{-1/2}$ gives $\mu_\Sigma = (c^{-1/2}I)_*\mu_\Sigma * \rho'$, and applying $(c^{1/2}I)_*$ and (b) with $A = c^{1/2}I$, $\mu_{c\Sigma} = (c^{1/2}I)_*\mu_\Sigma = \mu_\Sigma * (c^{1/2}I)_*\rho'$. Uniqueness and the formula for the increments are Lemma 3.3(2)–(3).

(3). First, the transform of a rotation-invariant law ν is radial: if $|\omega| = |\xi|$, the reflection Q in the hyperplane $(\omega - \xi)^\perp$ is orthogonal and symmetric and carries ω to ξ, so $\widehat{\nu}(\xi) = \widehat{\nu}(Q^{\mathsf{T}}\omega) = \widehat{Q_*\nu}(\omega) = \widehat{\nu}(\omega)$. With $\vartheta(u) := \widehat{\nu}(\sqrt{u}\, e_1)$, continuous on $[0,\infty)$, this is $\widehat{\nu}(\omega) = \vartheta(|\omega|^2)$; and for $\Sigma \in \mathrm{PSD}(d)$, $\Sigma^{1/2}$ being symmetric, $\widehat{(\Sigma^{1/2})_*\nu}(\omega) = \widehat{\nu}(\Sigma^{1/2}\omega) = \vartheta(|\Sigma^{1/2}\omega|^2) = \vartheta(\langle\omega, \Sigma\omega\rangle)$.

Now let the family be given; $(\mathrm{GL}(d), W)$ is a wedge datum, the ray order being invariant under every subgroup of $\mathrm{GL}(d)$ (Definition 3.1). By (2), μ_I is self-decomposable, by (a) it is not δ_0, and by (b) with $A = Q \in \mathrm{O}(d)$ at $\Sigma = I$, where $Q \cdot I = I$, it is rotation invariant. Let ϑ be its function as above. On $\mathrm{PD}(d)$, $\Sigma^{1/2} \in \mathrm{GL}(d)$ and $\Sigma^{1/2} \cdot I = \Sigma$, so (b) with $A = \Sigma^{1/2}$ gives $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ and $\widehat{\mu}_\Sigma(\omega) = \vartheta(\langle\omega, \Sigma\omega\rangle)$; this is the orbit form $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ of Proposition 4.5, obtained here from (b) alone. For $\Sigma \in \mathrm{PSD}(d)$, $\Sigma + \varepsilon I \in \mathrm{PD}(d)$ tends to Σ as $\varepsilon \downarrow 0$, and (c) gives $\widehat{\mu}_\Sigma(\omega) = \lim_{\varepsilon\downarrow 0} \vartheta(\langle\omega, (\Sigma + \varepsilon I)\omega\rangle) = \vartheta(\langle\omega, \Sigma\omega\rangle)$, the transform of $(\Sigma^{1/2})_*\mu_I$; so $\mu_\Sigma = (\Sigma^{1/2})_*\mu_I$ there too.

Conversely, let μ_I be rotation invariant, self-decomposable and not δ_0, with function ϑ, and put $\mu_\Sigma := (\Sigma^{1/2})_*\mu_I$, so that $\widehat{\mu}_\Sigma(\omega) = \vartheta(\langle\omega, \Sigma\omega\rangle)$. Then μ_I is its own member at $\Sigma = I$, as $\vartheta(|\omega|^2) = \widehat{\mu}_I(\omega)$, and the conditions of (2) hold.

- (a), as $\vartheta(0) = 1$.
- (b) for every $A \in \mathrm{GL}(d)$, as $\widehat{A_*\mu_\Sigma}(\omega) = \vartheta(\langle A^{\mathsf{T}}\omega, \Sigma A^{\mathsf{T}}\omega\rangle) = \vartheta(\langle\omega, A\Sigma A^{\mathsf{T}}\omega\rangle) = \widehat{\mu}_{A\Sigma A^{\mathsf{T}}}(\omega)$.
- (c), as ϑ and $\Sigma \mapsto \langle\omega, \Sigma\omega\rangle$ are continuous.
- Each μ_Σ is self-decomposable by (1), as an image of μ_I.

By (2) these are the kernels from scale zero of a family, whose isotropic member is μ_I. Two families with the isotropic member μ_I have the kernels from scale zero $(\Sigma^{1/2})_*\mu_I$, by the first half, and so agree on the pairs by the uniqueness in (2). □

C.3 The step set of an orbit

The four proofs of this subsection decide which steps across shapes an orbit survives: the stable tail gives necessity for the stable orbits, the ordered densities give sufficiency for every profile, the stable order on $\mathrm{PSD}(d)$ carries the general linear cell, and the anisotropy budget excludes every non-Gaussian law beyond C_2.

Proof of Lemma 4.8. Normalisation. Write $E(\omega)$ for the bracket in (4.2) with its sign reversed, so $\widehat{\rho} = e^{-E}$, and E is real and even. Then $\rho(\mathbb{R}^d) = \widehat{\rho}(0) = 1$, and $|\widehat{\rho}| \le \rho(\mathbb{R}^d) = 1$ gives $E \ge 0$. Put $D := \psi_{H'} - \psi_H$, homogeneous of degree α, so that $E(\omega/R) = R^{-2}\langle\omega, C\omega\rangle + R^{-\alpha}D(\omega)$ for $R > 0$.

Fourier side. Let ϕ be smooth with compact support in $\mathbb{R}^d \setminus \{0\}$, real, and $\widehat{\phi}(\omega) := \int e^{i\langle\omega,x\rangle}\phi(x)\,dx$, a Schwartz function, so that $\phi(x) = (2\pi)^{-d}\int \widehat{\phi}(\omega)e^{-i\langle\omega,x\rangle}d\omega$. Put $\phi_R(x) := \phi(x/R)$, with $\widehat{\phi_R}(\omega) = R^d\widehat{\phi}(R\omega)$. Integrating the inversion formula against ρ (Fubini, $\widehat{\phi}$ integrable and ρ finite) and using that $\widehat{\rho}$ is even,

$$\int \phi_R\, d\rho = (2\pi)^{-d}\int \widehat{\phi_R}(\omega)\,\widehat{\rho}(\omega)\,d\omega = (2\pi)^{-d}\int \widehat{\phi}(\zeta)\,e^{-E(\zeta/R)}\,d\zeta.$$

Since $\int \widehat{\phi} = (2\pi)^d\phi(0) = 0$, the factor $e^{-E(\zeta/R)}$ may be replaced by $e^{-E(\zeta/R)} - 1$.

The limit. Let $R \ge 1$ and $s := E(\zeta/R) \ge 0$. Since $0 \le 1 - e^{-s} \le s$ and $R^\alpha E(\zeta/R) = R^{\alpha-2}\langle\zeta, C\zeta\rangle + D(\zeta)$ with $R^{\alpha-2} \le 1$,

$$R^\alpha\big|e^{-E(\zeta/R)} - 1\big| \le R^\alpha E(\zeta/R) \le F(\zeta) := \|C\|\,|\zeta|^2 + |D(\zeta)|,$$

$\|C\|$ the operator norm. As $|D(\zeta)| \le (\|H'\|^{\alpha/2} + \|H\|^{\alpha/2})|\zeta|^\alpha \le (\|H'\|^{\alpha/2} + \|H\|^{\alpha/2})(1 + |\zeta|^2)$ and $\widehat{\phi}$ is a Schwartz function, $|\widehat{\phi}|F$ is integrable, and it dominates $R^\alpha\widehat{\phi}(\zeta)(e^{-E(\zeta/R)} - 1)$ for $R \ge 1$.

Pointwise, since $\alpha < 2$, $R^\alpha E(\zeta/R) \to D(\zeta)$ and $E(\zeta/R) \to 0$, and $|e^{-s} - 1 + s| \le s^2$ for $|s| \le 1$ gives $R^\alpha \bigl|e^{-E(\zeta/R)} - 1 + E(\zeta/R)\bigr| \le R^\alpha E(\zeta/R) \cdot E(\zeta/R) \to 0$; so $R^\alpha(e^{-E(\zeta/R)} - 1) \to -D(\zeta)$, and dominated convergence gives

$$\lim_{R\to\infty} R^\alpha \int \phi_R \, d\rho = -(2\pi)^{-d} \int \widehat{\phi}(\zeta)\, D(\zeta)\, d\zeta.$$

Back to x. Put $f := n_{H'} - n_H$, even and continuous on $\mathbb{R}^d \setminus \{0\}$. By Lemma 4.6(1), applied to H' and to H, $D(\zeta) = \int (1 - \cos\langle\zeta, x\rangle) f(x)\, dx$, and $\iint |\widehat{\phi}(\zeta)|(1 - \cos\langle\zeta, x\rangle)(n_{H'} + n_H)(x)\, dx\, d\zeta = \int |\widehat{\phi}|(\psi_{H'} + \psi_H) < \infty$ by Tonelli. So Fubini applies, and the inversion formula at x and at $-x$ gives $\int \widehat{\phi}(\zeta)(1 - \cos\langle\zeta, x\rangle)\, d\zeta = (2\pi)^d \bigl[\phi(0) - \frac{1}{2}(\phi(x) + \phi(-x))\bigr] = -(2\pi)^d \frac{1}{2}(\phi(x) + \phi(-x))$. Hence

$$-(2\pi)^{-d} \int \widehat{\phi}\, D = \int f(x)\, \tfrac{1}{2}\bigl(\phi(x) + \phi(-x)\bigr)\, dx = \int f\phi \, dx,$$

f being even. This is (4.3).

The sign. If $\phi \ge 0$, the left-hand side of (4.3) is a limit of nonnegative numbers, so $\int f\phi \ge 0$. If $f(x_0) < 0$ for some $x_0 \ne 0$, then $f < 0$ on a ball around x_0 not containing 0, and a smooth $\phi \ge 0$, not identically 0, supported in that ball gives $\int f\phi < 0$. So $f \ge 0$. □

Proof of Proposition 4.13. Clause (1). By C2(iii) there is an infinitely divisible μ_I with triplet $(0, m(|x|)\, dx, 0)$. The density is even, so the odd part $i(\sin\langle\omega, x\rangle - \langle\omega, x\rangle \mathbf{1}_{\{|x|\le 1\}})$ of the integrand of C2(i), which is integrable, integrates to 0, and $\widehat{\mu}_I = e^{-\psi_I}$. The transform determines the law, and it is radial, so μ_I is rotation invariant. The transform of μ_Σ at ω is $e^{-\psi_I(\Sigma^{1/2}\omega)}$, and the substitution $x = \Sigma^{1/2}y$ gives $\psi_I(\Sigma^{1/2}\omega) = \int (1 - \cos\langle\omega, x\rangle)\, n_\Sigma(x)\, dx$. Also $|\Sigma^{1/2}y|^2 \wedge 1 \le \max(1, \|\Sigma\|)(|y|^2 \wedge 1)$, so $\int (|x|^2 \wedge 1)\, n_\Sigma(x)\, dx < \infty$.

Clause (2): reduction to one direction. Substitute $x = \Sigma^{1/2}y$:

$$n_\Sigma(\Sigma^{1/2}y) = (\det\Sigma)^{-1/2} m(|y|), \qquad n_{\Sigma'}(\Sigma^{1/2}y) = (\det\Sigma')^{-1/2} m(\langle y, M^{-1}y\rangle^{1/2}),$$

so the ordering says $m(|y|) \le (\det M)^{-1/2} m(\langle y, M^{-1}y\rangle^{1/2})$ for every $y \ne 0$. Put $\lambda := \lambda_{\min}(M)^{1/2}$, $D := (\det M)^{1/2}$, and let v be a unit eigenvector of M for $\lambda_{\min}(M)$. On the sphere $|y| = r$, $\langle y, M^{-1}y\rangle \le r^2/\lambda^2$, with equality at $y = rv$; as m is nonincreasing, the right-hand side is least there. So the ordering is $D\, m(r) \le m(r/\lambda)$ for every $r > 0$, that is, multiplying by r^d,

$$k(r/\lambda) \ge (D/\lambda^d)\, k(r) \qquad (r > 0).$$

The least eigenvalue is at least 1. $D \ge \lambda^d$, since $\det M \ge \lambda_{\min}(M)^d$. Suppose $\lambda < 1$ and pick r with $k(r) > 0$. The display gives $k(r/\lambda) \ge k(r)$ with $r/\lambda > r$, so, k being nonincreasing, $k(r/\lambda) = k(r)$, and by iteration $k(r\lambda^{-n}) = k(r)$ for every $n \ge 1$. As $r\lambda^{-n} \to \infty$ and k is nonincreasing, $k = k(r) > 0$ on $[r, \infty)$, and then $\int_{|x|>r} m(|x|)\, dx = |S^{d-1}| \int_r^\infty k(s)\, s^{-1} ds = \infty$, against the integrability of m. So $\lambda \ge 1$, and the display is $D/\lambda^d \le R(\lambda)$, which is the stated inequality; $M \ge I$ gives $\Sigma' - \Sigma = \Sigma^{1/2}(M - I)\Sigma^{1/2} \ge 0$.

The increment. $f := n_{\Sigma'} - n_\Sigma \ge 0$ is even and at most $n_{\Sigma'}$, so $f(x)\, dx$ satisfies the conditions of C2 by clause (1). C2(iii) gives an infinitely divisible ρ with triplet $(0, f(x)\, dx, 0)$, and the odd part of the integrand vanishing as in clause (1), $\widehat{\rho} = \exp[-\int (1 - \cos\langle\omega, x\rangle) f(x)\, dx] = \widehat{\mu}_{\Sigma'}/\widehat{\mu}_\Sigma$, both integrals being finite.

Clause (3). Reflexivity, transitivity and antisymmetry are those of the pointwise order of the densities and of the Loewner order, which contains the relation. Covariance: $|(A\Sigma A^\mathsf{T})^{-1/2}x| = |\Sigma^{-1/2}A^{-1}x|$, so $n_{A\Sigma A^\mathsf{T}}(x) = |\det A|^{-1} n_\Sigma(A^{-1}x)$, and the ordering of the pair $(A\Sigma A^\mathsf{T}, A\Sigma' A^\mathsf{T})$ is that of (Σ, Σ') at $A^{-1}x$. A ray has $M = \lambda^2 I$, $\det M = \lambda^{2d} \le \lambda^{2d} R(\lambda)^2$. A rank-deficient step has $\lambda_{\min}(M) = 1 < \det M$, $M = I + \Sigma^{-1/2} P \Sigma^{-1/2}$ having the eigenvalue 1 and one above 1, while the inequality at $\lambda = 1$ reads $\det M \le R(1)^2 = 1$. For $k = c\, r^{-\alpha}$, $k(r/\lambda)/k(r) = \lambda^\alpha$

for every r. If $k(0+)$ is finite, it is positive, k being nonincreasing and not 0. Then $R(\lambda) \le \lim_{r\downarrow 0} k(r/\lambda)/k(r) = k(0+)/k(0+) = 1$, while $R \ge 1$ because $k(r/\lambda) \ge k(r)$ for $\lambda \ge 1$. So $R \equiv 1$, and the inequality becomes $\det M \le \lambda_{\min}(M)^d$. It holds only when every eigenvalue of M equals $\lambda_{\min}(M)$, that is $M = \lambda_{\min}(M) I$, a ray. □

Proof of Proposition 4.20. (1), the order. On $\mathrm{PD}(d)$ the relation is $\preceq_\alpha$: a definite matrix is not 0, and the second case is contained in the third, every ray $\Sigma \le c\,\Sigma$, $c \ge 1$, being a step of $\preceq_\alpha$ by Proposition 4.7(3) with $\lambda = \sqrt{c}$. If Σ is definite, every Σ' with $\Sigma \le_\alpha \Sigma'$ is definite. If Σ is singular and not 0, those Σ' are its multiples $c\Sigma$, $c \ge 1$. The properties a wedge datum requires of its order follow, whatever G is.

- Reflexivity is the case $c = 1$.
- Antisymmetry: the relation is contained in the Loewner order, as $\Sigma' - 0 \ge 0$, $c\Sigma - \Sigma = (c-1)\Sigma \ge 0$, and $\preceq_\alpha$ is contained in it (Proposition 4.7(1)).
- Transitivity: from $\Sigma = 0$ there is nothing to prove, from a nonzero singular Σ the relation runs along its multiples, and on $\mathrm{PD}(d)$ it is the transitivity of $\preceq_\alpha$ (Proposition 4.7(2)), the rays being steps of $\preceq_\alpha$.
- Invariance: $A \cdot 0 = 0$, $A \cdot (c\Sigma) = c\,A \cdot \Sigma$, and $\mathrm{GL}(d)$ preserves $\mathrm{PD}(d)$ and $\preceq_\alpha$ (Proposition 4.7(2)).
- The rays: for $0 \le c \le c'$, $c\Sigma \le_\alpha c'\Sigma$ by the first case if $c = 0$ and by the second, with the factor c'/c, otherwise.

For $\alpha \le \beta < 2$ only the third case changes: with $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$, $\Sigma \preceq_\alpha \Sigma'$ gives $\lambda_{\min}(M) \ge 1$ (Proposition 4.7(1)), so $\lambda_{\min}(M)^{d+\beta} \ge \lambda_{\min}(M)^{d+\alpha} \ge \det M$.

(1), the chains. From 0 to Σ' take $\gamma(s) := s\Sigma'$, and from $\Sigma \ne 0$ to $c\Sigma$ take $\gamma(s) := (1 + s(c-1))\Sigma$; along both, $\gamma(s) \le_\alpha \gamma(t)$ for $s \le t$ by the rays. For $\Sigma \preceq_\alpha \Sigma'$ in $\mathrm{PD}(d)$ write $M = U\,\mathrm{diag}(m_1, \dots, m_d)U^{\mathsf{T}}$ with $U \in \mathrm{O}(d)$, put $M^s := U\,\mathrm{diag}(m_1^s, \dots, m_d^s)U^{\mathsf{T}}$, and take the chain of Theorem 3.5(3)(c), $\gamma(s) := \Sigma^{1/2}M^s\Sigma^{1/2}$. It is continuous and definite, with $\gamma(0) = \Sigma$ and $\gamma(1) = \Sigma'$. Its points are ordered by the criterion (b) of Proposition 4.7(1). For $x = \Sigma^{1/2}y \ne 0$ put $z := U^{\mathsf{T}}y$; then

$$\langle x, \gamma(s)^{-1}x\rangle = \sum_i m_i^{-s} z_i^2 =: Q_s, \qquad \det\gamma(s) = \det\Sigma\,(\det M)^s,$$

so that $n_{\gamma(s)}(x) = c_{d,\alpha}^{-1}(\det\Sigma)^{-1/2}(\det M)^{-s/2}Q_s^{-(d+\alpha)/2}$. Let $s \le t$ and $q := \lambda_{\min}(M)$. Every m_i is at least q, so $m_i^{-t} \le q^{-(t-s)}m_i^{-s}$ and $Q_t \le q^{-(t-s)}Q_s$. And $\det M \le q^{d+\alpha}$ gives $(\det M)^{(t-s)/2} \le q^{(t-s)(d+\alpha)/2}$. Together, $(\det M)^{-s/2}Q_s^{-(d+\alpha)/2} \le (\det M)^{-t/2}Q_t^{-(d+\alpha)/2}$, which is $n_{\gamma(s)}(x) \le n_{\gamma(t)}(x)$. So $\gamma(s) \preceq_\alpha \gamma(t)$.

(1), the thresholds. The value $\varepsilon = 0$ is the first case, and $\Sigma \le_\alpha 0$ is impossible for $\Sigma \ne 0$. For $\varepsilon > 0$ and $t > 0$ the two relations are steps of $\preceq_\alpha$, with relative steps $M = \Sigma/\varepsilon$ and $M = t\Sigma^{-1}$, so that

$$\lambda_{\min}(\Sigma/\varepsilon) = \lambda_1/\varepsilon, \quad \det(\Sigma/\varepsilon) = \det\Sigma/\varepsilon^d, \qquad \lambda_{\min}(t\Sigma^{-1}) = t/\lambda_d, \quad \det(t\Sigma^{-1}) = t^d/\det\Sigma.$$

Condition (c) of Proposition 4.7(1) then reads

$$\lambda_1^{d+\alpha} \ge \varepsilon^\alpha \det\Sigma \qquad \text{and} \qquad t^\alpha \ge \lambda_d^{d+\alpha}/\det\Sigma,$$

which are the two equivalences. The ratio is $((\lambda_d/\lambda_1)^{d+\alpha})^{1/\alpha}$.

(2). By Proposition 4.13(1), μ_I is rotation invariant with $\widehat{\mu}_I = e^{-\psi_I} > 0$, and as in the proof of Proposition 3.4(3), $\mu_\Sigma := (\Sigma^{1/2})_*\mu_I$ has $\widehat{\mu}_\Sigma(\omega) = F(\langle\omega, \Sigma\omega\rangle)$ with $F(u) := \widehat{\mu}_I(\sqrt{u}\,e_1)$

continuous and positive. So the transforms have no zeros, and (a)–(c) of Lemma 3.3(2) hold for $G = \mathrm{GL}(d)$: (a) as $F(0) = 1$, (b) as $\widehat{A_*\mu_\Sigma}(\omega) = F(\langle\omega, A\Sigma A^{\mathsf{T}}\omega\rangle)$, and (c) as F is continuous; $\mu_I \neq \delta_0$ because ψ_I is not identically 0, k being positive near 0. By Lemma 3.3(3) it remains to give each pair $\Sigma \leq_\alpha \Sigma'$ an increment. From 0: $\mu_{\Sigma'}$. Along a ray, $\Sigma' = c\Sigma$ with $c > 1$: by Proposition 4.13(2)–(3) the ray $I \leq cI$ has an increment, $\mu_{cI} = \mu_I * \rho$, and applying $(\Sigma^{1/2})_*$, with $\Sigma^{1/2}\sqrt{c} = (c\Sigma)^{1/2}$, gives $\mu_{c\Sigma} = \mu_\Sigma * (\Sigma^{1/2})_*\rho$. On $\mathrm{PD}(d)$, with $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$ and $q := \lambda_{\min}(M) \geq 1$, the inequality $\det M \leq q^{d+\alpha}$, condition (c) of Proposition 4.7(1), and $R(q^{1/2})^2 \geq q^\alpha$ give $\det M \leq q^d R(q^{1/2})^2$, the step set of Proposition 4.13(2), which supplies the increment. For the stable profile $k(r) = c_{d,\beta}^{-1} r^{-\beta}$, $k(|x|)|x|^{-d}$ is $n_I(x)$ of (4.1) at index β, so $\mu_I = S_\beta(I)$ by Lemma 4.6(1)–(2), and $R(\lambda) = \lambda^\beta \geq \lambda^\alpha$ by Proposition 4.13(3); for the profiles of Proposition E.1(1) of index β, $R(\lambda) = \lambda^\beta$ by that clause. □

Proof of Proposition 4.11, clause (1). Infinite divisibility, and the Lévy measure. Let $L_t := e^{-tB}$ and $\mu = (L_t)_*\mu * \rho_t$. By C3, μ and every ρ_t are infinitely divisible. Let (A_μ, ν, γ) be the generating triplet of μ (C2(i)). As $-I \in \mathrm{O}(d)$, μ is symmetric, and uniqueness (C2(ii)) applied to the reflected triplet $(A_\mu, (-I)_*\nu, -\gamma)$ gives $\gamma = 0$ and $(-I)_*\nu = \nu$, so that $\widehat{\mu}(\omega) = \exp[-\frac{1}{2}\langle\omega, A_\mu\omega\rangle - \int(1 - \cos\langle\omega, x\rangle)\,\nu(dx)]$. Then $\nu \neq 0$, since otherwise μ is the Gaussian law $N(0, A_\mu)$. For $L \in \mathrm{GL}(d)$ this form, read at $L^{\mathsf{T}}\omega$, shows that $L_*\mu$ has the triplet $(LA_\mu L^{\mathsf{T}}, L_*\nu, 0)$. The exponents of a convolution add, so uniqueness applied to $\mu = (L_t)_*\mu * \rho_t$ gives $\nu = (L_t)_*\nu + \nu_t$ with ν_t the Lévy measure of ρ_t: hence $(L_t)_*\nu \leq \nu$ for every $t > 0$. Applied to $Q_*\mu = \mu$, $Q \in \mathrm{O}(d)$, it gives $Q_*\nu = \nu$.

Radial smoothing. Fix a smooth η on $\mathbb{R}$ with $0 \leq \eta \leq 1$, $\eta = 1$ on $[3/4, 5/4]$ and $\eta = 0$ off $(1/2, 3/2)$, and put $\nu_\eta(U) := \int \eta(s)\,\nu(s^{-1}U)\,ds$ for Borel U, the average of the dilates $(sI)_*\nu$. It is invariant under $\mathrm{O}(d)$, null at the origin and not 0, and $(L_t)_*\nu_\eta \leq \nu_\eta$, because L_t commutes with sI. Where $\eta(s) \neq 0$, $s < 3/2$ and $s^2|x|^2 \wedge 1 \leq \frac{9}{4}(|x|^2 \wedge 1)$, so $\int(|y|^2 \wedge 1)\,\nu_\eta(dy) \leq \frac{9}{4}\int\eta\int(|x|^2 \wedge 1)\,\nu(dx) < \infty$, and ν_η is finite off every ball about the origin. Substituting $r = s|x|$ and exchanging the integrals, its image under $y \mapsto |y|$ is $K(r)\,dr$ on $(0, \infty)$, with

$$K(r) := \int \eta(r/|x|)\,|x|^{-1}\,\nu(dx).$$

For r in a compact subset of $(0, \infty)$ only x with $|x|$ in a compact subset of $(0, \infty)$ contribute, a set of finite ν-measure, so K is finite there and may be differentiated under the integral. By polar coordinates the measure with density $h(y) := K(|y|)/(|S^{d-1}|\,|y|^{d-1})$ has the same radial image $K(r)\,dr$, and it is invariant under $\mathrm{O}(d)$. A measure invariant under $\mathrm{O}(d)$, null at the origin and finite off every ball about it is determined by its radial image: on each shell $\{|x| > \varepsilon\}$ its restriction m is finite and invariant, and averaging $\widehat{m}(|\omega|\theta)$ over unit vectors θ against σ_{d-1} and exchanging the integrals gives $\widehat{m}(\omega) = |S^{d-1}|^{-1}\int\Phi(|\omega|, r)\,(|\cdot|_*m)(dr)$ with $\Phi(s, r) := \int_{S^{d-1}} e^{isr\langle e_1,\theta\rangle}\,\sigma_{d-1}(d\theta)$. So ν_η has the density h; no Haar measure on $\mathrm{O}(d)$ is needed.

The infinitesimal inequality. Write $M(r) := K(r)/r^{d-1}$, so that $h(y) = M(|y|)/|S^{d-1}|$. The linear change of variables $z = L_t y$, with $|\det L_t|^{-1} = e^{t\,\mathrm{tr}\,B}$, turns $(L_t)_*\nu_\eta \leq \nu_\eta$ into $e^{t\,\mathrm{tr}\,B}M(|e^{tB}z|) \leq M(|z|)$ for almost every z, hence for every $z \neq 0$: the set where it fails is open, both sides being continuous off the origin, and null. Take $z = rv$ with v a unit eigenvector of $\mathrm{sym}\,B$ for λ. Then $\frac{d}{dt}|e^{tB}rv|$ at $t = 0$ is $r\langle v, Bv\rangle = r\lambda$, and a function of $t \geq 0$ that is largest at $t = 0$ has a nonpositive right derivative there:

$$\mathrm{tr}\,B\;M(r) + \lambda\,r\,M'(r) \leq 0 \qquad (r > 0).$$

Since $\det e^{-tB} = e^{-t\,\mathrm{tr}\,B} \to 0$, $\mathrm{tr}\,B > 0$. As $\nu_\eta \neq 0$ there is r_0 with $M(r_0) > 0$.

The budget. If $\lambda = 0$ the inequality gives $M(r_0) \leq 0$, which is excluded. Otherwise put $\kappa := \mathrm{tr}\,B/\lambda$; then $(r^\kappa M(r))' = r^{\kappa-1}(\mathrm{tr}\,B\,M(r) + \lambda r M'(r))/\lambda$, which is ≤ 0 if $\lambda > 0$ and ≥ 0 if $\lambda < 0$. If $\lambda < 0$, then $\kappa < 0$ and $M(r) \geq M(r_0)r_0^\kappa r^{-\kappa} \geq M(r_0)r_0^\kappa$ for $r \geq \max(r_0, 1)$, so that $K(r) = r^{d-1}M(r)$ is bounded below there by a positive constant and $\int_0^\infty(r^2 \wedge 1)K(r)\,dr =$

$\int(|y|^2\wedge 1)\,\nu_\eta(dy)$ is infinite, which is excluded. So $\lambda > 0$, and $M(r) \geq M(r_0)r_0^\kappa r^{-\kappa}$ for $0 < r \leq r_0$, so that with $r_1 := \min(r_0, 1)$

$$\int_0^{r_1} r^2 K(r)\,dr \geq M(r_0)\,r_0^\kappa \int_0^{r_1} r^{d+1-\kappa}\,dr,$$

which is infinite if $\kappa \geq d+2$. Hence $\kappa < d+2$, that is $\operatorname{tr} B < (d+2)\lambda$. □

C.4 The theorem

The theorem assembles the three subsections above along the chain of cones. Its first paragraph derives the properties of the orders $\preceq_C$ and $\preceq_C^+$ that it needs from the causal curves themselves. Theorem 4.3(4) gives them too, but Theorem 4.3 spends C4 through Lemma C.1(2), and the direct derivation keeps the theorem independent of C4.

Proof of Theorem 4.21. The orders. The proof reads of $\preceq_C$ and $\preceq_C^+$ only what the causal curves give directly. By facts (c) and (a) of the proof of Proposition 4.12: for $X \in C$ the curve $t \mapsto e^{tX}$ is C-causal from I to e^X, so $I \preceq_C e^X$; as $(\log c)I \in C$ for $c \geq 1$, $\Sigma \preceq_C c\Sigma$ for $\Sigma \in \mathrm{PD}(d)$; and if $C \subseteq \mathrm{PSD}(d)$, a C-causal curve γ has $\gamma' = \gamma^{1/2}(\gamma^{-1/2}\gamma'\gamma^{-1/2})\gamma^{1/2} \geq 0$ almost everywhere, so it is Loewner-monotone and $\preceq_C^+$ is contained in the Loewner order. A C-causal curve is C'-causal for every cone C' containing C, so $\preceq_C^+ \subseteq \preceq_{C'}^+$. Two C-causal curves γ_1, γ_2 with $\gamma_1(1) = \gamma_2(0)$ run in succession at double speed, $\gamma_1(2t)$ for $t \leq 1/2$ and $\gamma_2(2t-1)$ after, form a C-causal curve: its entries are absolutely continuous, each being its value at 0 plus the integral of the derivatives of γ_1 and γ_2 run at double speed, and its relative derivative is twice that of γ_1 or γ_2, which lies in the cone C. So $\preceq_C$ is transitive, and with the rays so is $\preceq_C^+$. For $A \in \mathrm{GL}(d)$, $A\gamma A^\mathsf{T}$ is C-causal with γ, C being $\mathrm{O}(d)$-invariant: its relative derivative is $Q(\gamma^{-1/2}\gamma'\gamma^{-1/2})Q^\mathsf{T}$ with $Q := (A\gamma A^\mathsf{T})^{-1/2}A\gamma^{1/2}$ orthogonal. So $\preceq_C^+$ is reflexive, transitive and $\mathrm{GL}(d)$-invariant on $\mathrm{PSD}(d)$ and contains the rays. Its antisymmetry is that of Theorem 4.3(4), and the causal curves give it directly. (i) Every nonzero $X \in C$ has $\operatorname{tr} X > 0$, by the last sentence of Lemma C.1(4), as $I \in C$. (ii) Let γ be C-causal. By fact (b) of the proof of Proposition 4.12, $\det\gamma$, a polynomial in the absolutely continuous entries of γ, is absolutely continuous on $[0,1]$, and at almost every t Jacobi's formula gives $(\det\gamma)' = \det\gamma\,\operatorname{tr} Y$ with $Y := \gamma^{-1/2}\gamma'\gamma^{-1/2} \in C$, so $(\det\gamma)' \geq 0$ by (i); being the integral of its derivative, $\det\gamma$ is nondecreasing. Now let $\Sigma \preceq_C \Sigma'$ and $\Sigma' \preceq_C \Sigma$, along C-causal curves γ_1 from Σ to Σ' and γ_2 from Σ' to Σ. Then $\det\Sigma \leq \det\Sigma' \leq \det\Sigma$, so $\det\gamma_1$, nondecreasing with equal values at 0 and 1, is constant, and its derivative $\det\gamma_1\operatorname{tr} Y_1$ vanishes almost everywhere. As $\det\gamma_1 > 0$, $\operatorname{tr} Y_1 = 0$ almost everywhere, and so $Y_1 = 0$ almost everywhere by (i). Hence $\gamma_1' = \gamma_1^{1/2}Y_1\gamma_1^{1/2} = 0$ almost everywhere, γ_1 is constant by fact (a), and $\Sigma = \Sigma'$. (iii) Let $\Sigma \preceq_C^+ \Sigma'$ and $\Sigma' \preceq_C^+ \Sigma$. Only 0 lies below 0, since $0 = c\,\Sigma''$ with $c \geq 1$ forces $\Sigma'' = 0$ and 0 is not definite; so if one of the two is 0, so is the other. Otherwise both are nonzero, and both are singular or both are definite: a matrix above a nonzero singular Σ is a multiple $c\,\Sigma$, the third case of the definition needing Σ definite, and a matrix above a definite one is definite. If both are singular, $\Sigma' = c\,\Sigma$ and $\Sigma = c'\Sigma'$ with $c, c' \geq 1$, so $cc' = 1$ and $c = 1$. If both are definite, each of the two relations is $\preceq_C$, as $\Sigma \preceq_C c\Sigma$ for $c \geq 1$, and (ii) gives $\Sigma = \Sigma'$. So $\preceq_C^+$ is a partial order, and $(\mathrm{GL}(d), \preceq_C^+)$ is a wedge datum (Definition 3.1), $\mathrm{GL}(d)$ being a closed subgroup containing the dilations. A family along one order is restricted to a smaller order by the argument of Lemma 3.3(1), which reads of the two orders only that one contains the other and that the smaller is invariant under the smaller group, which every order restricted to here is; and a function that agrees on the pairs $\Sigma \preceq_C^+ \Sigma'$ with a $(\mathrm{GL}(d), \preceq_C^+)$-cascade family is one, as (K1)–(K4) read it only on pairs, which $\preceq_C^+$ being transitive and $\mathrm{GL}(d)$-invariant keeps among themselves.

(1). Let $0 \leq \alpha < \beta \leq \infty$. A positive semidefinite X has $\lambda_{\min}(X) \geq 0$, so $(d+\beta)\lambda_{\min}(X) \geq (d+\alpha)\lambda_{\min}(X) \geq \operatorname{tr} X$ when $\beta < \infty$, and $C_\alpha \subseteq C_\beta$ (trivially if $\beta = \infty$). Strictness: for $\beta < \infty$, $X := \operatorname{diag}(1, \ldots, 1, 1+\beta)$ has $\operatorname{tr} X = d+\beta$ and $\lambda_{\min}(X) = 1$, so it lies in C_β and not in C_α;

for $\beta = \infty$, $X := e_1 e_1^{\mathsf{T}} \in \mathrm{PSD}(d)$ has $\lambda_{\min} = 0 < 1 = \mathrm{tr}$, so it lies in no C_α, $\alpha < \infty$. A C_α-causal curve is C_β-causal, so $\preceq_{C_\alpha} \subseteq \preceq_{C_\beta}$, and $I \preceq_{C_\beta} e^X$ for the X just found. But $I \not\preceq_{C_\alpha} e^X$: by Proposition 4.12(1), $I \preceq_{C_0} e^X$ would make $e^X = cI$, hence $X = (\log c)I \in C_0$, and for $\alpha > 0$, $I \preceq_{C_\alpha} e^X$ would give $e^{(d+\alpha)\lambda_{\min}(X)} = \lambda_{\min}(e^X)^{d+\alpha} \geq \det e^X = e^{\mathrm{tr}\, X}$, hence $X \in C_\alpha$. So $\preceq_{C_\alpha} \subsetneq \preceq_{C_\beta}$. For $\alpha < \infty$ and $0 \neq X \in C_\alpha$, $\mathrm{tr}\, X > 0$ and $\lambda_{\min}(X) \geq \mathrm{tr}\, X/(d+\alpha) > 0$. The union: a nonzero X in some C_α, $\alpha < 2$, has $\lambda_{\min}(X) > 0$ and so $(d+2)\lambda_{\min}(X) > (d+\alpha)\lambda_{\min}(X) \geq \mathrm{tr}\, X$; conversely $(d+2)\lambda_{\min}(X) > \mathrm{tr}\, X \geq d\lambda_{\min}(X)$ gives $\lambda_{\min}(X) > 0$, so $X \geq 0$, and $X \in C_\alpha$ with $\alpha := \mathrm{tr}\, X/\lambda_{\min}(X) - d \in [0, 2)$. The matrix $\mathrm{diag}(1, \ldots, 1, 3)$ lies in C_2, with equality, and not in the union.

(2), the kernels from scale zero. Let a $(\mathrm{GL}(d), \preceq_C^+)$-cascade family be given. The ray order is contained in $\preceq_C^+$ (its two cases are two of the three), so the family restricts to a $(\mathrm{GL}(d), \mathrm{rays})$-family with the same kernels from scale zero, and Proposition 3.4(3) gives $\mu_\Sigma = (\Sigma^{1/2})_* \mu_I$ with μ_I rotation invariant, self-decomposable and not δ_0, and $\widehat{\mu}_\Sigma(\omega) = F(\langle \omega, \Sigma\omega \rangle)$ with $F(u) = \widehat{\mu}_I(\sqrt{u}\, e_1)$ continuous, $F(0) = 1$. F has no zero (Proposition 3.4(1)) and is real (μ_I is invariant under $-I$), so $F > 0$, and $g := -\log F$ is continuous, $g(0) = 0$, $g \geq 0$ (as $|F| \leq 1$), and $g \not\equiv 0$ (as $\mu_I \neq \delta_0$). For $X \in C$, $I \preceq_C^+ e^X$, and (K1) at $0 \preceq_C^+ I \preceq_C^+ e^X$ gives $\mu_{e^X} = \mu_{I,e^X} * \mu_I$, that is

$$e^{-g(\langle \omega, e^X \omega \rangle)} = e^{-g(|\omega|^2)}\, \widehat{\mu}_{I,e^X}(\omega) \qquad (\omega \in \mathbb{R}^d). \tag{C.1}$$

(2)(a). Let $X \in C$ have an eigenvalue $\lambda < 0$ with unit eigenvector v, and $m := e^\lambda < 1$. At $\omega = uv$, (C.1) and $|\widehat{\mu}_{I,e^X}| \leq 1$ give $g(mu^2) \geq g(u^2)$, so $g(x) \leq g(mx) \leq \cdots \leq g(m^n x) \to g(0) = 0$ for every $x \geq 0$, against $g \not\equiv 0$.

(2)(b). Let $0 \neq X \in C$ be singular and $P := e^X - I$, positive semidefinite with $\mathrm{rank}\, P = \mathrm{rank}\, X$, since its eigenvalues $e^{x_i} - 1$ vanish exactly where those of X do; so $0 < \mathrm{rank}\, P < d$. (C.1) is (3.2) with $\Sigma = I$ and $\rho = \mu_{I,I+P}$, and Theorem 3.7 gives $g(u) = bu$ with $b = g(1)$, $b > 0$ as $g \not\equiv 0$. So $\widehat{\mu}_\Sigma(\omega) = e^{-b\langle \omega, \Sigma\omega \rangle}$ for every Σ. For $\Sigma \preceq_C^+ \Sigma'$, (K1) at $0 \preceq_C^+ \Sigma \preceq_C^+ \Sigma'$ gives $\widehat{\mu}_{\Sigma'} = \widehat{\mu}_{\Sigma,\Sigma'}\, \widehat{\mu}_\Sigma$, and $\widehat{\mu}_\Sigma$ has no zeros, so $\widehat{\mu}_{\Sigma,\Sigma'} = e^{-b\langle \omega, (\Sigma' - \Sigma)\omega \rangle}$, the transform of the affine Gaussian kernel ($\Sigma \leq \Sigma'$ in the Loewner order, as $C \subseteq \mathrm{PSD}(d)$): the family is the restriction of the affine Gaussian scale space. Conversely, the affine Gaussian scale space with constant $b > 0$ is an affine cascade family (Theorem 3.6), that is a $(\mathrm{GL}(d), \mathrm{Loewner})$-family; $\preceq_C^+$ is contained in the Loewner order as $C \subseteq \mathrm{PSD}(d)$, so its restriction is a $(\mathrm{GL}(d), \preceq_C^+)$-family, and so is every function that agrees with it on the pairs.

(2)(c), the number $\beta(C)$. Now $C \subseteq \mathrm{PSD}(d)$ has no nonzero singular member, so its nonzero members are positive definite. On them $f(X) := \mathrm{tr}\, X/\lambda_{\min}(X) - d$ is continuous, nonnegative and invariant under positive scaling. The slice $\{X \in C : \mathrm{tr}\, X = 1\}$ is closed, and bounded, since a positive semidefinite X with $\mathrm{tr}\, X = 1$ has entries in $[-1, 1]$; it contains I/d, and every $X \in C \setminus \{0\}$ has the multiple $X/\mathrm{tr}\, X$ in it. So f attains on the slice a largest value $\beta(C) \geq 0$, which is its largest value on $C \setminus \{0\}$. For $\alpha \in [0, \infty)$, $C \subseteq C_\alpha$ means $(d+\alpha)\lambda_{\min}(X) \geq \mathrm{tr}\, X$, that is $\alpha \geq f(X)$, for every nonzero $X \in C$, which is $\alpha \geq \beta(C)$.

(2)(c), the stable orbits. If $\gamma \geq \beta(C)$, then $C \subseteq C_\gamma$, so every C-causal curve is C_γ-causal and $\preceq_C^+ \subseteq \preceq_{C_\gamma}^+$; by Proposition 4.12(1), $\preceq_{C_\gamma}^+$ is the order $\leq_\gamma$ of Proposition 4.20(1), and that proposition's clause (2), read with its α equal to γ and the stable law of index γ, gives a $(\mathrm{GL}(d), \leq_\gamma)$-family with kernels $(\Sigma^{1/2})_* S_\gamma(I)$, which restricts. Conversely, if these are the kernels of a $(\mathrm{GL}(d), \preceq_C^+)$-family and $X \in C$, then $\mu_{e^X} = (e^{X/2})_* S_\gamma(I) = S_\gamma(e^X)$ (Lemma 4.6(2)) is $S_\gamma(I) * \mu_{I,e^X}$, and Proposition 4.7(1), (a)⇒(c), with $\Sigma = I$ and $\Sigma' = e^X$, gives $\lambda_{\min}(e^X)^{d+\gamma} \geq \det e^X$, that is $X \in C_\gamma$ (Proposition 4.12(1)). So $C \subseteq C_\gamma$ and $\gamma \geq \beta(C)$.

(2)(c), the non-Gaussian families. If $\beta(C) < 2$, take any index γ with $\beta(C) \leq \gamma < 2$ and $\gamma > 0$, for instance $\gamma := \max(\beta(C), 1)$. The stable orbit of index γ is a family, and $S_\gamma(I)$ is not Gaussian, its transform $e^{-|\omega|^\gamma}$ having modulus e^{-t^γ} at $\omega = te_1$, which is e^{-at^2} for no $a \geq 0$. Conversely let a family have a non-Gaussian μ_I, and let $0 \neq X \in C$ and $t \geq 0$. By (K1), as in (C.1) with $tX \in C$ in place of X, and the form of the kernels, $(e^{tX/2})_* \mu_I = \mu_{e^{tX}} = \mu_I * \mu_{I,e^{tX}}$;

applying $(e^{-tX/2})_*$, $\mu_I = (e^{-tB})_*\mu_I * (e^{-tB})_*\mu_{I,e^{tX}}$ with $B := X/2$, so μ_I is B-self-decomposable in the sense of Proposition 4.10. B is positive definite, so $e^{-tB} \to 0$, and μ_I is rotation invariant and not Gaussian; Proposition 4.11(1) gives $\operatorname{tr} B < (d+2)\lambda_{\min}(B)$, that is $f(X) < 2$. As the largest value of f is attained, $\beta(C) < 2$. If $\beta(C) \geq 2$, then every family has a Gaussian μ_I. Its image under $x \mapsto \langle e_1, x\rangle$ is a Gaussian law on $\mathbb{R}$, of some variance $v \geq 0$, so $e^{-g(t^2)} = |\widehat{\mu}_I(te_1)| = e^{-vt^2/2}$ and $g(u) = bu$ with $b := v/2$, and $b > 0$ as $g \not\equiv 0$; the end of (2)(b) shows that the family is the restricted affine Gaussian scale space; these exist, by the converse there, which used only $C \subseteq \mathrm{PSD}(d)$.

(3). By Proposition 4.12(1), $\Sigma \preceq_{C_0} \Sigma'$ on $\mathrm{PD}(d)$ means $\Sigma' = c\Sigma$ with $c \geq 1$, so by the definition of $\preceq^+$ the order $\preceq^+_{C_0}$ is the ray order, and Proposition 3.4(3) is the description of $\mathfrak{F}_0$. For $\alpha \leq \beta$, $\preceq^+_{C_\alpha} \subseteq \preceq^+_{C_\beta}$ by (1), and a family restricts; so $\mathfrak{F}_\beta \subseteq \mathfrak{F}_\alpha$. For $0 \leq \alpha < \infty$, C_α satisfies the hypotheses of (2)(c), as its nonzero members are definite by (1), and $\beta(C_\alpha) = \alpha$, since $f \leq \alpha$ on C_α with equality at $\operatorname{diag}(1, \ldots, 1, 1+\alpha)$. The clauses of (3) follow, range by range.

- $\alpha = 0$: $\mathfrak{F}_0$ is described above.
- $0 < \alpha < 2$: the stable clause is (2)(c), and the profile clause is Proposition 4.20(2), its order $\leq_\alpha$ being $\preceq^+_{C_\alpha}$.
- $2 \leq \alpha < \infty$: (2)(c) with $\beta(C_\alpha) \geq 2$ gives the Gaussians.
- $\alpha = \infty$: $C_\infty = \mathrm{PSD}(d)$ contains $e_1 e_1^{\mathsf{T}}$, so (2)(b) gives the Gaussians.

Strictness for $\alpha < \beta \leq 2$: let $\gamma := \alpha$ if $\alpha > 0$ and $\gamma := \beta/2$ if $\alpha = 0$, so $0 < \gamma < \beta$ and $\gamma \geq \alpha$. By (2)(c) with $\beta(C_\alpha) = \alpha$, $S_\gamma(I) \in \mathfrak{F}_\alpha$; and $S_\gamma(I) \notin \mathfrak{F}_\beta$, by (2)(c) with $\beta(C_\beta) = \beta > \gamma$ if $\beta < 2$, and because $\mathfrak{F}_2$ contains only Gaussian laws if $\beta = 2$. □

D The orientation-preserving group

Covariance under the orientation-preserving matrices $\mathrm{GL}^+(d)$ alone is a weaker requirement than covariance under $\mathrm{GL}(d)$: it leaves out the reflections. For $d \geq 2$ it gives the same families. An order invariant under $\mathrm{GL}^+(d)$ is invariant under $\mathrm{GL}(d)$, and the $(\mathrm{GL}^+(d), W)$-cascade families of kernels and of operators are the $(\mathrm{GL}(d), W)$-cascade families (Proposition D.1). So the orientation-preserving group adds no row to Table 2, as §4.2 states. On the line the reflection matters, and clause (3) gives a family of $\mathrm{GL}^+(1)$ that is not one of $\mathrm{GL}(1)$.

Proposition D.1 (covariance under $\mathrm{GL}^+(d)$ is covariance under $\mathrm{GL}(d)$)**.** *Let $d \geq 1$ and $\mathrm{GL}^+(d) := \{A \in \mathrm{GL}(d) : \det A > 0\}$.*

1. (The orders.) *$\mathrm{GL}^+(d)$ is a closed subgroup of $\mathrm{GL}(d)$ containing the dilations. For all $\Sigma, \Sigma' \in \mathrm{PSD}(d)$ there is a $B \in \mathrm{GL}(d)$ with $\det B = -1$, $B\Sigma B^{\mathsf{T}} = \Sigma$ and $B\Sigma' B^{\mathsf{T}} = \Sigma'$. Consequently every relation on $\mathrm{PSD}(d)$ that is invariant under $\Sigma \mapsto A\Sigma A^{\mathsf{T}}$ for every $A \in \mathrm{GL}^+(d)$ is invariant under it for every $A \in \mathrm{GL}(d)$, and $(\mathrm{GL}^+(d), W)$ is a wedge datum (Definition 3.1) if and only if $(\mathrm{GL}(d), W)$ is one.*
2. (The families, $d \geq 2$.) *Let $d \geq 2$ and let $(\mathrm{GL}^+(d), W)$ be a wedge datum. A family of probability measures indexed by P_W is a $(\mathrm{GL}^+(d), W)$-cascade family of kernels if and only if it is a $(\mathrm{GL}(d), W)$-cascade family of kernels; and a family of operators indexed by P_W is a $(\mathrm{GL}^+(d), W)$-cascade family of operators if and only if it is a $(\mathrm{GL}(d), W)$-cascade family of operators. In particular, for W the Loewner order, the $(\mathrm{GL}^+(d), W)$-cascade families of kernels are the affine Gaussian scale spaces* (3.1)*, $b > 0$.*

3. (The line.) *Let $d = 1$ and let W be the Loewner order, which on $\mathrm{PSD}(1) = [0, \infty)$ is the usual order. The point masses $\mu_{s,t} := \delta_{\sqrt{t}-\sqrt{s}}$, $0 \le s \le t$, form a $(\mathrm{GL}^+(1), W)$-cascade family of kernels that is not a $(\mathrm{GL}(1), W)$-cascade family.*

So for $d \ge 2$ every statement made in §4 for $(\mathrm{GL}(d), W)$ holds verbatim for $(\mathrm{GL}^+(d), W)$. Two facts carry the proposition. An orientation-reversing matrix acts on a given pair as an orientation-preserving one does, through the joint stabilizer of clause (1). And a law invariant under $\mathrm{SO}(d)$ is invariant under $\mathrm{O}(d)$ when $d \ge 2$, because $\mathrm{SO}(d)$ is transitive on spheres. The second fact fails on the line, and clause (3) shows that the conclusion fails with it (Remark 4.24). A group that contains $\mathrm{SO}(d)$ but not $\mathrm{O}(d)$ and is not transitive on $\mathrm{PD}(d)$, such as the orientation-preserving similarities, is part of problem 4 of §5.1.

Sketch. (1). Put $T := \Sigma + \Sigma'$, and for $u \ne 0$ let $H_u := I - 2\langle u, u\rangle^{-1} u u^{\mathsf{T}}$, the reflection in $u^{\perp}$, of determinant -1; it commutes with every symmetric matrix of which u is an eigenvector. If T is singular, a $u \ne 0$ with $Tu = 0$ has $\Sigma u = \Sigma' u = 0$, and $B := H_u$ fixes both. Otherwise let $R := T^{1/2}$ and u an eigenvector of $M := R^{-1}\Sigma R^{-1}$, which is also one of $R^{-1}\Sigma' R^{-1} = I - M$; then $B := R H_u R^{-1}$ fixes both and has $\det B = -1$. For $\det A < 0$, the matrix $AB \in \mathrm{GL}^+(d)$ acts on the pair as A does, which carries invariance from $\mathrm{GL}^+(d)$ to $\mathrm{GL}(d)$. Both groups are closed and contain the dilations, and the other conditions of a wedge datum do not involve the group, so $(\mathrm{GL}^+(d), W)$ and $(\mathrm{GL}(d), W)$ are wedge data together.

(2). The conditions (K1), (K2) and (K4) of Definition 3.1 do not involve the group, so only (K3) for $\det A < 0$ is to be shown. For $d \ge 2$ the orthogonal matrices of determinant 1 act transitively on each sphere, so (K3) at $(0, I)$ makes μ_I rotation invariant, $\widehat{\mu}_I(\omega) = F(|\omega|^2)$. Then (K3) with $A = \Sigma^{1/2}$ and (K2) give $\widehat{\mu}_\Sigma(\omega) = F(\langle \omega, \Sigma\omega\rangle)$ on all of $\mathrm{PSD}(d)$. Along the ray of I put $f(s,t) := \widehat{\mu}_{sI,tI}(\omega)$. By (K2) it is continuous, and by (K1) $f(t,t) = 1$ and $f(r,t) = f(s,t)\, f(r,s)$. At a first zero $t^* > 0$ of $f(0,\cdot)$, continuity at (t^*, t^*) gives an $s < t^*$ with $f(s,t^*) \ne 0$, and then $f(0,t^*) = f(s,t^*)\, f(0,s) \ne 0$, a contradiction. So no transform vanishes. The increments are then the quotients $\widehat{\mu}_{\Sigma'}/\widehat{\mu}_\Sigma$, and (K3) follows for every $A \in \mathrm{GL}(d)$ by the uniqueness of the transform. For operators, Lemma A.1(1), (2) and (4) carry the statement over. For the Loewner order the families of kernels are the affine cascade families of Definition 3.2, which Theorem 3.6 identifies with the affine Gaussian scale spaces.

(3). $\mathrm{GL}^+(1) = (0, \infty)$ acts by $a \cdot s = a^2 s$, and the point masses satisfy (K1) to (K4) by inspection. The matrix $-1 \in \mathrm{GL}(1)$ fixes every s, while $(-1)_* \delta_1 = \delta_{-1} \ne \mu_{0,1}$. □

E Two further results on the step set of an orbit

Two results complete the picture of §4.6. The first keeps the stable cone for profiles that are stable only near the origin. The second shows, for the Matérn orbit of index 1, the mechanism by which a step across shapes fails.

E.1 Profiles stable at the origin

A profile that is stable at the origin keeps the stable cone however it is tempered at large radii.

Proposition E.1 (profiles stable at the origin keep the stable cone)**.** *Let $d \ge 1$ and $0 < \alpha < 2$, let $j : (0, \infty) \to [0, \infty)$ be nonincreasing with $j(0+) := \lim_{r \downarrow 0} j(r)$ in $(0, \infty)$, and put $k(r) := r^{-\alpha} j(r)$.*

1. (The stable cone.) *k satisfies the hypotheses of Proposition 4.13, and its dilation ratio is $R(\lambda) = \lambda^\alpha$ for every $\lambda \ge 1$. So for the orbit (μ_Σ) of Proposition 4.13(1), with Lévy densities n_Σ, and for $\Sigma, \Sigma' \in \mathrm{PD}(d)$ with $M := \Sigma^{-1/2}\Sigma'\Sigma^{-1/2}$, $n_\Sigma \le n_{\Sigma'}$ everywhere if and only if $\lambda_{\min}(M)^{d+\alpha} \ge \det M$, which is condition (c) of Proposition 4.7; and then $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ an infinitely divisible probability measure.*

2. (Exponential moments.) *If $\beta > 0$ and $\int_1^\infty e^{\beta r} r^{-1-\alpha} j(r)\, dr < \infty$, then $\int e^{\langle \theta, x\rangle}\, \mu_\Sigma(dx) < \infty$ for every $\Sigma \in \mathrm{PD}(d)$ and every $\theta \in \mathbb{R}^d$ with $|\Sigma^{1/2}\theta| \le \beta$.*

3. (Two families.) *(a) For $\ell > 0$, $j(r) := e^{-r/\ell}$ satisfies the hypotheses, and those of (2) with $\beta = 1/\ell$. (b) The function*

$$j(r) := \frac{\alpha/2}{\Gamma(1-\alpha/2)} \int_0^\infty (4\pi t)^{-d/2}\, t^{-1-\alpha/2}\, e^{-1/(4t) - r^2 t}\, dt$$

satisfies the hypotheses, and those of (2) for every $0 < \beta < 1$; the law μ_I of Proposition 4.13(1) for $k = r^{-\alpha} j$ has $\widehat{\mu}_I(\omega) = \exp[-((1+|\omega|^2)^{\alpha/2} - 1)]$, so its orbit has $\widehat{\mu}_\Sigma(\omega) = \exp[-((1+\langle \omega, \Sigma\omega\rangle)^{\alpha/2} - 1)]$, which at $\Sigma = \ell^2 I$ is $\exp[-((1+\ell^2|\omega|^2)^{\alpha/2} - 1)]$.

4. (The Matérn is the endpoint $\alpha = 0$.) *For $\gamma > 0$, $k_0(r) := \gamma \int_0^\infty (4\pi t)^{-d/2}\, t^{-1}\, e^{-1/(4t) - r^2 t}\, dt$ satisfies the hypotheses of Proposition 4.13, with $k_0(0+) = \gamma\, \pi^{-d/2} \Gamma(\frac{d}{2})$ finite, and the law of Proposition 4.13(1) for k_0 is the Matérn law of index γ, $\widehat{\mu}_I(\omega) = (1+|\omega|^2)^{-\gamma}$ (Proposition 4.4(4)). So its dilation ratio is $R \equiv 1$ and its ordered-density step set is the rays.*

For these profiles the dilation ratio, and with it the ordered-density step set, sees only the profile at the origin. The stable factor $r^{-\alpha}$ sets it, and any nonincreasing tempering leaves it unchanged. So the orbits of clause (3) have the stable orbit's freedom across shapes, with the anisotropy gained per step of Proposition 4.7(3), and they have exponential moments, as the Matérn law does and the stable law does not. The Matérn exponent $\gamma \log(1+u)$ is the limit of $(2\gamma/\alpha)((1+u)^{\alpha/2} - 1)$ as $\alpha \downarrow 0$, and clause (4) says what it lacks: its profile is the same subordination of the heat kernel at $\alpha = 0$, bounded at the origin, hence the rays. That a positive increment of a tempered orbit forces ordered densities is open.

The two families of clause (3) belong to two classes of the literature. Rosiński calls a law on $\mathbb{R}^d$ *tempered α-stable* if it is infinitely divisible without Gaussian part and its Lévy measure is $r^{-\alpha-1} q(r,u)\, dr\, \sigma(du)$ in polar coordinates, with σ a finite measure on the sphere and each $q(\cdot, u)$ completely monotone with $q(\infty, u) = 0$. The law is *proper* if moreover $q(0+, u) = 1$ ([53], Def. 2.1, p. 679). The law μ_I of Proposition 4.13(1) for $k = r^{-\alpha} j$ has the Lévy measure $r^{-\alpha-1} j(r)\, dr\, \sigma(du)$, with σ the surface measure. In clause (3)(a), $j(r) = e^{-r/\ell}$, so μ_I is proper tempered α-stable, with the exponential tempering that Rosiński calls tilting (p. 679). So is every $\mu_\Sigma = (\Sigma^{1/2})_* \mu_I$ of its orbit: the substitution $s = r|\Sigma^{1/2}u|$, $v = \Sigma^{1/2}u/|\Sigma^{1/2}u|$ keeps the polar form, with $q(s,v) = \exp(-s|\Sigma^{-1/2}v|/\ell)$ in the new direction v, and with σ replaced by the image of $|\Sigma^{1/2}u|^\alpha\, \sigma(du)$ under $u \mapsto v$. Clause (1) requires less of j, only that it be nonincreasing, and covers more than these laws, the stable law itself (j constant) among them.

Clause (3)(b) is the relativistic α-stable law. Ryznar defines it for $0 < \alpha < 2$ by the exponent $(m^{2/\alpha} + |\omega|^2)^{\alpha/2} - m$ ([54], p. 2, eq. (3)), with $m > 0$ (p. 1). This extends the case $\alpha = 1$, $\sqrt{|\omega|^2 + m^2} - m$, of Carmona, Masters and Simon ([8], p. 120, eq. (I.6)). At $m = 1$ it is the exponent of μ_I in clause (3)(b). Ryznar's Lemmas 1 and 2 (pp. 3–4), the subordination and the Lévy density, are stated for $d \ge 2$ (p. 2). The proof below makes the same subordination computation, and it holds for every $d \ge 1$. On the orbit, $\mu_{\ell^2 I}$ is the law at time ℓ^α of the process with $m = \ell^{-\alpha}$. The other members are not rotation invariant, so they lie outside Ryznar's family.

The law of clause (3)(b) is not tempered α-stable for this α. For a rotation-invariant Lévy measure, $q(\cdot, u)$ in the polar form is a positive constant multiple $c(u)\, j$ of j for σ-almost every u, and this j is not completely monotone. It is strictly decreasing with $j(0+) - j(r) = o(r)$ as $r \downarrow 0$, because $d + \alpha > 1$: the difference is of order $r^{d+\alpha}$ for $d + \alpha < 2$, $r^2 \log(1/r)$ at $d + \alpha = 2$ and r^2 above. A completely monotone $q = \int_{(0,\infty)} e^{-rs}\, Q(ds)$ with $Q \ne 0$ finite has instead $(q(0+) - q(r))/r \to \int s\, Q(ds) > 0$. The tempering sits in the subordinator. The law on $[0,\infty)$ with Laplace exponent $(1+u)^{\alpha/2} - 1$ has the Lévy measure $c_\alpha s^{-1-\alpha/2} e^{-s}\, ds$ of the proof below,

so it is proper tempered $\alpha/2$-stable, and μ_I is the heat kernel subordinated by it (Ryznar's Lemma 1, p. 3).

Sketch. (1). $k = r^{-\alpha}j$ is nonincreasing and integrable as required, and $k(r/\lambda)/k(r) = \lambda^\alpha j(r/\lambda)/j(r) \geq \lambda^\alpha$, with ratio tending to λ^α as $r \downarrow 0$. So $R(\lambda) = \lambda^\alpha$, and Proposition 4.13(2) gives the cone, which is condition (c) of Proposition 4.7(1).

(2). The compound Poisson laws π_ε of the Lévy measure restricted to $\{|x| > \varepsilon\}$ have transforms tending to $\widehat{\mu}_I$ as $\varepsilon \downarrow 0$, so they converge to μ_I weakly by Lévy's continuity theorem. By symmetry their exponential moments are $\exp \int_{|x|>\varepsilon} (\cosh\langle\theta, x\rangle - 1)\, m(|x|)\, dx$, at most $\exp \int (\cosh\langle\theta, x\rangle - 1)\, m(|x|)\, dx$, finite under the hypothesis, and the weak limit keeps the bound (truncate $e^{\langle\theta,x\rangle}$ at N, then let $N \to \infty$).

(3). (a) is a direct check. (b) By the Gamma integral, $j(0+)$ is finite and positive, and $j(r) \leq C_\eta e^{-(1-\eta)r}$ for every $\eta \in (0, 1)$. With h_s the heat kernel with transform $e^{-s|\omega|^2}$, the identity

$$c_\alpha \int_0^\infty (1 - e^{-us})\, s^{-1-\alpha/2} e^{-s}\, ds = (1+u)^{\alpha/2} - 1, \qquad c_\alpha := \frac{\alpha/2}{\Gamma(1-\alpha/2)},$$

and Tonelli show that $c_\alpha \int_0^\infty h_s\, s^{-1-\alpha/2} e^{-s}\, ds$ is the Lévy density of the law, and the substitution $s = |x|^2 t$ shows that it is $|x|^{-d-\alpha} j(|x|)$.

(4). The same subordination with $s^{-1}e^{-s}$, Frullani's integral as in the proof of Proposition 4.4(4), and Tonelli give the exponent $\gamma \log(1 + |\omega|^2)$ and the profile k_0, bounded at the origin by the Gamma integral and by $\gamma C e^{-r/2}$ at large radii, so Proposition 4.13(3) gives the rays. □

E.2 The Matérn orbit across shapes

A step of full rank widens every direction, so the subspace argument behind Proposition 4.4(3) does not reach it. For the Matérn orbit of index 1 it fails anyway off the rays: this is Proposition 4.17 at index 1, with the increment named. A sign computation at the origin (Remark E.3) shows the mechanism.

Proposition E.2 (in the Matérn orbit every step across shapes fails)**.** *Let $d \geq 2$ and let $(\mu_\Sigma)_{\Sigma \in \mathrm{PD}(d)}$ be the Matérn orbit of index 1, $\widehat{\mu}_\Sigma(\omega) = (1 + \langle\omega, \Sigma\omega\rangle)^{-1}$ (Proposition 4.4(4)). Let $\Sigma \leq \Sigma'$ in $\mathrm{PD}(d)$ with $\Sigma' \neq \Sigma$. Then there is a positive finite measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$ if and only if $\Sigma' = \lambda^2\Sigma$ for some $\lambda > 1$; and then ρ is the probability measure $\lambda^{-2}\delta_0 + (1 - \lambda^{-2})\mu_{\Sigma'}$.*

Proof. By Proposition 4.17 at $\gamma = 1$, which needs neither $d \geq 2$ nor $\Sigma \leq \Sigma'$, a positive finite measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$ exists if and only if $\Sigma' = \lambda^2\Sigma$ for some $\lambda > 1$, and then ρ is unique. It remains to name it. Let $\Sigma' = \lambda^2\Sigma$ with $\lambda > 1$, and put $\tau := \lambda^{-2}\delta_0 + (1-\lambda^{-2})\mu_{\Sigma'}$, a probability measure since $0 < \lambda^{-2} < 1$. For $\omega \in \mathbb{R}^d$ write $u := \langle\omega, \Sigma\omega\rangle \geq 0$, so that $\langle\omega, \Sigma'\omega\rangle = \lambda^2 u$. Since $\lambda^{-2}(1 + \lambda^2 u) + 1 - \lambda^{-2} = 1 + u$,

$$\widehat{\mu}_\Sigma(\omega)\, \widehat{\tau}(\omega) = \frac{1}{1+u}\Big(\lambda^{-2} + \frac{1-\lambda^{-2}}{1+\lambda^2 u}\Big) = \frac{1}{1+\lambda^2 u} = \widehat{\mu}_{\Sigma'}(\omega).$$

A finite measure is determined by its transform, so $\mu_{\Sigma'} = \mu_\Sigma * \tau$, and by the uniqueness $\rho = \tau$. □

Remark E.3 (the mechanism). A second argument for the "only if" half of Proposition E.2 shows why a step off the rays fails: near the origin the increment would have to remove mass from the directions the step widens least. The density of $\mu_{\Sigma'}$ is $K'(x) = (\det \Sigma')^{-1/2} \mathcal{G}(\Sigma'^{-1/2} x)$, with

$$\mathcal{G}(y) := \int_0^\infty e^{-t} (4\pi t)^{-d/2} e^{-|y|^2/4t}\, dt$$

the Green's function of $1-\Delta$, smooth off the origin. The increment would have the transform $1-\langle\omega, P\omega\rangle\widehat{K'}(\omega)$, $P := \Sigma' - \Sigma$, so off the origin it is the function $\nabla \cdot P\nabla K'$. In the coordinates $y = \Sigma'^{-1/2}x$, with $P' := \Sigma'^{-1/2}P\Sigma'^{-1/2}$, the substitution $t = r^2 s$ and the Gamma integral give, uniformly in the direction $\widehat{y}$,

$$r^d\, \nabla \cdot P'\nabla \mathcal{G}(r\widehat{y}) \to \mathcal{B}_d\big(d\,\langle \widehat{y}, P'\widehat{y}\rangle - \operatorname{tr} P'\big), \qquad \mathcal{B}_d = \tfrac{1}{2}\pi^{-d/2}\Gamma(\tfrac{d}{2}) > 0.$$

If P' is not a multiple of I, the bracket is negative near a least eigenvector, so the increment is negative on an open cone near the origin, which a positive measure does not allow. So $P' = \kappa I$, which is $\Sigma' = \lambda^2\Sigma$.

The same computation measures the failure. The would-be increment is not even a finite signed measure.

Proposition E.4 (how much a step across shapes fails: a logarithmic negative mass)**.** *Let $d \geq 2$ and let $(\mu_\Sigma)_{\Sigma\in \mathrm{PD}(d)}$ be the Matérn orbit of index 1 (Proposition 4.4(4)), with K_Σ the density of μ_Σ. Let $\Sigma, \Sigma' \in \mathrm{PD}(d)$ with $P := \Sigma' - \Sigma \in \mathrm{PSD}(d)$ and $P \neq 0$, put $P' := \Sigma'^{-1/2} P\, \Sigma'^{-1/2}$, and let ρ be the linear functional on Schwartz functions*

$$\rho(\phi) := (2\pi)^{-d} \int \widehat{\phi}(\omega)\, \frac{1 + \langle \omega, \Sigma\omega\rangle}{1 + \langle \omega, \Sigma'\omega\rangle}\, d\omega,$$

with $\widehat{\phi}(\omega) = \int e^{i\langle\omega,x\rangle}\phi(x)\,dx$; it is $\int \phi\, d\rho$ whenever $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ a finite signed measure.

1. (Off the origin.) *$q := \nabla \cdot P\nabla K_{\Sigma'}$ is smooth on $\mathbb{R}^d \setminus \{0\}$, and $\rho(\phi) = \int \phi\, q\, dx$ for every smooth ϕ with compact support in $\mathbb{R}^d \setminus \{0\}$.*

2. (The density near the origin.) *Write $x = \Sigma'^{1/2} y$ and $y = r\widehat{y}$ with $r > 0$, $|\widehat{y}| = 1$, and put $\mathcal{B}_d := \frac{1}{2}\pi^{-d/2}\Gamma(\frac{d}{2})$. Then, uniformly in $\widehat{y}$ as $r \downarrow 0$,*

$$(\det \Sigma')^{1/2}\, r^d\, q(\Sigma'^{1/2} r\widehat{y}) \to \mathcal{B}_d\big(d\langle \widehat{y}, P'\widehat{y}\rangle - \operatorname{tr} P'\big).$$

3. (The negative mass diverges logarithmically.) *For $0 < a < 1$ let $E_a := \{x : a < |\Sigma'^{-1/2}x| < 1\}$, and write $s_\pm := \max(\pm s, 0)$. Then, as $a \downarrow 0$,*

$$\int_{E_a} q_-\, dx = \mathcal{B}_d\, \Theta\, \log(1/a) + o(\log(1/a)), \qquad \Theta := \int_{S^{d-1}} \big(d\langle \widehat{y}, P'\widehat{y}\rangle - \operatorname{tr} P'\big)_-\, \sigma(d\widehat{y}),$$

with σ the surface measure on the unit sphere S^{d-1}, and the same holds for q_+ with the same Θ. Moreover $\Theta > 0$ if and only if Σ' is not a multiple of Σ, and then there is no finite signed measure ρ with $\mu_{\Sigma'} = \mu_\Sigma * \rho$.

4. (The plane.) *If $d = 2$ and $p_1 \geq p_2$ are the eigenvalues of P', then $\mathcal{B}_2\Theta = (p_1 - p_2)/\pi$. For the step from $\Sigma = t^2 I$ to $\Sigma' = t^2\,\mathrm{diag}(1+\varepsilon, 1)$, $t, \varepsilon > 0$, it is $\varepsilon/(\pi(1+\varepsilon))$.*

Off the origin the would-be increment is a smooth function, homogeneous of degree $-d$ to leading order, with an angular factor of mean zero. Its positive and negative parts both carry a mass that diverges like $\mathcal{B}_d\Theta\log(1/a)$ at the origin, where Θ measures how unevenly the step widens the directions and vanishes exactly on the rays. So the failure is logarithmic in the resolution. Since Θ is homogeneous of degree one in P', a smaller step only scales it down.

Sketch. (1) is the first half of Remark E.3, after Proposition E.2, with $q = \nabla\cdot P\nabla K_{\Sigma'}$, and *(2)* is its uniform limit, through the Matérn densities of Proposition 4.4(4). *(3).* In polar coordinates $\int_{E_a} q_-\, dx = \int_a^1 F(r)\, dr/r$ with $F(r)$ the sphere integral of $(r^d q)_-$ in the coordinates of (2); by (2), $F(r) \to \mathcal{B}_d\Theta$ as $r \downarrow 0$, which gives the logarithm. The same holds for q_+, because the bracket has mean zero over the sphere. If a finite signed measure were the increment, its total variation would be at least $\int_{E_a} q_-\, dx$ for every a. *(4).* In the plane the bracket is $(p_1 - p_2)\cos 2\varphi$ in an eigenbasis of P', whose negative part integrates to $2(p_1 - p_2)$, and $\mathcal{B}_2 = 1/(2\pi)$. □

Remark E.5 (the failure on a grid, measured). On a grid the frequencies are bounded, and the lattice increment of a step across shapes is a finite signed measure. A script deposited with the paper (`check-grid-matern-cross-shape.py`) computes it by the fast Fourier transform on a periodic grid of $n \times n$ points. It discretizes $-\nabla \cdot S\nabla$ by minus the generator of a random walk on the eight neighbours, with symbol $D_S(\theta) = a(2-2\cos\theta_1)+c(2-2\cos\theta_2)+b_+(2-2\cos(\theta_1+\theta_2))+b_-(2-2\cos(\theta_1-\theta_2))$, where $a = S_{11}-|S_{12}|$, $c = S_{22}-|S_{12}|$ and $b_\pm = \max(\pm S_{12}, 0)$; for the diagonal matrices below it is the five-point stencil. The lattice Matérn law of index γ has the symbol $(1+D_S)^{-\gamma}$, and the step is from $t^2 I$ to $t^2\,\mathrm{diag}(1+\varepsilon, 1)$, with t the aperture in grid spacings and $n = 64t$. In the computation the negative mass grows linearly in $\log t$; this lattice law is a numerical finding and is not proved. At $\gamma = 1$ and $\varepsilon = 0.01$, 0.1, 0.5 the least-squares slopes against $\log t$ over $t \in \{4, 8, 16, 32\}$ are 0.0030, 0.0283 and 0.1064. The heuristic comparison is with 0.0032, 0.0289 and 0.1061, the values $\varepsilon/(\pi(1+\varepsilon))$ of Proposition E.4(4), which are the rates of its clause (3) if the shells are cut off at the grid scale, $a \approx 1/t$. At $\gamma = 2$ the slopes are 0.0060, 0.0544 and 0.1768. The same heuristic, applied to the $|x|^{-2}$ term of the increment, gives the rates 0.0063, 0.0553 and 0.1777. That term is read off the degree-0 limit h of its transform (the proof of Proposition 4.15), and the same script computes the rates. The grid agrees with them within 4 %, as at index 1. The first-order term in P, γ times the index-1 values, is not the right comparison: at $\varepsilon = 0.5$ it gives 0.2122. The index-2 rate is computed and not proved.

F Further results on the smaller groups

The statements of this appendix fill the rows of the smaller groups in Table 2, which §4.9 summarizes. They show what a smaller group gains: families tied to one frame, a ray column whose isotropic member no longer fixes the family, and a stable order whose families lie strictly between their neighbours. The proofs close each subsection.

F.1 Frames

The affine image of a product family cascades along its own frame's axes, and only the Gaussian loses its frame.

Proposition F.1 (frame families). *Let $d \geq 1$ and, for $i = 1, \dots, d$, let $(\nu^{(i)}_{s,s'})_{0\leq s\leq s'}$ be probability measures on $\mathbb{R}$ with $\nu^{(i)}_{s,s} = \delta_0$ and $\nu^{(i)}_{s,s''} = \nu^{(i)}_{s',s''} * \nu^{(i)}_{s,s'}$ for $s \leq s' \leq s''$. For $s \leq s'$ in $[0,\infty)^d$, coordinatewise, put $\mu^{\otimes}_{s,s'} := \nu^{(1)}_{s_1,s'_1} \otimes \cdots \otimes \nu^{(d)}_{s_d,s'_d}$, and for $A \in \mathrm{GL}(d)$ put $\mu_{A;s,s'} := A_*\mu^{\otimes}_{s,s'}$, the* frame family.

1. (Cascade in the frame, and covariance.) $\widehat{\mu}_{A;s,s'}(\omega) = \prod_i \widehat{\nu}^{(i)}_{s_i,s'_i}\big((A^{\mathsf{T}}\omega)_i\big)$; $\mu_{A;s,s} = \delta_0$ *and* $\mu_{A;s,s''} = \mu_{A;s',s''} * \mu_{A;s,s'}$ *for $s \leq s' \leq s''$; and $B_*\mu_{A;s,s'} = \mu_{BA;s,s'}$ for $B \in \mathrm{GL}(d)$.*

2. (The Gaussian frame family.) *If $b \geq 0$ and $\nu^{(i)}_{s,s'}$ is the centred Gaussian of variance $2b(s'^2 - s^2)$ for every i, then $\mu_{A;0,s}$ is the centred Gaussian of covariance $2b\,A\,\mathrm{diag}(s_1^2, \dots, s_d^2)A^{\mathsf{T}}$, and so depends on (A, s) only through $A\,\mathrm{diag}(s^2)A^{\mathsf{T}}$.*

3. (Only the Gaussian loses its frame.) *Let $d \geq 2$, and suppose $\mu_{A;0,s}$ depends on $(A, s) \in \mathrm{GL}(d) \times (0,\infty)^d$ only through $A\,\mathrm{diag}(s^2)A^{\mathsf{T}}$. Then there is a $b \geq 0$ with $\nu^{(i)}_{0,t}$ the centred Gaussian of variance $2bt^2$ for every i and every $t > 0$ (the point mass δ_0 when $b = 0$).*

Clauses (2) and (3) say that the frame drops out exactly when every factor is the same Gaussian, with one constant b for every axis. The core of (3) is Kelker's lemma (Remark 3.14; [31], Lemma 5, p. 423): with A orthogonal and equal scales, the hypothesis makes the product law of the factors rotation invariant.

The converse holds in a fixed frame, under weak hypotheses: covariance under the positive diagonal matrices only, with no rotation and no reflection, and steps along one axis at a time.

Proposition F.2 (a fixed frame: the families are products)**.** *Let $d \geq 1$. For $s \in (0,\infty)^d$ write $D_s := \operatorname{diag}(s_1,\dots,s_d)$, and $\mathbf{1} := (1,\dots,1)$. Let $(\mu_s)_{s\in(0,\infty)^d}$ be probability measures on $\mathbb{R}^d$ with*

(D1) (covariance under the diagonal group) *$(D_\delta)_*\mu_s = \mu_{D_\delta s}$ for all $\delta, s \in (0,\infty)^d$;*

(D2) (a cascade along each axis) *for every $i \leq d$ and every $\lambda > 1$ there is a probability measure $\rho_{i,\lambda}$ on $\mathbb{R}^d$ with $\mu_{\mathbf{1}+(\lambda-1)e_i} = \mu_{\mathbf{1}} * \rho_{i,\lambda}$.*

Then there are self-decomposable laws $\nu_1,\dots,\nu_d$ on $\mathbb{R}$ (in the sense of (2.2) at $d=1$) with

$$\mu_s = (s_1)_*\nu_1 \otimes \cdots \otimes (s_d)_*\nu_d \qquad \text{for every } s \in (0,\infty)^d,$$

where $(s_i)_*$ is the image under $x \mapsto s_i x$ on $\mathbb{R}$. Conversely, for any self-decomposable laws $\nu_1,\dots,\nu_d$ on $\mathbb{R}$ this family satisfies (D1) and (D2), and it cascades along the coordinatewise order: $\mu_{s'} = \mu_s * (\rho^{(1)}_{s_1,s'_1} \otimes \cdots \otimes \rho^{(d)}_{s_d,s'_d})$ for $s \leq s'$, with probability measures $\rho^{(i)}_{s_i,s'_i}$ on $\mathbb{R}$.

So the non-Gaussian anisotropic families that cascade at a fixed orientation are exactly the products of d independent self-decomposable laws, one per axis, and the factors may differ. By Proposition F.1(3) the non-Gaussian factors then fix the frame. A product family gains anisotropy along its frame's axes as fast per step as the Gaussian does and cannot turn, while the stable orbits turn, with a smaller gain per step (Proposition 4.7). Turning alone therefore does not exclude non-Gaussian kernels. The proof of the product form uses the characteristic function of Lemma 3.8.

Proof of Proposition F.1. Clause (1). The transform of a product measure is the product of the transforms, and $\widehat{A_*\mu}(\omega) = \widehat{\mu}(A^{\mathsf{T}}\omega)$. The cascade is coordinatewise multiplication of transforms, and $(BA)_* = B_*A_*$ gives the covariance.

Clause (2). By clause (1), $\widehat{\mu}_{A;0,s}(\omega) = \exp[-b\sum_i s_i^2 (A^{\mathsf{T}}\omega)_i^2] = \exp[-b\langle\omega, A\operatorname{diag}(s^2)A^{\mathsf{T}}\omega\rangle]$.

Clause (3). For $Q \in \mathrm{O}(d)$, $Q\operatorname{diag}(\mathbf{1})Q^{\mathsf{T}} = I$, so the hypothesis makes the product law $\mu_{0,\mathbf{1}}^{\otimes}$ rotation invariant: its transform is $f(|\omega|^2)$ with f continuous, real and $f(0)=1$, and each factor has the transform $t \mapsto f(t^2)$. As $d \geq 2$, $\omega = \sqrt{x}\,e_1 + \sqrt{y}\,e_2$ gives $f(x+y) = f(x)f(y)$. Then $f > 0$, and $G := -\log f$ is continuous, nonnegative and additive, so $G(u) = bu$ by the last paragraph of the proof of Lemma 3.10, the same Cauchy equation on which Theorem 3.6 rests. So each $\nu^{(i)}_{0,1}$ is the centred Gaussian of variance $2b$, and $A = \operatorname{diag}(s)^{-1}$ in the hypothesis gives the variance $2bs_i^2$ at s. □

Proof of Proposition F.2 (sketch). (D1) gives $\mu_s = (D_s)_*\mu_{\mathbf{1}}$; put $\varphi := \widehat{\mu}_{\mathbf{1}}$.

A step along axis i is a blur along axis i. With $H := e_i^{\perp}$, (D2) gives $\varphi(\lambda x e_i + \eta) = \varphi(xe_i+\eta)\,\widehat{\rho}_{i,\lambda}(xe_i+\eta)$. At $x=0$, $\widehat{\rho}_{i,\lambda} = 1$ on a neighbourhood of 0 in H, so by Lemma 3.8, in its neighbourhood form, $\widehat{\rho}_{i,\lambda}(xe_i+\eta) = \widehat{\rho}_{i,\lambda}(xe_i)$. The same identity at $\eta = 0$ eliminates $\widehat{\rho}_{i,\lambda}$:

$$\varphi(\lambda x e_i + \eta)\,\varphi(xe_i) = \varphi(xe_i+\eta)\,\varphi(\lambda x e_i) \qquad (x \in \mathbb{R},\ \eta \in H,\ \lambda > 1).$$

For $x_0 \neq 0$ and x strictly between 0 and x_0 take $\lambda := x_0/x$. Letting $x \to 0$, continuity and $\varphi(0) = 1$ give $\varphi(x_0e_i + \eta) = \varphi(\eta)\,\varphi(x_0e_i)$, which splits off axis i.

The product. Repeating over the axes, $\varphi(\omega) = \prod_i \varphi(\omega_i e_i)$, so $\mu_{\mathbf{1}}$ is the product of its marginals ν_i. The image of (D2) under the i-th coordinate is $(\lambda)_*\nu_i = \nu_i * \rho'$, which is self-decomposability on the line.

Converse. D_δ acts coordinatewise, and self-decomposability of each factor gives the increments, a convolution of product measures being the product of the convolutions. This is Proposition F.1(1) in the standard frame. □

F.2 The ray column

Along rays a family of any group is a continuous covariant field of self-decomposable laws (Proposition 3.4(2)), and the bookkeeping of Lemma 3.3 serves every cell below. For the similarities the parameter set is a $(d-1)$-parameter set of orbits, and the isotropic member no longer determines the family.

Proposition F.3 (similarity covariance along rays: the families, and the parameter set)**.** *Let $d \geq 1$, let $\mathrm{S}(d) := \{cQ : c > 0,\ Q \in \mathrm{O}(d)\}$ be the similarities, and let W be the ray order. For $\Sigma \in \mathrm{PSD}(d) \setminus \{0\}$ write $\sigma(\Sigma) \in \mathbb{R}^d$ for its eigenvalues in nonincreasing order divided by $\operatorname{tr}\Sigma$, and put $\Delta_d := \{\sigma \in \mathbb{R}^d : \sigma_1 \geq \cdots \geq \sigma_d \geq 0,\ \sigma_1 + \cdots + \sigma_d = 1\}$.*

1. (The parameter set.) *$(\mathrm{S}(d), W)$ is a wedge datum. Two matrices $\Sigma, \Sigma' \in \mathrm{PSD}(d) \setminus \{0\}$ lie in one orbit of $\mathrm{S}(d)$, acting by $A \cdot \Sigma = A\Sigma A^{\mathsf{T}}$, exactly when $\sigma(\Sigma) = \sigma(\Sigma')$, and each orbit contains exactly one matrix $\operatorname{diag}(\sigma)$ with $\sigma \in \Delta_d$; so the orbits in $\mathrm{PSD}(d) \setminus \{0\}$ are indexed by Δ_d, a simplex of dimension $d-1$. The stabilizer of $\Sigma \neq 0$ in $\mathrm{S}(d)$ is $\{Q \in \mathrm{O}(d) : Q\Sigma = \Sigma Q\}$, and it is $\mathrm{O}(d)$ exactly when Σ is a multiple of I.*

2. (The families.) *The kernels from scale zero of the $(\mathrm{S}(d), W)$-cascade families of kernels are the maps of Proposition 3.4(2) with $G = \mathrm{S}(d)$. In such a family every μ_Σ is symmetric, $(-I)_*\mu_\Sigma = \mu_\Sigma$, and invariant under the stabilizer of Σ, and $\mu_{cQ\operatorname{diag}(\sigma)Q^{\mathsf{T}}} = (\sqrt{c}\,Q)_*\mu_{\operatorname{diag}(\sigma)}$ for $c > 0$, $Q \in \mathrm{O}(d)$ and $\sigma \in \Delta_d$: the family is determined by its kernels at the matrices $\operatorname{diag}(\sigma)$, $\sigma \in \Delta_d$.*

3. (The isotropic member does not determine the family.) *In such a family $\mu_{sI} = (\sqrt{s}\,I)_*\mu_I$ for $s \geq 0$, and μ_I is rotation invariant, self-decomposable and not δ_0. Conversely, let μ_I be such a law. Then (i) $\Sigma \mapsto (\Sigma^{1/2})_*\mu_I$ and (ii) $\Sigma \mapsto \nu_\Sigma := ((\det\Sigma)^{1/(2d)}I)_*\mu_I$ are the kernels from scale zero of $(\mathrm{S}(d), W)$-cascade families with isotropic member μ_I, and (ii) is also the family of kernels from scale zero of an $(\mathrm{S}(d), \text{Loewner})$-cascade family. For $d \geq 2$ the two differ at every $\Sigma \in \mathrm{PD}(d)$ that is not a multiple of I.*

4. (The isotropic members of the similarity row.) *Let W' be an $\mathrm{S}(d)$-invariant partial order on $\mathrm{PSD}(d)$ that contains the rays and is contained in the Loewner order. The isotropic members μ_I of the $(\mathrm{S}(d), W')$-cascade families of kernels are exactly the rotation-invariant self-decomposable laws other than δ_0.*

So the cell (similarities, rays) is large: a continuous field of symmetric self-decomposable laws over the shapes, covariant under similarities. The kernel at a shape is constrained pointwise only by its stabilizer, besides self-decomposability and continuity where eigenvalues meet, and it need not be elliptical. The isotropic member is any rotation-invariant self-decomposable law other than δ_0, as in the general linear row, but for $d \geq 2$ it no longer fixes the family. Clause (4) says that the isotropic members are the same in every column of the similarity row, the Loewner column included, while in the general linear row the Loewner order leaves the Gaussian alone. Which choices of kernels at the matrices $\operatorname{diag}(\sigma)$ extend to a continuous field is problem 4 of §5.1.

For the positive diagonal group the parameter set is larger still: an orbit is one frame's own shapes, and the stabilizer of I is trivial, so μ_I need carry no symmetry.

Proposition F.4 (diagonal covariance along rays: the families, and the parameter set)**.** *Let $d \geq 1$, let $\mathrm{D}_+(d)$ be the group of diagonal matrices with positive diagonal entries, and let W be the ray order. For $\Sigma \in \mathrm{PSD}(d)$ write $\Delta(\Sigma) := \operatorname{diag}(\Sigma_{11}, \ldots, \Sigma_{dd})$, for $s \in (0,\infty)^d$ write $D_s := \operatorname{diag}(s_1, \ldots, s_d)$, and put $\mathrm{Corr}(d) := \{\Theta \in \mathrm{PD}(d) : \Theta_{11} = \cdots = \Theta_{dd} = 1\}$.*

1. (The parameter set.) *$(\mathrm{D}_+(d), W)$ is a wedge datum. Every $\Sigma \in \mathrm{PD}(d)$ has the trivial stabilizer in $\mathrm{D}_+(d)$, and is $D \cdot \Theta = D\Theta D$ for exactly one pair $D \in \mathrm{D}_+(d)$, $\Theta \in \mathrm{Corr}(d)$,*

namely $D = \Delta(\Sigma)^{1/2}$. So the orbits in $\mathrm{PD}(d)$ are indexed by $\mathrm{Corr}(d)$, a nonempty open subset of the affine space of symmetric matrices with unit diagonal, of dimension $d(d-1)/2$; the orbit of I is the set of positive definite diagonal matrices.

2. (The families on the frame's shapes.) *The kernels from scale zero of the $(\mathrm{D}_+(d), W)$-cascade families of kernels are the maps of Proposition 3.4(2) with $G = \mathrm{D}_+(d)$. On the orbit of I they are $\mu_{D_s^2} = (D_s)_*\mu_I$, $s \in (0,\infty)^d$, with μ_I self-decomposable and not δ_0. Conversely, for every self-decomposable $\mu_I \neq \delta_0$ the map $\Sigma \mapsto (\Delta(\Sigma)^{1/2})_*\mu_I$ is the family of kernels from scale zero of a $(\mathrm{D}_+(d), W)$-cascade family.*

3. (The Loewner order: products.) *The kernels from scale zero of a $(\mathrm{D}_+(d), \mathit{Loewner})$-cascade family of kernels are, on the orbit of I,*

$$\mu_{D_s^2} = (s_1)_*\nu_1 \otimes \cdots \otimes (s_d)_*\nu_d, \qquad s \in (0,\infty)^d,$$

with self-decomposable laws $\nu_1, \ldots, \nu_d$ on $\mathbb{R}$, not all δ_0. Conversely, for such laws the map $\Sigma \mapsto (\Delta(\Sigma)^{1/2})_*(\nu_1 \otimes \cdots \otimes \nu_d)$ is the family of kernels from scale zero of a $(\mathrm{D}_+(d), \mathit{Loewner})$-cascade family.

4. (Rays only: no product.) *Let $d \geq 2$ and $\Theta \in \mathrm{Corr}(d)$ not diagonal. The centred Gaussian law $N(0,\Theta)$ is self-decomposable and is not of the form $\nu_1 \otimes \cdots \otimes \nu_d$. So the $(\mathrm{D}_+(d), W)$-family of (2) with $\mu_I = N(0,\Theta)$ is the restriction of no $(\mathrm{D}_+(d), \mathit{Loewner})$-family.*

5. (Off the frame's shapes.) *Let $d \geq 2$ and μ_I rotation invariant, self-decomposable and not δ_0. The maps $\Sigma \mapsto (\Delta(\Sigma)^{1/2})_*\mu_I$ and $\Sigma \mapsto (\Sigma^{1/2})_*\mu_I$ are the kernels from scale zero of two $(\mathrm{D}_+(d), W)$-cascade families with the same isotropic member, and they differ at every $\Sigma \in \mathrm{PD}(d)$ that is not diagonal.*

So the cell (diagonal, rays) is not empty. On the orbit of I it is Proposition 4.5 without rotations: the frame images of one self-decomposable law, any such law. Clauses (3) and (4) locate the product form of Proposition F.2. It is forced by the steps along single axes, which the Loewner order has and the ray order lacks, and not by the group. Off the orbit of I the ray order links nothing across the correlation matrices. The continuous covariant fields over $\mathrm{Corr}(d)$ are part of problem 4 of §5.1.

Proof of Proposition F.3 (sketch). (1). $\mathrm{S}(d)$ is closed in $\mathrm{GL}(d)$, as $AA^{\mathsf{T}} = (AA^{\mathsf{T}})_{11}\, I$ is a closed condition and an invertible A satisfying it is a similarity. A similarity multiplies the eigenvalues and the trace by the same factor, so σ is constant on orbits, and the spectral theorem gives the converse. Δ_d is the convex hull of the affinely independent points $k^{-1}(e_1 + \cdots + e_k)$, $1 \leq k \leq d$. A similarity fixing $\Sigma \neq 0$ has $c = 1$ by the traces, and a Σ commuting with every orthogonal matrix is a multiple of I, by the reflections and permutations of the proof of Lemma A.4.

(2) is Proposition 3.4(2) with (b) of Lemma 3.3(2) at $A = -I$, at the stabilizer, and at $A = \sqrt{c}\,Q$.

(3). μ_I is invariant under the stabilizer $\mathrm{O}(d)$ of I, as in Lemma A.5(5). Family (i) is Proposition 3.4(3) restricted by Lemma 3.3(1). For (ii), det is monotone along the Loewner order, so with $\eta := (\det\Sigma)^{1/(2d)} \leq \eta' := (\det\Sigma')^{1/(2d)}$ self-decomposability at η/η' gives the increment, and Lemma 3.3(3) gives the family. For $d \geq 2$, if the two kernels agree at $\Sigma \in \mathrm{PD}(d)$, then with $t := (\det\Sigma)^{1/(2d)}$ the matrix $t^{-1}\Sigma^{1/2}$ fixes μ_I, which is rotation invariant and not δ_0, so it is orthogonal by Lemma A.5(2), and $\Sigma = t^2 I$.

(4). A family along W' restricts to the rays, and the family (ii) of (3) restricts to W', which is contained in the Loewner order. □

Proof of Proposition F.4 (sketch). (1). $(D\Sigma D)_{ii} = s_i^2\Sigma_{ii}$ forces $D = I$ on a stabilizer, and $\Theta := \Delta(\Sigma)^{-1/2}\Sigma\Delta(\Sigma)^{-1/2}$ is the unique correlation matrix with $\Delta(\Sigma)^{1/2}\cdot\Theta = \Sigma$. $\mathrm{Corr}(d)$ is the

intersection of the affine space with the open set of matrices whose quadratic form is positive on the compact unit sphere.

(2). The first sentence is Proposition 3.4(2), and (b) of Lemma 3.3(2) at $A = D_s$ gives $\mu_{D_s^2} = (D_s)_*\mu_I$. Conversely $\Delta(D\Sigma D)^{1/2} = D\,\Delta(\Sigma)^{1/2}$ gives (b) for $\Sigma \mapsto (\Delta(\Sigma)^{1/2})_*\mu_I$, and (a), (c) and self-decomposability are immediate.

(3). On the orbit of I the axis steps $I \le D^2_{\mathbf{1}+(\lambda-1)e_i}$ are Loewner steps, so (D1) and (D2) of Proposition F.2 hold and the product form follows. Conversely $\Sigma \le \Sigma'$ gives $\Sigma_{ii} \le \Sigma'_{ii}$, and self-decomposability of each factor gives a product increment.

(4). $N(0,\Theta)$ is self-decomposable, and a product with unit diagonal would force $\Theta = I$.

(5). If the two kernels agree at $\Sigma \in \mathrm{PD}(d)$, then with $D := \Delta(\Sigma)^{1/2}$ the matrix $D^{-1}\Sigma^{1/2}$ fixes μ_I, which is rotation invariant and not δ_0, so it is orthogonal by Lemma A.5(2), and $\Sigma = D^2$ is diagonal. □

F.3 The diagonal group along the stable order

On the frame's shapes the stable order's steps are the diagonal exponents of the cone $\mathfrak{k}_\alpha$, so the isotropic member must be operator-self-decomposable for each of them. The axis steps that force products along the Loewner order are rank-deficient and absent.

Proposition F.5 (diagonal covariance along the stable order: the families on the frame's shapes)**.** *Let $d \ge 2$ and $0 < \alpha < 2$. Let $\mathrm{D}_+(d)$, D_s, $\Delta(\Sigma)$ and $\mathrm{Corr}(d)$ be as in Proposition F.4, let $\le_\alpha$ be the stable order on $\mathrm{PSD}(d)$ of Proposition 4.20(1), and use B-self-decomposability as in Proposition 4.10. Write $\mathcal{D}_\alpha$ for the set of diagonal matrices $\mathrm{diag}(b)$, $b \in \mathbb{R}^d$, with $(d+\alpha)\min_i b_i \ge b_1 + \cdots + b_d$; these are the diagonal matrices of the cone $\mathfrak{k}_\alpha$ of Proposition 4.10.*

1. (The datum.) *$(\mathrm{D}_+(d), \le_\alpha)$ is a wedge datum, and its orbits in $\mathrm{PD}(d)$ are those of Proposition F.4(1), indexed by $\mathrm{Corr}(d)$. On the orbit of I, $D_s^2 \le_\alpha D_{s'}^2$ if and only if $u := (s'_1/s_1, \ldots, s'_d/s_d)$ satisfies $(\min_i u_i)^{d+\alpha} \ge u_1 \cdots u_d$.*

2. (On the frame's shapes.) *The kernels from scale zero of a $(\mathrm{D}_+(d), \le_\alpha)$-cascade family of kernels are, on the orbit of I, $\mu_{D_s^2} = (D_s)_*\mu_I$, $s \in (0,\infty)^d$, where $\mu_I \ne \delta_0$ is B-self-decomposable for every $B \in \mathcal{D}_\alpha$; in particular μ_I is self-decomposable.*

3. (Two sources of families.) *(a) For self-decomposable laws $\nu_1, \ldots, \nu_d$ on $\mathbb{R}$, not all δ_0, the map $\Sigma \mapsto (\Delta(\Sigma)^{1/2})_*(\nu_1 \otimes \cdots \otimes \nu_d)$ is the family of kernels from scale zero of a $(\mathrm{D}_+(d), \le_\alpha)$-cascade family of kernels. (b) For a profile k as in Proposition 4.13 with $R(\lambda) \ge \lambda^\alpha$ for every $\lambda \ge 1$, and μ_I the law of that proposition's clause (1), so is $\Sigma \mapsto (\Sigma^{1/2})_*\mu_I$; in particular for $\mu_I = S_\beta(I)$, $\alpha \le \beta < 2$.*

4. (Strictly between its neighbours.) *(a) Let $|r| < 1$ with $r^2 > 4(1+\alpha)/(2+\alpha)^2$, and let $\Theta_r \in \mathrm{Corr}(d)$ have the entries $(\Theta_r)_{12} = (\Theta_r)_{21} = r$ and agree with I elsewhere. The centred Gaussian law $N(0,\Theta_r)$ is self-decomposable and is not B-self-decomposable for $B := I + \alpha\, e_1 e_1^{\mathsf{T}} \in \mathcal{D}_\alpha$. So the $(\mathrm{D}_+(d), \text{rays})$-family of Proposition F.4(2) with $\mu_I = N(0,\Theta_r)$ is the restriction of no $(\mathrm{D}_+(d), \le_\alpha)$-family. (b) $S_\alpha(I)$ is not of the form $\nu_1 \otimes \cdots \otimes \nu_d$. So the family of (3)(b) with $\mu_I = S_\alpha(I)$ is the restriction of no $(\mathrm{D}_+(d), \text{Loewner})$-family. (c) Let σ_α be the law on $\mathbb{R}$ with transform $e^{-|\tau|^\alpha}$ (Lemma 4.6(2) at $d = 1$). The map $\Sigma \mapsto (\Sigma^{1/2})_* S_\alpha(I) * (\Delta(\Sigma)^{1/2})_*(\sigma_\alpha \otimes \cdots \otimes \sigma_\alpha)$ is the family of kernels from scale zero of a $(\mathrm{D}_+(d), \le_\alpha)$-cascade family that is the restriction neither of a $(\mathrm{GL}(d), \le_\alpha)$-family nor of a $(\mathrm{D}_+(d), \text{Loewner})$-family.*

So on the frame's shapes the cell (diagonal, stable order) lies strictly between its two neighbours in the diagonal row: every restricted (diagonal, Loewner) family belongs to it, every family of it restricts to the ray cell, neither inclusion is an equality, and the cell is not the

union of the restrictions of the (diagonal, Loewner) cell and the general linear cell. The condition on the isotropic member sits between self-decomposability, which the ray cell requires, and the product form, which the Loewner order's axis steps force. Whether every such law is the isotropic member of a family asks for an extension off the frame's shapes, where $\leq_\alpha$ relates matrices of different correlation orbits. That question and the families off the frame's shapes are part of problem 4 of §5.1.

Proof of Proposition F.5 (sketch). (1). Proposition 4.20(1) makes $(\mathrm{D}_+(d), \leq_\alpha)$ a wedge datum, and the orbits are those of Proposition F.4(1). The relative step of D_s^2 and $D_{s'}^2$ is D_u^2, and condition (c) of Proposition 4.7(1) is the stated inequality squared.

(2). (K3) of Definition 3.1 with $A = D_s$ gives the frame images. For $B = \mathrm{diag}(b) \in \mathcal{D}_\alpha$ and $u_i := e^{tb_i}$, (1) gives $I \leq_\alpha D_u^2 = e^{2tB}$, and (K1) at $0 \leq_\alpha I \leq_\alpha D_u^2$ is B-self-decomposability in the sense of Proposition 4.10; $B = I$ gives self-decomposability.

(3). (a) restricts the families of Proposition F.4(3) by Lemma 3.3(1), $\leq_\alpha$ being contained in the Loewner order. (b) restricts those of Proposition 4.20(2), the profiles of Proposition 4.13 included, to $\mathrm{D}_+(d)$.

(4). (a) An increment for e^{-tB} would make $\Theta_r - e^{-tB}\Theta_r e^{-tB}$ positive semidefinite, hence $B\Theta_r + \Theta_r B$, whose principal 2×2 block has the negative determinant $4(1+\alpha) - (2+\alpha)^2 r^2$. (b) $e^{-|\omega|^\alpha} = \prod_i e^{-|\omega_i|^\alpha}$ fails at $e_1 + e_2$, the transform being that of Lemma 4.6(2). (c) The convolution of the two families of (3) is a family by Lemma 3.3(3); its μ_I is not rotation invariant, which the isotropic member of a general linear family is (Lemma A.5(5)), and not a product. □

G One linear flow

The families covariant under one group $\{e^{sB}\}$ that cascade along its curve are the operator-self-decomposable laws. They lie outside the scheme, the curve not being an order on $\mathrm{PSD}(d)$, and they are the row "One linear flow" below the others in Table 2. The anisotropy of the exponents B of the rotation-invariant non-Gaussian ones is bounded by Proposition 4.11 in §4.6.

The families of one flow are characterized as the ray families are.

Proposition G.1 (the families of one linear flow)**.** *Let $d \geq 1$ and let B be a real $d \times d$ matrix. Let $(\mu_t)_{t\in\mathbb{R}}$ be probability measures on $\mathbb{R}^d$ with*

(F1) (covariance under the flow) $(e^{sB})_*\mu_t = \mu_{t+s}$ *for all $s, t \in \mathbb{R}$;*

(F2) (a cascade along the flow) *for every $u > 0$ there is a probability measure ρ_u on $\mathbb{R}^d$ with $\mu_u = \mu_0 * \rho_u$.*

1. (The characterization.) *Then μ_0 is B-self-decomposable (Proposition 4.10), $\mu_t = (e^{tB})_*\mu_0$ for every t, and the family cascades between any two of its members: $\mu_t = \mu_s * (e^{sB})_*\rho_{t-s}$ for $s < t$. Conversely, for every B-self-decomposable law μ_0 the family $\mu_t := (e^{tB})_*\mu_0$ satisfies (F1) and (F2).*

2. (The origin.) *If every eigenvalue of B has positive real part, that is if $e^{tB} \to 0$ as $t \to -\infty$, then $\mu_t \to \delta_0$ weakly as $t \to -\infty$.*

3. (The curve of apertures.) *For $\Sigma_0 \in \mathrm{PD}(d)$ put $\Sigma(t) := e^{tB}\Sigma_0 e^{tB^\mathsf{T}}$. Then $\Sigma(s) \leq \Sigma(t)$ for all $s \leq t$ in the Loewner order if and only if $B\Sigma_0 + \Sigma_0 B^\mathsf{T} \in \mathrm{PSD}(d)$; and $\Sigma(t) \to 0$ as $t \to -\infty$ if and only if $e^{tB} \to 0$.*

4. (The Gaussian rides exactly the monotone curves.) *For $b > 0$ the family $\mu_t := N(0, 2b\,\Sigma(t))$ satisfies (F1), and it satisfies (F2) if and only if $B\Sigma_0 + \Sigma_0 B^\mathsf{T} \in \mathrm{PSD}(d)$.*

5. (Two kinds of member.) *If* $B = \operatorname{diag}(b_1, \dots, b_d)$ *with every* $b_i > 0$*, then every product* $\nu_1 \otimes \cdots \otimes \nu_d$ *of self-decomposable laws on* $\mathbb{R}$ *is* B*-self-decomposable. If* $d \geq 2$*,* $0 < \alpha < 2$ *and* $(d+\alpha)\lambda_{\min}(\operatorname{sym} B) \geq \operatorname{tr} B$*, then* $S_\alpha(I)$ *is* B*-self-decomposable and is not a product law in any orthonormal frame.*

For contracting B these are Urbanik's operator-self-decomposable laws [62]. Sato and Yamazato characterize them as the limit laws of the processes of Ornstein–Uhlenbeck type, in which a process with independent increments replaces the Brownian motion ([59], p. 73). Jurek writes each as the law of $\int_0^\infty e^{-tB}\, dY(t)$ for a process Y with stationary independent increments and $\mathbf{E}\log(1+|Y(1)|) < \infty$ ([28], p. 385). Clause (1) for contracting B is close to Hazod and Siebert's characterization of self-decomposability along a contracting group by continuous hemigroups ([22], Thm. 2.14.21, (i)⇔(v), p. 383). Their hemigroups are indexed by $t \geq 0$ and continuous, which (F1) and (F2) do not ask. By clause (1), clause (4) is Jurek's description of the exponents of a centred Gaussian law of covariance S, the B with $BS + SB^{\mathsf{T}} \geq 0$ in this paper's sign ([29], Example, p. 1057). With $B = I$ the proposition is the ray cascade of Proposition 4.5 without the rest of the affine group, and so without rotation invariance. With B diagonal it contains the products of Proposition F.2 along every curve of the frame at once, and clause (5) shows that it is larger than both: in the plane $S_\alpha(I)$ rides the curves of $B = \operatorname{diag}(1, b)$ exactly for $1/(1+\alpha) \leq b \leq 1+\alpha$ (Proposition 4.10). For vision the row is shape drift locked to scale: the aperture's shape is a fixed function of its scale, set by B, and does not adapt to the image. Only for the Gaussian is the label $\Sigma(t)$ the kernel's covariance, and then a cascade along the curve is exactly a Loewner-monotone curve (clause (4)). With Propositions F.2 and 4.11 this separates two obstructions. Frame families take the Gaussian's anisotropy per step and cannot turn. Rotation-invariant non-Gaussian laws turn, with the skew part of B free, and have a bounded anisotropy budget, while the Gaussian has neither limit.

Proof. Clause (1) is the argument of Proposition 4.5 with λI replaced by e^{sB}. (F1) at $t = 0$ gives $\mu_s = (e^{sB})_*\mu_0$. (F2) reads $(e^{uB})_*\mu_0 = \mu_0 * \rho_u$, and applying $(e^{-uB})_*$ gives $\mu_0 = (e^{-uB})_*\mu_0 * (e^{-uB})_*\rho_u$. For $s < t$, $\mu_t = (e^{sB})_*(\mu_0 * \rho_{t-s}) = \mu_s * (e^{sB})_*\rho_{t-s}$. The converse is the same computation read backwards.

Clause (2). $\int f\, d\mu_t = \int f(e^{tB}x)\, \mu_0(dx) \to f(0)$ by dominated convergence.

Clause (3). With $C := B\Sigma_0 + \Sigma_0 B^{\mathsf{T}}$ and $v(t) := e^{tB^{\mathsf{T}}}\omega$, $\frac{d}{dt}\langle \omega, \Sigma(t)\omega\rangle = \langle v(t), Cv(t)\rangle$, so $C \in \mathrm{PSD}(d)$ makes every $t \mapsto \langle \omega, \Sigma(t)\omega\rangle$ nondecreasing, and conversely their derivatives at 0 are the $\langle \omega, C\omega\rangle$. Also $\Sigma(t) = M(t)M(t)^{\mathsf{T}}$ with $M(t) := e^{tB}\Sigma_0^{1/2}$, whose squared entries are bounded by the diagonal of $\Sigma(t)$, so $\Sigma(t)$ tends to 0 exactly when e^{tB} does.

Clause (4). $(e^{sB})_*N(0, 2b\Sigma(t)) = N(0, 2b\Sigma(t+s))$, which is (F1). The Gaussian transforms have no zero, so an increment of $N(0, 2b\Sigma_0)$ to $N(0, 2b\Sigma(u))$ is unique, with the transform $e^{-b\langle \omega, (\Sigma(u)-\Sigma_0)\omega\rangle}$. That is a characteristic function exactly when $\Sigma(u) \geq \Sigma_0$: a negative eigenvalue of $\Sigma(u) - \Sigma_0$ would give it modulus above 1. So (F2) holds exactly when $\Sigma_0 \leq \Sigma(u)$ for every $u > 0$. By (F1)'s transport $\Sigma(s+u) = e^{sB}\Sigma(u)e^{sB^{\mathsf{T}}}$, and as a congruence preserves the Loewner order, that is the monotonicity of the whole curve, which clause (3) characterizes.

Clause (5). For diagonal B, e^{-uB} acts coordinatewise and each factor is self-decomposable. The second sentence's first half is Proposition 4.10; for the second half, $S_\alpha(I)$ is rotation invariant (Lemma 4.6(2)), so a product form in one orthonormal frame is one in the standard frame. Its factors then have the transforms $e^{-|t|^\alpha}$, and at $e_1 + e_2$ the product gives $e^{-2} = e^{-2^{\alpha/2}}$, which forces $\alpha = 2$. □

H The similarities with the Loewner order

The statements of this appendix are summarized in §4.9, under "The similarities with the Loewner order", and in the cell of Table 2. They concern families of kernels from scale zero

covariant under the similarities whose Loewner steps have infinitely divisible increments, or positive ones when there is no Gaussian part. The question throughout is how far from rotation invariance the non-Gaussian part of a kernel can be. The appendix answers it in three steps. The first subsection takes the part to be elliptical, an image of one law, and bounds its eccentricity. The second takes a general part, compares it with its limit at the thinnest apertures, and shows which hypotheses bound its shape and which do not. The last subsection shows that every family of the scheme has the infinitely divisible increments that the first two assume.

H.1 An elliptical non-Gaussian part

The first two statements take the non-Gaussian part to be elliptical.

Proposition H.1 (similarity-covariant families with a stable part)**.** *Let $d \geq 2$ and $0 < \alpha < 2$. For maps $h : \mathrm{PD}(d) \to \mathrm{PD}(d)$ and $C : \mathrm{PD}(d) \to \mathrm{PSD}(d)$ put $\mu_\Sigma := N(0, C(\Sigma)) * S_\alpha(h(\Sigma))$, with $N(0, C)$ the centred Gaussian law of covariance C.*

1. (The stable parts are ordered.) *If $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ for a positive finite measure ρ, then $h(\Sigma) \preceq_\alpha h(\Sigma')$.*

2. (Bounded shape.) *Suppose that $h(cQ\Sigma Q^{\mathsf{T}}) = c\,Q\,h(\Sigma)\,Q^{\mathsf{T}}$ for $c > 0$ and $Q \in \mathrm{O}(d)$, that h extends continuously to $\mathrm{PSD}(d) \setminus \{0\}$ with values in $\mathrm{PD}(d)$, and that every Loewner step $\Sigma \leq \Sigma'$ in $\mathrm{PD}(d)$ has an increment as in (1). Then $h(I) = h_0 I$ with $h_0 > 0$, and with $c_1 := \lambda_{\max}(h(e_1 e_1^{\mathsf{T}})) > 0$,*

$$\frac{\lambda_{\max}(h(\Sigma))}{\lambda_{\min}(h(\Sigma))} \leq \Big(\frac{h_0}{c_1}\Big)^{\alpha} \qquad \text{for every } \Sigma \in \mathrm{PD}(d).$$

3. (A stable part that is not rotation invariant.) *Let $b \geq 0$ and $\kappa > 0$ with $\alpha\kappa^2 - (d+1)\kappa - 1 \geq 0$, and put $h(\Sigma) := \Sigma + \kappa\,\mathrm{tr}(\Sigma) I$ and $C(\Sigma) := 2b\Sigma$ for $\Sigma \in \mathrm{PSD}(d)$, with $S_\alpha(0) := \delta_0$. For every $\Sigma \leq \Sigma'$ in $\mathrm{PSD}(d)$ there is a probability measure $\rho_{\Sigma,\Sigma'}$ with $\mu_{\Sigma'} = \mu_\Sigma * \rho_{\Sigma,\Sigma'}$, and the family $(\rho_{\Sigma,\Sigma'})$ satisfies (K1), (K2) and (K4) of Definition 3.2, and (K3) for every similarity $A = cQ$, $c > 0$, $Q \in \mathrm{O}(d)$. Its stable part $S_\alpha(h(\Sigma))$ is rotation invariant only when Σ is a multiple of I.*

In clause (3) the non-Gaussian part carries the shape of Σ, and $b = 0$ is allowed, so no Gaussian part is needed for that. Clause (2)'s bound tends to 1 as $\alpha \to 0$. For clause (3)'s family that bound is $((1 + d\kappa)/(1 + \kappa))^\alpha$, and the eccentricity of the stable part is at most $(1 + \kappa)/\kappa$, its value at rank one. The condition on κ is sufficient, not sharp. Without the extension of h to rank one with values in $\mathrm{PD}(d)$ clause (2) fails (Proposition H.6).

Sketch. (1). The increment has the transform (4.2) with $\frac{1}{2}(C(\Sigma') - C(\Sigma))$ for its matrix C, $H = h(\Sigma)$ and $H' = h(\Sigma')$, so Lemma 4.8 orders the Lévy densities, which is $h(\Sigma) \preceq_\alpha h(\Sigma')$ by Proposition 4.7(1).

(2). Equivariance under $\mathrm{O}(d)$ gives $h(I) = h_0 I$, as in the proof of Lemma A.4. Scale so that $\lambda_{\max}(\Sigma) = 1$, and sandwich $Q\,\mathrm{diag}(1, \varepsilon, \dots, \varepsilon)Q^{\mathsf{T}} \leq \Sigma \leq I$ with Qe_1 a top eigenvector. Let $\eta_1 \leq \cdots \leq \eta_d$ be the eigenvalues of $h(\Sigma)$. By (1) and the inclusion of $\preceq_\alpha$ in the Loewner order, letting $\varepsilon \to 0$ gives $Q\,h(e_1 e_1^{\mathsf{T}})\,Q^{\mathsf{T}} \leq h(\Sigma)$, so $\eta_d \geq c_1$. For $h(\Sigma) \preceq_\alpha h_0 I$ the relative step $M = h_0 h(\Sigma)^{-1}$ has $\lambda_{\min}(M) = h_0/\eta_d$ and $\det M = h_0^d/(\eta_1 \cdots \eta_d)$, so condition (c) of Proposition 4.7(1) reads

$$\prod_i \frac{\eta_i}{\eta_d} \geq \Big(\frac{\eta_d}{h_0}\Big)^{\alpha}, \qquad \text{whence} \qquad \frac{\eta_d}{\eta_1} \leq \Big(\frac{h_0}{\eta_d}\Big)^{\alpha} \leq \Big(\frac{h_0}{c_1}\Big)^{\alpha},$$

the left-hand product being at most η_1/η_d.

(3). Let $P := \Sigma' - \Sigma$ with $\Sigma \neq 0$, and put $H := h(\Sigma)$, $\tau := \operatorname{tr}\Sigma$ and $u := \operatorname{tr}(P)/\tau$. The relative step $M = H^{-1/2}h(\Sigma')H^{-1/2} = I + H^{-1/2}PH^{-1/2} + \kappa\operatorname{tr}(P)H^{-1}$ has $\lambda_{\min}(M) \geq 1+\kappa u/(1+\kappa)$ and $\operatorname{tr} M \leq d+(1+d\kappa)u/\kappa$, because $\kappa\tau I \leq H \leq (1+\kappa)\tau I$. With $A := (1+d\kappa)/\kappa$ and $B := \kappa/(1+\kappa)$ these bounds read $\operatorname{tr} M \leq d + Au$ and $\lambda_{\min}(M) \geq 1 + Bu$, and the condition on κ, $(d+\alpha)\kappa^2 \geq (1+d\kappa)(1+\kappa)$, is $A \leq (d+\alpha)B$. The inequality of arithmetic and geometric means for the eigenvalues of M, and Bernoulli's inequality, give

$$\det M \leq \Big(\frac{\operatorname{tr} M}{d}\Big)^d \leq \Big(1 + \frac{d+\alpha}{d}\,\frac{\kappa u}{1+\kappa}\Big)^d \leq \lambda_{\min}(M)^{d+\alpha},$$

which is condition (c). So the densities are ordered, and Proposition 4.7(1) gives the stable increment; the Gaussian increment is $N(0, 2bP)$. The axioms are read off the telescoping transforms. □

For an elliptical part of any self-decomposable profile, infinitely divisible increments order the parts, and the bound on the shape is read off the dilation ratio.

Proposition H.2 (similarity-covariant families with an elliptical part: bounded shape, and shape-blind at index 0)**.** *Let $d \geq 2$, let k be as in Proposition 4.13 and right-continuous, with dilation ratio R, and write $\nu_H := (H^{1/2})_*\nu_I$, $H \in \mathrm{PD}(d)$, for the orbit of the law ν_I of Proposition 4.13(1), with Lévy densities n_H. For maps $h : \mathrm{PD}(d) \to \mathrm{PD}(d)$ and $C : \mathrm{PD}(d) \to \mathrm{PSD}(d)$ put $\mu_\Sigma := N(0, C(\Sigma)) * \nu_{h(\Sigma)}$.*

1. (Infinitely divisible increments order the parts.) *If $\Sigma, \Sigma' \in \mathrm{PD}(d)$ and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ infinitely divisible, then $C(\Sigma) \leq C(\Sigma')$ and $n_{h(\Sigma)} \leq n_{h(\Sigma')}$ everywhere: the step $h(\Sigma) \to h(\Sigma')$ is in the step set of Proposition 4.13(2).*

2. (Bounded shape.) *Suppose $h(cQ\Sigma Q^{\mathsf{T}}) = c\,Q\,h(\Sigma)\,Q^{\mathsf{T}}$ for $c > 0$ and $Q \in \mathrm{O}(d)$, that h extends continuously to $\mathrm{PSD}(d) \setminus \{0\}$ with values in $\mathrm{PD}(d)$, and that every Loewner step $\Sigma \leq \Sigma'$ in $\mathrm{PD}(d)$ has an infinitely divisible increment. Then $h(I) = h_0 I$ with $h_0 > 0$, and with $c_1 := \lambda_{\max}(h(e_1 e_1^{\mathsf{T}}))$,*

$$\frac{\lambda_{\max}(h(\Sigma))}{\lambda_{\min}(h(\Sigma))} \leq R\big((h_0/c_1)^{1/2}\big)^2 \qquad \text{for every } \Sigma \in \mathrm{PD}(d).$$

3. (Index 0: shape-blind.) *If $k(0+)$ is finite, h is equivariant as in (2) (no extension needed), and every Loewner step has an infinitely divisible increment, then $h(\Sigma)$ is a multiple of I for every Σ.*

4. (Logarithmic exponent, no Gaussian part.) *Let instead the non-Gaussian parts be the orbit (ν_H) of a law satisfying the hypothesis of Proposition 4.15, let $C \equiv 0$ and h be equivariant as in (2), and suppose every Loewner step has a positive finite increment. Then $h(\Sigma)$ is a multiple of I for every Σ.*

Without a Gaussian part, clause (4) gets shape-blindness from positive increments alone. With a Gaussian part a positive increment need not order the parts, so clauses (1)–(3) require infinitely divisible increments. A computation, Remark H.3, suggests that this hypothesis cannot simply be dropped; it is not a proof. Proposition H.9 discharges the hypothesis for every family of the scheme.

Sketch. (1). Both kernels are infinitely divisible with triplets $(C(\cdot), n_{h(\cdot)}(x)\,dx, 0)$, and by the uniqueness C2(ii) the triplet of ρ adds to that of μ_Σ; so $C(\Sigma) \leq C(\Sigma')$ and $n_{h(\Sigma)} \leq n_{h(\Sigma')}$ almost everywhere. In the coordinates $x = h(\Sigma)^{1/2}y$, an open set of positive measure near each radius and the right-continuity of k give $D\,m(r) \leq m(r/\lambda)$ for every $r > 0$, where $m(r) = k(r)r^{-d}$, D^2

is the determinant of the relative step M and λ^2 its least eigenvalue; as $\langle y, M^{-1}y\rangle^{1/2} \le |y|/\lambda$ and m is nonincreasing, that is the ordering everywhere.

(2) is the proof of Proposition H.1(2), with the step set of Proposition 4.13(2) in place of the stable cone, R being nondecreasing, and $h(I) = h_0 I$ as in the proof of Lemma A.4.

(3). $R \equiv 1$ (Proposition 4.13(3)), so every Loewner step moves h along a ray, and $\Sigma \le tI$ makes $h(\Sigma)$ a multiple of $h(tI) = th_0 I$.

(4). The positive increment of $\Sigma \le tI$ is an increment of the orbit (ν_H) from $h(\Sigma)$ to $th_0 I$, and Proposition 4.15 makes them multiples. □

Remark H.3 (a Gaussian part and a positive increment, measured). In the plane take $C(\Sigma) = 2b\Sigma$, the Matérn orbit of index 1 as the elliptical part, and $h(\Sigma) = \Sigma + \kappa \operatorname{tr}(\Sigma) I$ with $\kappa = 2$. Consider the step from $\operatorname{diag}(1, 0.3)$ by $\operatorname{diag}(0.05, 0.55)$, for which $h(\Sigma')$ is not a multiple of $h(\Sigma)$. A script deposited with the paper computes its would-be increment by the fast Fourier transform. It samples the transform of the increment on the frequency grid of a spatial square of side $60\,\lambda_{\max}(h(\Sigma'))^{1/2} \approx 132$ centred at the origin, at two resolutions, 1024^2 and 2048^2 points (spacings 0.129 and 0.065, frequencies up to π over the spacing, about 24 and 49 in each coordinate). At $b = 2$ the computed density is nonnegative to rounding: its minimum is -10^{-17} of its maximum at both resolutions, as for the ray steps that calibrate the computation. At $b = 0.1$ it is negative, with negative mass $1.4 \cdot 10^{-2}$ at 2048^2 points. At 1024^2 points the Gaussian factor is still 0.05 at the largest frequency. These are numerical findings, and positivity at $b = 2$ is not proved. If that increment is positive, it is not infinitely divisible, since its Lévy measure would be the signed $n_{h(\Sigma')} - n_{h(\Sigma)}$. The computation thus suggests that the Gaussian part smooths away the obstruction of Proposition 4.15 on this full-rank step. By Proposition H.9 these kernels are the kernels from scale zero of no (similarities, Loewner)-cascade family, so the example shows why the axioms require continuity along the order.

H.2 A general non-Gaussian part

For a non-Gaussian part with no orbit structure the same sandwich works in the order of Lévy measures, and in the order of exponents when there is no Gaussian part.

Proposition H.4 (the sandwich for a general similarity-covariant part)**.** *Let $d \ge 2$. For each $\Sigma \in \mathrm{PD}(d)$ let μ_Σ be a symmetric infinitely divisible probability measure on $\mathbb{R}^d$ with generating triplet $(C(\Sigma), L_\Sigma, 0)$, and put $\psi_\Sigma(\omega) := \int (1 - \cos\langle \omega, x\rangle)\, L_\Sigma(dx)$; so $\mu_\Sigma = N(0, C(\Sigma)) * \nu_\Sigma$, where the* non-Gaussian part *ν_Σ has $\widehat{\nu}_\Sigma = e^{-\psi_\Sigma}$. Suppose the family is covariant under similarities, $(\sqrt{c}\,Q)_*\mu_\Sigma = \mu_{cQ\Sigma Q^\mathsf{T}}$ for $c > 0$, $Q \in \mathrm{O}(d)$ and $\Sigma \in \mathrm{PD}(d)$, which is (K3) of Definition 3.2 for $A = \sqrt{c}\,Q$. Put $\Sigma_\varepsilon := \operatorname{diag}(1, \varepsilon, \dots, \varepsilon)$ for $0 < \varepsilon \le 1$.*

1. (Covariance.) *$C(cQ\Sigma Q^\mathsf{T}) = c\,Q\,C(\Sigma)\,Q^\mathsf{T}$, $L_{cQ\Sigma Q^\mathsf{T}} = (\sqrt{c}\,Q)_* L_\Sigma$ and $\psi_{cQ\Sigma Q^\mathsf{T}}(\omega) = \psi_\Sigma(\sqrt{c}\,Q^\mathsf{T}\omega)$. In particular L_I and ψ_I are rotation invariant.*

2. (Infinitely divisible increments.) *If $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ infinitely divisible, then $C(\Sigma) \le C(\Sigma')$, $L_\Sigma \le L_{\Sigma'}$ and $\psi_\Sigma \le \psi_{\Sigma'}$.*

3. (Positive increments without a Gaussian part.) *If $C \equiv 0$ and $\mu_{\Sigma'} = \mu_\Sigma * \rho$ with ρ a positive finite measure, then ρ is a probability measure, $\widehat{\rho} = e^{-(\psi_{\Sigma'} - \psi_\Sigma)}$, and $\psi_\Sigma \le \psi_{\Sigma'}$.*

4. (The rank-one limit.) *Suppose every Loewner step $\Sigma \le \Sigma'$ in $\mathrm{PD}(d)$ has an infinitely divisible increment. Then $L_*(B) := \lim_{\varepsilon \downarrow 0} L_{\Sigma_\varepsilon}(B)$ exists for every Borel set B bounded away from the origin, and L_* extends to a Lévy measure with $L_* \le L_{\Sigma_\varepsilon}$ for every ε; and $\psi_*(\omega) := \lim_{\varepsilon \downarrow 0} \psi_{\Sigma_\varepsilon}(\omega)$ exists for every ω. Both are invariant under every $R \in \mathrm{O}(d)$ with $Re_1 = \pm e_1$. If instead $C \equiv 0$ and every Loewner step has a positive finite increment, ψ_* exists and has the same invariance.*

5. (The sandwich.) *Let* $\Sigma \in \mathrm{PD}(d)$ *with* $\lambda_{\max}(\Sigma) = 1$, *and let* $Q \in \mathrm{O}(d)$ *have as its first column a unit eigenvector of* Σ *for the eigenvalue* 1. *Under the first hypothesis of (4),*

$$Q_* L_* \le L_\Sigma \le L_I \qquad \text{and} \qquad \psi_*(Q^{\mathsf{T}}\omega) \le \psi_\Sigma(\omega) \le \psi_I(\omega) \quad (\omega \in \mathbb{R}^d);$$

under the second, the inequalities between the exponents hold.

The rank-one limit needs no continuity hypothesis: the Lévy measures along Σ_ε decrease as $\varepsilon \downarrow 0$, so the limit exists by monotonicity alone. The analogue of the extension of h to rank one in the elliptical statements is therefore a hypothesis on the size of the limit, and the limit may be 0 (Proposition H.6). For the families of the scheme the standing hypotheses and the first hypothesis of (4) are theorems (Proposition H.9(1)).

Sketch. (1). The symmetry of μ_Σ and the uniqueness C2(ii) make L_Σ symmetric, so the transform collapses to $\exp[-\frac{1}{2}\langle\omega, C(\Sigma)\omega\rangle - \psi_\Sigma(\omega)]$ as in the proof of Lemma 4.6(2). The image under $A = \sqrt{c}\,Q$, which (K3) of Definition 3.2 identifies with $\mu_{cQ\Sigma Q^{\mathsf{T}}}$, has the triplet $(AC(\Sigma)A^{\mathsf{T}}, A_* L_\Sigma, 0)$, and uniqueness gives the identities.

(2). The triplets of a convolution add, as in the proof of Proposition H.2(1). *(3).* The transform of μ_Σ is positive, so $\widehat{\rho} = e^{-(\psi_{\Sigma'} - \psi_\Sigma)}$ and $|\widehat{\rho}| \le 1$.

(4). $\varepsilon \mapsto L_{\Sigma_\varepsilon}(B)$ is nondecreasing and bounded by $L_I(B)$ on sets bounded away from the origin, so it converges. On a shell $\{|x| \ge \delta\}$ the limit is countably additive, since for disjoint B_n the terms $L_{\Sigma_\varepsilon}(B_n)$ are dominated by the summable $L_I(B_n)$, and it extends to a Lévy measure below every L_{Σ_ε}. The invariance holds because $R\Sigma_\varepsilon R^{\mathsf{T}} = \Sigma_\varepsilon$ when $Re_1 = \pm e_1$.

(5). $Q\Sigma_\varepsilon Q^{\mathsf{T}} \le \Sigma \le I$ for $\varepsilon \le \lambda_{\min}(\Sigma)$, and (2) or (3) with (1) and (4) gives the inequalities. □

The shape of a general part is measured by one of three functionals, and each bound is the sandwich read in one of them.

Proposition H.5 (bounded shape under a comparable rank-one limit)**.** *For a measure* L *on* $\mathbb{R}^d \setminus \{0\}$, *a function* $\psi \ge 0$ *on* $\mathbb{R}^d$, $s > 0$ *and* $H \in \mathrm{PD}(d)$, *put*

$$\mathrm{rs}(L) := \inf\{K \ge 1 : Q_* L \le K\, L \text{ for every } Q \in \mathrm{O}(d)\},$$

$$\mathrm{es}(\psi) := \inf\{K \ge 1 : \psi(Q\omega) \le K\, \psi(\omega) \text{ for every } Q \in \mathrm{O}(d),\ \omega \in \mathbb{R}^d\},$$

$$\mathcal{M}_s(L) := \int (|x|^2 \wedge s^2)\, \frac{xx^{\mathsf{T}}}{|x|^2}\, L(dx), \qquad T_s(L) := \operatorname{tr} \mathcal{M}_s(L) = \int (|x|^2 \wedge s^2)\, L(dx),$$

$$\mathrm{ecc}(H) := \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)},$$

the rotational spread, *the* exponent spread, *the* moment matrix at scale s *and the eccentricity, the infimum of the empty set being* ∞, *and* $\operatorname{cond} M := \lambda_{\max}(M)/\lambda_{\min}(M) \in [1, \infty]$ *for* $M \in \mathrm{PSD}(d) \setminus \{0\}$. *Let the family be as in Proposition H.4.*

1. (The functionals.) *For* $c > 0$ *and* $Q \in \mathrm{O}(d)$, $\mathrm{rs}((cQ)_* L) = \mathrm{rs}(L)$, $\mathrm{es}(\psi(cQ\,\cdot)) = \mathrm{es}(\psi)$ *and* $\mathcal{M}_s((cQ)_* L) = c^2 Q\, \mathcal{M}_{s/c}(L)\, Q^{\mathsf{T}}$. $\mathrm{rs}(L) = 1$ *exactly when* L *is rotation invariant,* $\mathrm{es}(\psi) = 1$ *exactly when* ψ *is, and* $\mathrm{es}(\psi_L) \le \mathrm{rs}(L)$ *for the exponent* $\psi_L(\omega) := \int (1 - \cos\langle\omega, x\rangle)\, L(dx)$; *the exponent spread of a sum is at most the largest exponent spread of its terms. For the orbit* (ν_H) *of a profile as in Proposition 4.13(1),* $\mathrm{es}(\psi_{\nu_H}) \le 4\,\mathrm{ecc}(H)$; *for the stable orbit,* $\mathrm{es}(\psi_{S_\alpha(H)}) = \mathrm{ecc}(H)^{\alpha/2}$.

2. (Lévy level.) *Suppose every Loewner step in* $\mathrm{PD}(d)$ *has an infinitely divisible increment and* $L_* \ge \theta L_I$ *for some* $\theta \in (0, 1]$. *Then for every* $\Sigma \in \mathrm{PD}(d)$, *with* $t := \lambda_{\max}(\Sigma)$,

$$\theta\, L_{tI} \le L_\Sigma \le L_{tI}, \qquad \mathrm{rs}(L_\Sigma) \le 1/\theta, \qquad \mathrm{es}(\psi_\Sigma) \le 1/\theta.$$

3. (Exponent level.) *Suppose every Loewner step in* $\mathrm{PD}(d)$ *has an infinitely divisible increment, or that* $C \equiv 0$ *and every Loewner step has a positive finite increment; and that* $\psi_* \geq \theta\psi_I$ *for some* $\theta \in (0,1]$. *Then* $\theta\psi_{tI} \leq \psi_\Sigma \leq \psi_{tI}$ *and* $\mathrm{es}(\psi_\Sigma) \leq 1/\theta$ *for every* $\Sigma \in \mathrm{PD}(d)$, $t := \lambda_{\max}(\Sigma)$.

4. (Moments, scale by scale.) *Suppose every Loewner step in* $\mathrm{PD}(d)$ *has an infinitely divisible increment, and* $L_I \neq 0$. *For* $s > 0$ *the limit* $\mathcal{M}_s^* := \lim_{\varepsilon\downarrow 0} \mathcal{M}_s(L_{\Sigma_\varepsilon})$ *exists and is* $\mathrm{diag}(a_s, b_s, \dots, b_s)$; *put* $\theta_s := d\min(a_s, b_s)/T_s(L_I) \in [0,1]$. *Then for every* $\Sigma \in \mathrm{PD}(d)$ *and* $s > 0$, *with* $t := \lambda_{\max}(\Sigma)$, $\mathrm{cond}\,\mathcal{M}_s(L_\Sigma) \leq 1/\theta_{s/\sqrt{t}}$; *so* $\mathrm{cond}\,\mathcal{M}_s(L_\Sigma) \leq 1/\inf_{r>0}\theta_r$ *for all* Σ *and* s.

The exponent spread is the functional in which the shape is measured. It is defined for every symmetric infinitely divisible law, it is the condition number for a Gaussian, it is $\mathrm{ecc}(H)^{\alpha/2}$ on the stable orbit and at most $4\,\mathrm{ecc}(H)$ on the elliptical orbit of every nonincreasing profile, so it extends the eccentricity that the elliptical statements bound, and it equals 1 exactly for the rotation-invariant parts. It is also the only one of the three that positive increments see when there is no Gaussian part (clause (3)). The hypotheses are sufficient, not necessary: the shape-blind family $((\det\Sigma)^{1/(2d)})_*\nu_I$ has $L_* = 0$ and spread 1.

Sketch. (1). The rotational and the exponent spread are unchanged under conjugation and dilation, and $\mathcal{M}_s((cQ)_*L) = c^2 Q\,\mathcal{M}_{s/c}(L)\,Q^{\mathsf{T}}$ by the substitution $x \mapsto cQx$. If $Q_*L \leq L$ for every Q, then $L = Q_*(Q^{-1})_*L \leq Q_*L$, so equality holds. For the orbit of a profile, $\psi_{\nu_H}(\omega) = g(\langle\omega, H\omega\rangle)$ with g nondecreasing, as k is nonincreasing in the substitution of Proposition 4.13(1), and $g(4u) \leq 4g(u)$ from $1 - \cos 2\tau \leq 4(1-\cos\tau)$; so $g(cu) \leq 4c\,g(u)$ for $c \geq 1$, which gives $4\,\mathrm{ecc}(H)$. For the stable orbit the exponent is $\langle\omega, H\omega\rangle^{\alpha/2}$ (Lemma 4.6(2)).

(2), (3). After scaling to $t = 1$, Proposition H.4(5) gives $\theta L_I \leq Q_*L_* \leq L_\Sigma \leq L_I$, and, for every $Q' \in \mathrm{O}(d)$, $Q'_*L_\Sigma \leq L_I \leq \theta^{-1}L_\Sigma$; the same for exponents.

(4). $\mathcal{M}_s$ is monotone in the Lévy measure, so the limit exists; its invariance under the Q' with $Q'e_1 = \pm e_1$ makes it $\mathrm{diag}(a_s, b_s, \dots, b_s)$, by the argument of the proof of Lemma A.4, and $\mathcal{M}_s(L_I) = \frac{1}{d}T_s(L_I)\,I$. The sandwich gives $Q\mathcal{M}_s^*Q^{\mathsf{T}} \leq \mathcal{M}_s(L_\Sigma) \leq \frac{1}{d}T_s(L_I)\,I$. □

Without a hypothesis on the rank-one limit the shape need not be bounded, already for elliptical stable parts.

Proposition H.6 (without a nondegenerate rank-one limit the shape is unbounded)**.** *Let* $d \geq 2$, $0 < \alpha < 2$, $\frac{1}{d+\alpha} \leq \gamma < \frac{1}{d}$ *and* $p := 1 - d\gamma$, *so that* $0 < p \leq \frac{\alpha}{d+\alpha}$. *For* $\Sigma \in \mathrm{PSD}(d)$ *put* $h(\Sigma) := (\det\Sigma)^\gamma\,\Sigma^p$, *which is* 0 *when* Σ *is singular. For* $b \geq 0$ *put* $\mu_\Sigma := N(0, 2b\Sigma) * S_\alpha(h(\Sigma))$, *with* S_α *and* n_H *as in Lemma 4.6 and* $S_\alpha(0) := \delta_0$; *write* $L_\Sigma := n_{h(\Sigma)}(x)\,dx$ *and* ψ_Σ *for the Lévy measure and the exponent of the non-Gaussian part,* $\Sigma \in \mathrm{PD}(d)$, *and use the notation of Proposition H.5.*

1. (A cascade.) h *is continuous on* $\mathrm{PSD}(d)$, $h(cQ\Sigma Q^{\mathsf{T}}) = c\,Q\,h(\Sigma)\,Q^{\mathsf{T}}$ *for* $c > 0$ *and* $Q \in \mathrm{O}(d)$, *and* $n_{h(\Sigma)} \leq n_{h(\Sigma')}$ *for every* $\Sigma \leq \Sigma'$ *in* $\mathrm{PD}(d)$. *For every* $\Sigma \leq \Sigma'$ *in* $\mathrm{PSD}(d)$ *there is an infinitely divisible probability measure* $\rho_{\Sigma,\Sigma'}$ *with* $\mu_{\Sigma'} = \mu_\Sigma * \rho_{\Sigma,\Sigma'}$, *and the family* $(\rho_{\Sigma,\Sigma'})$ *satisfies (K1), (K2) and (K4) of Definition 3.2, and (K3) for every similarity* $A = cQ$.

2. (Unbounded shape.) *For* $\Sigma \in \mathrm{PD}(d)$, $\mathrm{ecc}(h(\Sigma)) = \mathrm{ecc}(\Sigma)^p$, $\mathrm{es}(\psi_\Sigma) = \mathrm{ecc}(\Sigma)^{p\alpha/2}$, $\mathrm{rs}(L_\Sigma) \geq \mathrm{ecc}(\Sigma)^{p\alpha/2}$, *and for every* $s > 0$

$$\mathrm{cond}\,\mathcal{M}_s(L_\Sigma) \geq \frac{G_1}{d\,G_0}\,\mathrm{ecc}(\Sigma)^{p\alpha/2},$$

$$G_0 := \int_{\mathbb{R}^d} (z_d^2 \wedge 1)\,|z|^{-d-\alpha}\,dz, \quad G_1 := \int_{\mathbb{R}^d} (|z|^2 \wedge 1)\,|z|^{-d-\alpha}\,dz,$$

both finite and positive. The rank-one limit of Proposition H.4(4) is $L_* = 0$.

3. (A nonzero rotation-invariant rank-one limit is not enough.) *Let* $0 < \beta < \alpha$*, let* $S'_\beta(H)$ *and* n'_H *be the law and the Lévy density of Lemma 4.6 for the index* β*, with* $S'_\beta(0) := \delta_0$*, and put* $\mu'_\Sigma := \mu_\Sigma * S'_\beta(\operatorname{tr}(\Sigma)\, I)$*, with Lévy measure* $L'_\Sigma = (n_{h(\Sigma)} + n'_{\operatorname{tr}(\Sigma)I})(x)\, dx$ *and exponent* ψ'_Σ*. The family* (μ'_Σ) *has all the properties of (1); its rank-one limit is* $L'_* = n'_I(x)\, dx$*, nonzero and rotation invariant; for every* $s > 0$*,* $\theta_s > 0$ *in Proposition H.5(4); and for every* $\Sigma \in \mathrm{PD}(d)$

$$\operatorname{es}(\psi'_\Sigma) \geq \operatorname{ecc}(\Sigma)^{p\alpha/2}, \qquad \sup_{s>0} \operatorname{cond} \mathcal{M}_s(L'_\Sigma) \geq \frac{G_1}{d\, G_0} \operatorname{ecc}(\Sigma)^{p\alpha/2}.$$

So under similarity covariance with a cascade along the whole Loewner order, the non-Gaussian part need not stay within a bounded neighbourhood of isotropy, with or without a Gaussian part. The extension of h to rank one with values in $\mathrm{PD}(d)$ cannot be dropped: here h extends by 0 and the shape is unbounded. It is not necessary either, since the gauge $(\det \Sigma)^{1/d} I$, the endpoint $\gamma = 1/d$, also vanishes at rank one and has spread 1. The family interpolates between the stable orbit, $\gamma = 0$, which has no rank-one step, and that shape-blind gauge. Clause (3) shows that a nonzero rank-one limit, even a rotation-invariant one, is not enough: what Proposition H.5 needs is comparability with L_I.

Sketch. The Loewner–Heinz inequality ([4], Thm. V.1.9, p. 115). For $0 < p < 1$, $\Sigma^p = \kappa_p^{-1} \int_0^\infty (I - s(\Sigma + sI)^{-1}) s^{p-1}\, ds$, and inversion reverses the order on $\mathrm{PD}(d)$, so $\Sigma \leq \Sigma'$ gives $\Sigma^p \leq \Sigma'^p$ and $\Sigma'^{-p} \leq \Sigma^{-p}$.

(1). $\det h(\Sigma) = \det \Sigma$, so by (4.1)

$$n_{h(\Sigma)}(x) = c_{d,\alpha}^{-1} (\det \Sigma)^{((d+\alpha)\gamma - 1)/2} \langle x, \Sigma^{-p} x \rangle^{-(d+\alpha)/2},$$

and along a Loewner step both factors increase, the first because $(d+\alpha)\gamma \geq 1$. So Proposition 4.7(1) gives the stable increments, and the axioms follow as in the proof of Proposition H.1(3). For $\gamma < 1/(d+\alpha)$ the display gives $n_{h(I+\tau e_1 e_1^\mathsf{T})}(e_2) < n_{h(I)}(e_2)$, and Lemma 4.8 excludes a positive increment, so the lower bound on γ is needed.

(2). The eigenvalues of $h(\Sigma)$ are $(\det \Sigma)^\gamma \sigma_i^p$, σ_i those of Σ, so $\operatorname{ecc}(h(\Sigma)) = \operatorname{ecc}(\Sigma)^p$, and Proposition H.5(1) gives the spreads. For the moments let $H \in \mathrm{PD}(d)$ have unit eigenvectors $v_\pm$ for its eigenvalues $\eta_+ := \lambda_{\max}(H)$ and $\eta_- := \lambda_{\min}(H)$, and substitute $x = H^{1/2} y$, under which $n_H(x)\, dx = c_{d,\alpha}^{-1} |y|^{-d-\alpha}\, dy$ and $\langle v_\pm, x \rangle = \eta_\pm^{1/2} \langle v_\pm, y \rangle$. A rotation taking v_- to e_d and the dilation $y = s\eta_-^{-1/2} z$ give the upper bound, and $|x|^2 \leq \eta_+ |y|^2$ with the average $\frac{1}{d}$ of $\langle v_+, y \rangle^2 / |y|^2$ over the sphere the lower:

$$\langle v_-, \mathcal{M}_s(n_H) v_- \rangle \leq c_{d,\alpha}^{-1} \eta_-^{\alpha/2} s^{2-\alpha} G_0, \qquad \langle v_+, \mathcal{M}_s(n_H) v_+ \rangle \geq \frac{c_{d,\alpha}^{-1}}{d}\, \eta_+^{\alpha/2} s^{2-\alpha} G_1.$$

Their ratio is at most $\operatorname{cond} \mathcal{M}_s(n_H)$, which gives the bound with $H = h(\Sigma)$. Along Σ_ε the densities tend to 0 off a null set, so $L_* = 0$ by dominated convergence, in the sense of Proposition H.4(4).

(3). The gauge $\operatorname{tr}(\Sigma) I$ is equivariant and monotone, so the second factor cascades too, and $L'_* = n'_I(x)\, dx$. At large $|\omega|$ the index-α term dominates the exponent, and at small s it dominates the moments, which gives the two bounds. □

The rotational spread is not equivalent to the other two functionals, and exact shape-blindness at index 0 fails for a general part. Both are shown by a part laid along lines.

Proposition H.7 (a part laid along lines)**.** *Let* $d \geq 2$*, let* $k : (0, \infty) \to [0, \infty)$ *be nonincreasing and not identically* 0 *with* $\int_0^1 r k(r)\, dr + \int_1^\infty k(r)\, dr/r < \infty$*, let* $\ell(dr) := k(|r|)\, dr/|r|$ *on* $\mathbb{R} \setminus \{0\}$*,*

and let $\overline{\sigma}$ be the normalised surface measure on S^{d-1}. For $\Sigma \in \mathrm{PSD}(d)$ and $q_\Sigma(u) := \langle u, \Sigma u\rangle$ let L_Σ be the measure on $\mathbb{R}^d \setminus \{0\}$ with

$$\int f\, dL_\Sigma := \int_{S^{d-1}} \overline{\sigma}(du) \int_{\mathbb{R}} f\big(r\, q_\Sigma(u)^{1/2} u\big)\, \ell(dr)$$

for Borel $f \geq 0$ on $\mathbb{R}^d$ with $f(0) = 0$, and for $b \geq 0$ let $\mu_\Sigma := N(0, 2b\Sigma) * \nu_\Sigma$, ν_Σ the law with triplet $(0, L_\Sigma, 0)$.

1. (A cascade.) *L_Σ is a symmetric Lévy measure; for $\Sigma \in \mathrm{PD}(d)$ it is the image of $\overline{\sigma}(du)\, 2k(s\, q_\Sigma(u)^{-1/2})\, ds/s$ under $(u, s) \mapsto su$, a polar profile nonincreasing in s and at most $2k(0+)$; $L_{cQ\Sigma Q^\intercal} = (\sqrt{c}\, Q)_* L_\Sigma$; and $L_\Sigma \leq L_{\Sigma'}$ for $\Sigma \leq \Sigma'$. Hence for every $\Sigma \leq \Sigma'$ in $\mathrm{PSD}(d)$ there is an infinitely divisible probability measure $\rho_{\Sigma,\Sigma'}$ with $\mu_{\Sigma'} = \mu_\Sigma * \rho_{\Sigma,\Sigma'}$, and the family $(\rho_{\Sigma,\Sigma'})$ satisfies (K1), (K2), (K4) of Definition 3.2 and (K3) for every similarity.*

2. (Three functionals.) *For every $\Sigma \in \mathrm{PD}(d)$ and $s > 0$, $\mathrm{es}(\psi_\Sigma) \leq 4d(d+2)$ and $\operatorname{cond} \mathcal{M}_s(L_\Sigma) \leq d + 2$, while $\mathrm{rs}(L_{\Sigma_\varepsilon}) \to \infty$ as $\varepsilon \downarrow 0$. The rank-one limit L_* is at least the image of $\overline{\sigma}(du)\, 2k(s/|u_1|)\, ds/s$ (the set $u_1 = 0$ is $\overline{\sigma}$-null), so it charges every open cone.*

3. (Not shape-blind at index 0.) *If $\Sigma \in \mathrm{PD}(d)$ is not a multiple of I, then L_Σ, and so ν_Σ, is not rotation invariant. This holds in particular when $k(0+) < \infty$, so that every polar profile of every L_Σ is bounded at the origin.*

The Lévy measure is laid along lines, and in direction u it is the one-dimensional profile dilated by $q_\Sigma(u)^{1/2}$. The exponent and the moments stay within a bounded neighbourhood of isotropy, while the rotational spread does not, although the rank-one limit is nonzero and charges every direction. Clause (3) holds even under infinitely divisible increments: a general part with polar profiles bounded at the origin need not be shape-blind. For this family the index at the origin is $2k(0+)$ in every direction for every Σ, and the shape is carried at positive radii.

Sketch. (1). Splitting the line integral at 0 and substituting $s = rq^{1/2}$ gives the polar form, nonincreasing in s because k is, and $q_\Sigma \leq q_{\Sigma'}$ for $\Sigma \leq \Sigma'$ orders the measures. C2(iii) gives the laws and increments, and the axioms (K1)–(K4) of Definition 3.2, (K3) for similarities, follow from the transforms as in the proof of Proposition H.4(1).

(2). The exponent is $\psi_\Sigma(\omega) = \int \overline{\sigma}(du)\, \psi_\ell(q_\Sigma(u)^{1/2}\langle \omega, u\rangle)$, where ψ_ℓ, the exponent of the line profile, is even, nondecreasing on $[0, \infty)$ and satisfies $\psi_\ell(a\tau) \geq \frac{a^2}{4}\psi_\ell(\tau)$ for $0 < a \leq 1$, from $\psi_\ell(2\tau) \leq 4\psi_\ell(\tau)$ and the monotonicity of ψ_ℓ. Let $\Sigma \in \mathrm{PD}(d)$, $t := \lambda_{\max}(\Sigma)$ and v a unit eigenvector for it, so that $t\langle v, u\rangle^2 \leq q_\Sigma(u) \leq t$ on the sphere. Then $\psi_\Sigma(R\omega) \leq \psi_\ell(t^{1/2}|\omega|)$ for every $R \in \mathrm{O}(d)$. For $\omega \neq 0$, $w := \omega/|\omega|$ and $a := |\langle v, u\rangle|\, |\langle w, u\rangle| \leq 1$, the fourth moments of the normalised surface measure, $\int \langle v, u\rangle^2 \langle w, u\rangle^2\, \overline{\sigma}(du) = \frac{1 + 2\langle v, w\rangle^2}{d(d+2)}$, give

$$\psi_\Sigma(\omega) \geq \int \overline{\sigma}(du)\, \psi_\ell\big(a\, t^{1/2}|\omega|\big) \geq \tfrac{1}{4}\psi_\ell(t^{1/2}|\omega|) \int \langle v, u\rangle^2 \langle w, u\rangle^2\, \overline{\sigma}(du) \geq \frac{\psi_\ell(t^{1/2}|\omega|)}{4d(d+2)},$$

which is the bound on $\mathrm{es}(\psi_\Sigma)$, the functionals being those of Proposition H.5. In the same way $\langle y, \mathcal{M}_s(L_\Sigma) y\rangle = \int \overline{\sigma}(du)\, \langle y, u\rangle^2 G_s(q_\Sigma(u))$, with $G_s(\tau) := \int (\tau r^2 \wedge s^2)\, \ell(dr)$, lies between $G_s(t)|y|^2/(d(d+2))$ and $G_s(t)|y|^2/d$, so $\operatorname{cond} \mathcal{M}_s(L_\Sigma) \leq d + 2$. For the rotational spread fix r_0 with $k(r_0) > 0$, the shell $r_0/4 < |x| < r_0/2$ and, for $q_2 \in (0, \frac{1}{4}]$, the cap U of directions with $u_1^2 < q_2/2$ and $u_2^2 > \frac{1}{4}$; let R exchange e_1 and e_2. For $\varepsilon < q_2/2$, $q_{\Sigma_\varepsilon} < q_2$ on U and $q_{\Sigma_\varepsilon} > \frac{1}{4}$ on RU, so on the shell the sector over U carries at most $2\overline{\sigma}(U) I(q_2)$ and the sector over RU at least $2\overline{\sigma}(U) I(\frac{1}{4}) > 0$, where $I(q) := \int_{r_0/4}^{r_0/2} k(sq^{-1/2})\, ds/s \leq \log 2 \cdot k(\frac{r_0}{4} q^{-1/2})$ tends to 0 with q,

k tending to 0 at infinity. So the rotational spread is unbounded. Finally $q_{\Sigma_\varepsilon}(u) \geq u_1^2$ puts the image of $\overline{\sigma}(du)\, 2k(s/|u_1|)\, ds/s$ below every L_{Σ_ε}, hence below L_*; and a nonempty open cone contains a sector $\{x : cr_0/2 < |x| < cr_0,\ x/|x| \in V\}$ with V open, $\overline{\sigma}(V) > 0$ and $|u_1| > c$ on V, which every L_{Σ_ε} charges with mass at least $\overline{\sigma}(V)k(r_0)\log 2$.

(3). Let $\lambda_{\min}(\Sigma) < m'' < m' < \lambda_{\max}(\Sigma)$. If L_Σ were rotation invariant, a cone of directions around the largest eigenvector, on which $q_\Sigma > m'$, and its image under the reflection exchanging the largest and the least eigenvectors, on which $q_\Sigma < m''$, would carry equal mass on every shell. As k is nonincreasing, this gives $\int_a^b k(s/\sqrt{m'})\, ds/s \leq \int_a^b k(s/\sqrt{m''})\, ds/s$ for $0 < a < b$, the reverse of the pointwise inequality between the integrands. So $k(r) = k(\lambda r)$ for almost every r, with $\lambda := (m'/m'')^{1/2} > 1$, and then $k = 0$. □

What survives of shape-blindness at index 0 for a general part is shape-blindness at the origin.

Proposition H.8 (index 0 for a general part: shape-blind at the origin)**.** *Let the family be as in Proposition H.4, and suppose L_I is the Lévy measure of an isotropic origin datum $(0, k_d)$ in dimension d (Definition 2.8) with $c := k_d(0+) < \infty$. Let $\overline{\sigma}$ be the normalised surface measure on S^{d-1}, and for a Borel set $A \subseteq S^{d-1}$ and $0 < \varepsilon < 1$ put $K_A(\varepsilon) := \{x : \varepsilon < |x| \leq 1,\ x/|x| \in A\}$.*

1. (Infinitely divisible increments: the index at the origin.) *If every Loewner step in* PD(d) *has an infinitely divisible increment, then for every $\Sigma \in$ PD(d) and every Borel $A \subseteq S^{d-1}$*

$$\lim_{\varepsilon \downarrow 0} \frac{L_\Sigma(K_A(\varepsilon))}{\log(1/\varepsilon)} = c\,\overline{\sigma}(A).$$

2. (The exponent, to within a bounded term.) *If every Loewner step has an infinitely divisible increment, or if $C \equiv 0$ and every Loewner step has a positive finite increment, then for every $\Sigma \in$ PD(d) every limit point of $\psi_\Sigma(\omega) - \psi_I(\omega)$ as $|\omega| \to \infty$ lies in $[\frac{c}{2}\log\lambda_{\min}(\Sigma), \frac{c}{2}\log\lambda_{\max}(\Sigma)]$.*

3. (The blow-up.) *Under either hypothesis of (2), let $\Sigma \in$ PD(d) and suppose* (D)*: for $\overline{\sigma}$-almost every $u \in S^{d-1}$ the limit $\Delta_\Sigma(u) := \lim_{r\to\infty}\big(\psi_\Sigma(ru) - \psi_I(ru)\big)$ exists. Then Δ_Σ is $\overline{\sigma}$-almost everywhere equal to a constant.*

Under infinitely divisible increments, with or without a Gaussian part, the Lévy measure of every part has the isotropic member's logarithmic index c at the origin, spread uniformly over directions (clause (1)). Under positive increments alone, and without a Gaussian part, the exponents are sandwiched in the same way (clause (2)), and the blow-up of Proposition 4.15 works for a general part under (D) (clause (3)). Condition (D) for a general part with polar profiles bounded at the origin requires the second term of the exponent's expansion, which needs more than boundedness, and is open.

Sketch. (1). $\lambda_{\min}(\Sigma)I \leq \Sigma \leq \lambda_{\max}(\Sigma)I$, so by Proposition H.4(2) L_Σ lies between $L_{tI} = (\sqrt{t})_* L_I$ at the two values of t. By (2.4) of Definition 2.8, $L_{tI}(K_A(\varepsilon)) = \overline{\sigma}(A)\int_\varepsilon^1 k_d(rt^{-1/2})\, dr/r$, and $k_d \to c$ at the origin.

(2). The same sandwich for the exponents, with $\psi_{tI}(\omega) - \psi_I(\omega) \to \frac{c}{2}\log t$ by Lemma 2.9(2).

(3). For the step $\Sigma \leq tI$, $t := \lambda_{\max}(\Sigma)$, let ρ be its increment under the second hypothesis. Under the first, let ρ be the non-Gaussian factor of its increment: the law with triplet $(0, \nu', 0)$ (C2(iii)), where $\nu = L_{tI} - L_\Sigma$ is the Lévy measure of the increment (C2(ii)) and ν' its symmetrization. In both cases $\widehat{\rho} = e^{-(\psi_{tI} - \psi_\Sigma)}$, so $0 < \widehat{\rho} \leq 1$, and $\widehat{\rho}(\omega/\varepsilon) \to t^{-c/2} e^{\Delta_\Sigma(\omega/|\omega|)}$ by Lemma 2.9(2) and (D). On the measure side $\int \phi(x/\varepsilon)\, \rho(dx) \to \phi(0)\rho(\{0\})$ by dominated convergence, and Fourier inversion makes the limit of the transforms the constant $\rho(\{0\})$ almost everywhere. For an elliptical part of a profile bounded at the origin (D) holds by Lemma 2.9(2), and this gives Proposition H.2(4) again. □

H.3 The families of the scheme

The statements above assume that every Loewner step has an infinitely divisible increment. For the families of the scheme this is a theorem, which discharges that hypothesis everywhere above.

Proposition H.9 (similarity cascades: infinitely divisible increments are a theorem of the axioms)**.** *Let $d \geq 2$, let G be a closed subgroup of $\mathrm{GL}(d)$ containing the similarities cQ, $c > 0$, $Q \in \mathrm{O}(d)$, and let $(\mu_{\Sigma,\Sigma'})$ be a $(G, \text{Loewner})$-cascade family of kernels (Definition 3.1(2)), with kernels from scale zero $\mu_\Sigma := \mu_{0,\Sigma}$.*

1. (Infinitely divisible kernels and increments.) *Every $\mu_{\Sigma,\Sigma'}$ is infinitely divisible. For every $\Sigma \in \mathrm{PD}(d)$, μ_Σ is a symmetric infinitely divisible law with generating triplet $(C(\Sigma), L_\Sigma, 0)$, where $C(\Sigma) \in \mathrm{PSD}(d)$ and L_Σ is a symmetric Lévy measure; $(\sqrt{c}\,Q)_*\mu_\Sigma = \mu_{cQ\Sigma Q^\intercal}$ for $c > 0$ and $Q \in \mathrm{O}(d)$; and every Loewner step $\Sigma \leq \Sigma'$ in $\mathrm{PD}(d)$ has the infinitely divisible increment $\mu_{\Sigma,\Sigma'}$, $\mu_{\Sigma'} = \mu_\Sigma * \mu_{\Sigma,\Sigma'}$.*

2. (The general part.) *Consequently $(\mu_\Sigma)_{\Sigma\in\mathrm{PD}(d)}$ is a family as in Proposition H.4, and the hypothesis "every Loewner step in $\mathrm{PD}(d)$ has an infinitely divisible increment" holds. So Proposition H.4(1)–(5), Proposition H.5(1)–(4) and Proposition H.8(1)–(3) hold for it under their remaining hypotheses alone, which are:*

   - *the comparability $L_* \geq \theta L_I$ in Proposition H.5(2);*
   - *$\psi_* \geq \theta\psi_I$ in (3) there;*
   - *$L_I \neq 0$ in (4) there;*
   - *in Proposition H.8, that L_I is the Lévy measure of an isotropic origin datum $(0, k_d)$ with $k_d(0+) < \infty$, with (D) in its clause (3).*

3. (Elliptical parts.) *If moreover $\mu_\Sigma = N(0, C(\Sigma)) * \nu_{h(\Sigma)}$ for every $\Sigma \in \mathrm{PD}(d)$, with k, (ν_H), h and C as in Proposition H.2, then that proposition's clauses (2) and (3) hold under their remaining hypotheses alone (in (2): h equivariant and extending continuously to $\mathrm{PSD}(d) \setminus \{0\}$ with values in $\mathrm{PD}(d)$; in (3): $k(0+) < \infty$ and h equivariant), with or without a Gaussian part.*

So for every family of the scheme at (similarities, Loewner), with or without a Gaussian part, the shape is bounded whenever the rank-one limit is comparable to the isotropic member, and at index 0 the part is shape-blind at the origin. Only the infinite divisibility is discharged. The comparability conditions, $L_I \neq 0$, the origin datum and (D) remain hypotheses. The statements above keep their hypothesis because they concern families of kernels from scale zero with no increments between other pairs and no continuity; replacing it by the cascade axioms could only narrow them.

Proof. (1). $(G, \text{Loewner})$ is a wedge datum (Definition 3.1), and by Theorem 3.5(3)(a) it has the path condition, so by Theorem 3.5(2) every $\mu_{\Sigma,\Sigma'}$ and every $\mu_\Sigma = \mu_{0,\Sigma}$ is infinitely divisible. The similarity $-I$ fixes every Σ, so (K3) makes μ_Σ symmetric, and the uniqueness C2(ii) makes its drift 0 and its Lévy measure symmetric. (K3) at $(0, \Sigma)$ with $A = \sqrt{c}\,Q$ is the covariance, and (K1) at $0 \leq \Sigma \leq \Sigma'$ gives the increment.

(2) and *(3).* By (1) the family is as in Proposition H.4, and every Loewner step has an infinitely divisible increment, which is the hypothesis of the named clauses of Propositions H.4, H.5, H.8 and H.2. □

I Verification

This appendix gives the detail behind §1.1: what is proved, which statements rest on which cited fact, the standard theorems the proofs take from Mathlib, the review of the formal statements, and where the formal development is.

What is proved. The theorems are machine-checked: the single step and the rigidity, on kernels and on operators (Theorems 3.7, 3.6 and 3.13), the infinitely divisible increments (Theorem 3.5), the orders of cone fields (Theorem 4.3), and the theorem on the general linear group along the chain of cones (Theorem 4.21), whose complete proof Appendix C collects. So is every other definition, lemma and proposition of §§3–4 and of the appendices. A proof marked as a sketch shows the decisive steps and leaves routine verifications out. Twenty statements have a sketch only: Propositions 4.4, 4.18, 4.25, A.3, D.1, E.1(2)–(4), E.4, F.2, F.3, F.4, F.5, H.1, H.2, H.4, H.5, H.6, H.7 and H.8, and Lemmas 4.14 and A.5. Their complete proofs are in the LaTeX sources deposited with this version (see the code availability statement at the end), whose index, `records/affine/INDEX-complete-proofs.md`, names, for each statement number, the file and the place of its complete proof. A sketched statement is machine-checked like every other.

The cited facts, statement by statement. The statements whose formal proofs rest on the five cited facts of §1.1 are these, section by section.

- §2. C2, its clauses (ii) and (iii) at $d = 1$: the isotropic classification, Theorem 2.5, restated from the isotropic paper.
- §3. C1: Theorem 3.5. No other statement of §3 or of Appendix A rests on one.
- §4. C2: in §4.6, Lemma 4.6, Proposition 4.7, for the increments of its clauses (1) and (2), and Propositions 4.10, 4.11, 4.13, 4.19 and 4.20; in §4.9, Proposition 4.25; and Theorem 4.21 and Proposition 4.22. C3: Proposition 4.11, for its clause (1), and through it Theorem 4.21. C1: Proposition 4.22, and Proposition 4.25, for its clause (1). C5: Proposition 4.22(2). C4: in §4.4, Lemma C.1, for its clause (2) and the products in its clause (5); and through it Theorem 4.3, for its clauses (1), (2) and (4).
- The supplementary appendices. C2: Proposition E.1 (Appendix E); Proposition F.5 (Appendix F); Proposition G.1 (Appendix G); and the eight propositions of Appendix H, H.1, H.2, H.4, H.5, H.6, H.7, H.8 and H.9. C1: Proposition H.9.

These are machine-checked on Lean's logical axioms alone, with no cited fact:

- the isotropic families read on the multiples of the identity (Proposition A.3);
- the passage between operators and kernels (Lemmas A.1 and A.2);
- the rigidity on kernels and on operators (Theorems 3.6 and 3.13), with the single step and the lemmas behind it, and where continuity enters it (Proposition 3.11);
- the dictionary between invariant orders and step sets (Proposition 4.2), and the stable orders read in the Loewner order (Proposition 4.12).

Standard theorems. The proofs use the standard theorems of measure theory, Fourier analysis, weak convergence and elementary analysis (Fekete's lemma on superadditive functions among them) without citing them, as they use Fubini's theorem. Three of them are named here because they are limit theorems of probability, like the cited facts: Prokhorov's theorem,

Lévy's continuity theorem on $\mathbb{R}^d$ and the portmanteau theorem. The proof of Lemma A.1 uses the first, to obtain the kernel of an operator as a weak limit. The proof of Theorem 3.5 uses the other two, to show that the small steps of a cascade form a null array. All three are proved in Mathlib, and the formal proofs take them from it, whereas they take each cited fact as an assumption.

Review. Each formal statement was read against the statement printed here, in search of one that is weaker than printed, holds vacuously, or is true only by a convention of the proof assistant, and the review found none. It found that each cited fact states what its page supports, with the departures recorded in §2.4. For each theorem named under "What is proved", a formal example that rests on no cited fact shows that the hypotheses can be met.

Where to look. The formal development, its ledger of the cited facts with the page-verified citation of each, and the LaTeX sources of the statements and their proofs, in which the declaration behind each numbered statement is named in a `\lean` tag on that statement and the complete proof of every sketched statement is given, are in the repository accompanying this paper, `https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-affine`, at its tag `v0.1`, as the code availability statement at the end says. The toolchain is pinned in-tree: Lean 4 `v4.31.0`, Mathlib `v4.31.0`, the shared library `ScaleSpaceCore` at its tag `v0.5.0`. The build configuration requires these two packages and no other. `lake build` in `Formalization/` reproduces the verification, and `lake env lean CIAxiomGuard.lean` prints the axioms of every named declaration, to be read against the file `blueprint/trust-boundary.txt` beside it. In that file the five cited facts are these axioms, each in the namespace `AffineHemigroup`.

C1	`khintchine_null_array`	C2(iii)	`levy_khintchine_converse`
C2(i)	`levy_khintchine_exists`	C3	`operatorSelfDecomposable_infinitelyDivisible`
C2(ii)	`levy_khintchine_unique`	C4	`gelfand_naimark_logMajorized`
		C5	`sato_polar_criterion`

On the declarations of Theorems 3.6, 3.13, 3.5, 4.3 and 4.21 and of Proposition 4.22, it prints Lean's three logical axioms (propositional extensionality, `propext`; the axiom of choice, `Classical.choice`; and the soundness of quotient types, `Quot.sound`) and names of the cited facts, and nothing else:

```
'AffineHemigroup.affine_rigidity' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.affine_rigidity_operators' depends on axioms: [propext,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.gw_family_infinitelyDivisible' depends on axioms: [propext,
 AffineHemigroup.khintchine_null_array,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.causalLE_order' depends on axioms: [propext,
 AffineHemigroup.gelfand_naimark_logMajorized,
 Classical.choice,
 Quot.sound]
'AffineHemigroup.wedge_axis_row' depends on axioms: [propext,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_exists,
 AffineHemigroup.levy_khintchine_unique,
 AffineHemigroup.operatorSelfDecomposable_infinitelyDivisible,
```

```
 Classical.choice,
 Quot.sound]
'AffineHemigroup.wedge_axis_cell' depends on axioms: [propext,
 AffineHemigroup.khintchine_null_array,
 AffineHemigroup.levy_khintchine_converse,
 AffineHemigroup.levy_khintchine_exists,
 AffineHemigroup.levy_khintchine_unique,
 AffineHemigroup.sato_polar_criterion,
 Classical.choice,
 Quot.sound]
```

Declarations

The author's contribution is stated in the next section, with the use of AI.

Competing interests. The author declares no competing interests.

Funding. This work was carried out independently and received no funding.

Data and code availability. No data were generated or analysed. The scripts that draw Figures 1, 2 and 3 and compute the values in Remarks E.5 and H.3, with their output, are deposited with this version of the paper. So are the LaTeX sources in which the statements and their proofs were developed, which hold the complete proofs of the statements whose proofs are sketched here, and the formal development in Lean 4 that machine-checks formal counterparts of the statements named as such in §1.1, with its ledger of the cited facts. All of these are in the repository `https://github.com/danielfagerstrom/spatial-hemigroup-scale-space-affine`, at its tag `v0.1`. It is the public verification export of this version, taken from the development repository, which is private, and it is deposited with this version of the paper under the DOI printed on its first page.

Author contribution and use of AI

This article was written in close collaboration with generative AI, beyond editing assistance, at every stage.

The programme, the question, the decisions of scope and the organization of the material are mine, except that a standing rule lets the agents narrow a statement to what its proof gives. The prose, statements, proofs and code, the formal development in Lean that machine-checks the statements listed in §1.1 among them, are the agents' work, mostly unattended at first and later under my section-by-section review and decisions, and Proposition 4.22 began in a report by one of the AI referees named below.

Agents with no part in the writing reviewed it blind throughout, and an agent acting as librarian read the cited results at page anchors. The complete manuscript then went through three rounds of external review by AI systems: two by Gemini 4.0 Argon, a system of another vendor, and one by five referees on Claude Fable 5.1. Each finding was checked before use (the trust base, §1.1), and the reviews are deposited verbatim with the plan that records what was done with each. Beyond me, no human has reviewed the paper, and the formal statements in Lean have been reviewed by AI agents only, under review protocols that I set up.

The division of labour, with its derivation and dates, is in the process account deposited with the paper (`notes/PROCESS.md`). Over thirty-one to thirty-seven archived days to 4 November 2026 the agents' words outnumber those I typed by between twenty-two and thirty-nine to one

(computed 2026-11-04). The agents' count is a lower bound: it leaves out code, LaTeX and proofs, the external reviewers' own text, the subagents of two cloud sessions, whose traces are not held, and any work in sessions the archive does not hold.

Responsibility, errors included, is mine alone.

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